

Lecture Notes in Microeconomics

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Chapter 1

Economic method and basic tools

1.1 What microeconomics is

Microeconomics studies how individuals and firms make decisions when resources are scarce. It analyzes:

- consumption choices,
- production decisions,
- price formation,
- the effects of public policies.

Key idea

Microeconomics uses simplified models to isolate the fundamental mechanisms of economic decisions.

Example

Airfares and consumer choice

An airline observes that, when airfares rise, many students give up their weekend away and choose to stay in town. Microeconomics studies exactly this kind of behavior: how agents respond to constraints, incentives, and prices.

1.2 Scarcity and trade-offs

Resources are limited, wants are potentially unlimited. Every choice involves giving something up: this is the trade-off.

1.2.1 Opportunity cost

Definition 1.1 (Opportunity cost).

The opportunity cost is the value of the best alternative forgone.

Example**University or work.**

Explicit costs (fees, books): €9,000. Opportunity cost (forgone wages): €54,000 over 3 years.
Total economic cost: €63,000.

Example**Free concert**

You have a free ticket for a concert, but to go you must give up an evening of work paying 80 euros. Even though the ticket is free, the opportunity cost of attending is 80 euros.

Economic intuition

“Free” always has a cost. A “free” concert ticket costs the wage of the shift you give up.

1.3 Production possibility frontier (PPF)

The PPF represents the maximum combinations of two goods (or activities) that can be produced given the available resources and technology. Points *on* the frontier are efficient; points *inside* it are attainable but inefficient (resources are wasted); points *outside* it are currently unattainable.

Linear example with two goods A and B :

$$B = 10 - 0.5A.$$

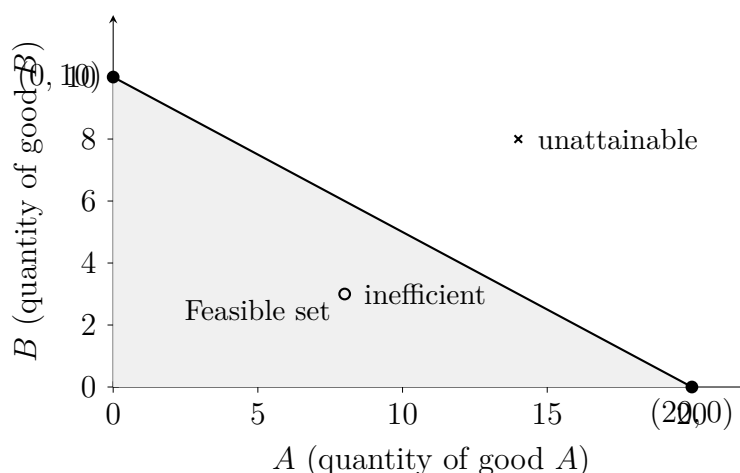


Figure 1.1: Production possibility frontier $B = 10 - 0.5A$. The slope -0.5 is the opportunity cost: producing one more unit of A requires giving up 0.5 units of B .

- Vertical intercept: 10 (everything in B , nothing in A).
- Horizontal intercept: 20 (everything in A , nothing in B).
- Slope: -0.5 (opportunity cost of A in terms of B).

Key idea

The slope of the PPF (in absolute value) measures the opportunity cost: how many units of B must be sacrificed to produce one more unit of A . With diminishing returns in transformation, the PPF is concave (increasing opportunity cost).

1.4 Sunk costs

Definition 1.2 (Sunk cost).

A sunk cost is a cost that has already been incurred and cannot be recovered.

Common pitfall

Sitting through a boring film “because I paid for the ticket” is the sunk cost fallacy. Decide on future costs/benefits.

Example

Annual gym membership

You have already paid 400 euros for the gym, but today you are tired and do not feel like going. Those 400 euros are a cost already incurred and not recoverable: they should not affect today’s decision, which depends only on future costs and benefits.

1.5 Marginal analysis

Optimal decisions are made by comparing marginal benefits and marginal costs.

Optimum when $MB = MC$

Profit example:

$$\pi(q) = 50q - 5q^2 - 100$$

$$\pi'(q) = 50 - 10q = 0 \Rightarrow q = 5$$

Economic intuition

As long as marginal benefit exceeds marginal cost, it pays to increase production.

Example

Hours of study

A student decides how many hours to devote to preparing for an exam. The first hours of study greatly improve preparation, but after many hours fatigue reduces the marginal benefit. The optimal choice is reached when the marginal benefit of one more hour equals the marginal cost in terms of effort and time.

1.6 Basic mathematical tools

See Appendix A for the complete rules of differentiation.

Derivative

The derivative measures the marginal change: $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$.

For $U(x, y)$:

$$\frac{\partial U}{\partial x}, \quad \frac{\partial U}{\partial y}$$

These tools will be used for:

- marginal utilities,
- MRS,
- marginal product,
- MRTS.

1.7 Exercises

Exercise 1.1.

Compute the opportunity cost on the PPF $B = 20 - A$.

Exercise 1.2.

Compute the derivative of $f(x) = 4x^2 + 3x$.

Exercise 1.3.

A firm has $\pi(q) = 60q - 6q^2 - 100$. Compute the optimal quantity.

Exercise 1.4.

Explain why a sunk cost is irrelevant for a future decision.

Exercise 1.5.

Integrated case: PPF $B = 12 - 0.25A$. Find the intercepts and interpret the slope.

Part I

Consumer Theory

Chapter 2

Preferences, utility, and the marginal rate of substitution

2.1 The consumer's problem

The consumer chooses the bundle that maximizes his or her well-being. Before introducing prices and income, we describe preferences over $\mathbb{R}_+^2$.

Example

Choosing between food and leisure

A consumer can spend income on dinners out or save it for other expenses and more free time. Even before introducing prices, we need to understand which combinations of goods and activities he or she prefers.

2.2 Bundles and preference relations

A bundle is $(x, y) \in \mathbb{R}_+^2$.

Notation:

- $A \succ B$
- $A \succeq B$
- $A \sim B$

Example

Netflix and Spotify

Suppose a consumer is comparing two bundles:

- bundle A: Netflix subscription + no Spotify subscription
- bundle B: no Netflix + Spotify subscription

The preference relation describes whether A is preferred to B, B is preferred to A, or the consumer is indifferent.

2.3 Axioms of rational preferences

Definition 2.1 (Axioms of preferences).

Let $\succsim$ be a preference relation on $X \subseteq \mathbb{R}_+^2$. The axioms are:

1. **Completeness:** $\forall A, B \in X, A \succsim B$ or $B \succsim A$.
2. **Transitivity:** $\forall A, B, C \in X$, if $A \succsim B$ and $B \succsim C$, then $A \succsim C$.
3. **Monotonicity:** if A has at least as much x and y as B , and more of at least one, then $A \succ B$.
4. **Convexity:** if $A \sim B$, then $\lambda A + (1 - \lambda)B \succsim A$ for $\lambda \in (0, 1)$.

Proposition 2.1. *If preferences satisfy the 4 axioms and are continuous, there exists a continuous $U : X \rightarrow \mathbb{R}$ such that $A \succsim B \iff U(A) \geq U(B)$.*

Proof idea

Completeness and transitivity make $\succsim$ a complete preorder (all bundles are comparable, but distinct bundles can be indifferent). Continuity (an additional technical assumption) guarantees the representation by a continuous U ; monotonicity and convexity determine its shape.

Economic intuition

Without transitivity, “money pump”: apple $\succ$ banana $\succ$ orange $\succ$ apple $\rightarrow$ endless trades = poverty.

2.4 Ordinal utility and cardinal utility

Definition 2.2.

Ordinal utility represents preferences up to increasing monotonic transformations: $V = f(U)$, $f' > 0$.

Analytical example

$U = xy$, $V = 2xy + 5$, $W = \ln(xy)$ represent the *same* preferences.

Key idea

Rankings matter, not absolute values. $U = 10 > U = 5$ does not mean “twice as good”.

Example

Same ranking, different numbers

If two students rank the same films in the same order, one may use scores from 1 to 10 and the other from 1 to 100. The numbers change, but the ordering stays identical. The same holds for ordinal utility: what matters is the ranking of bundles, not the absolute numerical value.

2.5 Indifference curves

$$U(x, y) = \bar{U}, \quad (x, y) \in \mathbb{R}_+^2$$

Properties

1. **Negative slope** (monotonicity).
2. **They do not intersect** (transitivity).
3. **North-east is better** (monotonicity).
4. **Convexity** (convexity of preferences).

Example

Pizza and beer

An indifference curve collects all combinations of pizza and beer that give the consumer the same level of satisfaction. If one combination contains more pizza and more beer than another, it generally lies on a higher indifference curve.

2.6 Types of preferences

2.6.1 Cobb-Douglas

$$U = x^\alpha y^\beta$$

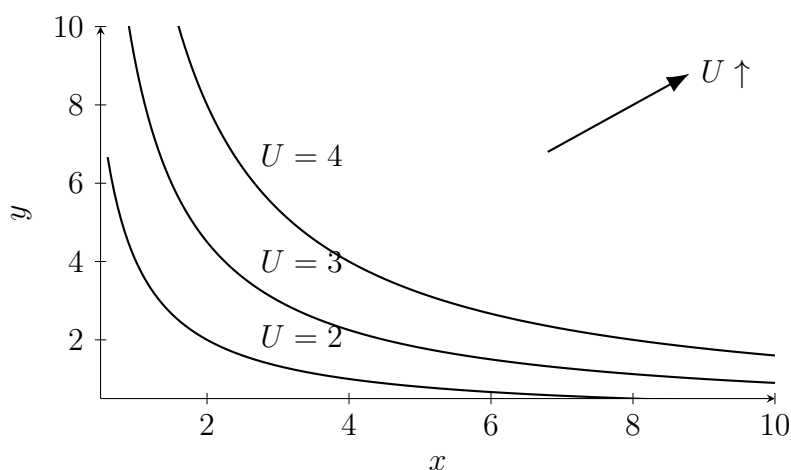


Figure 2.1: Cobb–Douglas indifference curves: convex to the origin (imperfect substitutability).

Example

Meals out and cinema

A Cobb–Douglas preference can describe a consumer who wants both meals out and entertainment. The two goods are only partially substitutable: more of one can make up for less of the other, but not perfectly.

2.6.2 Perfect substitutes

$$U = ax + by$$

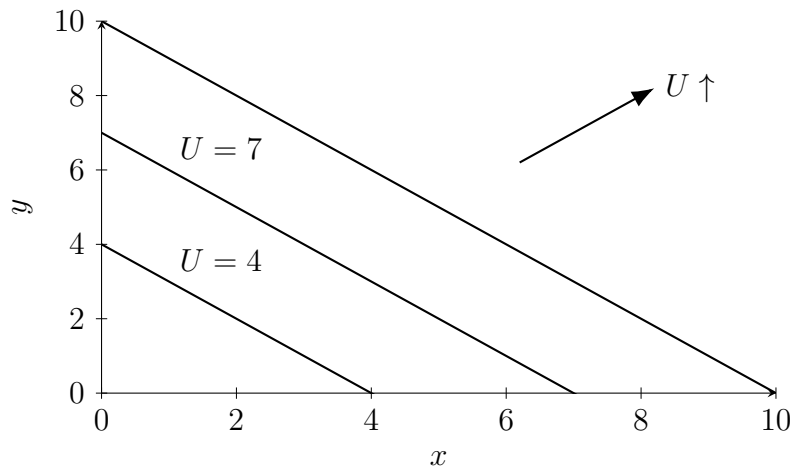


Figure 2.2: Perfect substitutes: linear indifference curves (constant MRS).

Example

Two identical brands of water

If two brands of mineral water are perceived as identical, the consumer may regard them as perfect substitutes. In this case only the total “equivalent” quantity of the two goods matters.

2.6.3 Perfect complements

$$U = \min\{x, y\}$$

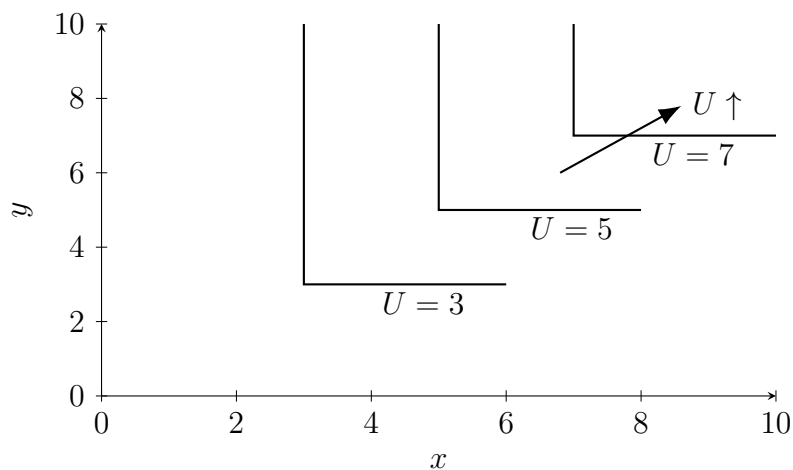


Figure 2.3: Perfect complements: L-shaped curves (goods in fixed proportions).

Example

Left shoes and right shoes

A left shoe and a right shoe are perfect complements: having many extra left shoes, without additional right shoes, does not increase utility.

2.7 Marginal utilities and MRS

$$MU_x = \frac{\partial U}{\partial x}, \quad MU_y = \frac{\partial U}{\partial y}$$

Definition 2.3.

$$\text{MRS}(x, y) = -\frac{dy}{dx}\bigg|_{dU=0} = \frac{MU_x}{MU_y}$$

Cobb-Douglas example $U = \sqrt{xy}$:

$$MU_x = \frac{1}{2}\sqrt{\frac{y}{x}}, \quad \text{MRS} = \frac{y}{x}$$

Economic intuition

$\text{MRS}=2 \implies$ willing to give up 2 beers for 1 pizza while keeping utility constant.

Key idea

At the optimum: $\text{MRS} = \frac{p_x}{p_y}$.

Example

Urban mobility: Uber and the bus

If a consumer is willing to give up two bus rides to get one Uber ride while keeping satisfaction unchanged, his or her MRS at that point equals 2. The MRS thus measures the subjective trade-off between the two goods.

2.8 Exercises

Exercise 2.1.

Compute the MRS for $U = x^2y$.

Exercise 2.2.

Verify that $U = xy$ and $V = 2xy + 5$ represent the same preferences.

Exercise 2.3.

Explain why convexity implies a diminishing MRS.

Exercise 2.4.

Given $B = 12 - 0.25A$: (i) find the intercepts, (ii) interpret the slope as an opportunity cost.

Exercise 2.5.

Explain in your own words why monotonicity implies that indifference curves slope downward.

Exercise 2.6.

For $U(x, y) = x^{0.4}y^{0.6}$ compute MU_x , MU_y , and the MRS.

Exam essentials

1. explain scarcity, trade-offs, opportunity cost, and sunk costs with examples;
2. read and interpret the slope and intercepts of a line (PPF and, shortly, the budget constraint);
3. compute derivatives and partial derivatives in simple cases;
4. define preferences and axioms (completeness, transitivity, monotonicity, convexity);
5. draw indifference curves for Cobb–Douglas, substitutes, and complements;
6. compute MU_x , MU_y , and $MRS = \frac{MU_x}{MU_y}$;
7. interpret the MRS as the willingness to trade goods at constant utility.

Chapter 3

Budget constraint and optimal choice

3.1 The economic problem

The consumer has preferences described by a utility function $U(x, y)$ and the corresponding indifference curves. He or she would like to consume as much as possible of both goods, but has limited income and must pay market prices.

The fundamental questions are:

- which bundles (x, y) are *affordable*?
- which bundle *will* the consumer *choose* among the affordable ones?

These two questions are answered, respectively, by the consumer's *budget constraint* and *optimal choice*.

Example

Student with a monthly budget

A student has a limited monthly budget and must decide how to divide it among books, transport, food, and leisure. Even though he or she would like to consume more of all goods, available income limits the affordable bundles.

3.2 Budget constraint

Let:

- $p_x > 0$: unit price of good x ;
- $p_y > 0$: unit price of good y ;
- $M > 0$: the consumer's disposable income;
- $(x, y) \in \mathbb{R}_+^2$: quantities purchased of the two goods.

Definition 3.1 (Budget constraint).

The consumer's total expenditure cannot exceed disposable income:

$$p_x x + p_y y \leq M.$$

The set of bundles that satisfy the constraint is called the *budget set*:

$$B(M, p_x, p_y) = \{(x, y) \in \mathbb{R}_+^2 : p_x x + p_y y \leq M\}.$$

Definition 3.2 (Budget line).

The *budget line* is the set of bundles that exactly exhaust income:

$$p_x x + p_y y = M.$$

Solving it for y we obtain the explicit form:

$$y = \frac{M}{p_y} - \frac{p_x}{p_y} x.$$

Intercepts and slope

From the explicit form we easily obtain:

- **vertical intercept:** $y = M/p_y$ (when $x = 0$);
- **horizontal intercept:** $x = M/p_x$ (when $y = 0$);
- **slope:** $-\frac{p_x}{p_y}$.

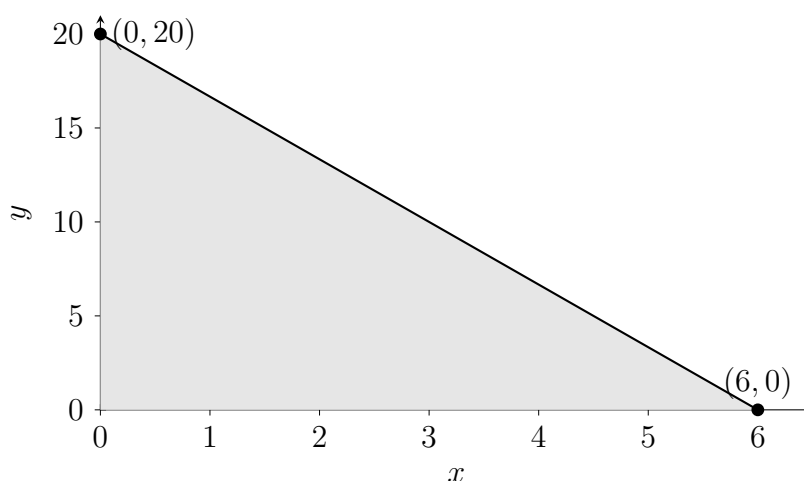


Figure 3.1: Budget line and budget set for $M = 60$, $p_x = 10$, $p_y = 3$.

The budget line represents the combinations that exhaust income; the area below it is the budget set, i.e., the set of affordable bundles.

Economic intuition

The slope $-\frac{p_x}{p_y}$ measures the *market rate of substitution*: how many units of y I must give up to buy 1 more unit of x .

Example

Coffee and sandwiches at lunch break

If a worker has 15 euros for the lunch break, he or she can buy different combinations of coffee and sandwiches. The budget constraint lists all the combinations that can be bought with that income and those prices.

3.3 Budget set

Definition 3.3 (Budget set).

The budget set is the set of all bundles affordable to the consumer:

$$B(M, p_x, p_y) = \{(x, y) \in \mathbb{R}_+^2 : p_x x + p_y y \leq M\}.$$

Remark 3.1.

Geometrically, $B(M, p_x, p_y)$ is a triangle in the first quadrant with vertices $(0, 0)$, $(M/p_x, 0)$, and $(0, M/p_y)$. Its frontier is the budget line; the interior contains the bundles that do not exhaust income.

Proposition 3.1 (Why all income is spent at the optimum). *If preferences are monotonic (“more is better”), then at an optimal bundle*

$$p_x x + p_y y = M.$$

Proof idea

If at a supposed optimum we had $p_x x + p_y y < M$, some income would be left unspent. Since $p_x, p_y > 0$, we could increase at least one of the two goods while staying within the constraint, obtaining a strictly preferred bundle by monotonicity: a contradiction.

Proposition 3.2 (Property of the budget set). *For every $M > 0$ and $p_x, p_y > 0$ the budget set $B(M, p_x, p_y)$ is convex.*

Proof idea

If (x_1, y_1) and (x_2, y_2) are affordable, then $p_x x_i + p_y y_i \leq M$ for $i = 1, 2$. For every $\lambda \in [0, 1]$ we have

$$p_x(\lambda x_1 + (1 - \lambda)x_2) + p_y(\lambda y_1 + (1 - \lambda)y_2) = \lambda(p_x x_1 + p_y y_1) + (1 - \lambda)(p_x x_2 + p_y y_2) \leq M,$$

so the convex combination also belongs to $B(M, p_x, p_y)$.

3.4 Changes in income and prices

3.4.1 Changes in income

If income rises from M to $M' > M$ (with constant prices), the new budget line is

$$y = \frac{M'}{p_y} - \frac{p_x}{p_y}x.$$

The slope is unchanged, while the intercepts increase: the line shifts outward in parallel.

If $M'' < M$, the line shifts inward in parallel.

An increase in income produces a parallel outward shift, while a decrease shifts it inward; the market trade-off stays unchanged.

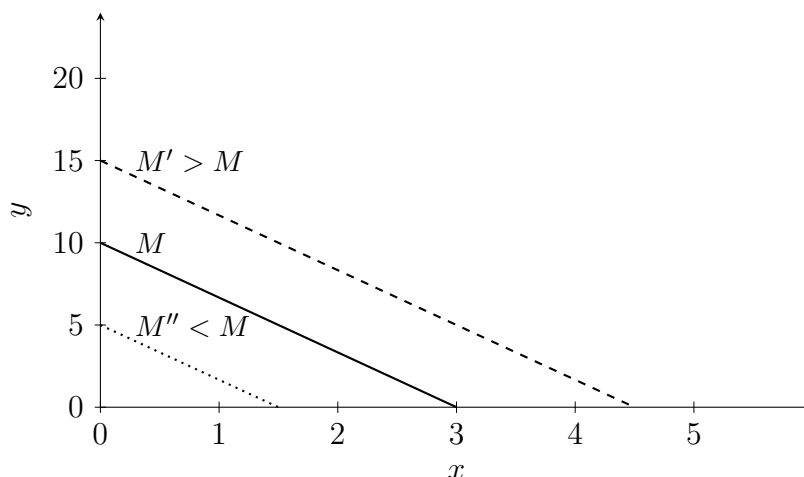


Figure 3.2: Changes in income: the budget line shifts in parallel (slope unchanged).

Key idea

Income changes the *size* of the budget set but not the market trade-off p_x/p_y .

Example

Scholarship

If a student receives a scholarship, his or her disposable income increases. With prices unchanged, the budget line shifts outward in parallel: he or she can afford more of both goods.

3.4.2 Changes in prices

If only p_x rises (with M and p_y constant), the vertical intercept M/p_y stays unchanged, while the horizontal intercept M/p_x falls and the line rotates inward around the point $(0, M/p_y)$.

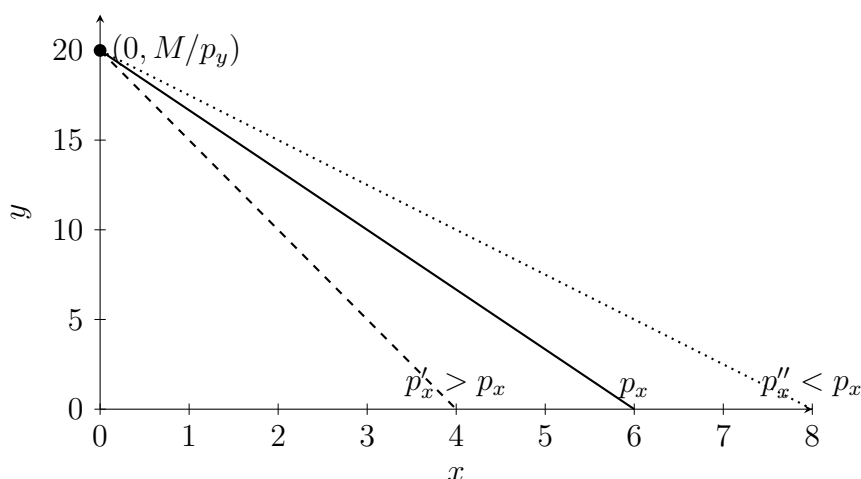


Figure 3.3: Change in p_x with M and p_y fixed: the line rotates around the vertical intercept.

A change in the price of one good rotates the budget line around the intercept of the good whose price has not changed.

If p_x falls, the line rotates outward around the same point.

Changes in p_y produce analogous rotations around the horizontal intercept $(M/p_x, 0)$.

Example

Discount on public transport passes

If the price of a public transport pass falls, the consumer can buy more of it with the same income, or free up resources for other goods. In the graph, the constraint rotates outward around the intercept of the other good.

3.4.3 Proportional inflation

Proposition 3.3 (Homogeneity of degree zero). *If we simultaneously multiply (M, p_x, p_y) by the same factor $\lambda > 0$, the budget set does not change:*

$$B(\lambda M, \lambda p_x, \lambda p_y) = B(M, p_x, p_y).$$

Proof idea

The condition $\lambda p_x x + \lambda p_y y \leq \lambda M$ is equivalent (dividing by λ) to $p_x x + p_y y \leq M$.

Economic intuition

Proportional inflation (prices and income growing by the same proportion) does not change real consumption possibilities: only *relative* prices and income matter.

3.5 From feasibility to choice

So far we have described *which bundles are affordable*. We now introduce the decision problem: *given what is affordable, which bundle maximizes utility?*

3.6 The consumer's optimal choice

The consumer's problem is a constrained maximization:

$$\max_{x, y \geq 0} U(x, y) \quad \text{subject to} \quad p_x x + p_y y = M.$$

3.6.1 Graphical solution and the tangency condition

Graphically, the consumer looks for the highest indifference curve that intersects the budget set. At an interior optimum (x^*, y^*) the indifference curve is tangent to the budget line.

Definition 3.4 (Tangency condition).

At an interior optimum

$$\text{MRS}(x^*, y^*) = \frac{p_x}{p_y},$$

that is,

$$\frac{MU_x(x^*, y^*)}{MU_y(x^*, y^*)} = \frac{p_x}{p_y}.$$

Equivalently:

$$\frac{MU_x}{p_x} = \frac{MU_y}{p_y},$$

i.e., the marginal utility per euro is the same for both goods.

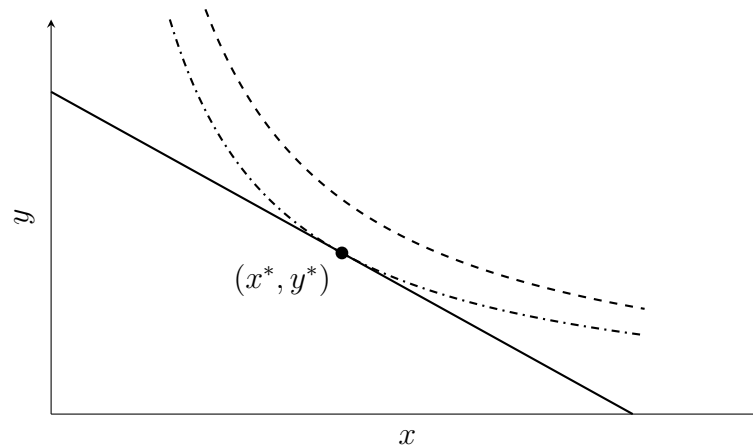


Figure 3.4: Interior optimum: the indifference curve is tangent to the budget line.

The interior optimum occurs at the point where the highest attainable indifference curve is tangent to the budget line: at that point $MRS = p_x/p_y$.

Economic intuition

If $MU_x/p_x > MU_y/p_y$, 1 euro spent on x yields more utility than on y : it pays to substitute x for y until the two marginal utilities per euro are equal.

Key idea

Three possible cases:

1. **Interior optimum:** $x^* > 0$, $y^* > 0$ and $MRS(x^*, y^*) = \frac{p_x}{p_y}$.
2. **Corner on x :** $y^* = 0$ and the optimum lies on the x axis (typically at $(M/p_x, 0)$ if preferences make it worthwhile).
3. **Corner on y :** $x^* = 0$ and the optimum lies on the y axis (typically at $(0, M/p_y)$).

Tangency is a necessary condition only in the interior case; with corner solutions one must directly evaluate the admissible extreme bundles.

Example

Choosing between Uber and the bus

A consumer chooses how many Uber rides and how many bus passes to buy. The optimum depends on:

- preferences (convenience vs savings),
- prices,
- income.

The optimal bundle is the one that maximizes utility among all affordable ones.

3.7 The substitution method

A first method for solving the problem is to substitute the constraint into the utility function.

Substitution method

1. Solve the constraint for one variable, for example $y = \frac{M - p_x x}{p_y}$.
2. Substitute into $U(x, y)$ to obtain $\tilde{U}(x) = U\left(x, \frac{M - p_x x}{p_y}\right)$.
3. Maximize $\tilde{U}(x)$ with respect to x with $\tilde{U}'(x) = 0$ and use the SOC.
4. Obtain y^* from the budget constraint.

3.8 The Lagrange method

The standard method in microeconomics uses the Lagrange multiplier.

Lagrangian

Define

$$\mathcal{L}(x, y, \lambda) = U(x, y) - \lambda(p_x x + p_y y - M).$$

The first-order conditions are

$$\frac{\partial \mathcal{L}}{\partial x} = MU_x - \lambda p_x = 0, \quad \frac{\partial \mathcal{L}}{\partial y} = MU_y - \lambda p_y = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda} = M - p_x x - p_y y = 0.$$

From the first two equations we immediately obtain the tangency condition:

$$\frac{MU_x}{MU_y} = \frac{p_x}{p_y}.$$

Remark 3.2 (Interpretation of λ).

From the condition $MU_x = \lambda p_x$ it follows that $\lambda = MU_x/p_x$: λ measures the marginal utility of income, i.e., of 1 additional euro.

Analytical example

Consider $U(x, y) = \ln x + 2 \ln y$ with $x, y > 0$ and constraint $p_x x + p_y y = M$. The FOCs imply

$$\frac{MU_x}{MU_y} = \frac{(1/x)}{(2/y)} = \frac{y}{2x} = \frac{p_x}{p_y} \Rightarrow y = \frac{2p_x}{p_y} x.$$

Substituting into the constraint, we find that the consumer spends 1/3 of income on x and 2/3 on y (shares determined by the “weights” of the function).

3.9 The Cobb–Douglas solution

Consider a Cobb–Douglas utility function

$$U(x, y) = x^\alpha y^\beta, \quad \alpha, \beta > 0.$$

Proposition 3.4 (Cobb–Douglas optimal choice). *Given the constraint $p_x x + p_y y = M$, the interior solution is*

$$x^* = \frac{\alpha}{\alpha + \beta} \frac{M}{p_x}, \quad y^* = \frac{\beta}{\alpha + \beta} \frac{M}{p_y}.$$

Proof idea

We compute the marginal utilities: $MU_x = \alpha x^{\alpha-1} y^\beta$, $MU_y = \beta x^\alpha y^{\beta-1}$. The condition $MU_x/MU_y = p_x/p_y$ implies

$$\frac{\alpha}{\beta} \frac{y}{x} = \frac{p_x}{p_y} \Rightarrow y = \frac{\beta}{\alpha} \frac{p_x}{p_y} x.$$

Substituting into the constraint and solving yields the expressions for x^* and y^* .

Economic intuition

With Cobb–Douglas preferences, the consumer spends a fixed share of income on each good: $\alpha/(\alpha + \beta)$ on x , $\beta/(\alpha + \beta)$ on y .

3.10 Corner solutions

So far we have considered interior optima. In some cases (especially with perfect substitutes) the optimum lies at an intercept of the budget line.

Definition 3.5 (Corner solution).

A corner solution occurs if at the optimum the consumer chooses $x^* = 0$ or $y^* = 0$, i.e., all income is spent on a single good.

In other words, when the subjective trade-off differs systematically from the market trade-off, the optimum may lie at an extreme of the budget constraint.

Analytical example

Perfect substitutes

If $U(x, y) = ax + by$ with $a, b > 0$, then $MRS = a/b$ is constant. Comparing MRS with p_x/p_y :

- if $\frac{a}{b} > \frac{p_x}{p_y}$ the optimum is $(M/p_x, 0)$ (only x);
- if $\frac{a}{b} < \frac{p_x}{p_y}$ the optimum is $(0, M/p_y)$ (only y);
- if $\frac{a}{b} = \frac{p_x}{p_y}$ every bundle on the line is optimal.

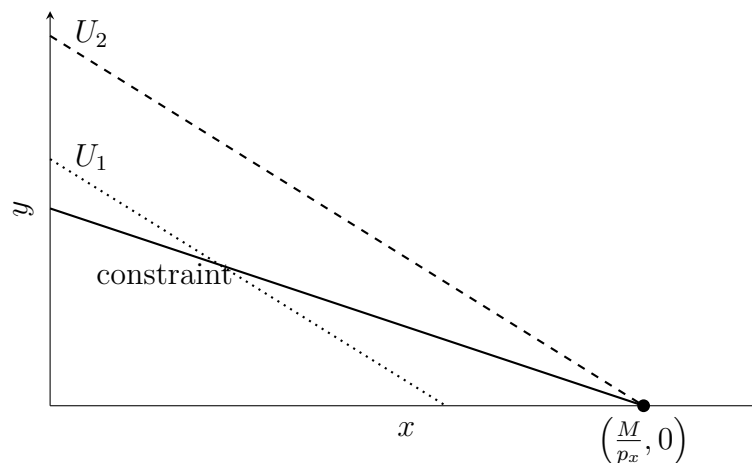


Figure 3.5: Corner solution (perfect substitutes): the highest attainable indifference curve touches the constraint at the extreme $(\frac{M}{p_x}, 0)$.

Analytical example

Perfect complements: fixed proportions

If $U(x, y) = \min\{ax, by\}$, the consumer wants the goods in fixed proportion. At an optimum

$$ax^* = by^*.$$

Proof idea

If $ax < by$, increasing y does not change U ; if $ax > by$, increasing x does not change U . To avoid “waste” the optimum requires equality at the kink: $ax = by$. Substituting $y = \frac{a}{b}x$ into the constraint yields the closed-form solution:

$$x^* = \frac{bM}{bp_x + ap_y}, \quad y^* = \frac{aM}{bp_x + ap_y}.$$

Key idea

The tangency condition holds for interior optima. With corner solutions one must directly compare utility at the admissible vertices.

Example

Tap water and bottled water

If the consumer regards tap water and bottled water as near-perfect substitutes, but tap water is much cheaper, he or she may choose a corner solution, spending everything on the relatively cheaper good.

3.11 Exercises

Exercise 3.1.

Given $M = 60$, $p_x = 10$, $p_y = 3$:

1. write the budget constraint and find the intercepts and slope;

2. determine whether the bundles $(3, 10)$, $(2, 8)$, and $(5, 15)$ are affordable;
3. draw the budget set.

Exercise 3.2.

Show graphically and analytically the effect of an increase in income from M to $1.5M$ at fixed prices. What happens to the intercepts and the slope?

Exercise 3.3.

For $U(x, y) = x^{0.5}y^{0.5}$, $p_x = 10$, $p_y = 5$, $M = 100$:

1. solve the problem with the substitution method;
2. solve the same problem with the Lagrange method;
3. verify that in both cases you obtain $x^* = 5$, $y^* = 10$.

Exercise 3.4.

For $U(x, y) = 2x + 3y$, $p_x = 5$, $p_y = 4$, $M = 100$:

1. compute MRS and p_x/p_y ;
2. determine whether the solution is interior or a corner;
3. find the optimal bundle.

Exam essentials

1. write and interpret the budget constraint, budget line, and budget set;
2. analyze graphically the effect of changes in income and prices;
3. formulate the consumer's maximization problem;
4. use the substitution and Lagrange methods in simple cases;
5. apply the closed-form formula for Cobb–Douglas functions;
6. recognize when the solution is interior or a corner.

Chapter 4

Individual demand, comparative statics, and elasticity

4.1 Marshallian demand (individual demand)

In the previous chapter we solved the consumer's problem:

$$\max_{x,y \geq 0} U(x,y) \quad \text{s.t.} \quad p_x x + p_y y = M,$$

obtaining, for each triple (p_x, p_y, M) , an optimal bundle (x^*, y^*) .

Definition 4.1 (Marshallian demand).

The *Marshallian* (or *ordinary*) *demand* is the function that maps prices and income into the optimal quantity chosen:

$$x^D = x^D(p_x, p_y, M), \quad y^D = y^D(p_x, p_y, M).$$

Economic intuition

Marshallian demand is simply the **optimal choice** “viewed as a function” of the parameters: if prices or income change, the constraints change and hence the optimum changes.

Analytical example

Cobb–Douglas

If $U(x,y) = x^\alpha y^\beta$ with $\alpha, \beta > 0$, the interior optimal choice is:

$$x^D(p_x, p_y, M) = \frac{\alpha}{\alpha + \beta} \frac{M}{p_x}, \quad y^D(p_x, p_y, M) = \frac{\beta}{\alpha + \beta} \frac{M}{p_y}.$$

Remark 4.1 (Homogeneity of degree zero).

For every $\lambda > 0$:

$$x^D(\lambda p_x, \lambda p_y, \lambda M) = x^D(p_x, p_y, M),$$

i.e., only *relative* prices and income matter.

Example**Demand for pizza**

The Marshallian demand for pizza indicates how many pizzas a consumer buys for each combination of the price of pizza, the price of other goods, and income. It is not a mere preference: it is the actual optimal choice, given the constraints.

4.2 Demand curve and comparative statics

Definition 4.2 (Demand curve).

The *demand curve* for good x represents the relationship between p_x and x^D holding p_y and M fixed.

Key idea

- If p_x changes $\Rightarrow$ **movement along** the demand curve for x .
- If M or p_y changes $\Rightarrow$ **shift** of the demand curve for x .

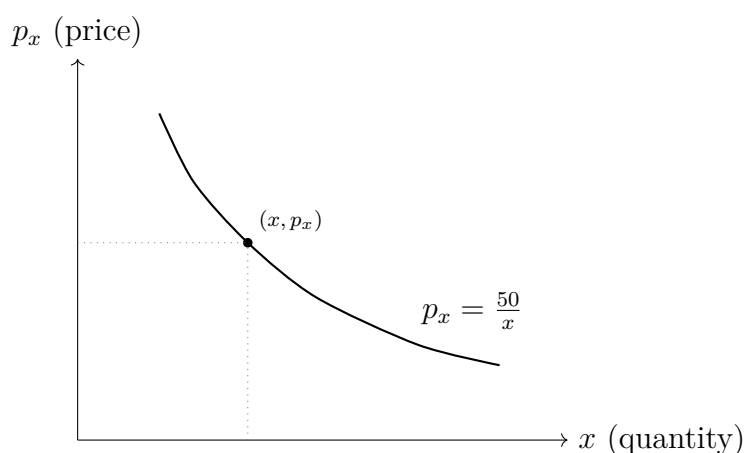


Figure 4.1: Demand curve (Marshall's convention): price p_x on the vertical axis and quantity x on the horizontal axis.

Example**Streaming subscriptions**

If the price of Netflix rises, a consumer may reduce the number of months purchased: this is a movement along the demand curve. If instead income rises, the entire demand curve may shift.

4.2.1 Price of the other good: substitutes and complements

Definition 4.3 ((Gross) substitutes and complements).

Let $x^D = x^D(p_x, p_y, M)$ be the Marshallian demand. The sign of

$$\frac{\partial x^D}{\partial p_y}$$

determines the relationship between the goods:

- **Substitutes** (gross) if $\frac{\partial x^D}{\partial p_y} > 0$;
- **Complements** (gross) if $\frac{\partial x^D}{\partial p_y} < 0$.

Remark 4.2 (Gross vs net).

The definition adopted here, based on the derivative of the Marshallian demand, identifies *gross* substitutes/complements. The *net* (or Hicksian) relationship is instead based on the derivative of the compensated demand, $\partial x^c/\partial p_y$, and is theoretically more robust because it is symmetric: $\partial x^c/\partial p_y = \partial y^c/\partial p_x$ (cf. the Slutsky equation, next section). Gross and net relationships may in principle differ in sign because of the income effect: if the income effect is strong enough, two goods that are net substitutes may appear as gross complements (and vice versa).

Economic intuition

If y becomes more expensive and x replaces it, I buy more x (substitutes). If x and y “go together”, the price increase of y also reduces the use of x (complements).

Example

Coffee and tea

If the price of tea rises and consumers buy more coffee, the two goods behave as substitutes. If instead the price of coffee capsules rises and the demand for compatible machines also falls, the two goods behave as complements.

4.3 ICC, Engel curves, and types of goods

Definition 4.4 (Income-consumption curve (ICC)).

The *income-consumption curve* (ICC) is the locus of optimal bundles as income M varies, holding prices p_x and p_y fixed.

Economic intuition

The ICC is the “path” of the optimum in (x, y) space when the budget constraint shifts in parallel (at constant prices) as income varies.

Definition 4.5 (Engel curve).

The *Engel curve* describes the relationship between income and quantity demanded at given prices:

$$x^D = x^D(M; p_x, p_y).$$

By convention, M is shown on the vertical axis and x on the horizontal axis.

Definition 4.6 (Normal and inferior goods). • x is a **normal good** if

$$\frac{\partial x^D}{\partial M} > 0.$$

- x is an **inferior good** if

$$\frac{\partial x^D}{\partial M} < 0.$$

Definition 4.7 (Necessities and luxury goods).

A normal good:

- is a **necessity** if the income elasticity satisfies $0 < \varepsilon_M < 1$;
- is a **luxury** if $\varepsilon_M > 1$.

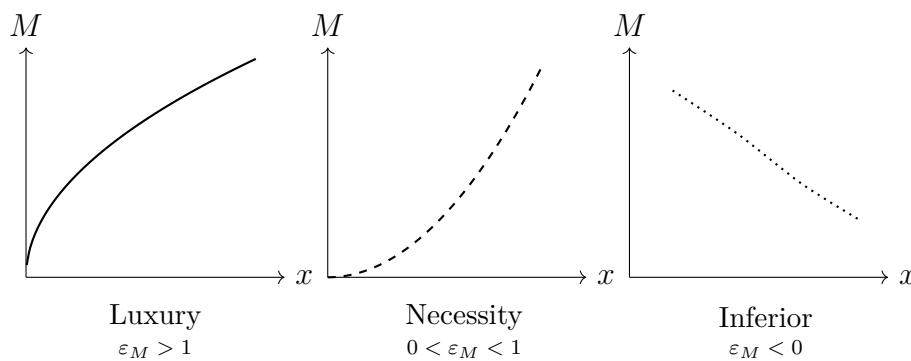


Figure 4.2: Engel curves (M on the vertical axis, x on the horizontal axis): luxury good (concave: x grows more than proportionally in M), necessity (convex: x grows less than proportionally in M), inferior good (decreasing).

Example

Necessities and luxury goods

When a household's income grows:

- spending on bread and pasta increases a little,
- spending on international travel may increase a lot.

This is why the former are often necessities and the latter luxury goods.

4.4 Compensated (Hicksian) demand and the Slutsky equation

Definition 4.8 (Hicksian (compensated) demand).

The Hicksian demand $x^c(p_x, p_y, \bar{U})$ is the quantity of good x that minimizes the expenditure needed to reach a given utility level $\bar{U}$:

$$\min_{x, y \geq 0} p_x x + p_y y \quad \text{s.t.} \quad U(x, y) \geq \bar{U}.$$

Key idea

- Marshallian $x^D(\cdot)$: maximizes utility with fixed income (M).
- Hicksian $x^c(\cdot)$: minimizes expenditure with fixed utility ($\bar{U}$).

Definition 4.9 (Substitution effect (SE) and income effect (IE)).

- **Substitution effect (SE)**: change in x due to the change in relative prices, at *constant utility*. Direction: **always opposite** to Δp_x .

- **Income effect (IE):** change in x due to the change in real purchasing power. Direction: depends on the sign of $\frac{\partial x^D}{\partial M}$ (normal vs inferior good).

Slutsky equation

$$\frac{\partial x^D}{\partial p_x} = \underbrace{\frac{\partial x^c}{\partial p_x}}_{\text{substitution effect}} - \underbrace{x^D \frac{\partial x^D}{\partial M}}_{\text{income effect}}.$$

Remark 4.3.

The substitution effect $\frac{\partial x^c}{\partial p_x}$ is always negative (for monotonic and convex preferences), while the sign of the income effect depends on the nature of the good (normal or inferior).

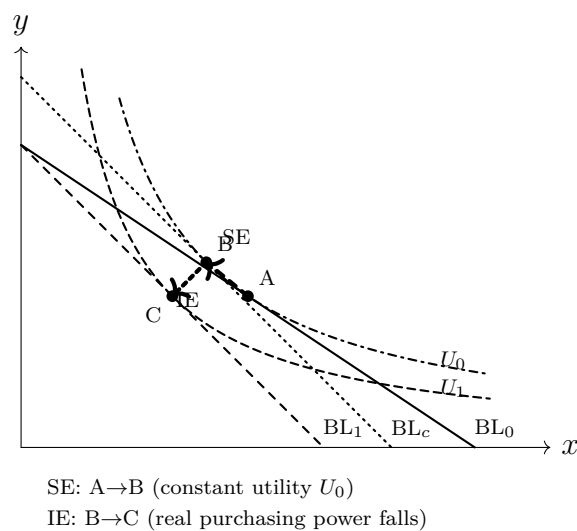


Figure 4.3: Slutsky decomposition: p_x rises. SE (A→B) at constant utility; IE (B→C) due to the loss of real purchasing power.

Remark 4.4 (Giffen goods).

A Giffen good is a case (theoretically possible but rare) in which

$$\frac{\partial x^D}{\partial p_x} > 0,$$

i.e., a price increase leads to an increase in quantity demanded. This happens when the income effect (for a *strongly inferior* good with a large budget share) dominates the substitution effect in absolute value.

Example**Rise in the price of gasoline**

When the price of gasoline rises:

- some people use the train or the bus more (**substitution effect**),
- their real purchasing power falls (**income effect**).

The total change in the demand for gasoline is the sum of these two effects.

4.5 Price-consumption curve and graphical derivation of demand

Definition 4.10 (Price-consumption curve (PCC)).

The *price-consumption curve* (PCC) is the locus of optimal bundles as p_x varies, holding p_y and M fixed.

Economic intuition

The PCC is the “path” of the optimum in (x, y) space when the budget constraint rotates as p_x varies. Projecting the optimal points onto (x, p_x) you obtain the demand curve.

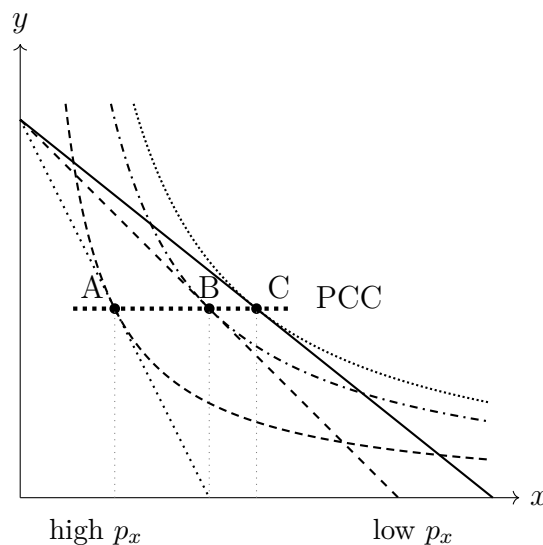


Figure 4.4: Price-consumption curve (PCC): as p_x falls, the horizontal intercept increases and the optimum moves along the PCC.

Example**Price of smartphones**

If the price of smartphones falls progressively, consumers’ optimal bundles shift. The PCC collects these optimal bundles; projecting them yields the demand curve.

4.6 Elasticity of demand

Derivatives such as $\frac{\partial x^D}{\partial p_x}$ measure changes in *physical units* (kg, liters, pieces). To compare different markets and goods we use a *percentage* measure: elasticity.

Definition 4.11 (Price elasticity of demand (point)).

The price elasticity of demand is defined as

$$\varepsilon_P = \frac{\partial x^D}{\partial p_x} \cdot \frac{p_x}{x^D}.$$

Typically $\varepsilon_P < 0$ (law of demand). For classification the absolute value $|\varepsilon_P|$ is often used.

Definition 4.12 (Classification (in absolute value)). • **Inelastic** if $|\varepsilon_P| < 1$;

- **Unit elastic** if $|\varepsilon_P| = 1$;
- **Elastic** if $|\varepsilon_P| > 1$.

Definition 4.13 (Income elasticity).

The income elasticity of demand is

$$\varepsilon_M = \frac{\partial x^D}{\partial M} \cdot \frac{M}{x^D}.$$

- $\varepsilon_M > 0$: normal good;
- $\varepsilon_M < 0$: inferior good;
- $0 < \varepsilon_M < 1$: typically a necessity;
- $\varepsilon_M > 1$: typically a luxury good.

Definition 4.14 (Cross-price elasticity (between x and y)).

The cross-price elasticity of demand is

$$\varepsilon_{xy} = \frac{\partial x^D}{\partial p_y} \cdot \frac{p_y}{x^D}.$$

- $\varepsilon_{xy} > 0$: substitute goods;
- $\varepsilon_{xy} < 0$: complementary goods;
- $\varepsilon_{xy} \approx 0$: nearly independent goods.

Key idea

For small percentage changes:

$$\Delta x\% \approx \varepsilon_P \cdot \Delta p_x\%.$$

Economic intuition

If $|\varepsilon_P|$ is large, quantity responds strongly to price: demand is sensitive. If it is small, quantity responds little.

Example**Cigarettes and essential medicines**

For goods such as cigarettes or some essential medicines, quantity demanded responds little to price: demand tends to be relatively inelastic. For goods such as travel or restaurants, the response can be much stronger.

4.7 Elasticity and total revenue

Definition 4.15 (Total revenue).

The *total revenue* from selling good x at unit price p_x is

$$TR(p_x) = p_x \cdot x(p_x).$$

where $x(p_x)$ is the quantity demanded at price p_x (given income and other prices).

Proposition 4.1 (Elasticity–revenue relationship). *Let ε_P be the price elasticity of demand:*

$$\varepsilon_P = \frac{dx}{dp_x} \frac{p_x}{x}.$$

Then:

$$\frac{dTR}{dp_x} = x(1 + \varepsilon_P).$$

In particular:

- if $|\varepsilon_P| > 1$ (elastic demand), then $p_x \uparrow \Rightarrow TR \downarrow$;
- if $|\varepsilon_P| < 1$ (inelastic demand), then $p_x \uparrow \Rightarrow TR \uparrow$;
- if $|\varepsilon_P| = 1$, then $\frac{dTR}{dp_x} = 0$ (revenue locally maximized).

Proof idea

Since $TR(p_x) = p_x x(p_x)$, by the product rule:

$$\frac{dTR}{dp_x} = x + p_x \frac{dx}{dp_x}.$$

Factoring out x and using the definition of elasticity:

$$\frac{dTR}{dp_x} = x \left(1 + \frac{dx}{dp_x} \frac{p_x}{x} \right) = x(1 + \varepsilon_P).$$

Remark 4.5.

For an ordinary demand $dx/dp_x < 0$, hence $\varepsilon_P < 0$.

Example**Cinema tickets**

If the demand for tickets is elastic, a price increase may sharply reduce the number of moviegoers and lower total revenue. If instead demand is inelastic, the price increase may raise revenue.

4.8 Exercises

Exercise 4.1.

Marshallian demand: $x^D(p, M) = 20 - 2p + 0.1M$. For $p = 5$ and $M = 100$:

1. compute x^D ;
2. compute $\frac{\partial x^D}{\partial p}$ and $\frac{\partial x^D}{\partial M}$;
3. use Slutsky to derive $\frac{\partial x^c}{\partial p}$;
4. interpret the SE and IE economically.

Exercise 4.2.

Linear demand: $Q = 100 - 2P$.

1. compute the point elasticity $\varepsilon(P)$;
2. evaluate ε at $P = 10$ and $P = 40$ and classify (elastic/inelastic);
3. find the price that maximizes total revenue.

Exercise 4.3.

Cross-price elasticity: the demand for pizza is $x^D(p_x, p_y, M) = 20 - 2p_x + p_y + 0.01M$.

1. compute $\frac{\partial x^D}{\partial p_y}$ and interpret (substitutes or complements);
2. compute ε_{xy} at $p_y = 3$, $p_x = 5$, $M = 100$.

Exam essentials

1. define Marshallian demand and the demand curve;
2. distinguish a movement along the curve from a shift of the curve;
3. use ICC/Engel curves to classify goods: normal/inferior and necessity/luxury;
4. explain and represent the SE and IE and read the graphical decomposition;
5. write and interpret the Slutsky equation (signs and cases);
6. compute and interpret price, income, and cross-price elasticities;
7. link price elasticity to total revenue and pricing decisions (TR is maximized when $|\varepsilon| = 1$);
8. recognize when Giffen-type behavior can arise (necessary conditions).

Questions and exercises

Multiple-choice questions

1. Which statement is correct?

- ☐ A The opportunity cost always coincides with the observed monetary expenditure.
- ☐ B An unrecoverable cost (*sunk cost*) should affect the future optimal choice.
- ☐ C The opportunity cost is the value of the best alternative forgone.
- ☐ D Marginal analysis concerns only discrete (0/1) choices.

2. For a differentiable function $f(x)$, a sufficient condition for an interior local maximum at x^* is:

- ☐ A $f'(x^*) = 0$ and $f''(x^*) > 0$;
- ☐ B $f'(x^*) = 0$ and $f''(x^*) < 0$;
- ☐ C $f'(x^*) > 0$ and $f''(x^*) = 0$;
- ☐ D $f'(x^*) < 0$ and $f''(x^*) = 0$.

3. **Transitivity** of the relation $\succeq$ means that:

- ☐ A for every A, B , $A \succeq B$ or $B \succeq A$;
- ☐ B if $A \succeq B$ and $B \succeq C$, then $A \succeq C$;
- ☐ C if $A \succeq B$, then $A \succ B$;
- ☐ D there exists a function U that represents the preferences.

4. Under monotonicity (more is better) and rationality, a correct property of indifference curves is:

- ☐ A they can intersect without creating contradictions;
- ☐ B they always have a positive slope;
- ☐ C they are typically downward sloping (negative slope);
- ☐ D they are always straight lines.

5. At an interior consumer optimum:

☐ A $\frac{MU_x}{MU_y} = \frac{p_y}{p_x};$

☐ B $\frac{MU_x}{p_x} = \frac{MU_y}{p_y};$

☐ C $MU_x = MU_y$ always;

☐ D necessarily $x^* = 0$ or $y^* = 0$.

Exercises and open questions

1. Consider the functions $U(x, y) = xy$ and $V(x, y) = 2xy + 10$.

- Show (formally) whether they represent the same preferences.
- List four typical properties of indifference curves under monotonicity and convexity and explain each in one line.
- Compute the MRS for $U(x, y) = xy$ and interpret it economically.

2. Let $U(x, y) = x^{0.5}y^{0.5}$, $p_x = 10$, $p_y = 5$, $M = 100$.

- Solve for the optimum with the substitution method.
- Solve the same problem with the Lagrangian and interpret λ .
- Verify the tangency condition $MRS = p_x/p_y$ at the optimal bundle.

3. Explain clearly (at most 10 lines) the differences between:

- the PCC and the demand curve for x ;
- the ICC and the Engel curve.

Then answer:

- If along the ICC the optimal quantity of x falls as M increases, what type of good is x ?
- If, in the standard graph with M on the vertical axis and x on the horizontal axis, the Engel curve of x appears *concave* (i.e., the function $M(x)$ is concave), what can be said about the income elasticity ε_M ?

Part II

Producer Theory

Chapter 5

Technology and Production

5.1 The producer's problem

The consumer chooses the bundle that maximizes utility given a budget constraint. The firm faces a mirror-image problem: given the goal of producing a certain output, it must figure out *how* to combine the inputs available to it.

Key idea

Just as in the consumer's problem the constraint was the **budget**, in the producer's problem the constraint is **technology**. The firm wants to produce as much as possible given what is technically feasible, or to produce a given output at minimum cost.

The fundamental questions are:

- how many units of output can be obtained from a given combination of inputs?
- which combination of inputs minimizes the cost of producing a given output?

Definition 5.1 (Technology).

Technology describes the technical constraints that determine how inputs can be transformed into output.

Definition 5.2 (Production function).

The production function gives the maximum quantity of output obtainable from each combination of inputs:

$$Q = f(L, K)$$

where L is labor, K is capital and Q is the output produced.

Example

Call center

Output is the number of calls handled per day. L = number of operators; K = workstations, lines and software. If the number of workstations is fixed, adding operators does not proportionally increase the calls handled: technology creates a *bottleneck*.

5.2 Short run and long run

Definition 5.3 (Short run).

The short run is the time horizon in which at least one factor of production is fixed (typically capital K). The firm can vary only the variable factors.

Definition 5.4 (Long run).

The long run is the time horizon in which all factors of production are variable. The firm can freely choose the combination (L, K) .

Remark 5.1.

The distinction is not about a fixed length of time (“3 months vs 3 years”). It depends on the nature of the industry: a bank can change its number of branches over a different time frame than a power plant changing its installed capacity.

5.3 Marginal product and diminishing marginal returns

Definition 5.5 (Marginal product of labor).

$$MPL = \frac{\partial f(L, K)}{\partial L}$$

It measures how much output increases when labor increases by one unit, holding capital fixed.

Definition 5.6 (Marginal product of capital).

$$MPK = \frac{\partial f(L, K)}{\partial K}$$

It measures how much output increases when capital increases by one unit, holding labor fixed.

Definition 5.7 (Diminishing marginal returns).

In the short run (with K fixed) there are diminishing marginal returns when each additional unit of labor generates a progressively smaller increase in output:

$$\frac{\partial MPL}{\partial L} = \frac{\partial^2 f}{\partial L^2} < 0.$$

Example

A restaurant kitchen

L = cooks; K = stoves and kitchen space (fixed in the short run). The first additional cook raises production a lot. The tenth, in a crowded kitchen, contributes much less. Each extra cook generates a smaller benefit: diminishing marginal returns.

5.4 Total, average and marginal product

The three fundamental short-run curves are closely related to each other.

Definition 5.8 (Total, average and marginal product). • **Total product:** $TP = Q = f(L, \bar{K})$.

- **Average product:** $APL = Q/L$ (output per unit of labor).

- **Marginal product:** $MPL = \partial Q / \partial L$ (contribution of the last unit of L).

Key idea

The relationship between MPL and APL is analogous to that between MC and AC for costs:

- if $MPL > APL$, then APL is increasing;
- if $MPL < APL$, then APL is decreasing;
- if $MPL = APL$, then APL is at its maximum.

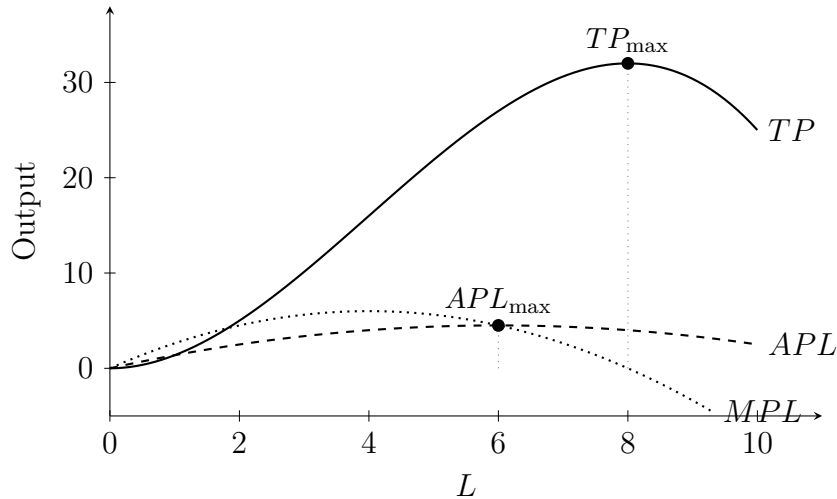


Figure 5.1: Total, average and marginal product for $f(L, \bar{K}) = 1.5L^2 - L^3/8$. TP reaches its maximum where $MPL = 0$; APL is at its maximum where $MPL = APL$.

Economic intuition

The logic of a grade average: if the next grade is above the average, the average rises. If it is below, the average falls. The average reaches its maximum when it equals the marginal grade.

5.5 Returns to scale

Returns to scale describe what happens to output when **all** inputs increase proportionally. It is a long-run property.

Definition 5.9 (Returns to scale).

Compare $f(tL, tK)$ with $t \cdot f(L, K)$ for $t > 1$:

- **Increasing returns** if $f(tL, tK) > t f(L, K)$;
- **Constant returns** if $f(tL, tK) = t f(L, K)$;
- **Decreasing returns** if $f(tL, tK) < t f(L, K)$.

Remark 5.2 (Cobb–Douglas).

For $f(L, K) = AL^\alpha K^\beta$:

$$f(tL, tK) = A(tL)^\alpha (tK)^\beta = t^{\alpha+\beta} AL^\alpha K^\beta = t^{\alpha+\beta} f(L, K).$$

Hence: increasing returns if $\alpha + \beta > 1$, constant if $\alpha + \beta = 1$, decreasing if $\alpha + \beta < 1$.

Example

E-commerce platform

Doubling warehouses, software and staff can more than double the orders handled thanks to standardized processes, fixed costs spread over more sales and more efficient logistics. This amounts to increasing returns.

5.6 Exercises

Exercise 5.1.

Let $Q = L^{0.5}K^{0.5}$.

- Compute MPL and MPK .
- Classify the returns to scale.
- Verify that MPL is decreasing in L (with K fixed).

Exercise 5.2.

A firm has $Q = 6L^2 - L^3$ (short run, K fixed).

- Compute MPL and APL .
- Verify that APL is at its maximum when $MPL = APL$.
- Find the level of L at which $MPL = 0$.

Exercise 5.3.

Given $Q = L^{0.7}K^{0.4}$, classify the returns to scale and compute MPL and MPK .

Exercise 5.4.

A firm doubles all its inputs and observes that output goes from 100 to 180. What type of returns to scale does it observe? Provide a possible economic explanation.

Exam essentials

- define technology and the production function;
- distinguish the short run from the long run;
- compute and interpret MPL , MPK and APL ;
- explain diminishing marginal returns and illustrate them graphically;
- describe the relationship between MPL and APL (when APL rises/falls/is at its maximum);
- classify returns to scale using the definition and the Cobb–Douglas formula ($\alpha + \beta$).

Chapter 6

Isoquants, MRTS and Types of Technology

6.1 Isoquants

Definition 6.1 (Isoquant).

An isoquant is the set of input combinations (L, K) that produce the same level of output:

$$\{(L, K) \in \mathbb{R}_+^2 : f(L, K) = \bar{Q}\}.$$

Economic intuition

Isoquants are the analogue of the consumer's **indifference curves**, but they represent levels of **output** instead of utility. Similarly, an isoquant farther from the origin corresponds to a higher level of output.

Properties of isoquants

The properties follow directly from the assumptions on technology (analogous to the axioms on consumer preferences):

1. **Negative slope:** to keep Q constant, if L increases K must decrease (and vice versa), provided both factors have a positive marginal product.
2. **They do not intersect:** the production function assigns a single output to each bundle (if they intersected, the same input bundle would produce two different outputs, a contradiction).
3. **Convex to the origin:** reflects the fact that the MRTS is diminishing (substitutability between factors falls as more of one of them is used).
4. **Higher isoquants to the north-east:** more input produces more output.

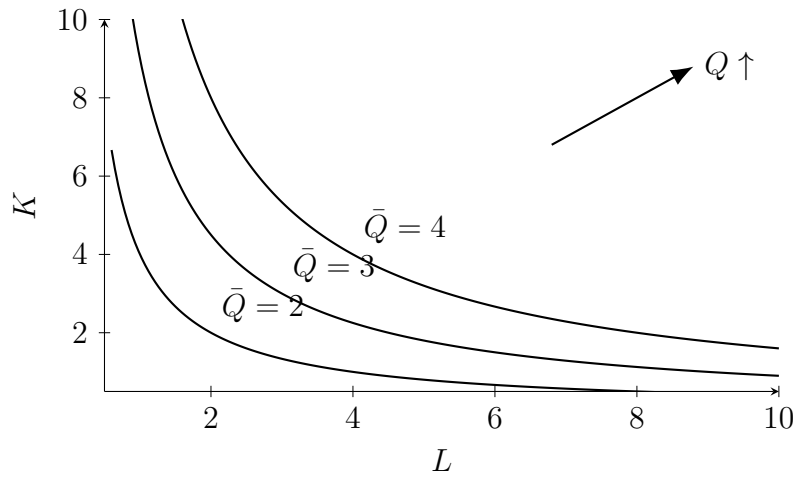


Figure 6.1: Cobb–Douglas isoquants $Q = \sqrt{LK}$: convex to the origin, analogous to indifference curves.

6.2 Marginal rate of technical substitution (MRTS)

Definition 6.2 (MRTS).

Along an isoquant $dQ = MPL dL + MPK dK = 0$, hence:

$$MRTS_{LK} = -\left. \frac{dK}{dL} \right|_{dQ=0} = \frac{MPL}{MPK}.$$

The $MRTS_{LK}$ measures how many units of capital can be given up when labor increases by one unit, *holding output constant*. It is the slope (in absolute value) of the isoquant.

Economic intuition

$MRTS = 2$ means: by increasing L by one unit, we can reduce K by about two units without changing production. The firm is willing to substitute 2 units of capital with 1 unit of labor.

Proposition 6.1 (Diminishing MRTS). *Along a convex isoquant, $MRTS_{LK}$ decreases as L increases. As more labor (and less capital) is used, each additional unit of L produces less and less output (MPL falls) and each unit of K given up becomes more and more valuable (MPK rises):*

$$MRTS_{LK} = \frac{MPL}{MPK} \downarrow \quad \text{as } L \text{ increases.}$$

6.3 Types of technology

6.3.1 Cobb–Douglas (general case)

For $Q = AL^\alpha K^\beta$:

$$MRTS_{LK} = \frac{\alpha K}{\beta L}.$$

Isoquants are convex to the origin (as in the figure above). The two factors are substitutes, but not perfect ones: the more labor is used, the more capital becomes relatively scarce and valuable.

6.3.2 Perfect substitutes

$$Q = aL + bK \quad (a, b > 0).$$

Isoquants are straight lines with slope $-a/b$. The $MRTS = a/b$ is constant: the technology substitutes L for K always in the same proportion, regardless of the current mix.

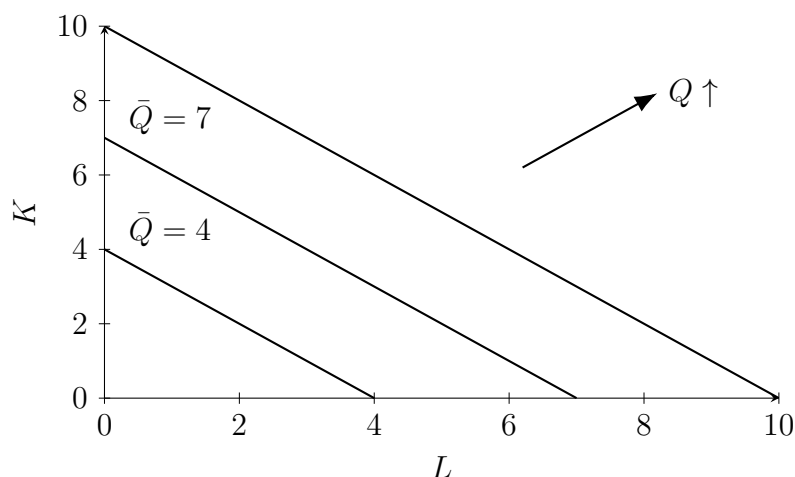


Figure 6.2: Perfect substitutes: linear isoquants, constant $MRTS$.

Example

Two identical machines

If two lathe models produce the same quantity per hour of use, they are perfect technical substitutes: the firm can switch from one to the other without any change in output.

6.3.3 Perfect complements (fixed proportions)

$$Q = \min\{aL, bK\} \quad (a, b > 0).$$

Inputs must be used in the fixed proportion $L/K = b/a$. Isoquants are L-shaped: using more of one factor without increasing the other does not increase output.

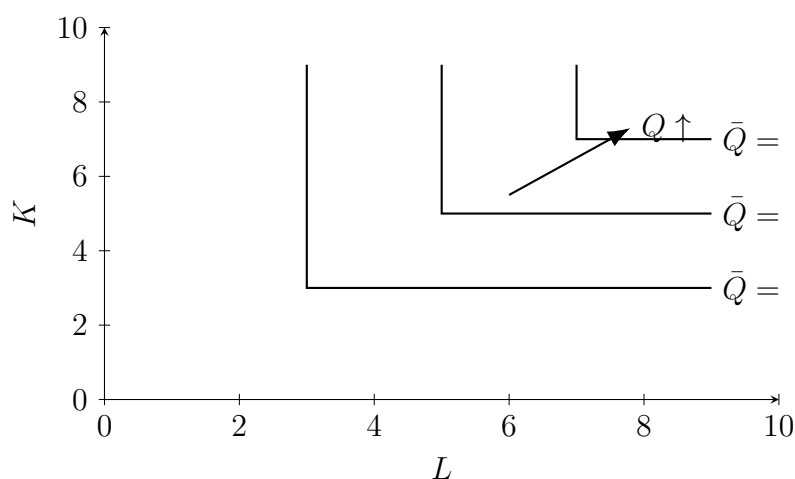


Figure 6.3: Perfect complements: L-shaped isoquants, fixed proportions.

Example**Driver and truck**

Exactly 1 driver and 1 truck are needed for each delivery shift. Adding a second driver without a second truck (or vice versa) does not increase the number of deliveries.

6.4 Exercises

Exercise 6.1.

For $Q = L^{0.6}K^{0.4}$, compute $MRTS_{LK}$ at the point $(L, K) = (16, 9)$ and interpret it.

Exercise 6.2.

Explain in 4–5 lines why two isoquants cannot intersect.

Exercise 6.3.

For each of the following production functions, sketch the isoquants and state whether the factors are perfect complements, perfect substitutes or an intermediate case:

- a) $Q = 3L + 5K$;
- b) $Q = \min\{2L, K\}$;
- c) $Q = L^{0.5}K^{0.5}$.

Exercise 6.4.

A firm with $Q = L^{0.5}K^{0.5}$ is at the point $(L, K) = (4, 9)$. Compute $MRTS_{LK}$ and give an economic interpretation of the result.

Exam essentials

1. define an isoquant and describe its four properties;
2. compute and interpret $MRTS_{LK} = MPL/MPK$;
3. explain why the $MRTS$ is diminishing along a convex isoquant;
4. recognize and draw isoquants for Cobb–Douglas, perfect substitutes and perfect complements;
5. relate the shape of the isoquant to the substitutability between factors.

Chapter 7

Costs of Production and Cost Curves

7.1 From technology to costs

We have studied technology: $Q = f(L, K)$, isoquants, $MRTS$, returns to scale. The next step is to understand how this technology translates into monetary costs.

Key idea

Just as the budget constraint delimits the consumer's choices, the cost structure delimits and guides the firm's decisions: prices, optimal scale, profitability.

Technology + factor prices $\Rightarrow$ cost function.

7.2 Fixed, variable and total costs

Definition 7.1 (Fixed, variable and total costs). • **Fixed costs** (FC): do not depend on the quantity produced Q . They exist even at $Q = 0$. Examples: rent, depreciation of machinery, management salaries.

- **Variable costs** ($VC(Q)$): increase with Q . Examples: raw materials, energy, direct labor.
- **Total costs**: $TC(Q) = FC + VC(Q)$.

Example

Barilla (Parma)

FC : the Pedrignano plant (capacity 1,000 t/day), depreciation of equipment, fixed energy contracts. VC : durum wheat, water, variable energy, on-call labor.

7.3 Average costs and marginal cost

Definition 7.2 (Average costs).

$$AC(Q) = \frac{TC(Q)}{Q}, \quad AFC(Q) = \frac{FC}{Q}, \quad AVC(Q) = \frac{VC(Q)}{Q}.$$

It always holds that: $AC = AFC + AVC$.

Definition 7.3 (Marginal cost).

$$MC(Q) = \frac{dTC(Q)}{dQ} = \frac{dVC(Q)}{dQ}.$$

Marginal cost measures the additional cost of producing one more unit.

Remark 7.1.

Since FC is constant, $\frac{dFC}{dQ} = 0$: marginal cost depends only on variable costs.

Economic intuition

Link with marginal productivity. If I add a worker (cost w), I produce MPL more units, hence:

$$MC \approx \frac{w}{MPL}.$$

When MPL falls (diminishing marginal returns), MC rises.

Example

Ferrari

Each car requires hundreds of hours of craft work. Adding one car to the production schedule means overtime and bottlenecks in the specialized departments: MC rises quickly. Ferrari deliberately chooses not to expand beyond the threshold at which MC starts to rise sharply, thereby keeping average costs under control and preserving the product's scarcity.

7.4 Shape of the cost curves and the relationship between MC and AC

AFC is always decreasing (a hyperbola: fixed costs are “spread” over more units). AVC and AC are typically U-shaped, which is linked to marginal returns: first increasing (economies of specialization), then diminishing (congestion).

Proposition 7.1 (MC crosses AC and AVC at their minima). *The following relationships hold:*

- if $MC < AC$, then AC is decreasing;
- if $MC > AC$, then AC is increasing;
- if $MC = AC$, then AC is at its minimum.

The same logic applies with AVC in place of AC .

Proof idea

$\frac{d(AC)}{dQ} = \frac{d}{dQ} \left(\frac{TC}{Q} \right) = \frac{MC \cdot Q - TC}{Q^2} = \frac{MC - AC}{Q}$. This is negative (AC decreasing) if and only if $MC < AC$, and positive (AC increasing) if $MC > AC$. At the minimum of AC the derivative is zero, hence $MC = AC$.

Economic intuition

It is the same logic as a grade average: if the next grade is below the average, the average falls; if it is above, it rises. The average is at its minimum when it equals the marginal grade.

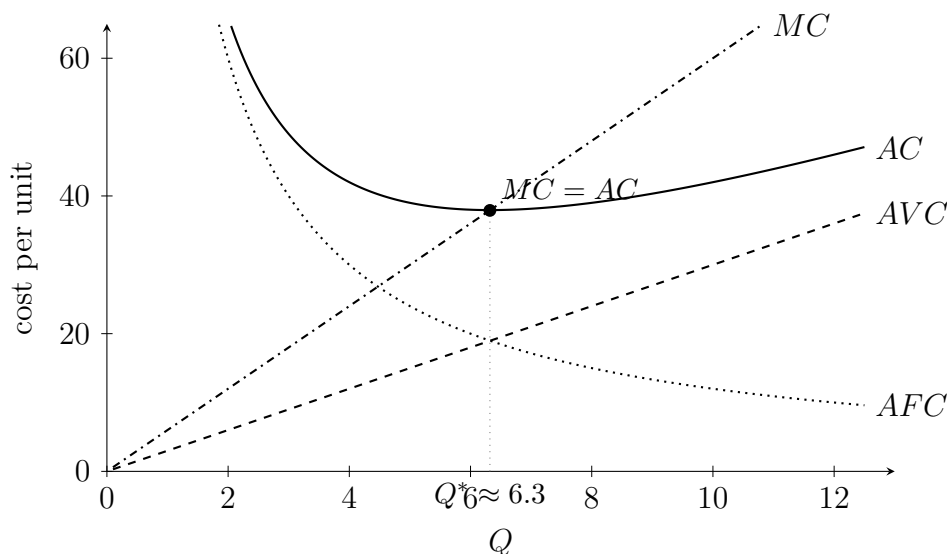


Figure 7.1: Cost curves for $TC(Q) = 120 + 3Q^2$. AFC is a decreasing hyperbola; AVC is linear and increasing; AC is U-shaped with its minimum at $Q^* \approx 6.3$, where $MC = AC$.

Example**Digital startup and AFC**

A startup with fixed costs (servers, licenses, offices) of €100,000/month: if it serves 1,000 users, $AFC = €100/\text{user}$; if it serves 100,000, $AFC = €1/\text{user}$. This is why digital platforms aim for scale.

7.5 Long-run costs and economies of scale

In the long run all factors are variable. The firm chooses the combination (L, K) that minimizes the cost of producing Q :

$$C(Q) = \min_{L, K} \{wL + rK : f(L, K) \geq Q\}.$$

The long-run average cost curve $LAC(Q) = C(Q)/Q$ is typically U-shaped:

- at low quantities: **economies of scale** (LAC decreasing)—division of labor, specialization, fixed costs spread over more units;
- at high quantities: **diseconomies of scale** (LAC increasing)—coordination problems, bureaucracy.

The minimum point of LAC is called the **minimum efficient scale (MES)**.

Proposition 7.2 (Relationship between returns to scale and LAC). • *Increasing returns* $\Rightarrow$ *LAC decreasing (economies of scale)*;

- *Constant returns* $\Rightarrow$ *LAC constant*;
- *Decreasing returns* $\Rightarrow$ *LAC increasing (diseconomies of scale)*.

Remark 7.2 (LAC as an envelope).

The *LAC* curve is the *lower envelope* of all the short-run *AC* curves (each corresponding to a fixed level of K). For each Q the long-run firm can choose the optimal level of K , hence $LAC(Q) \leq AC_{SR}(Q)$ for every short-run curve.

Analytical example

From TC to all the curves

Let $TC(Q) = 120 + 3Q^2$.

- $FC = 120$, $VC(Q) = 3Q^2$.
- $AFC = 120/Q$; $AVC = 3Q$; $AC = 120/Q + 3Q$; $MC = 6Q$.
- Minimum of AC : $\frac{dAC}{dQ} = -120/Q^2 + 3 = 0 \Rightarrow Q^* = \sqrt{40} \approx 6.32$.
- Check: $MC(Q^*) = 6\sqrt{40} = AC(Q^*)$. ✓

Remark: AVC is linear and increasing; MC has twice the slope of AVC .

Analytical example

Shock to variable costs

Start from $TC(Q) = 120 + 3Q^2$. An energy shock raises variable costs by 20%:

$TC'(Q) = 120 + 3.6Q^2$.

- $MC'(Q) = 7.2Q$; $AVC'(Q) = 3.6Q$.
- FC remains unchanged at 120.
- New minimum of AC' : $Q^{**} = \sqrt{120/3.6} \approx 5.77 < Q^* \approx 6.32$.

With higher variable costs, the minimum point shifts to the left: it pays to produce less to minimize average cost per unit.

Example

The 2022 energy crisis in Emilia-Romagna

In 2022 the price of gas rose by more than 300%. Manufacturing firms in Emilia (Sassuolo ceramics, engineering, food) suffered a shock to VC . AVC and MC shifted upward; AFC remained unchanged. The hardest hit: *energy-intensive* firms facing elastic demand (they could not pass the cost on to consumers).

7.6 Exercises

Exercise 7.1.

Given $TC(Q) = 200 + 5Q + Q^2$:

- Identify FC and $VC(Q)$.

- b) Compute AFC , AVC , AC and MC .
- c) What is the shape of AVC ? (Linear, U-shaped, hyperbola?)
- d) Find the Q^* that minimizes AC and verify that $MC(Q^*) = AC(Q^*)$.

Exercise 7.2.

Consider $TC(Q) = F + cQ^2$ with $F > 0$, $c > 0$.

- a) Derive in symbolic form $Q^* = \sqrt{F/c}$.
- b) How does Q^* change if F increases? And if c increases? Give an economic interpretation.

Exercise 7.3 (True or False).

Justify each answer in two lines.

- a) If $MC < AC$, then AC is decreasing.
- b) MC crosses AC at the maximum point of AC .
- c) MC does not depend on FC .
- d) If $MC < AVC$, then AVC is decreasing.

Exercise 7.4.

For each of the following industries, do you expect a decreasing, constant or increasing LAC ? Briefly explain.

- a) Microchip production (huge investments in fabs; marginal cost close to zero).
- b) Individual professional services (lawyers, accountants).
- c) Large-scale retail chains (supermarkets and hypermarkets).

Exam essentials

1. distinguish FC , VC , TC and derive them from a cost function;
2. compute AFC , AVC , AC and MC from $TC(Q)$;
3. explain why MC crosses AC (and AVC) at their respective minima (using the derivative);
4. find the Q^* that minimizes AC and verify the condition $MC(Q^*) = AC(Q^*)$;
5. analyze the effects of a shock to FC or to VC on the cost curves;
6. relate LAC to returns to scale: describe economies and diseconomies of scale and the MES.

Chapter 8

Cost Minimization and Optimal Input Choice

8.1 Symmetry with the consumer

Key idea

The consumer **maximizes utility** given the budget constraint.
The firm **minimizes cost** given the production constraint (the isoquant).
Same mathematical structure, reversed direction.

The previous chapter gave us the cost curves “already solved”. Now we see *where* they come from: given an output target $\bar{Q}$, which (L, K) mix costs the least?

8.2 The cost-minimization problem

Definition 8.1 (Cost-minimization problem (LR)).

$$\min_{L, K \geq 0} C = wL + rK \quad \text{s.t.} \quad f(L, K) \geq \bar{Q}.$$

where $w > 0$ is the wage (cost per unit of labor) and $r > 0$ is the cost of capital (leasing fee, required rate of return).

Geometrically: find the point on the isoquant $f(L, K) = \bar{Q}$ that lies on the *lowest possible isocost line*.

Example

A logistics firm (Emilia-Romagna)

It must deliver 10,000 parcels per day ($\bar{Q} = 10,000$). It can do so with many couriers and few vans, or with many automated vans and few couriers. The optimal mix depends on w (cost of labor) and r (leasing cost of the vans).

8.3 Isocost lines

Definition 8.2 (Isocost line).

The isocost line is the set of combinations (L, K) that involve the same expenditure C :

$$wL + rK = C \quad \Rightarrow \quad K = \frac{C}{r} - \frac{w}{r}L.$$

- **Intercept on K :** C/r —how many units of capital can be bought by spending everything on K .
- **Intercept on L :** C/w —how many units of labor can be bought by spending everything on L .
- **Slope:** $-w/r$ —the price of labor relative to capital.

Shifts of the isocost line (analogy with the budget constraint):

- higher expenditure $\Rightarrow$ parallel isocost line farther from the origin;
- $w \uparrow \Rightarrow$ steeper isocost line (labor relatively more expensive);
- $r \uparrow \Rightarrow$ flatter isocost line.

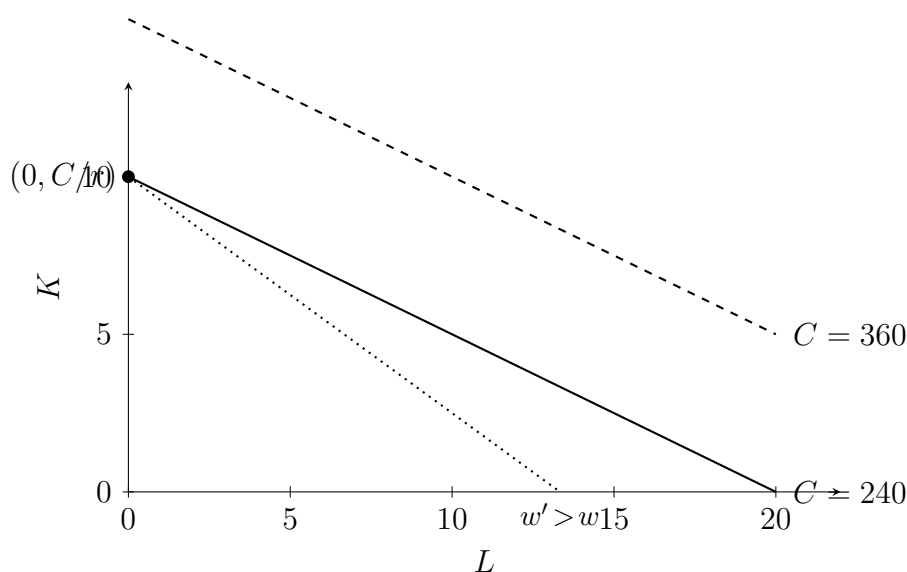


Figure 8.1: Isocost lines for $w = 12$, $r = 24$, $C = 240$ (solid) and $C = 360$ (dashed). The dotted line shows the effect of a wage increase to $w' = 18$: the line pivots inward on the L axis (the L intercept falls from 20 to ≈ 13.3), becoming steeper.

8.4 Tangency condition: $MRTS = w/r$

At an interior optimum, the isoquant and the isocost line are tangent: their slopes are equal.

Definition 8.3 (Optimality condition (long run)).

$$MRTS_{LK} = \frac{MPL}{MPK} = \frac{w}{r}.$$

Equivalent form (marginal product per euro spent):

$$\frac{MPL}{w} = \frac{MPK}{r}.$$

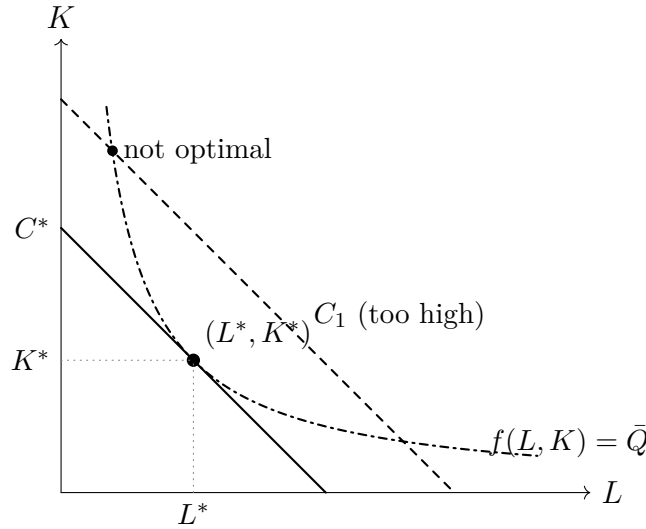


Figure 8.2: Cost minimization. The isocost line C^* is the lowest one that reaches the isoquant $\bar{Q}$. The isocost line $C_1 > C^*$ crosses the isoquant (feasible but not optimal: too much is spent). The tangency point (L^*, K^*) satisfies $MRTS = w/r$.

Economic intuition

If $\frac{MPL}{w} > \frac{MPK}{r}$: one euro spent on labor produces more output than one euro spent on capital. It pays to shift spending toward L (and reduce K) until the returns per euro are equalized.

Key idea

The producer's optimality condition is structurally identical to the consumer's:

$$\underbrace{\frac{MU_x}{p_x} = \frac{MU_y}{p_y}}_{\text{consumer}} \quad \longleftrightarrow \quad \underbrace{\frac{MPL}{w} = \frac{MPK}{r}}_{\text{producer}}.$$

8.5 Solution with Cobb–Douglas technology

For $Q = AL^\alpha K^\beta$, the tangency condition becomes:

$$\frac{\alpha K}{\beta L} = \frac{w}{r} \quad \Rightarrow \quad K = \frac{\beta}{\alpha} \cdot \frac{w}{r} \cdot L.$$

Proposition 8.1 (Cobb–Douglas minimization procedure). 1. Derive the ratio $K/L = (\beta/\alpha)(w/r)$ from the tangency condition.

2. Substitute $K = (\beta/\alpha)(w/r)L$ into the constraint $f(L, K) = \bar{Q}$ to find L^* .
3. Obtain K^* from the ratio.
4. Compute $C^* = wL^* + rK^*$.

Example 8.1 (Complete minimization).

Technology $Q = L^{0.6}K^{0.4}$; $w = 10$, $r = 20$; target $\bar{Q} = 50$.

Step 1—Optimal ratio:

$$\frac{K}{L} = \frac{0.4}{0.6} \cdot \frac{10}{20} = \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{3} \quad \Rightarrow \quad K = \frac{L}{3}.$$

Step 2—Output constraint ($\alpha + \beta = 1$, simplification):

$$50 = L^{0.6} \left(\frac{L}{3} \right)^{0.4} = L \cdot 3^{-0.4} \quad \Rightarrow \quad L^* = 50 \cdot 3^{0.4} \approx 77.6.$$

Step 3— $K^* = L^*/3 \approx 25.9$.

Step 4—Minimum cost: $C^* = 10 \cdot 77.6 + 20 \cdot 25.9 \approx 1,294$.

Remark 8.1.

With $r = 20 > w = 10$, capital is relatively expensive. The optimal mix is $K/L = 1/3$: the firm is labor-intensive.

8.6 Comparative statics: effects of changes in w and r

From the optimality condition $K/L = (\beta/\alpha)(w/r)$:

Shock	Effect on K/L	Interpretation
$w \uparrow$	$K/L \uparrow$	Substitutes K for L (automation)
$w \downarrow$	$K/L \downarrow$	Uses more labor, less capital
$r \uparrow$	$K/L \downarrow$	Capital more expensive, prefers L
$r \downarrow$	$K/L \uparrow$	Investment subsidy $\Rightarrow$ more K

Remark 8.2.

The minimum cost C^* *always rises* when w or r increase: the firm pays more to produce the same output. But the increase is *dampened* by the possibility of substituting away from the factor that has become more expensive.

Example

Amazon and warehouse automation

Over the last 10 years wages in the logistics sector have risen ($w \uparrow$): the optimal ratio K/L rises, and it pays to replace labor with robots. At the same time the cost of robots has fallen ($r \downarrow$): a reinforcing effect. In Amazon warehouses with robots (the Kiva system), productivity per m^2 has increased and the cost per unit shipped has fallen.

8.7 Exercises

Exercise 8.1.

With $w = 12$, $r = 24$, $C = 240$:

- Write the isocost line in the form $K = \dots$
- Intercepts and slope.
- If w rises to 18 (with r and C unchanged), how does the isocost line change?

Exercise 8.2.

A firm is at a point on the isoquant with $MPL = 8$, $MPK = 4$, $w = 16$, $r = 24$.

- Compute $MRTS$ and w/r . Are we at the optimum?
- If we are not at the optimum, in which direction should the firm move?
- What happens to MPL and MPK as the firm moves toward the optimum?

Exercise 8.3.

Technology $Q = 2L^{0.5}K^{0.5}$; $w = 8$, $r = 2$; target $\bar{Q} = 40$.

- Find the optimal K/L .
- Derive L^* and K^* .
- Compute C^* .
- Is the mix labor-intensive or capital-intensive? Why?

Exercise 8.4.

Starting from the previous exercise (with $w = 8$, $r = 2$), suppose w falls to 2.

- Compute the new K/L and the new quantities L^{**} , K^{**} .
- Compute C^{**} . Compare it with C^* : what do you notice?
- Give an economic interpretation of the result.

Exercise 8.5.

The government introduces a minimum wage that raises w from 10 to 14; $r = 20$; $Q = L^{0.7}K^{0.3}$.

- How does the optimal ratio K/L change?
- Which firms can adapt easily and which cannot?
- Does the minimum cost of producing the same $\bar{Q}$ rise or fall?

Exam essentials

1. set up the minimization problem: $\min wL + rK$ s.t. $f(L, K) \geq \bar{Q}$;
2. construct and interpret isocost lines (intercepts, slope $-w/r$, shifts);
3. derive and interpret the tangency condition $MRTS = w/r$ (and the equivalent form $MPL/w = MPK/r$);
4. solve the minimization problem with Cobb–Douglas technology (4-step recipe);
5. do comparative statics: effects of changes in w or r on the ratio K/L and on minimum cost;
6. relate the optimal choice of inputs to real business decisions (automation, offshoring).

Questions and exercises

Multiple-choice questions

1. Which statement about the production function $Q = f(L, K)$ is correct?
 - ☐ A The production function describes the minimum expenditure needed to produce Q .
 - ☐ B The production function gives the maximum quantity of output obtainable from each combination of inputs.
 - ☐ C In the short run all factors of production are variable.
 - ☐ D The production function is defined only for $L = K$.

2. For the Cobb–Douglas function $Q = AL^\alpha K^\beta$, returns to scale are increasing if and only if:
 - ☐ A $\alpha > \beta$;
 - ☐ B $\alpha + \beta = 1$;
 - ☐ C $\alpha + \beta > 1$;
 - ☐ D $\alpha \cdot \beta > 1$.

3. The average product of labor (APL) is at its **maximum** when:
 - ☐ A $MPL = 0$;
 - ☐ B $MPL > APL$;
 - ☐ C $MPL = APL$;
 - ☐ D $APL = 0$.

4. The AC curve reaches its **minimum** at the point where:
 - ☐ A $AFC = AVC$;
 - ☐ B $MC = AFC$;
 - ☐ C $MC = AC$;
 - ☐ D $\frac{dMC}{dQ} = 0$.

5. A firm faces a **shock to fixed costs** (for example, its rent doubles). Which curve remains unchanged?

- ☐ A AC .
- ☐ B AFC .
- ☐ C AFC and MC .
- ☐ D AVC and MC .

Exercises and open questions

1. Consider the production function $Q = L^{0.5}K^{0.5}$.

- a) Compute MPL and MPK and verify that both are decreasing in their respective inputs (holding the other fixed).
- b) Classify the returns to scale using the formal definition $f(tL, tK)$ vs $tf(L, K)$.
- c) Compute $MRTS_{LK}$ at the point $(L, K) = (4, 9)$ and give an economic interpretation of the result: how many units of K can the firm replace with one more unit of L without changing output?
- d) Explain in your own words why $MRTS_{LK}$ is diminishing along a convex isoquant.

2. Given $TC(Q) = 200 + 4Q + Q^2$:

- a) Identify FC and $VC(Q)$ and write the expressions for $AFC(Q)$, $AVC(Q)$, $AC(Q)$, $MC(Q)$.
- b) Find the level Q^* that minimizes AC and verify analytically that $MC(Q^*) = AC(Q^*)$.
- c) A shock raises variable costs by 50%: new $VC'(Q) = 1.5(4Q + Q^2)$. How do MC' , AVC' and the new Q^{**} that minimizes AC' change? Is Q^{**} larger or smaller than Q^* ?
- d) Explain in three lines why MC does not depend on FC and what this implies for the firm's short-run decisions.

3. Explain clearly (at most 12 lines) the analogies and differences between:

- a) indifference curve and isoquant;
- b) budget constraint and isocost line;
- c) the consumer's optimality condition ($MRS = p_x/p_y$) and the producer's optimality condition ($MRTS = w/r$).

Then answer: in the consumer's problem the free parameter is *income* M ; in the producer's long-run problem it is the *output target* $\bar{Q}$. How is this difference reflected in the geometric structure of the two problems?

Part III

Perfectly Competitive Markets

Chapter 9

Profit Maximization

9.1 Accounting profit and economic profit

In economics the word *profit* has a precise meaning, narrower than its everyday accounting use. The distinction is fundamental for the firm's decisions.

Definition 9.1 (Economic profit).

Economic profit is defined as:

$$\pi = TR - TC = P \cdot Q - C(Q)$$

where TC includes *all* opportunity costs: both explicit (actual monetary payments) and implicit (the value of the best alternative use of the resources committed to the firm).

Definition 9.2 (Explicit and implicit costs).

- **Explicit costs:** observable monetary payments—rents, employee wages, raw materials, interest on debt, depreciation.
- **Implicit (opportunity) costs:** the value of the best alternative use of the owner's own resources—the salary the entrepreneur would earn as an employee elsewhere, the return that the owner's capital would generate in its next best use, the cost of the time spent.

Accounting profit includes only explicit costs: $\pi_{\text{acc}} = TR - \text{explicit costs} \geq \pi_{\text{econ}}$.

Key idea

Economic profit measures the gain *in excess* of the best available alternative. Zero economic profit does not mean that the entrepreneur earns nothing: it means that they are earning exactly the **normal return**, i.e., what they would obtain in their best alternative.

Example 9.1 (Accounting vs. economic profit).

A pizzeria earns annual revenues of €140,000 with explicit costs of €95,000. The owner could earn €38,000 as an employee elsewhere; the owner's capital invested (€50,000) would have an alternative return of 4% (i.e., €2,000).

	Accounting profit	Economic profit
Revenues	€140,000	€140,000
Explicit costs	−€95,000	−€95,000
Alternative salary	—	−€38,000
Alternative return on capital	—	−€2,000
Profit	€45,000	€5,000

The owner earns €5,000 more than they would obtain in their best alternative. If revenues fell to €130,000, economic profit would become −€5,000 despite a still positive accounting profit (€35,000): it would pay to close and work as an employee.

Remark 9.1 (Implications for market dynamics).

The three situations have precise implications for entry and exit:

- $\pi > 0$: the resources in the firm earn more than their alternative $\Rightarrow$ new firms have an incentive to enter the market.
- $\pi = 0$: normal return; no net incentive to enter or exit.
- $\pi < 0$: the resources earn less than their alternative $\Rightarrow$ some firms will exit the market.

This logic underlies the long-run equilibrium in perfect competition (ch. 6).

9.2 The price-taking firm

Definition 9.3 (Perfect competition).

A market is **perfectly competitive** if it satisfies four conditions:

1. **Many buyers and sellers**: no agent is large enough to influence the market price.
2. **Homogeneous good**: the products of different firms are perfect substitutes.
3. **Perfect information**: all agents know prices and quality.
4. **Free entry and exit**: in the long run there are no barriers to entering the market.

Definition 9.4 (Price-taking firm).

A firm is a **price taker** if it takes the price P^* as given, regardless of the quantity produced. The demand curve the firm faces is horizontal: $P(Q) = P^*$ for every $Q \geq 0$.

Definition 9.5 (Total revenue and marginal revenue of the price-taking firm).

For a price-taking firm at price P :

$$TR(Q) = P \cdot Q \quad (\text{line through the origin with slope } P)$$

$$MR(Q) = \frac{dTR}{dQ} = P \quad (\text{constant, equal to the market price})$$

Remark 9.2 (Comparison with monopoly).

The result $MR = P$ is specific to perfect competition. Under monopoly, the firm must lower the price to sell one more unit, so $MR(Q) < P(Q)$ for every $Q > 0$. For the price-taking firm, each additional unit is sold at the same price P , and marginal revenue coincides with it.

9.3 The profit maximization problem

9.3.1 Formal setup

The price-taking firm chooses the quantity $Q \geq 0$ to maximize profit:

$$\max_{Q \geq 0} \pi(Q) = P \cdot Q - C(Q)$$

where $P > 0$ is the price given by the market and $C(Q)$ is the total cost function of the previous chapter, with $C(0) = FC \geq 0$.

9.3.2 First- and second-order conditions

Proposition 9.1 (First-order condition — FOC). *If $Q^* > 0$ is an interior profit maximum, then it satisfies the **first-order condition**:*

$$\left. \frac{d\pi}{dQ} \right|_{Q=Q^*} = P - MC(Q^*) = 0 \quad \implies \quad \boxed{P = MC(Q^*)}$$

Proposition 9.2 (Second-order condition — SOC). *For Q^* to be a maximum (and not a minimum) of profit, the **second-order condition** must hold:*

$$\left. \frac{d^2\pi}{dQ^2} \right|_{Q=Q^*} = - \left. \frac{dMC}{dQ} \right|_{Q=Q^*} < 0 \quad \iff \quad MC'(Q^*) > 0$$

Marginal cost must be **strictly increasing** at the optimal point.

Proof idea

With $\pi(Q) = PQ - C(Q)$ and P constant:

$$\pi'(Q) = P - C'(Q) = P - MC(Q), \quad \pi''(Q) = -C''(Q) = -MC'(Q).$$

The FOC $\pi'(Q^*) = 0$ gives $P = MC(Q^*)$. The SOC $\pi''(Q^*) < 0$ requires $MC'(Q^*) > 0$: the MC curve must be increasing at Q^* . If it were decreasing, we would be at a *minimum* of profit.

Global maximum and multiple roots of the FOC

The FOC is *necessary* for an interior optimum, but not always globally *sufficient*. For typical cubic costs ($C(Q) = aQ^3 + bQ^2 + cQ + d$), the MC curve is a parabola and the equation $P = MC(Q)$ can have **two roots**. Only the one with $MC'(Q^*) > 0$ is a local maximum; the other is a local minimum.

The *global* maximum requires an additional step: comparing $\pi(Q_{\text{loc}}^*)$ with $\pi(0) = -FC$ (shutdown). More precisely, the optimal candidate among the interior local maxima is the one with the highest profit, and this must then be compared with the value of not producing.

A *sufficient* condition for a unique interior global maximum is the global concavity of π , i.e., $MC'(Q) > 0$ for every $Q \geq 0$ (strictly convex cost functions). With increasing linear MC , as in $C(Q) = Q^2 + bQ + c$, this condition is always satisfied.

Economic intuition

The condition $P = MC$ is the **principle of marginal analysis** applied to production. The unit-by-unit reasoning is:

- $P > MC(Q)$: producing one more unit adds P to revenues and $MC(Q)$ to costs. Since $P > MC$, profit rises $\Rightarrow$ it pays to expand.
- $P < MC(Q)$: the last unit produced costs more than it brings in $\Rightarrow$ it pays to reduce output.
- $P = MC(Q)$: neither expanding nor contracting improves profit; profit is stationary. If MC is increasing, this is a maximum.

The firm expands as long as the additional revenue exceeds the additional cost; it stops when they are equal.

Analytical example**Discrete analysis with $P = 12$**

Unit	MC	$P - MC$	Produce?
1	3	+9	Yes
2	5	+7	Yes
3	8	+4	Yes
4	11	+1	Yes
5	12	0	Indifferent ($Q^* = 5$)
6	14	-2	No
7	17	-5	No

The first four units add positive net profit; the fifth leaves it unchanged. The sixth would cost 14 but bring in only 12: it does not pay. The discrete FOC is satisfied at $Q^* = 5$.

9.4 Profit as an area

Proposition 9.3 (Profit as the area of a rectangle). *For a price-taking firm producing Q^* with $P = MC(Q^*)$, profit can be written as:*

$$\pi(Q^*) = [P - AC(Q^*)] \cdot Q^*$$

In the (Q, P) plane, profit is the area of the rectangle with base Q^ and height $(P - AC(Q^*))$.*

Proof idea

By definition of average cost $AC(Q) = C(Q)/Q$, so $C(Q) = AC(Q) \cdot Q$. Therefore:

$$\pi(Q^*) = PQ^* - C(Q^*) = PQ^* - AC(Q^*) \cdot Q^* = [P - AC(Q^*)] \cdot Q^*.$$

Remark 9.3 (The three cases).

The sign of $P - AC(Q^*)$ distinguishes three situations:

- $P > AC(Q^*)$: the rectangle has positive area $\Rightarrow$ **positive profit**.
- $P = AC(Q^*)$: the rectangle has zero height $\Rightarrow$ **zero profit** (*break-even point*). This happens when $P = AC_{\min}$.
- $AVC_{\min} \leq P < AC(Q^*)$: the rectangle has negative area $\Rightarrow$ **loss**; but, as we will see, the firm may still find it worthwhile to produce in the short run.

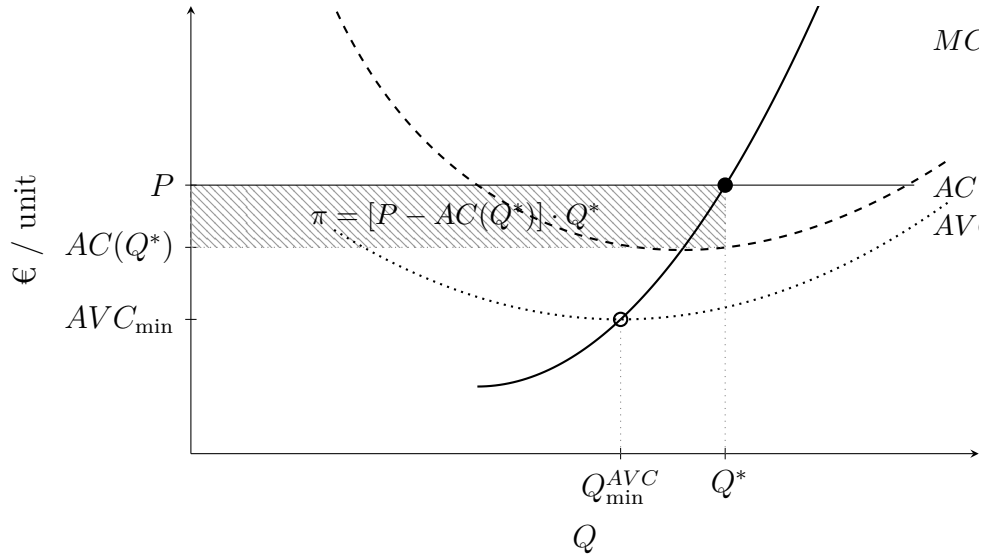


Figure 9.1: Profit as the area of the shaded rectangle. Cost $C(Q) = Q^3 - 6Q^2 + 15Q + 10$, $P = 12$, $Q^* = 2 + \sqrt{3} \approx 3.73$, $AC(Q^*) \approx 9.22$. Total profit $\pi \approx (12 - 9.22) \cdot 3.73 \approx 10.4$. The point $Q_{\min}^{AVC} = 3$ marks the minimum of AVC (shutdown point).

9.5 The shutdown condition

9.5.1 Setting up the problem

In the short run, the firm always has an alternative to producing: not producing ($Q = 0$). If it does not produce, profit is:

$$\pi(0) = P \cdot 0 - C(0) = -FC$$

The loss at $Q = 0$ is exactly equal to fixed costs: the firm bears them anyway. The question is: when does it pay to produce instead of shutting down?

9.5.2 Formal derivation

Proposition 9.4 (Shutdown condition — short run). *The firm finds it optimal to produce ($Q^* > 0$) if and only if:*

$$\boxed{P \geq AVC_{\min}}$$

The shutdown point corresponds to the minimum of the AVC curve.

Proof idea

The firm produces if and only if $\pi(Q^*) \geq \pi(0) = -FC$, i.e.:

$$\begin{aligned} PQ^* - C(Q^*) &\geq -FC \\ PQ^* &\geq C(Q^*) - FC = VC(Q^*) \\ P &\geq \frac{VC(Q^*)}{Q^*} = AVC(Q^*) \end{aligned}$$

Hence producing is profitable relative to shutting down if and only if the price covers average variable cost at the optimal quantity. Since for Q^* in the relevant range $AVC(Q^*) \geq AVC_{\min}$, the firm is willing to produce something if and only if $P \geq AVC_{\min}$. Below this threshold no $Q^* > 0$ satisfies the condition.

Key idea

Fixed costs do not affect the decision to produce in the short run because they are **sunk**: incurred and unrecoverable regardless of the choice. The relevant difference is:

$$\pi(Q^*) - \pi(0) = \underbrace{PQ^* - VC(Q^*)}_{\text{contribution margin}} \geq 0$$

which does not contain FC . Producing is preferable to not producing if and only if revenues cover *variable* costs; fixed costs do not enter the calculation.

Example**Restaurant with rent already paid**

A restaurant has already incurred the monthly fixed cost of €5,000 (rent). Two scenarios:

- *If it opens*: revenues €8,000, variable costs €4,000
 $\Rightarrow \pi = 8,000 - 4,000 - 5,000 = -1,000$.
- *If it stays closed*: $\pi = -5,000$.

It pays to open because $-1,000 > -5,000$. The rent is unrecoverable: it does not affect the decision. What matters is $PQ^* - VC(Q^*) = 8,000 - 4,000 = 4,000 > 0$: operating fully covers variable costs and leaves €4,000 as a contribution to fixed costs, reducing the loss from €5,000 to €1,000.

9.5.3 Shutdown in the long run

In the long run all costs are variable: there are no sunk costs. The firm can dismantle plants, not renew contracts, and exit the market freely.

Proposition 9.5 (Shutdown condition — long run). *In the long run, the firm produces if and only if:*

$$P \geq AC_{\min}$$

If $P < AC_{\min}$, the firm exits the market. At the break-even point $P = AC_{\min}$ the firm is indifferent between staying and exiting.

Remark 9.4.

The logic mirrors that of the short run but without sunk fixed costs. In the LR the cost “of

not producing” is not $-FC$ but zero (all resources can be reallocated to their alternative use). Hence the condition becomes: produce iff revenues cover *all* costs, fixed and variable, i.e., iff $P \geq AC_{\min}$.

9.6 The four zones and the short-run supply curve

Proposition 9.6 (The four zones). *Given $AVC_{\min}$ and $AC_{\min}$ (with $AVC_{\min} < AC_{\min}$), the firm’s optimal behavior can be summarized in four zones:*

Zone	Condition on P	SR decision	LR decision
1	$P < AVC_{\min}$	Shutdown ($Q = 0$)	Exits
2	$AVC_{\min} \leq P < AC_{\min}$	Produces ($\pi < 0$)	Exits
3	$P = AC_{\min}$	Produces ($\pi = 0$)	Indifferent
4	$P > AC_{\min}$	Produces ($\pi > 0$)	Stays

Remark 9.5.

In zone 2 the firm makes a loss but it is rational to produce: the loss $\pi(Q^*) = -[AC(Q^*) - P] \cdot Q^*$ is smaller in absolute value than the loss it would incur by not producing ($-FC$). Producing makes it possible to cover at least part of the fixed costs through the contribution margin $PQ^* - VC(Q^*) > 0$.

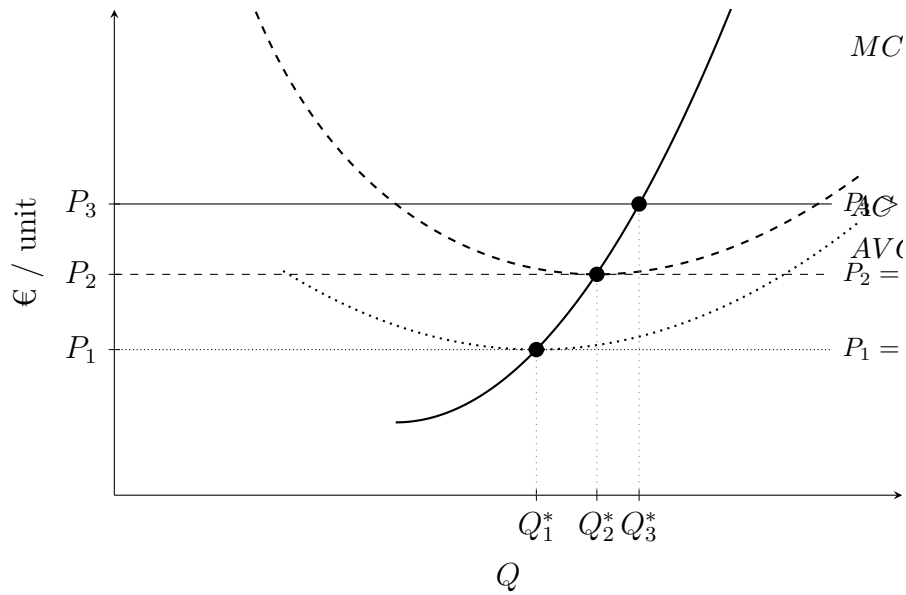


Figure 9.2: The four zones for $C(Q) = Q^3 - 6Q^2 + 15Q + 10$. $AVC_{\min} = 6$ (zone 1/2 boundary, SR shutdown); $AC_{\min} \approx 9.1$ (zone 2/3 boundary, LR break-even). The SR supply curve is the thick solid part of MC above P_1 .

Remark 9.6 (The short-run supply curve—a preview).

Proposition 9.4 implies that the individual SR supply curve is the *increasing* part of the MC curve above $AVC_{\min}$. Formally:

$$Q^s(P) = \begin{cases} Q : MC(Q) = P, MC'(Q) > 0 & \text{if } P \geq AVC_{\min} \\ 0 & \text{if } P < AVC_{\min} \end{cases}$$

This function is the subject of the next chapter.

9.7 Exercises

Exercise 9.1.

A firm has $C(Q) = Q^3 - 6Q^2 + 15Q + 10$.

- Compute MC , AVC and AC . Find $AVC_{\min}$ by minimizing analytically; verify that $MC = AVC$ at the minimum.
- For $P = 9$: solve the FOC $P = MC$, identify both roots, select the maximum using the SOC. Compute $\pi(Q^*)$ and $\pi(0)$. Does the firm produce in the SR?
- Find $AC_{\min}$ numerically by imposing $MC = AC$. Does the firm remain in the market in the LR with $P = 9$?
- Repeat for $P = 12$ and $P = 4$, classifying each case into the four zones of Proposition 9.6.

Exercise 9.2.

A firm has $C(Q) = Q^2 + 4Q + 25$ (with $FC = 25$).

- Compute MC , AVC , $AVC_{\min}$, AC and $AC_{\min}$. (*Careful*: the AVC of this cost is linear and strictly increasing; where is its minimum?)
- Derive the SR supply curve $Q^s(P)$ in analytical form, specifying its domain.
- For $P \in \{6, 10, 14, 18\}$: table with Q^* , π , SR decision and LR decision.
- At what exact price is the firm at the LR break-even point? Verify that at that point $P = MC = AC$.

Exercise 9.3 (Economic profit and the exit decision).

Martina runs a graphic design studio. Annual data: revenues €120,000; explicit costs (rent, software, collaborators) €74,000; alternative salary as an employee €36,000; capital invested €60,000, alternative rate of return 3%.

- Compute π_{acc} and π_{econ} .
- Should Martina continue the business?
- If revenues fell to €105,000, what would the optimal decision be?
- If the fixed cost (rent) were already paid and unrecoverable for the current year, and variable costs were €50,000 against revenues of €105,000, does it pay to open in the SR? Compute $\pi(\text{open})$ and $\pi(\text{closed})$ explicitly.

Exercise 9.4 (Fixed costs and the marginal decision).

A firm has $FC = 300$, $AVC_{\min} = 18$, $AC_{\min} = 30$. The current price is $P = 22$ and the optimal quantity (if it produces) is $Q^* = 15$ with $AC(Q^*) = 40$.

- Which zone is the firm in?
- Compute $\pi(Q^*)$ and $\pi(0)$. Is it correct to produce in the SR?

- c) By how much would Q^* change if fixed costs doubled (with unchanged MC and AVC)? Justify formally using the FOC.
- d) Explain in three lines why a firm with an accounting loss might still find it rational to keep producing in the short run.

Exam essentials

1. Distinguish accounting and economic profit; compute π_{econ} including all opportunity costs.
2. Set up the problem $\max_Q \{PQ - C(Q)\}$ and derive the FOC $P = MC(Q^*)$.
3. Check the SOC: $MC'(Q^*) > 0$ (maximum, not minimum, of profit).
4. With multiple roots of the FOC: select the one with increasing MC ; if necessary, compare profit values.
5. Compute profit as an area: $\pi = [P - AC(Q^*)] \cdot Q^*$.
6. Derive and apply the SR shutdown condition: $P \geq AVC_{\min}$ (fixed costs are sunk).
7. Distinguish SR ($P \geq AVC_{\min}$) from LR ($P \geq AC_{\min}$).
8. Classify the situation into the four zones of Proposition 9.6.

Chapter 10

Firm Supply and Market Supply

10.1 From optimal choice to the supply curve

In the previous chapter we solved the firm's problem for a given price P : find Q^* such that $P = MC(Q^*)$ with $MC''(Q^*) > 0$. Now we take a different conceptual step: we *let P vary* and study how Q^* changes.

Key idea

The individual supply curve is neither a technological relationship nor an arbitrary choice: it is the collection of all optimal quantities the firm chooses at every possible market price. It is the summary result of the profit maximization problem.

Definition 10.1 (Individual short-run supply curve).

The **individual short-run supply curve** is the function $Q^s : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ defined by:

$$Q^s(P) = \begin{cases} Q^* : MC(Q^*) = P, MC''(Q^*) > 0 & \text{if } P \geq AVC_{\min} \\ 0 & \text{if } P < AVC_{\min} \end{cases}$$

Equivalently, its **inverse form** is:

$$P^s(Q) = MC(Q) \quad \text{for } Q \geq Q_{\min}^{AVC}$$

where $Q_{\min}^{AVC}$ is the quantity that minimizes AVC .

Remark 10.1 (Direct form vs. inverse form).

The two forms describe the same curve:

- the **direct form** $Q^s(P)$ is useful in numerical exercises (given the price, find the quantity);
- the **inverse form** $P^s(Q) = MC(Q)$ is useful for graphs with P on the vertical axis and for computing producer surplus.

In inverse form, the supply curve *coincides* with MC in the relevant range.

Common pitfall —

The supply curve does **not** coincide with the whole MC curve. Below $AVC_{\min}$ the firm shuts down: those points of MC do not correspond to quantities actually supplied. Only the increasing part of MC above $AVC_{\min}$ constitutes the supply curve.

10.2 Analytical derivation

10.2.1 Step-by-step procedure

Proposition 10.1 (Checklist for the individual supply curve). *Given a cost function $C(Q) = FC + VC(Q)$:*

1. Compute $MC(Q) = C'(Q)$.
2. Compute $AVC(Q) = VC(Q)/Q$.
3. Find $AVC_{\min}$ by minimizing AVC (i.e., imposing $AVC'(Q) = 0$, or observing where $MC = AVC$).
4. Solve $P = MC(Q)$ for Q .
5. Select the branch with $MC'(Q) > 0$.
6. Impose the domain condition $P \geq AVC_{\min}$.

10.2.2 Case with quadratic cost (simple closed form)

Analytical example

Supply with $C(Q) = Q^2 + 4Q + 9$

1. $MC = 2Q + 4$.
2. $VC = Q^2 + 4Q$, so $AVC = Q + 4$.
3. AVC is strictly increasing ($AVC' = 1 > 0$), so $AVC_{\min} = \lim_{Q \rightarrow 0^+} AVC = 4$ (reached at the intercept).
4. Set $P = 2Q + 4$, hence $Q = (P - 4)/2$.
5. $MC' = 2 > 0$ always: unique solution, already on the increasing branch.
6. Shutdown condition: $P \geq 4$.

Supply curve:

$$Q^s(P) = \frac{P - 4}{2}, \quad P \geq 4.$$

Check at break-even: $AC_{\min}$ is found from $MC = AC$:

$$2Q + 4 = Q + 4 + \frac{9}{Q} \Rightarrow Q^2 = 9 \Rightarrow Q = 3, \quad AC(3) = 3 + 4 + 3 = 10.$$

Hence $AC_{\min} = 10$ and the LR break-even is at $P = 10$, $Q^* = 3$.

10.2.3 Case with cubic cost (multiple roots)

Analytical example

Supply with $C(Q) = Q^3 - 6Q^2 + 15Q + 10$

From the previous chapter: $MC = 3Q^2 - 12Q + 15$, $AVC_{\min} = 6$.

Setting $P = MC$:

$$3Q^2 - 12Q + 15 = P \Rightarrow Q^2 - 4Q + \frac{15 - P}{3} = 0 \Rightarrow Q = 2 \pm \sqrt{\frac{P - 3}{3}}$$

(the root is real for $P \geq 3$).

$MC'(Q) = 6Q - 12 > 0$ iff $Q > 2$, so we select the branch $Q = 2 + \sqrt{(P - 3)/3}$. With the shutdown condition $P \geq 6$:

$$Q^s(P) = 2 + \sqrt{\frac{P - 3}{3}}, \quad P \geq 6.$$

Check: $P = 6 \Rightarrow Q^s = 2 + \sqrt{1} = 3$ (shutdown point).

$P = 15 \Rightarrow Q^s = 2 + \sqrt{4} = 4$. ✓

10.3 Properties of the supply curve

10.3.1 Monotonicity: the supply curve is increasing

Proposition 10.2 (Law of supply). *The individual SR supply curve is strictly increasing:*

$$\frac{dQ^s}{dP} = \frac{1}{MC'(Q^s(P))} > 0.$$

Proof idea

The optimal quantity satisfies the identity $P \equiv MC(Q^s(P))$ for every $P \geq AVC_{\min}$. Differentiating both sides with respect to P :

$$1 = MC'(Q^s(P)) \cdot \frac{dQ^s}{dP} \Rightarrow \frac{dQ^s}{dP} = \frac{1}{MC'(Q^s(P))}.$$

The SOC guarantees $MC'(Q^s) > 0$, hence $dQ^s/dP > 0$. The slope of the supply curve is the inverse of the slope of MC at the optimal point.

Remark 10.2.

This result formalizes what the SOC already suggested: if MC is steeper (rises faster), quantity supplied responds less to price changes; if MC is flatter, the response is larger.

10.3.2 The envelope theorem and the indirect profit function

The envelope theorem applied to profit

Define the **indirect** (or maximized) **profit function**:

$$\Pi^*(P) = \max_{Q \geq 0} \{PQ - C(Q)\} = PQ^s(P) - C(Q^s(P)).$$

The **envelope theorem** states that the derivative of Π^* with respect to the parameter P is computed as if Q were held fixed at its optimal value:

$$\frac{d\Pi^*}{dP} = \frac{\partial}{\partial P} [PQ - C(Q)]_{Q=Q^s(P)} = Q^s(P).$$

Interpretation (Hotelling's Lemma for the competitive firm): the individual supply curve is the derivative of the indirect profit function with respect to price. In other words, a price increase dP raises maximized profit by $Q^s(P) \cdot dP$: each unit produced is worth dP more.

Why the direct proof works:

$$\frac{d\Pi^*}{dP} = Q^s + P \frac{dQ^s}{dP} - C'(Q^s) \frac{dQ^s}{dP} = Q^s + \underbrace{[P - MC(Q^s)]}_{=0 \text{ by FOC}} \frac{dQ^s}{dP} = Q^s(P).$$

The term in brackets vanishes thanks to the FOC: there is no need to know dQ^s/dP . This result has an important consequence: since $\Pi^*(P)$ is convex in P (a maximum of functions linear in P), its derivative $Q^s(P)$ is increasing—recovering the law of supply by another route.

10.4 Movements along the curve and shifts of the curve

Definition 10.2 (Movement along the supply curve).

A **movement along the curve** occurs when the price of the good P changes, leaving the cost function $C(Q)$ unchanged. Quantity supplied changes, but the *curve* stays the same.

Definition 10.3 (Shift of the supply curve).

A **shift of the curve** occurs when the cost function $C(Q)$ changes (because of a change in input prices w , r , or in technology). At every given price, the firm now supplies a different quantity.

Cause	Movement along S	Shift of S
Price of the good P	✓	
Input prices (w , r)		✓
Technological change		✓
Specific tax per unit of output		✓
Increase in fixed cost (SR)		— (no)

Remark 10.3 (Fixed costs do not shift individual SR supply).

In the short run, an increase in fixed cost changes neither MC nor AVC : the FOC $P = MC(Q^*)$

is unchanged. The optimal quantity does not change. Only the level of profit changes (by $-\Delta FC$). In the long run, however, a higher fixed cost implies a higher $AC_{\min}$, which may push some firms to exit the market, reducing *market* supply.

Example

Increase in labor costs

Suppose wages rise, changing MC from $2Q + 4$ to $2Q + 7$.

- Before: $Q^s = (P - 4)/2$.
- After: $Q^{s'} = (P - 7)/2$ (with $P \geq 7$ as the new shutdown threshold).

At every price (for example $P = 10$) the firm supplies less: $Q^s = 3$ becomes $Q^{s'} = 1.5$. The supply curve shifts to the *left*.

10.5 Market supply: horizontal aggregation

10.5.1 The aggregation principle

In perfect competition all firms face the same price P . Each chooses its own optimal quantity $Q_i^s(P)$ independently of the others. The total market quantity is the sum of the individual quantities.

Definition 10.4 (Market supply).

With n firms indexed by $i = 1, \dots, n$, the **market supply curve** is:

$$Q_{\text{mkt}}^s(P) = \sum_{i=1}^n Q_i^s(P)$$

Aggregation is **horizontal**: at each price P we add up the quantities (not the prices).

Remark 10.4.

Aggregation is horizontal because, in the (Q, P) plane with P on the vertical axis, “fixing a price and adding up the quantities” corresponds to moving horizontally from one curve to the other. Prices are not added up.

10.5.2 Identical firms

Proposition 10.3 (Aggregation with n identical firms). *If all firms have the same supply curve $Q^s(P)$ and their number is n :*

$$Q_{\text{mkt}}^s(P) = n \cdot Q^s(P).$$

The inverse form of market supply is obtained by substituting $Q_{\text{mkt}} = nQ^s$: if $P = P^s(Q^s)$, then $P = P^s(Q_{\text{mkt}}/n)$.

Analytical example

10 identical firms with $Q_i^s = (P - 4)/2$

$$Q_{\text{mkt}}^S = 10 \cdot \frac{P - 4}{2} = 5(P - 4) = 5P - 20, \quad P \geq 4.$$

Inverse form: $P = Q_{\text{mkt}}/5 + 4$.

At $P = 10$: a single firm supplies $Q_i^s = 3$; the market supplies $Q^S = 30$. The market curve is flatter ($dQ^S/dP = 5$ versus $dQ_i^s/dP = 1/2$: slope $\times 10$).

10.5.3 Heterogeneous firms and kinks

When firms differ in costs, different shutdown thresholds imply that not all of them produce at the same prices. The market curve then has a **kink** (a point where the slope changes) or a **jump**.

Proposition 10.4 (Kink in the market curve). *Suppose there are two groups: n_A firms of type A with threshold P_A and n_B firms of type B with threshold $P_B > P_A$. Then:*

$$Q_{\text{mkt}}^S(P) = \begin{cases} n_A \cdot Q_A^s(P) & \text{if } P_A \leq P < P_B \\ n_A \cdot Q_A^s(P) + n_B \cdot Q_B^s(P) & \text{if } P \geq P_B \end{cases}$$

If type-B firms enter producing zero output, $Q_B^s(P_B) = 0$ (as when AVC_B is minimized at $Q_B = 0$), a **kink** forms at $P = P_B$: the curve remains continuous but becomes flatter because new producers enter. If instead they enter with $Q_B^s(P_B) > 0$ (minimum of AVC_B at a positive quantity, as with U-shaped average costs), market supply has a **jump** of size $n_B Q_B^s(P_B)$.

Analytical example

50 firms of type A and 30 of type B

Type A: $MC_A = 2Q_A$, $AVC_{\min}^A = 0$, so $Q_A^s = P/2$ for $P > 0$.

Type B: $MC_B = 4Q_B$, $AVC_{\min}^B = 0$, so $Q_B^s = P/4$ for $P > 0$.

Since both types have $AVC_{\min} = 0$, they enter immediately for $P > 0$:

$$Q_{\text{mkt}}^S = 50 \cdot \frac{P}{2} + 30 \cdot \frac{P}{4} = 25P + 7.5P = 32.5P.$$

Inverse form: $P = Q_{\text{mkt}}^S/32.5$.

With heterogeneous firms entering at different prices: if type B instead had $MC_B = 10 + 4Q_B$ (so $AVC_B = 10 + 2Q_B$, with $AVC_{\min}^B = 10$ at $Q_B = 0$), then $Q_B^s = (P - 10)/4$ for $P \geq 10$ and:

$$Q_{\text{mkt}}^S(P) = \begin{cases} 25P & \text{if } 0 < P < 10 \\ 25P + 7.5(P - 10) = 32.5P - 75 & \text{if } P \geq 10 \end{cases}$$

with a kink at $P = 10$ ($Q_{\text{mkt}}^S = 250$): the curve is continuous and the slope dQ^S/dP rises from 25 to 32.5.

Economic intuition

The kink signals a change in the composition of the market: at low prices, only the most efficient firms (type A) produce; once the threshold P_B is passed, the less efficient firms (type B) enter as well, and total supply grows faster because two groups respond simultaneously to the price increase.

10.6 Elasticity of supply

10.6.1 Definition and computation

Definition 10.5 (Price elasticity of supply).

$$\varepsilon_S = \frac{dQ^s}{dP} \cdot \frac{P}{Q^s} = \frac{\% \Delta Q^s}{\% \Delta P} \geq 0.$$

It measures the percentage responsiveness of quantity supplied to a percentage change in price.

Proposition 10.5 (Elasticity and the shape of the curve). 1. For a linear supply curve $Q^s = aP + b$ with $a > 0$:

$$\varepsilon_S = a \cdot \frac{P}{aP + b} = \frac{aP}{Q^s}.$$

Elasticity varies along the curve even though the slope is constant.

2. If $b = 0$ (line through the origin, $Q^s = aP$):

$$\varepsilon_S = a \cdot \frac{P}{aP} = 1 \quad \forall P.$$

Every straight-line supply curve through the origin has constant unit elasticity.

3. Vertical curve ($Q^s = \bar{Q}$): $\varepsilon_S = 0$ (perfectly inelastic supply).

4. Horizontal curve ($P = \bar{P}$): $\varepsilon_S \rightarrow +\infty$ (perfectly elastic supply).

ε_S	Name	Shape of the curve
0	Perfectly inelastic	Vertical
$0 < \varepsilon_S < 1$	Inelastic	Steep
1	Unit elastic	Line through the origin
$\varepsilon_S > 1$	Elastic	Flat
$\rightarrow +\infty$	Perfectly elastic	Horizontal

Analytical example

Computing elasticity with $Q^s = 3P - 12$ ($P \geq 4$)

$dQ^s/dP = 3$.

At $P = 8$: $Q^s = 12$, $\varepsilon_S = 3 \cdot 8/12 = 2$ (elastic).

At $P = 12$: $Q^s = 24$, $\varepsilon_S = 3 \cdot 12/24 = 1.5$ (elastic, but less so).

In general: $\varepsilon_S = 3P/(3P - 12) = P/(P - 4)$. As P increases, $\varepsilon_S \rightarrow 1$ (the curve becomes less elastic moving upward, tending to the behavior of a line through the origin).

Remark 10.5 (Slope and elasticity are not the same thing).

The slope of $Q^s(P)$ is constant for a linear curve, but elasticity varies because it also depends on the ratio P/Q^s . Two curves with the same slope but different intercepts have different elasticities. Confusing slope and elasticity is a frequent mistake to avoid.

10.6.2 Determinants of elasticity

Key idea

The elasticity of supply depends on how easy it is to expand production.

- **High elasticity:** marginal cost rising only slowly; inputs abundant and easy to obtain; unused production capacity; possibility of rapid entry by new firms.
- **Low elasticity:** marginal cost rising steeply; specialized and scarce inputs; plants operating at full capacity; long lead times for building new capacity.

Proposition 10.6 (Short vs. long run). *The elasticity of supply is generally **greater in the long run** than in the short run, for two reasons:*

1. *In the LR all inputs are variable: the long-run MC curve is flatter than the SR one (greater flexibility in the use of inputs).*
2. *In the LR firms can enter or exit the market: a price change induces changes both in each firm's quantity and in the number of active firms.*

10.7 Exercises

Exercise 10.1.

A firm has $C(Q) = 3Q^2 + 6Q + 12$.

- a) Compute MC , AVC , $AVC_{\min}$, AC and $AC_{\min}$.
- b) Derive the supply curve $Q^s(P)$ in analytical form.
- c) Build the table for $P \in \{12, 18, 24, 30\}$: Q^* , π , SR decision, LR decision.
- d) Compute the elasticity of supply at $P = 18$ and $P = 30$. How does ε_S change as P increases?

Exercise 10.2.

The market has 60 firms of type A with $C_A = Q_A^2$ and 40 firms of type B with $C_B = 2Q_B^2$ (both with $AVC_{\min} = 0$).

- a) Derive $Q_A^s(P)$ and $Q_B^s(P)$.
- b) Write market supply $Q^S(P)$ and its inverse form.
- c) Compute Q^S for $P \in \{5, 10, 20\}$.
- d) If the number of type A firms rises to 120, how does the market curve change? Does the curve become flatter or steeper? Compute the new market ε_S at $P = 10$ in the original case and in the modified case.

Exercise 10.3.

The market has 80 firms of type A ($MC_A = 2Q_A$, $AVC_{\min}^A = 0$) and 20 firms of type B ($MC_B = Q_B$, $AVC_{\min}^B = 8$).

- a) Write the individual supplies $Q_A^s(P)$ and $Q_B^s(P)$.
- b) Write market supply $Q^s(P)$, distinguishing the two zones. At what value of P do type B firms enter?
- c) Compute Q^s for $P \in \{4, 8, 16\}$.
- d) Is the elasticity of market supply greater before or after that price? Justify using the formula for ε_S .

Exercise 10.4 (Envelope theorem).

A firm has $C(Q) = Q^2 + 4Q + 9$, so $Q^s(P) = (P - 4)/2$ and $\Pi^*(P) = P(P - 4)/2 - [(P - 4)/2]^2 - 4(P - 4)/2 - 9$.

- a) Simplify the expression for $\Pi^*(P)$ as a function of P .
- b) Compute $d\Pi^*/dP$ directly. Verify that it coincides with $Q^s(P)$.
- c) Compute $d^2\Pi^*/dP^2$. What is its sign? What does it imply about the convexity of Π^* and the monotonicity of Q^s ?

Exercise 10.5 (Elasticity and shifts).

A firm has supply curve $Q^s = 5P - 10$ for $P \geq 2$.

- a) Compute ε_S at $P = 4$ and $P = 10$.
- b) If an increase in variable costs shifts the curve to $Q^{s'} = 5P - 15$ (with $P \geq 3$), how does ε_S change at $P = 10$?
- c) If instead fixed costs increase (with unchanged MC), how does the supply curve change? Justify analytically.

Exam essentials

1. Define the individual SR supply curve as the increasing part of MC above $AVC_{\min}$; derive it by inverting the FOC $P = MC(Q)$.
2. Select the correct branch when $P = MC$ has two solutions (SOC: $MC' > 0$).
3. Distinguish a movement along the curve (change in P) from a shift (change in variable costs or technology).
4. Aggregate individual supplies horizontally to build market supply; handle the case with heterogeneous firms and kinks.
5. Explain why an increase in fixed costs does not shift the individual SR supply curve.
6. Compute the elasticity of supply $\varepsilon_S = (dQ^s/dP) \cdot (P/Q^s)$ and recognize the special cases (line through the origin: $\varepsilon_S = 1$; vertical: $\varepsilon_S = 0$; horizontal: $\varepsilon_S \rightarrow \infty$).
7. Explain why supply is more elastic in the long run than in the short run.

Chapter 11

Competitive Equilibrium, Welfare and Long-Run Adjustment

In the previous chapters we analyzed separately the demand side (the utility-maximizing consumer) and the supply side (the profit-maximizing price-taking firm, with its individual supply curve and its horizontal aggregation to the market). Now we bring them together: where the market demand curve intersects the market supply curve the **competitive equilibrium** is determined, i.e., the pair (P^*, Q^*) that the Walrasian model predicts as the spontaneous outcome of decentralized bargaining. We first study the short-run equilibrium (fixed number of firms), then its welfare effects, then comparative statics with respect to demand and supply shocks, and finally long-run adjustment with *free entry*, which drives economic profit to zero.

11.1 Short-run equilibrium

Definition 11.1 (Competitive equilibrium).

A *competitive equilibrium* is a pair (P^*, Q^*) such that

$$Q^D(P^*) = Q^S(P^*) = Q^*.$$

At the price P^* quantity demanded equals quantity supplied: the market *clears*, in the sense that there is neither excess demand nor excess supply.

In practice, to find the equilibrium we solve the system

$$\begin{cases} Q^D = Q^D(P) \\ Q^S = Q^S(P) \\ Q^D = Q^S \end{cases}$$

substituting the third condition into the first two and obtaining an equation in P .

Analytical example

Let $Q^D = 120 - 4P$ (inverse demand: $P_D = 30 - Q/4$) and $Q^S = 20P$ (inverse supply: $P_S = Q/20$). Imposing $Q^D = Q^S$:

$$120 - 4P = 20P \implies 24P = 120 \implies P^* = 5.$$

The equilibrium quantity is $Q^* = Q^D(5) = 100 = Q^S(5)$. ✓

11.1.1 The adjustment mechanism

The underlying question is: *why* should the market reach precisely (P^*, Q^*) ? The classic answer is the **Walrasian price adjustment mechanism**. If $P > P^*$ sellers cannot sell all their goods (excess supply) and lower the price; if $P < P^*$ buyers compete for the available quantities (excess demand) and the price rises. In both cases P tends to P^* and the process stops only where the market clears.

Example

Continuing with the baseline market of the previous box:

- $P = 3 < P^*$: $Q^D(3) = 108$, $Q^S(3) = 60$. **Excess demand** of 48 units: consumers compete, P rises toward 5.
- $P = 8 > P^*$: $Q^D(8) = 88$, $Q^S(8) = 160$. **Excess supply** of 72 units: producers compete, P falls toward 5.

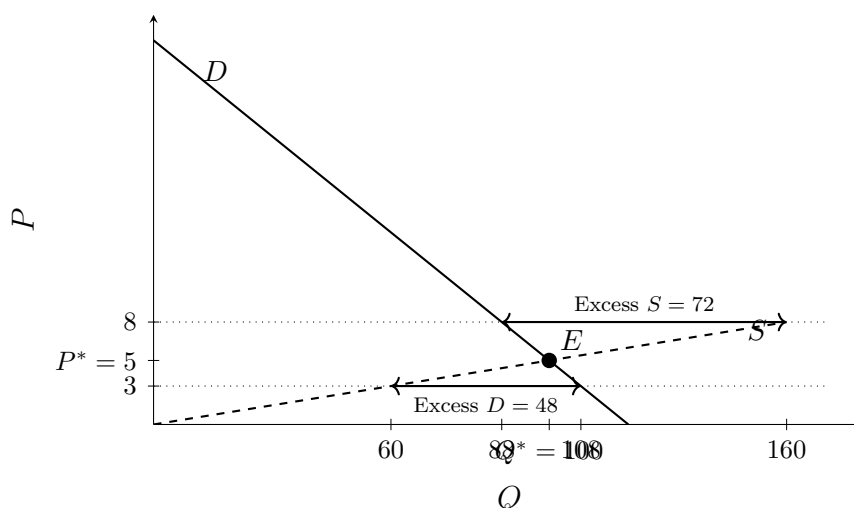


Figure 11.1: Walrasian equilibrium with excesses at out-of-equilibrium prices. Below P^* : excess demand; above P^* : excess supply.

11.1.2 Existence, uniqueness, stability

Under standard conditions— $Q^D(P)$ continuous and decreasing, $Q^S(P)$ continuous and increasing, with the two curves intersecting in the positive orthant—the equilibrium **exists and is unique**. The *stability* of the equilibrium requires that a small perturbation of P triggers forces that bring the price back toward P^* . Formally, the Walrasian stability condition is:

$$\left. \frac{d}{dP} [Q^S(P) - Q^D(P)] \right|_{P^*} > 0.$$

Economic intuition

The stability condition requires that, as soon as P moves upward from P^* , supply grows faster than demand (or demand grows less): excess supply then increases, pushing P down. If instead the supply curve were *backward-bending* and dominated the demand side, there could be multiple or unstable equilibria.

In the rest of the course we always assume the standard conditions: decreasing demand, increasing supply, a unique and stable equilibrium. The exceptions (Giffen goods, labor supplies that bend backward at high wages) are known in the literature but empirically rare in goods markets.

Remark 11.1 (Market clearing in practice).

The Walrasian mechanism is not just a theoretical abstraction. Several real-world institutions work exactly this way: an *auctioneer*, explicit or implicit, updates the price until demand and supply coincide. Examples include: the wholesale fish market (Chioggia, Rialto, Milan) with descending open-outcry auctions; the Italian Treasury auctions of BTPs and BOTs; the *opening auction* of stock exchanges, where the electronic system computes the price that maximizes the volume traded; the day-ahead electricity market (MGP, GME) with hourly prices. In markets without an institutional structure (supermarkets, services) market clearing happens more slowly and in a fragmented way: firms observe inventories and adjust prices and quantities. The long-run outcome is the same.

11.2 Consumer surplus, producer surplus and total welfare

The competitive equilibrium is not just a geometric meeting point: it is the configuration that realizes a *gain from trade* measurable in monetary units. To quantify it we introduce two measures: consumer surplus and producer surplus.

Definition 11.2 (Consumer surplus).

Consumer surplus (CS) is the difference between the amount that consumers would be willing to pay for the units purchased (*willingness to pay*, $WTP = P_D(q)$) and the amount actually paid at the market price P^* :

$$CS = \int_0^{Q^*} [P_D(q) - P^*] dq.$$

Geometrically it is the area between the inverse demand curve P_D and the horizontal line $P = P^*$.

Definition 11.3 (Producer surplus).

Producer surplus (PS) is the difference between the producers' revenue at price P^* and cumulative marginal cost:

$$PS = \int_0^{Q^*} [P^* - P_S(q)] dq.$$

Geometrically it is the area between the horizontal line $P = P^*$ and the inverse supply curve P_S .

Total welfare (*total surplus*) is the sum of the two: $TS = CS + PS$. For *linear* demand and supply, both CS and PS are triangular areas and are computed immediately with the formula $\text{base} \times \text{height}/2$.

Analytical example

For the baseline market ($Q^D = 120 - 4P$, $Q^S = 20P$, equilibrium $P^* = 5$, $Q^* = 100$) the vertical intercept of inverse demand is $P_{\max}^D = 30$. Then:

$$CS = \frac{1}{2} \cdot Q^* \cdot (P_{\max}^D - P^*) = \frac{1}{2} \cdot 100 \cdot 25 = 1,250,$$

$$PS = \frac{1}{2} \cdot Q^* \cdot P^* = \frac{1}{2} \cdot 100 \cdot 5 = 250, \quad TS = CS + PS = 1,500.$$

The asymmetry $CS \gg PS$ reflects the greater steepness of demand relative to supply: consumers capture a larger share of the gain from trade.

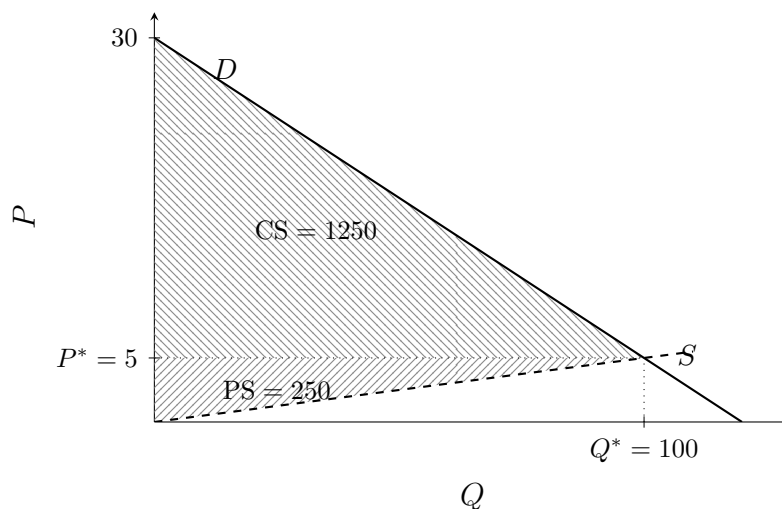


Figure 11.2: Consumer and producer surplus as triangular areas. Left hatching: CS (above P^* , below D). Right hatching: PS (below P^* , above S).

11.2.1 A discrete example: from individual WTP to aggregate surplus

Example 11.1.

Five consumers each want to buy one beer. Their willingness to pay is $WTP = \{10, 8, 6, 4, 2\}$ euros. The market price is $P = 5$ euros. Who buys?

Consumer	WTP	Buys?	Individual surplus
1	10	yes	$10 - 5 = 5$
2	8	yes	$8 - 5 = 3$
3	6	yes	$6 - 5 = 1$
4	4	no	—
5	2	no	—
Total CS			9 euros

Only those with $WTP \geq P$ buy; individual surplus is $WTP - P$. In the continuous limit, the sum becomes an integral and aggregate surplus is the area below the demand curve and above P^* .

The reasoning on the producer side is symmetric.

Example 11.2.

Five firms can each produce one beer at marginal costs $MC = \{1, 2, 3, 4, 5\}$ euros. At the price $P = 4$ euros only the four with $MC \leq 4$ produce, generating an individual surplus $P - MC$:

$$(4 - 1) + (4 - 2) + (4 - 3) + (4 - 4) = 6 \text{ euros.}$$

The fifth firm ($MC = 5$) does not produce. In the continuous limit, PS is the area above the supply curve and below P^* .

Common pitfall PS vs accounting profit

Producer surplus is *not* economic profit. PS equals revenue minus variable costs; it coincides with profit only if fixed costs are zero. A firm can have $PS > 0$ and at the same time $\pi < 0$ (fixed costs erode the surplus to the point of generating an accounting loss).

11.2.2 The First Welfare Theorem

Proposition 11.1 (First Fundamental Theorem of Welfare Economics). *In a perfectly competitive market (price taking, complete information, no externalities or public goods), the competitive equilibrium (P^*, Q^*) **maximizes total welfare** TS.*

Proof idea

At equilibrium, every unit traded satisfies two conditions: (i) a consumer with $WTP \geq P^*$ receives it, so it contributes positively to CS; (ii) a firm with $MC \leq P^*$ produces it, so it contributes positively to PS. All units for which $WTP \geq MC$ are traded; those with $WTP < MC$ are not. In other words, all possible gains from trade are exploited: there is no untraded unit that, if traded, would increase TS, nor a traded unit that, if eliminated, would increase it. Any deviation from (P^*, Q^*) —quantity $\tilde{Q} \neq Q^*$ or price $\tilde{P} \neq P^*$ —generates a welfare *deadweight loss* (DWL), i.e., an area of lost surplus.

Key idea

The competitive equilibrium is Pareto efficient: there are no reallocations that make someone better off without making someone else worse off. Every departure from equilibrium (taxes, administered prices, production quotas, tariffs) generates a **deadweight loss**. This motivates the whole analysis of the next two chapters, in which we will measure the DWL of different forms of public intervention.

11.3 Comparative statics

Comparative statics compares two equilibria—before and after an exogenous shock—without describing the adjustment path. It tells us the **direction** and the **size** of the changes in P^* and Q^* . The procedure is always the same:

1. identify the shock (does it hit demand or supply? in which direction?);
2. compute the new equilibrium by imposing $Q^D = Q^S$ on the post-shock curves;
3. compare (P^{**}, Q^{**}) with (P^*, Q^*) .

The summary scheme for the four simple shocks is:

Shock	Curve	Direction	ΔP^*	ΔQ^*
Increase in demand	D	$\rightarrow$	$\uparrow$	$\uparrow$
Decrease in demand	D	$\leftarrow$	$\downarrow$	$\downarrow$
Increase in supply	S	$\rightarrow$	$\downarrow$	$\uparrow$
Decrease in supply	S	$\leftarrow$	$\uparrow$	$\downarrow$

11.3.1 Demand shocks

Analytical example

Baseline market, plus an increase in income that shifts demand to $Q^{D'} = 140 - 4P$. New equilibrium:

$$140 - 4P = 20P \implies P^{**} = \frac{35}{6} \approx 5.83, \quad Q^{**} = \frac{350}{3} \approx 116.7.$$

Both price and quantity increase: $\Delta P^* > 0$, $\Delta Q^* > 0$, consistently with the summary table.

Remark 11.2 (Empirical example: face masks in March 2020).

Demand for FFP2/surgical masks rose sharply in March 2020 with the COVID-19 emergency. With production capacity rigid in the short run, the price of some categories went from about €0.50 to over €5 in a few weeks. This is the limiting case of a demand shock with very inelastic supply: the entire adjustment goes through the price.

11.3.2 Supply shocks

Analytical example

Same initial demand, but an increase in raw material costs shifts supply to $Q^{S'} = 20P - 80$. New equilibrium:

$$120 - 4P = 20P - 80 \implies P^{**} = \frac{25}{3} \approx 8.33, \quad Q^{**} = \frac{260}{3} \approx 86.7.$$

The price rises, the quantity falls.

Remark 11.3 (Empirical example: wheat in 2022).

Russia and Ukraine together supply about 30% of the world's wheat. The Russian invasion of Ukraine (February 2022) abruptly reduced the exportable share, pushing the FAO wheat index up by +60% in a few months. A negative supply shock, with relatively inelastic demand curves in the short run: prices rose sharply, quantity traded fell.

11.3.3 Simultaneous shocks and ambiguous effects

When *both* curves shift, one of the two effects (ΔP^* or ΔQ^*) has a definite sign, while the other depends on the relative sizes of the shocks.

Simultaneous shocks	ΔQ^*	ΔP^*	Ambiguous
Both to the right ($D \uparrow, S \uparrow$)	$\uparrow$ certain	?	P^*
Both to the left ($D \downarrow, S \downarrow$)	$\downarrow$ certain	?	P^*
Opposite directions ($D \uparrow, S \downarrow$)	?	$\uparrow$ certain	Q^*
Opposite directions ($D \downarrow, S \uparrow$)	?	$\downarrow$ certain	Q^*

11.3.4 Incidence of shocks and elasticity

The same demand shock has very different effects on P^* and Q^* depending on the elasticity of supply. The rule, valid also at the margin: a demand shock *is absorbed by the less elastic curve*.

Economic intuition

If supply is rigid (e.g. the short run in many markets, land in historic city centers), the whole weight of the demand shock falls on the price: ΔP large, ΔQ small. If supply is elastic (e.g. replicable digital services), the whole weight falls on quantities: ΔP small, ΔQ large. Symmetrically, a supply shock is absorbed by the less elastic demand curve.

A paradigmatic case study of simultaneous shocks in opposite directions is the **semiconductor crisis of 2020–22**. Demand for chips grew because of remote working (PCs, laptops, webcams), gaming (new consoles in November 2020) and the transition to electric vehicles. At the same time, supply contracted because of lockdowns in Taiwan and Southeast Asia, the Taiwanese drought of 2021 (chip production requires a lot of water) and the Renesas fire in Japan (March 2021). The documented result: +200% on spot prices for some categories, supply *lead times* going from about 12 to about 52 weeks, an estimated loss of about 7.7 million vehicles in global car production in 2021 (AlixPartners). Consistently with the table: P^* rising sharply, ΔQ^* with uncertain sign (falling for advanced chips, where the supply effect dominated, rising for some standard chips thanks to new capacity).

11.4 Long-run adjustment

So far we have assumed a *fixed number of firms*. In the long run, however, the number of firms is endogenous: if the industry earns more than average, new entrants have an incentive to set up firms; if it earns less, some firms exit. Economic profit acts as a signal.

Key idea

In the canonical model with *free entry*:

- $\pi > 0 \implies$ new firms enter $\implies S$ shifts to the right $\implies P$ falls;
- $\pi < 0 \implies$ some firms exit $\implies S$ shifts to the left $\implies P$ rises;
- $\pi = 0 \implies$ no incentive to enter or exit: **long-run equilibrium**.

11.4.1 The long-run equilibrium condition

Proposition 11.2 (Long-run equilibrium with free entry). *Suppose all firms have the same technology $C(q)$ and that entry and exit are free. In long-run equilibrium:*

$$P_{\text{LR}}^* = AC_{\min}, \quad \pi = 0,$$

and each active firm produces the quantity $q^* \in \arg \min_q AC(q)$, i.e., the efficient scale of production.

Proof idea

If $P > AC_{\min}$ the incumbent firms make $\pi > 0$: new firms enter until aggregate supply shifts to the right enough to bring the price back to $AC_{\min}$. If $P < AC_{\min}$ the active firms make $\pi < 0$: some exit, supply contracts, P rises up to $AC_{\min}$. The process stops only at $P = AC_{\min}$, where economic profit is zero and there is no longer any incentive either to enter or to exit. That each firm produces at the efficient scale follows from combining the FOC $P = MC(q^*)$ and $P = AC_{\min}$, which implies $MC(q^*) = AC(q^*)$; for U -shaped costs this happens exactly at the minimum of AC .

Analytical example

Let $C(q) = q^2 + 4q + 9$ for each firm. We compute the long-run equilibrium with demand $Q^D = 120 - 4P$.

Step 1—efficient scale and LR price. $MC(q) = 2q + 4$, $AC(q) = q + 4 + 9/q$. Imposing $MC = AC$:

$$2q + 4 = q + 4 + \frac{9}{q} \implies q = \frac{9}{q} \implies q^* = 3, \quad P_{\text{LR}}^* = AC(3) = 3 + 4 + 3 = 10.$$

Step 2—market quantity. $Q_{\text{mkt}}^* = Q^D(10) = 120 - 40 = 80$.

Step 3—number of firms. $n^* = Q_{\text{mkt}}^*/q^* = 80/3 \approx 26.7$, rounded to 26 or 27 firms.

Check $\pi = 0$: $\pi = P_{\text{LR}}^* \cdot q^* - C(q^*) = 10 \cdot 3 - (9 + 12 + 9) = 0$. ✓

11.4.2 Market and representative firm: SR and LR compared

It is instructive to place two graphs side by side: one for the market, which determines P^* , and one for the representative firm, which takes P^* as given. The internal consistency of the model requires $Q_{\text{mkt}}^* = n \cdot q^s(P^*)$.

Suppose that in the SR $n = 18$ firms operate, each with $C(q) = q^2 + 4q + 9$. Individual supply is $q^s(P) = (P - 4)/2$ for $P \geq 4$ and market supply is $Q_{\text{SR}}^S = 18 \cdot (P - 4)/2 = 9P - 36$. Setting it equal to demand: $120 - 4P = 9P - 36$, whence $P_{\text{SR}}^* = 12$, $Q_{\text{mkt}}^* = 72$. Each firm produces $q^s(12) = 4$, with $AC(4) = 4 + 4 + 9/4 = 10.25$ and individual profit $\pi = (12 - 10.25) \cdot 4 = 7$. The market is in *short-run* equilibrium, but with $\pi > 0$: in the long run entry is triggered.

In the long run entry brings the number of firms to $n \approx 27$: market supply becomes $Q^S = 27 \cdot (P - 4)/2 = 13.5P - 54$, and the new equilibrium moves to $P_{\text{LR}}^* = 10$, $Q_{\text{mkt}}^* = 80$. Each firm now produces $q^* = 3$ (the efficient scale, minimum of AC) with $AC(3) = 10 = P_{\text{LR}}^*$ and hence $\pi = 0$.

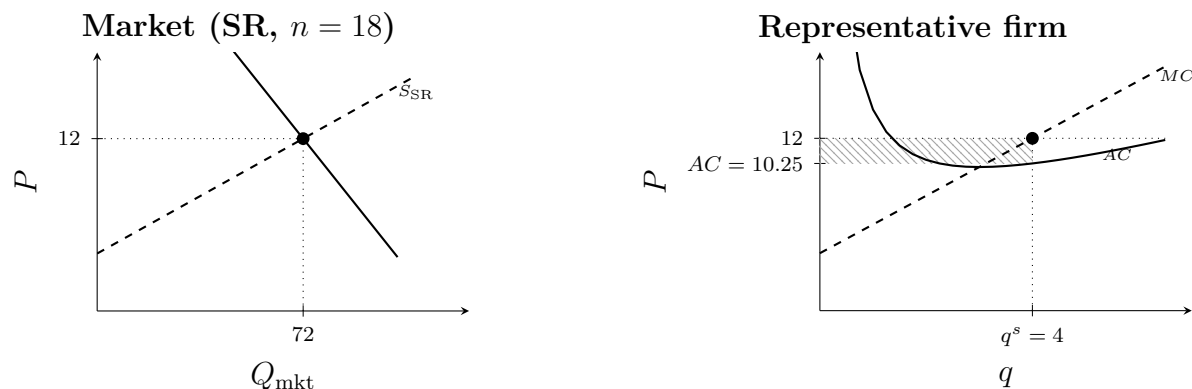


Figure 11.3: Short run with 18 firms. At $P_{\text{SR}}^* = 12$ each firm produces $q^s = 4$ and earns $\pi = 7$ (hatched area on the left of the firm's graph). The market is in SR equilibrium but with positive profits: entry.

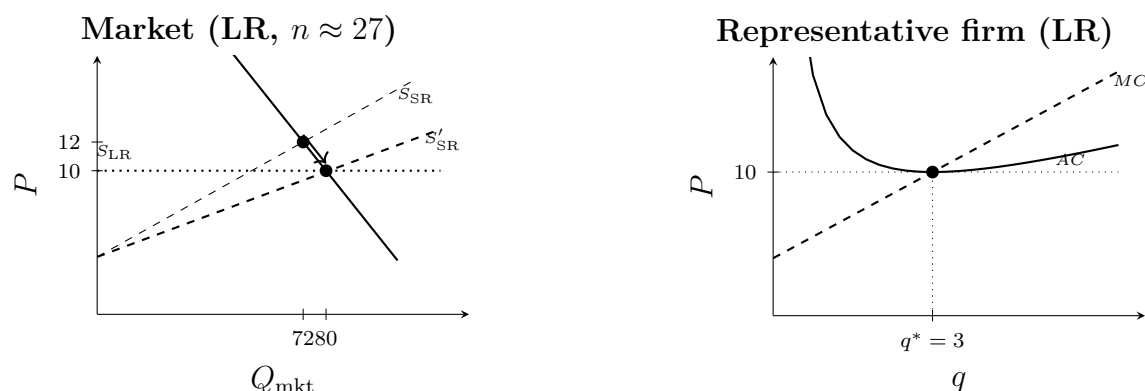


Figure 11.4: Long run: $n \approx 27$ firms, $P_{\text{LR}}^* = AC_{\min} = 10$, each firm operates at the efficient scale $q^* = 3$, $\pi = 0$. Long-run supply is horizontal (constant-cost industry, cf. next section).

11.4.3 Short, medium and long run: what adjusts

The concept of a *period* in microeconomics is not chronological but analytical: it indicates which variables are free to adjust. The calendar time associated with each period varies with the market considered.

Period	What is fixed	What adjusts
Market	n , K , output already produced	price P only
Short (SR)	n , K , technology	P , q and variable inputs
Medium	n , technology	P , q , K
Long (LR)	nothing	P , q , K , n , technology

The calendar time of the LR depends on the industry: a few days for fresh fish, a few months for digital services, 1–3 years for standard manufacturing, 5–15 years for quality wines, cargo ships or nuclear power plants. In the rest of the course we use only the two poles (SR vs LR) and omit the market period and the medium run for simplicity.

11.4.4 A demand shock solved in the SR and in the LR

Analytical example

We start from the LR equilibrium just computed: $P_{LR}^* = 10$, $Q^* = 80$, $n^* = 27$ firms, $q^* = 3$. Demand increases permanently to $Q^{D'} = 140 - 4P$.

Short run. The number of firms stays fixed at $n = 27$: market supply $Q^S = 13.5P - 54$. New equilibrium:

$$140 - 4P = 13.5P - 54 \implies 17.5P = 194 \implies P_{SR}^{**} \approx 11.09.$$

Firms make $\pi > 0$ because $P > AC_{\min} = 10$: entry is triggered.

Long run. With free entry the price returns to $P_{LR}^* = AC_{\min} = 10$ (unchanged). The market quantity becomes $Q_{LR}^{**} = 140 - 40 = 100$ and the number of firms rises to $n^{**} = 100/3 \approx 33$.

Key idea

In a constant-cost industry, a permanent demand shock has *transitory* effects on the price (only in the SR) and *permanent* effects on aggregate quantity and on the number of firms. The LR price is entirely determined by technology ($AC_{\min}$), not by demand.

11.5 The long-run supply curve

The shape of the LR supply curve depends on how input prices change as the size of the industry grows. Let $w(n)$ be the price of the key variable input when n firms are active. The LR price depends on w through $AC_{\min}(w)$:

$$P_{LR}^*(n) = AC_{\min}(w(n)), \quad \frac{dP_{LR}^*}{dn} = \frac{\partial AC_{\min}}{\partial w} \cdot \frac{\partial w}{\partial n}.$$

We distinguish three regimes.

11.5.1 Constant-cost industry

If $\partial w / \partial n = 0$ (the expansion of the industry does not change input prices), $AC_{\min}$ stays unchanged and the LR supply curve is **horizontal** at $P = AC_{\min}$ for every n . The LR elasticity is $\varepsilon_S^{LR} = \infty$: demand shocks are absorbed entirely by quantities, not by prices.

Typical examples: manufacturing industries that use widely available inputs (standard plastics, commodity electronic components). The expansion of a domestic industry does not change the world price of steel or oil.

11.5.2 Increasing-cost industry

If $\partial w / \partial n > 0$, the expansion of the industry drives up the price of specialized inputs, so $AC_{\min}$ rises. The LR curve is **increasing** (but less steep than S_{SR}), with $0 < \varepsilon_S^{LR} < \infty$.

Examples: oil extraction (ever deeper wells), quality wine growing (prime land), skill-intensive industries (rising wages of specialists).

Remark 11.4 (Urban restaurants as an increasing-cost industry).

Restaurants in historic city centers are a textbook case: many independent firms, modest barriers to entry (typical initial capital 50–150 thousand euros), a differentiated but highly substitutable product; annual opening rate $\approx 15\%$ and closing rate $\approx 13\%$, with 60% of new businesses closing within the first 3–5 years. When local demand grows (tourism, gentrification), entry intensifies competition for central commercial space, rents rise and $AC_{\min}$ rises: the equilibrium price of meals settles at a structurally higher level (typical of central Florence or Venice compared with less touristic cities).

11.5.3 Decreasing-cost industry

If $\partial w/\partial n < 0$ (external economies: shared infrastructure, technological spillovers, industrial clusters), $AC_{\min}$ falls as the industry grows and the LR curve is **decreasing**. It is a rare but not nonexistent case: semiconductors (Moore's law), photovoltaics (panel costs reduced by $\approx 90\%$ in 15 years), software with network economies.

Common pitfall Decreasing LR curve $\neq$ demand

A decreasing LR supply curve is not a demand curve. It is a locus of *long-run equilibria* parametrized by the size of the industry: for each level of n , $P_{LR}^*(n)$ is the price that drives economic profit to zero, given the corresponding $AC_{\min}$. The curve moves as n grows, not as a function of the price at which consumers buy.

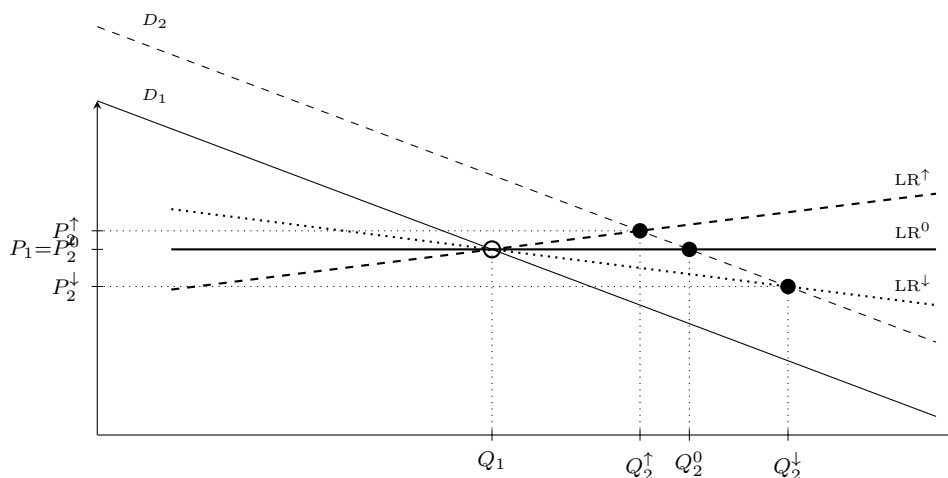


Figure 11.5: Comparison of the three types of LR supply curve after a demand shock ($D_1 \rightarrow D_2$). Constant costs (LR^0 , solid): the price does not change, the whole adjustment goes through quantities. Increasing costs ($LR^\uparrow$, dashed): $\Delta P > 0$, ΔQ dampened. Decreasing costs ($LR^\downarrow$, dotted): $\Delta P < 0$, ΔQ amplified.

Key idea

The shape of the LR supply curve is a structural property of the *input market*, not of the technology of individual firms. All firms have the same cost function $C(q)$; what distinguishes the three regimes is how w (and hence $AC_{\min}$) responds to the expansion of the industry.

11.6 Exercises

Exercise 11.1 (Short-run equilibrium).

Market with $Q^D = 60 - 4P$ and $Q^S = 16P$.

- Find (P^*, Q^*) .
- At $P = 1$: compute the excess demand/supply; indicate in which direction the price moves.
- At $P = 5$: same.
- Demand increases to $Q^{D'} = 80 - 4P$. Find the new equilibrium and comment on the signs of ΔP^* and ΔQ^* .

Exercise 11.2 (Long-run equilibrium).

All firms in the industry have $C(q) = q^2 + 6q + 25$. Market demand: $Q^D = 120 - 2P$.

- Find P_{LR}^* and the efficient scale q^* .
- Find Q_{mkt}^* and the number of firms n^* .
- Demand increases to $Q^{D'} = 160 - 2P$. Find the new LR equilibrium.
- How many firms entered in the transition from Q^D to $Q^{D'}$?

Exercise 11.3 (Qualitative analysis of shocks).

For each shock, indicate: (i) which curve shifts and in which direction; (ii) the effect on P^* and Q^* in the SR; (iii) what happens in the LR in a constant-cost industry.

- The government introduces a tax of 3 euros per unit produced.
- A new technology reduces variable costs by 20%.
- Consumer income falls (normal good).
- The price of a key input doubles.

Exercise 11.4 (Surplus, welfare and comparative statics).

Coffee market: $Q^D = 180 - 6P$ (inverse: $P_D = 30 - Q/6$) and $Q^S = 12P$ (inverse: $P_S = Q/12$).

- Find (P^*, Q^*) and compute CS, PS, TS.
- An increase in income shifts demand to $Q^{D'} = 216 - 6P$. Find the new equilibrium and recompute CS, PS, TS.
- Compute ΔCS , ΔPS , ΔTS . Who gets the larger share of the gain? Explain based on the relative slopes of D and S .

Exercise 11.5 (Demand shock in the LR).

In a constant-cost market demand increases by 30%. Describe the SR and LR adjustment, indicating what happens to P^* , Q_{mkt}^* , n^* , CS and PS.

Exercise 11.6 (Elasticity of supply and incidence of shocks).

Market with $Q^D = 300 - 10P$. Compare two scenarios for supply: (i) inelastic supply $Q^S = 100$ (fixed); (ii) elastic supply $Q^S = 40P$. In which of the two scenarios does a positive demand shock $\Delta Q^D = +50$ produce the larger increase in P^* ? Justify formally by computing the two new equilibria.

Exam essentials

1. Define a competitive equilibrium (P^*, Q^*) as a solution of $Q^D(P^*) = Q^S(P^*)$; compute it by solving the system.
2. Recognize excess demand/supply at out-of-equilibrium prices and predict the direction of Walrasian adjustment.
3. Compute CS and PS as triangular areas for linear demand and supply; obtain $TS = CS + PS$.
4. State and apply the First Fundamental Theorem of Welfare Economics (Prop. 11.1): the competitive equilibrium maximizes TS; any deviation generates a deadweight loss.
5. Apply comparative statics to demand, supply and simultaneous shocks; know the four cases in the table and recognize when an effect is ambiguous.
6. Derive the LR equilibrium with free entry: $P_{LR}^* = AC_{\min}$, q^* efficient scale, $n^* = Q_{\text{mkt}}^*/q^*$, check $\pi = 0$.
7. Distinguish constant-, increasing- and decreasing-cost industries by the sign of $\partial w/\partial n$ and recognize the shape of the LR supply curve in each case.

Chapter 12

Deadweight Loss and Administered Prices

The First Welfare Theorem (Prop. 11.1) states that the competitive equilibrium maximizes total welfare. Consequently, any intervention that moves the quantity traded away from Q^* generates a reduction in welfare, called **deadweight loss** (DWL). In this chapter we introduce the operational formula to compute it and apply it to two classic regulatory instruments: the *price ceiling* (maximum price) and the *price floor* (minimum price). In the next chapter we will see two other families of interventions—taxes and subsidies—that generate DWL with a similar logic.

The baseline model we will work with throughout the chapter is the market

$$Q^D = 80 - 2P \quad (\text{inverse: } P_D = 40 - Q/2), \quad Q^S = 3P \quad (\text{inverse: } P_S = Q/3),$$

whose equilibrium is obtained from $80 - 2P = 3P$, i.e., $P^* = 16$, $Q^* = 48$. The surpluses are $CS = \frac{1}{2} \cdot 48 \cdot (40 - 16) = 576$, $PS = \frac{1}{2} \cdot 48 \cdot 16 = 384$, $TS = 960$.

12.1 Deadweight loss

Definition 12.1 (Deadweight loss, DWL).

Given a market with equilibrium (P^*, Q^*) , the *deadweight loss* associated with a quantity traded $\tilde{Q} \neq Q^*$ is the reduction in total welfare relative to equilibrium:

$$DWL = TS(Q^*) - TS(\tilde{Q}) \geq 0.$$

The intuition is clear: at Q^* all efficient trades—those in which the consumer's willingness to pay exceeds the producer's marginal cost—are carried out. If $\tilde{Q} < Q^*$ is traded, for every unit between $\tilde{Q}$ and Q^* we have $P_D(Q) > P_S(Q)$: those units were worth more to consumers than it cost to produce them, and not producing them makes the gain from trade vanish.

Common pitfall DWL is not a transfer

DWL is not money passing from one party to another: it is value that *disappears*. It does not end up in the pockets of consumers, producers, or the State. Precisely for this reason it is the pure measure of inefficiency.

12.1.1 The triangle formula

With linear demand and supply curves, if the quantity traded falls from Q^* to $\tilde{Q} < Q^*$, the DWL area is a triangle with vertices $(\tilde{Q}, P_S(\tilde{Q}))$, (Q^*, P^*) , $(\tilde{Q}, P_D(\tilde{Q}))$:

$$\text{DWL} = \frac{1}{2} \cdot \underbrace{(Q^* - \tilde{Q})}_{\text{base}} \cdot \underbrace{[P_D(\tilde{Q}) - P_S(\tilde{Q})]}_{\text{height}} \quad (12.1)$$

The *base* measures how far the quantity traded has moved away from the optimum; the *height* measures the wedge between willingness to pay and marginal cost on the last unit traded. When the curves are not linear the triangle formula becomes only an approximation, and the integral definition is more reliable:

$$\text{DWL}(\tilde{Q}) = \int_{\tilde{Q}}^{Q^*} [P_D(q) - P_S(q)] dq.$$

Analytical example

In the reference market, suppose that because of some intervention the quantity traded becomes $\tilde{Q} = 30$. Then:

$$P_D(30) = 40 - 15 = 25, \quad P_S(30) = 30/3 = 10,$$

$$\text{DWL} = \frac{1}{2}(48 - 30)(25 - 10) = \frac{1}{2} \cdot 18 \cdot 15 = 135.$$

Check with the integral:

$$\int_{30}^{48} [(40 - q/2) - q/3] dq = \int_{30}^{48} (40 - 5q/6) dq = [40q - 5q^2/12]_{30}^{48} = 960 - 825 = 135. \quad \checkmark$$

Economic intuition

The height of the DWL triangle is $P_D(\tilde{Q}) - P_S(\tilde{Q})$: with steep (inelastic) curves it grows quickly as $\tilde{Q}$ moves away from Q^* ; with flat (elastic) curves it grows slowly. The policy implication is clear-cut: for the same deviation in quantity, interventions in more elastic markets produce less DWL.

12.2 Price ceiling

Definition 12.2 (Price ceiling).

A *price ceiling* (maximum price) is a price $\bar{P}$ set by law at or below which transactions are legal and above which they are prohibited: no seller may charge a price higher than $\bar{P}$.

The declared motivation is typically consumer protection: controlled rents (*rent control* in Stockholm, New York, Berlin), prices of essential medicines, prices of bread or grain in times of war or famine, regulated electricity tariffs for household use.

Remark 12.1 (A ceiling is binding only if $\bar{P} < P^*$).

If $\bar{P} \geq P^*$, the market operates freely at $P^* \leq \bar{P}$: the ceiling does not bite. The analytical consequences emerge only when $\bar{P} < P^*$, because at that price producers are not willing to sell the quantity Q^* .

12.2.1 Mechanism and excess demand

With $\bar{P} < P^*$: quantity demanded $Q^D(\bar{P})$ exceeds quantity supplied $Q^S(\bar{P})$, and *excess demand* arises. The market cannot allocate more than producers are willing to sell, so the quantity traded is the **short side**:

$$\tilde{Q} = \min\{Q^D(\bar{P}), Q^S(\bar{P})\} = Q^S(\bar{P}) < Q^*.$$

Analytical example

In the reference market, let $\bar{P} = 10 < 16 = P^*$. Then:

$$Q^D(10) = 60, \quad Q^S(10) = 30, \quad \tilde{Q} = 30, \quad \text{excess D} = 60 - 30 = 30.$$

At that price consumers would like to buy 60 units, but producers offer only 30. Thirty units must be allocated among sixty potential buyers: a *rationing problem* arises to which the market, by construction, cannot respond through the price.

12.2.2 Computing CS, PS and DWL

The areas are computed with reference to the effective price $\bar{P}$ and the quantity traded $\tilde{Q}$:

$$CS' = \int_0^{\tilde{Q}} [P_D(q) - \bar{P}] dq, \quad PS' = \int_0^{\tilde{Q}} [\bar{P} - P_S(q)] dq.$$

Analytical example

Continuing with $\bar{P} = 10$, $\tilde{Q} = 30$:

$$CS' = \int_0^{30} (40 - q/2 - 10) dq = \left[30q - q^2/4\right]_0^{30} = 900 - 225 = 675,$$

$$PS' = \frac{1}{2} \cdot 30 \cdot 10 = 150, \quad DWL = \frac{1}{2}(48 - 30)(25 - 10) = 135.$$

Check: $CS' + PS' + DWL = 675 + 150 + 135 = 960 = TS^*$. ✓

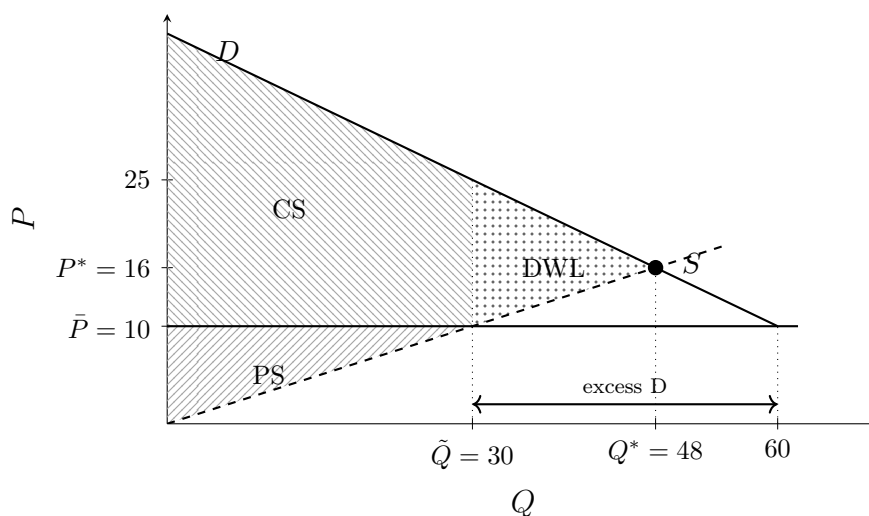


Figure 12.1: Price ceiling at $\bar{P} = 10$. NW hatching: CS (trapezoid with base $\tilde{Q} = 30$); NE hatching: PS (triangle below $\bar{P}$); dotted: DWL (triangle between $\tilde{Q}$ and Q^*).

	Equilibrium ($P^* = 16$)	Ceiling ($\bar{P} = 10$)	Δ
CS	576	675	+99
PS	384	150	−234
TS	960	825	−135
DWL	0	135	

12.2.3 Does CS really increase? The rationing problem

The previous calculation ($\Delta\text{CS} = +99$) contains a fundamental **hidden assumption**: that the 30 available units go to the consumers with the highest willingness to pay, i.e., those with $\text{WTP} \in [25, 40]$. In a market that cannot use the price as an allocation tool this assumption is far from obvious.

Key idea

With a price ceiling the market loses its natural mechanism of allocation through the price. The available units must be allocated *anyway*, but the criterion—queues, lottery, seller discretion, black market, administrative rationing—in no way guarantees that they reach those who value them most.

To illustrate the sensitivity of CS to the allocation criterion, we consider two extremes.

Efficient rationing. The 30 units go to the consumers with the highest WTP. They are the first 30 along the demand curve, with WTP between 25 and 40. They pay $\bar{P} = 10$ and obtain the CS computed above: $\text{CS} = 675$.

Random rationing. The 30 units are allocated to 30 buyers chosen at random among the 60 who would like to buy. The WTPs of the 60 are distributed uniformly between 10 (the last one who would buy at $\bar{P}$) and 40 (the first along demand); the average WTP is therefore $(40 + 10)/2 = 25$, and it remains so on average for any random sample of 30 of them. CS becomes $\text{CS} = 30 \cdot (25 - 10) = 450$, and $\Delta\text{CS} = -126$ relative to equilibrium.

	CS	ΔCS vs equilibrium
Competitive equilibrium	576	—
Ceiling, efficient rationing	675	+99
Ceiling, random rationing	450	−126

The consumers' gain from a price ceiling is therefore largely hypothetical: it exists only if rationing works well. With random allocation (typical in reality: queues, waiting lists, “first come, first served”), total CS can even *fall* below the equilibrium level. The triangular DWL, equal to 135, measures only the loss due to the lower quantity: with random rationing there is an additional misallocation loss equal to $675 - 450 = 225$ (PS remains 150).

12.2.4 Effects not captured: quality, black market, long run

The model with fixed curves and a single price systematically underestimates the distortions of a price ceiling. Among the most relevant effects not captured by the triangle analysis:

- **Quality reduction.** If the price cannot rise, producers can maintain quantity but lower quality (less maintenance in rent-controlled apartments, less freshness in food products, longer delivery times). The “implicit price per unit of quality” returns close to the market value. Documented evidence for *rent control*: Olsen (1972) and Sims (2007) find significant deterioration of the regulated housing stock.
- **Queues and search time.** “First come, first served” rationing turns the monetary price into a time price. Frank (2015) estimates that during the 1973 oil crisis the cost of time lost in gasoline lines exceeded the savings on the price.
- **Black market.** If the gap $P^* - \bar{P}$ is large, illegal transactions emerge that restore market prices, with additional costs of search, legal risk, and information fragmentation.
- **Contraction of long-run supply.** LR supply is typically more elastic than SR supply: over time, the ceiling discourages investment in new capacity. Diamond, McQuade and Qian (AER 2019) estimate that San Francisco’s rent control reduced the supply of regulated rentals by 15% and raised rents in non-regulated areas by 5%.

Key idea

The triangular DWL of the model is a *lower bound* on the social cost of a price ceiling. The real costs are systematically larger, especially in the long run, when supply too has time to contract.

12.2.5 Historical cases of price ceilings

Remark 12.2 (World War II: the *Office of Price Administration*).

In the United States, the OPA (1941–46) imposed ceilings on about 90% of consumer goods to contain *shortage* inflation and free up resources for the war effort. Ration coupons covered sugar, coffee, meat, gasoline, tires. The black market was significant (estimates of 7–15% of actual GDP). Once the war was over, prices rose by 20% in a few months, releasing all at once the supply backlog and pent-up demand.

Remark 12.3 (The 1973 oil crisis).

After the OPEC embargo, the gasoline price ceilings introduced by the Nixon administration in 1971 remained in force (average price ≈ 0.40 USD/gallon in 1973, ≈ 0.53 in 1974). The visible consequences were miles-long lines (waits of several hours at gas stations) and odd-even license plate rationing. Frank (2015) estimates that the costs of time lost in lines exceeded in value the savings on the price.

Remark 12.4 (Rent control: Stockholm, San Francisco, Berlin).

Three recent cases illustrate well the LR effects of rent control. In Stockholm, the average wait for a rent-controlled municipal apartment ranges from 9 to 20 years depending on the neighborhood. In San Francisco, Diamond-McQuade-Qian (AER 2019) document a 15% reduction in the supply of regulated rentals and +5% on non-regulated rents in the same area. In Berlin, the 2020 *Mietendeckel* reduced the rents of covered units by 7–11% but cut the supply of new listings by about 30%; the law was struck down by the German Constitutional Court in 2021.

12.3 Price floor

Definition 12.3 (Price floor).

A *price floor* (minimum price) is a price $\underline{P}$ set by law at or above which transactions are legal and below which they are prohibited: no transaction can take place at a price lower than $\underline{P}$.

The declared motivation is generally the protection of producers: the minimum wage in the labor market, minimum agricultural prices under the European CAP, minimum fees for professional services. As with the ceiling, a floor is binding only if $\underline{P} > P^*$.

12.3.1 Mechanism and excess supply

With $\underline{P} > P^*$ *excess supply* arises: producers want to sell $Q^S(\underline{P})$ but consumers are willing to buy only $Q^D(\underline{P}) < Q^S(\underline{P})$. The short side is now demand:

$$\tilde{Q} = \min\{Q^D(\underline{P}), Q^S(\underline{P})\} = Q^D(\underline{P}) < Q^*.$$

Analytical example

Let $\underline{P} = 25 > 16 = P^*$. Then

$$Q^D(25) = 30, \quad Q^S(25) = 75, \quad \tilde{Q} = 30, \quad \text{excess S} = 75 - 30 = 45.$$

Forty-five units remain unsold: in goods markets, they are accumulated surpluses (the historic CAP “butter mountains”, then bought up by the State); in the labor market, they are involuntary unemployment.

12.3.2 Computing CS, PS and DWL

Analytical example

With $\underline{P} = 25$, $\tilde{Q} = 30$:

$$CS = \frac{1}{2} \cdot 30 \cdot (40 - 25) = 225,$$

$$PS = \int_0^{30} (25 - q/3) dq = \left[25q - q^2/6 \right]_0^{30} = 750 - 150 = 600,$$

$$DWL = \frac{1}{2}(48 - 30)(25 - 10) = 135.$$

(Note that $P_S(30) = 10$, so the height of the DWL triangle is identical to the ceiling case.)
Check: $CS + PS + DWL = 225 + 600 + 135 = 960$. ✓

	Equilibrium ($P^* = 16$)	Floor ($\underline{P} = 25$)	Δ
CS	576	225	-351
PS	384	600	+216
TS	960	825	-135
DWL	0	135	

Key idea: Ceiling/floor symmetry

The price ceiling $\bar{P} = 10$ and the price floor $\underline{P} = 25$ produce the *same* $\tilde{Q} = 30$ and the *same* $DWL = 135$, because the welfare loss depends only on the deviation of quantity from equilibrium, not on which side of the market is rationed. What changes is the **distribution**: the ceiling redistributes from producers to consumers (with the limits discussed above), the floor from consumers to producers.

12.3.3 Main applications

Remark 12.5 (Minimum wage: empirical evidence).

The empirical debate on the minimum wage is one of the liveliest in labor economics. Card and Krueger (AER 1994) study the fast-food market in New Jersey after an increase in the minimum and find no significant effect on employment. Cengiz, Dube, Lindner and Zipperer (QJE 2019) use 138 US state-level changes and estimate that moderate increases (up to 59% of the median wage) do not reduce aggregate employment. Neumark and Wascher instead find negative effects, particularly on young and low-skilled workers. The apparent contradiction with the standard model has three complementary explanations: (i) *monopsony* in the labor market (an employer with market power already hires less than L^* ; a moderate minimum wage can increase employment); (ii) aggregate income effects; (iii) pass-through to final prices rather than to employment.

Remark 12.6 (CAP: guaranteed prices before 1992).

The European Common Agricultural Policy (1957–92) guaranteed minimum prices for cereals, milk, sugar and beef. If the market price fell below $\underline{P}$, the Community bought up the unsold output; import tariffs prevented world prices from pushing down the domestic price. The result was the famous “butter mountains” (over 1 Mt stockpiled in the 1980s), “wine lakes” (up to 3 billion liters), and millions of tonnes of wheat, poultry and sugar withdrawn from the market, at estimated costs of 10–15 billion euros per year. The MacSharry reform (1992) cut guaranteed prices and compensated farmers with per-hectare payments still tied to crops; the Fischler reform (2003) introduced decoupled direct payments (per hectare, independent of production): equivalent to lump-sum transfers, less distortionary but politically more resilient.

12.3.4 Long-run adjustment

As with the ceiling, the floor too produces LR effects more severe than SR ones. In the labor market: in the SR supply is relatively inelastic (it is hard to change jobs quickly) and the floor is mainly a transfer from employers to the workers who remain employed. In the LR labor supply becomes more elastic (choices of education, migration, participation), and employers invest in automation, substituting capital for expensive labor. The LR reduction in employment is structural, and the DWL grows over time. A similar logic applies to rent control: in the SR the housing stock is almost fixed, but in the LR nobody builds new apartments subject to regulation.

12.4 Application: the 2022–23 energy price caps

The most relevant recent case of a large-scale price ceiling is the European response to the 2022 gas shock. Russia’s invasion of Ukraine in February 2022 multiplied the European wholesale gas

price by about ten in a few months: over 300 €/MWh in August 2022, against 20–25 €/MWh before the crisis. Three typical responses:

- **Italy:** “Mercato Tutelato” (protected market) with frozen prices for vulnerable households, at a cost of over 60 billion euros in 2022–23.
- **France:** *bouclier tarifaire* with a cap at +4% for households, for about 45 billion euros.
- **Germany:** *Gaspreisbremse* with a maximum price of 12 cents/kWh for up to 80% of *baseline* consumption, for about 200 billion euros.

Read through the model of this chapter: CS was protected relative to the free market with x10 prices; the PS of suppliers (Eni, Enel, etc.) was limited but offset by state subsidies; the fiscal cost was financed by windfall-profit taxation and public debt; the DWL manifested itself mainly in the under-incentive to use energy efficiently and in the reduced relevance of price signals for investment in renewables.

Key idea

Temporary price caps can be justified as a stabilization tool when the shock is considered transitory and the market cannot readjust quickly. If prolonged into the long run, however, they block precisely the incentives for energy efficiency and renewables that would be needed to prevent the next crisis: a specific form of the short-run/long-run tension seen above.

12.5 When is price control justified?

The competitive model leads to an apparently clear-cut conclusion: ceilings and floors generate DWL, hence they reduce total welfare. In practice, however, many countries use them. Four situations in which intervention can be rational:

1. **Non-competitive market.** If one of the parties has market power (monopoly, monopsony, cartel), the equilibrium price is not P^* but a distorted level. A moderate ceiling can *increase* efficiency by bringing the price back toward marginal cost. We will study this formally in the chapter on monopoly.
2. **Externalities or merit goods.** If the good generates social benefits undervalued by the market (vaccines, education, decent housing), the market price under-incentivizes consumption; a well-sized subsidy (or, for goods with social costs, a floor) can correct the distortion.
3. **Distributional constraints.** If society considers the market allocation unacceptable (lack of access to essential goods for the poor, wages below the subsistence threshold), it can consciously choose to bear a DWL in exchange for a gain in equity.
4. **Temporary shocks.** In crises (war, pandemic, sudden scarcity), the market price would rise to politically unsustainable levels; a temporary ceiling accompanied by explicit rationing may be preferable.

Key idea

DWL analysis does not say “never intervene”. It says: *measure* the cost of the intervention and compare it with the benefit (correction of a market failure, equity, stability). Intervention is justified when the benefit exceeds the DWL—an assessment that is ultimately political, not technical.

12.6 Exercises

Exercise 12.1 (Price ceiling).

Market with $Q^D = 80 - 2P$, $Q^S = 3P$. Equilibrium: $P^* = 16$, $Q^* = 48$, $TS = 960$. The government imposes $\bar{P} = 12$.

- Determine $Q^D(\bar{P})$, $Q^S(\bar{P})$ and the quantity traded $\tilde{Q}$. Is there excess demand or excess supply?
- Compute $P_D(\tilde{Q})$ and $P_S(\tilde{Q})$.
- Compute CS, PS and DWL in the presence of the ceiling.
- Compare with equilibrium: who gains, who loses, and how does the identity $\Delta CS + \Delta PS + DWL = 0$ close?

Exercise 12.2 (Price floor).

Same market. The government imposes $\underline{P} = 22$.

- Determine $Q^D(\underline{P})$, $Q^S(\underline{P})$ and the quantity traded $\tilde{Q}$. Excess demand or excess supply?
- Compute $P_D(\tilde{Q})$ and $P_S(\tilde{Q})$.
- Compute CS, PS and DWL in the presence of the floor.
- Compare the DWL with that of Exercise 1. Why do they coincide?

Exercise 12.3 (True/false with justification).

State whether each statement is true or false, justifying your answer.

- “A price ceiling always helps all consumers.”
- “A price floor in the labor market always reduces unemployment.”
- “If the DWL is zero, there has been no distributional effect.”
- “A price ceiling at a level $\bar{P} > P^*$ has no effect on the market.”

Exercise 12.4 (Minimum wage).

Low-skill labor market: demand $L^D = 600 - 10w$, supply $L^S = 20w$, with w the hourly wage in euros.

- Find (w^*, L^*) .
- The government introduces $\underline{w} = 24$. How many workers are employed? How many are involuntarily unemployed?

- c) Compute the surplus of employed workers, the surplus of firms and the DWL in the new situation.
- d) Does the minimum wage improve workers' welfare in the aggregate? Decompose it into *insiders* (those who remain employed) and *outsiders* (those who lose their job).

Exercise 12.5 (Elasticity and the size of the DWL).

For the same deviation of quantity from equilibrium, does a price ceiling produce a larger DWL in a market with very elastic or very inelastic supply? Justify with a graphical argument based on the height of the triangle.

Exercise 12.6 (DWL in the long run).

Explain why the DWL of a price ceiling (or floor) tends to grow in the transition from the short to the long run. Give two distinct reasons, linking them to the response of supply and demand.

Exam essentials

1. Define the DWL as $TS(Q^*) - TS(\tilde{Q})$; compute it with the triangle formula (12.1) for linear curves.
2. Identify, given $\bar{P}$ or $\underline{P}$, the short side of the market and the quantity traded $\tilde{Q}$.
3. Compute CS, PS and DWL in the two cases (ceiling and floor) and verify the conservation identity $CS + PS + DWL = TS^*$.
4. Recognize the ceiling/floor symmetry for a given $\tilde{Q}$ and explain that the difference lies in distribution, not in the efficiency loss.
5. Apply the reasoning on rationing (efficient vs random) to qualify the conclusions about the CS of a ceiling.
6. Recognize the effects not captured by the model: quality, black market, contraction of LR supply; interpret the triangular DWL as a lower bound on the real social cost.
7. List the four cases in which a price intervention can be rationally justified.

Chapter 13

Taxes and Subsidies

Continuing the analysis of administered prices, in this chapter we study two other instruments through which the State intervenes in the market: *taxes* and *subsidies*. Unlike ceilings and floors, which act directly on the price and generate rationing, taxes introduce a *wedge* between the price paid by the buyer and the price received by the seller; subsidies are their mirror image. In both cases, the quantity traded moves away from Q^* and a deadweight loss arises. The chapter's reflection is threefold: *how* the intervention distorts, *how much* it costs in terms of DWL, and *who* really bears its economic burden.

The reference market is the same as in the previous chapter: $Q^D = 80 - 2P$, $Q^S = 3P$, equilibrium $P^* = 16$, $Q^* = 48$, $TS = 960$.

13.1 Three types of tax

Definition 13.1 (Three classes of indirect taxes). • *Specific tax (excise tax)*: a fixed amount t for each unit traded. Examples: excise duties on gasoline (euros/liter), tobacco (euros/pack).

- *Ad valorem tax*: a rate τ applied to the price; the amount collected varies with the price. Examples: VAT at 22%, percentage tariffs.
- *Lump-sum tax*: a fixed amount independent of quantity and price. Examples: license fees, flat-rate wealth taxes.

Key idea

From the efficiency standpoint, there is only one key difference: a lump-sum tax does not change MC , so it does not alter the choice of Q^* and generates $DWL = 0$; specific and ad valorem taxes change MC (and consequently the supply curve), distort Q^* and generate $DWL > 0$.

Remark 13.1 (The British Poll Tax).

The efficiency of a lump-sum tax is maximal, but its political acceptability may be nil. In 1989 the Thatcher government introduced the *Community Charge*, a fixed tax per adult independent of income and wealth. Analytically highly efficient ($DWL = 0$), it was perceived as unfair and regressive (a poor citizen paid the same as a rich one). After violent protests it was abolished in 1991. Moral: technical Pareto efficiency is not the only criterion for evaluating public policies, and the efficiency–equity *trade-off* returns in almost all *public finance* choices.

For the rest of the chapter we work almost exclusively with specific taxes, because they expose the incidence mechanism more transparently. The ad valorem tax is analytically equivalent (we will see this at the end).

13.2 Specific tax

13.2.1 Effect on the supply curve and the new equilibrium

Consider a specific tax t levied on the producer. For each unit sold at the price P_d (paid by the buyer), the producer pays t to the State and keeps $P_s = P_d - t$. It will agree to produce only if $P_d - t \geq MC(Q)$, i.e.,

$$P_d \geq MC(Q) + t.$$

The supply curve, expressed as a function of the price paid by the buyer, **shifts vertically upward by t** : to produce the same quantity, producers now require t more.

Analytical example

With $t = 5$ in the reference market, the new supply is $Q_t^S(P_d) = 3(P_d - 5) = 3P_d - 15$.
Imposing $Q^D = Q_t^S$:

$$80 - 2P_d = 3P_d - 15 \implies 95 = 5P_d \implies P_d^* = 19, P_s^* = 14, Q_t^* = 42.$$

Check: $Q^D(19) = 42 = Q_t^S = 3 \cdot 14 = 42$. ✓

13.2.2 The areas: CS, PS, tax revenue, DWL

Unlike a ceiling or floor, with a tax a fourth area emerges besides CS, PS and DWL: the *tax revenue* rectangle $T = t \cdot Q_t^*$. This area is a transfer to the State, not a net loss: it must be distinguished from the DWL triangle, which instead is value that vanishes.

Analytical example

With $t = 5$, $P_d^* = 19$, $P_s^* = 14$, $Q_t^* = 42$:

$$CS = \frac{1}{2} \cdot 42 \cdot (40 - 19) = 441, \quad PS = \frac{1}{2} \cdot 42 \cdot 14 = 294,$$

$$T = 5 \cdot 42 = 210, \quad DWL = \frac{1}{2}(48 - 42) \cdot 5 = 15.$$

Check of the conservation identity:

$$CS + PS + T + DWL = 441 + 294 + 210 + 15 = 960 = TS^*. \checkmark$$

The changes relative to the initial equilibrium are:

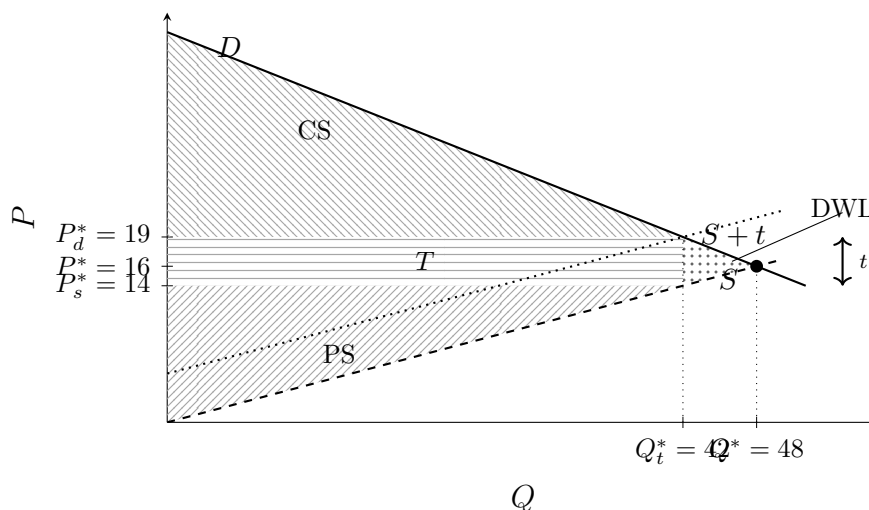


Figure 13.1: Specific tax $t = 5$. NW hatching: CS; horizontal: revenue $T = t \cdot Q_t^*$; NE hatching: PS; dotted: DWL. The tax shifts supply from S to $S + t$ (dotted line).

	Equilibrium	Tax $t = 5$	Δ
CS	576	441	-135
PS	384	294	-90
Revenue T	0	+210	+210
DWL	0	+15	+15
Net TS	960	945	-15

Note the decomposition: consumers lose 135, producers 90; the total private loss is 225. Of these 225, 210 end up in the coffers of the State as a transfer and 15 vanish as DWL.

13.3 Tax incidence

A tax is *legally* levied on one party (the producer, in our example), but the market redistributes its burden between buyers and sellers through price adjustments. The distinction between who remits the tax to the State and who economically *bears* it is the heart of incidence analysis.

Definition 13.2 (Tax incidence).

Tax incidence measures how the economic burden of the tax is distributed between consumers and producers, in terms of the change in the prices paid and received:

$$\text{Cons. share} = \frac{\Delta P_d}{t} = \frac{P_d^* - P^*}{t}, \quad \text{Prod. share} = \frac{|\Delta P_s|}{t} = \frac{P^* - P_s^*}{t}.$$

By construction the two shares sum to 1.

Analytical example

In our example: $\Delta P_d = 19 - 16 = 3$, $|\Delta P_s| = 16 - 14 = 2$. Hence

$$\text{Cons. share} = \frac{3}{5} = 60\%, \quad \text{Prod. share} = \frac{2}{5} = 40\%.$$

Consumers bear 60% of the tax, even though it is formally the producer who remits it to the State.

13.3.1 The Equivalence Principle

Proposition 13.1 (Equivalence Principle). *Given a specific tax t , the final equilibrium (P_d^*, P_s^*, Q_t^*) and all the areas (CS , PS , T , DWL) are identical regardless of whether the tax is legally levied on the seller or on the buyer.*

Proof idea

Taxing the *seller* (the narrative case), the new supply is $Q_t^S(P_d) = Q^S(P_d - t)$. Taxing the *buyer*, they pay P_s to the seller and t to the State: their demand, as a function of what the seller receives, becomes $Q_t^D(P_s) = Q^D(P_s + t)$. In both cases the equilibrium relationship is the same: $P_d^* - P_s^* = t$ and $Q^D(P_d^*) = Q^S(P_s^*)$. The equations are identical, hence so is the solution. In particular, in our market with $t = 5$ on the buyer:

$$Q_t^D = 80 - 2(P_s + 5) = 70 - 2P_s; \text{ equilibrium } 70 - 2P_s = 3P_s \Rightarrow P_s^* = 14, P_d^* = 19, \\ Q_t^* = 42 \text{—identical to the previous case.}$$

Key idea

Tax incidence does not depend on who remits the tax to the State, but only on the elasticities of demand and supply at equilibrium. The legal formulation of the tax (on the seller or on the buyer) is economically irrelevant; the market automatically “shifts” the burden of the tax along the slopes of the curves.

13.3.2 Incidence and elasticity: the general formula

For small changes, the definitions of elasticity give $\Delta P_d \approx \frac{\Delta Q/Q^*}{\varepsilon_D} \cdot P^*$ and $\Delta P_s \approx \frac{\Delta Q/Q^*}{\varepsilon_S} \cdot P^*$. The ratio in absolute value is

$$\frac{\Delta P_d}{|\Delta P_s|} = \frac{\varepsilon_S}{|\varepsilon_D|} = \frac{\varepsilon_S}{-\varepsilon_D}.$$

Combined with $\Delta P_d + |\Delta P_s| = t$, this yields the compact incidence formula:

$$\boxed{\text{Cons. share} = \frac{\varepsilon_S}{\varepsilon_S - \varepsilon_D}, \quad \text{Prod. share} = \frac{-\varepsilon_D}{\varepsilon_S - \varepsilon_D}.} \quad (13.1)$$

(Recall $\varepsilon_D < 0$ and $\varepsilon_S > 0$; the signs ensure positive shares that sum to 1.)

Analytical example

In our market, at the equilibrium $(P^*, Q^*) = (16, 48)$:

$$\varepsilon_D = \frac{dQ^D}{dP} \cdot \frac{P}{Q} = -2 \cdot \frac{16}{48} = -\frac{2}{3}, \quad \varepsilon_S = 3 \cdot \frac{16}{48} = 1.$$

Cons. share = $1/(1 - (-2/3)) = 1/(5/3) = 3/5 = 60\%$, consistent with the point calculation.

✓

Economic intuition

Whoever has the *steeper* (more inelastic) curve bears the larger share. If D is nearly vertical, the consumer “has no alternatives”: they cannot reduce the quantity appreciably as the price rises, and absorb almost all of the tax. Conversely, if S is nearly vertical (fixed supply, like urban land), the producer absorbs almost all of the tax.

The four extreme cases are particularly instructive:

- $\varepsilon_D = 0$ (perfectly inelastic demand): **100%** on consumers. The quantity does not change and the whole adjustment goes through the gross price.
- $\varepsilon_S \rightarrow \infty$ (perfectly elastic supply): **100%** on consumers. Supply is available at a single price, which does not fall; the tax is added entirely on top.
- $\varepsilon_S = 0$ (perfectly inelastic supply): **100%** on producers. This is the classic case of land taxation (Henry George’s argument): land can be neither produced nor destroyed, and a tax on land falls entirely on the owners.
- $\varepsilon_D \rightarrow -\infty$ (perfectly elastic demand): **100%** on producers. Consumers have perfect substitutes and accept no price increase; the entire burden falls on sellers.

13.3.3 Applications: excise duties and the tax wedge

Remark 13.2 (Excise duties on tobacco and gasoline).

Both markets have highly inelastic demand (physiological dependence for tobacco, low short-run substitutability for gasoline) and very elastic supply (production is easily expandable). For tobacco: $|\varepsilon_D| \approx 0.3\text{--}0.5$ for adults (up to ≈ 1 for adolescents, who are more price-sensitive); $\varepsilon_S \approx 5$ in the long run. Consumer share: $5/(5 + 0.4) \approx 93\%$. In Italy the excise duty on tobacco accounts for about 58% of the retail price, and Bank of Italy studies find almost complete *pass-through* to the consumer. A typical normative lesson: taxing inelastic goods is efficient—minimum DWL for a given revenue—but tends to be regressive, because inelastic goods weigh more in the basket of low-income households.

Remark 13.3 (The tax wedge on labor).

The Italian labor market is a textbook example of direct application of incidence theory. For a gross salary of 30,000 euros, a typical breakdown is: labor cost to the firm $\approx 40,000$ euros (of which $\approx 10,000$ are employer INPS social security contributions); gross pay 30,000 euros (of which $\approx 6,000$ are IRPEF income tax and local surcharges and $\approx 3,000$ employee INPS contributions); net pay to the worker $\approx 21,000$ euros. The total wedge is $\approx 19,000$ euros, about 47% of the labor cost. In microeconomic terms: $P_d^* = \text{cost to the firm} = 40\text{k}$, $P_s^* = \text{net to the worker} = 21\text{k}$, wedge $t = 19\text{k}$. The real incidence depends on the elasticities: with labor supply inelastic in the SR (hard to change jobs in a short time) the worker bears almost all of the wedge;

in the LR supply is more elastic (participation, migration, education choices) and the incidence is redistributed. Italy has one of the highest wedges in the OECD (47% against an average of 35%), with estimated effects of reducing labor force participation by 3–5 percentage points.

13.4 DWL and the tax rate: the quadratic relationship

Proposition 13.2 (DWL grows with the square of the tax rate). *With linear demand and supply curves, the deadweight loss from a specific tax t satisfies*

$$\text{DWL}(t) \propto t^2.$$

In particular, doubling the tax rate quadruples the DWL.

Proof idea

When t doubles, the reduction in quantity $\Delta Q = Q^* - Q_t^*$ doubles (by the linearity of the curves), and so does the height of the triangle $P_D(\tilde{Q}) - P_S(\tilde{Q})$, which is exactly t . The DWL is $\frac{1}{2} \cdot \Delta Q \cdot t \propto t \cdot t = t^2$.

Analytical example

In the reference market:

t	Q_t^*	$Q^* - Q_t^*$	DWL
2.5	45	3	3.75
5	42	6	15
10	36	12	60
20	24	24	240

The ratio $\text{DWL}(t)/\text{DWL}(2.5)$ is exactly $(t/2.5)^2$. Doubling t from 5 to 10 raises the DWL from 15 to 60, quadrupling it.

Key idea

The quadratic relationship implies that, to raise a given revenue $\bar{T}$, it is more efficient to impose many small taxes on different bases than a single high tax on a single base. The “*broad base, low rate*” principle is one of the cornerstones of the theory of optimal taxation. Moreover, the DWL, proportional to t^2 , grows faster than revenue (proportional to t for small t): the “fiscal productivity” decreases as the tax rate increases.

13.5 Ad valorem tax

An ad valorem tax collects a share τ of the gross price P_d : the producer keeps $(1 - \tau)P_d$ and agrees to produce only if $(1 - \tau)P_d \geq MC(Q)$. Unlike the specific tax, which shifts the supply curve, the ad valorem tax *rotates* it: the vertical distance between the original S and S_τ increases linearly with the price (small at low prices, large at high prices).

Proposition 13.3 (Specific/ad valorem equivalence). *For every specific tax t that produces an equilibrium with price P_d^* , there exists an ad valorem rate $\tau^* = t/P_d^*$ that generates the same final equilibrium. The areas CS , PS , revenue and DWL are identical in the two cases.*

Analytical example

With $t = 5$ we found $P_d^* = 19$, $P_s^* = 14$, $Q_t^* = 42$. The equivalent ad valorem rate is $\tau^* = 5/19 \approx 26.3\%$. Check: the producer keeps $(1 - \tau^*) \cdot 19 = 0.737 \cdot 19 \approx 14 = P_s^*$. ✓

Remark 13.4 (VAT and pass-through).

Italian VAT at 22% is an ad valorem tax on the consumer: on a net price P_s of 100 euros, the consumer pays $P_d = P_s(1 + \tau) = 122$ euros. The effective pass-through depends on relative elasticity. For goods with inelastic demand (subscriptions, energy), the producer passes almost all of the VAT on to the consumer; for goods with elastic demand, the producer may absorb part of it. The two practical advantages of the ad valorem tax over the specific tax are: (i) automatic indexation to inflation (if prices rise by 10%, nominal revenue rises proportionally without the need to update the law); (ii) greater equity when the market is heterogeneous (a uniform rate produces revenue that increases with the price).

13.6 Tariffs and import quotas

States typically restrict imports with two instruments:

- **Tariff:** a specific or ad valorem tax on imported goods. Formally identical to a specific tax, and it is analyzed with the tools already seen.
- **Import quota:** a fixed cap $\bar{Q}$ on the quantity that can enter the country. Whoever obtains the import license captures a *rent*.

Analytical example

In the reference market, with domestic demand $Q^D = 80 - 2P$ and supply from domestic producers $Q^S = 3P$, the equilibrium without restrictions is $P^* = 16$, $Q^* = 48$ (with $CS = 576$ and $PS = 384$). Consider two alternative interventions that limit the quantity traded to 36 units:

- Specific tariff $t = 10$ per unit.
- Quota $\bar{Q} = 36$.

In both cases the price paid by consumers rises to $P_d = 22$ (from $80 - 2P_d = 36$), while the price at which domestic producers would sell the 36th unit is P_s such that $3P_s = 36$, i.e., $P_s = 12$: the wedge of $t = 10$ per unit separates the two prices. The quantity traded falls to 36, with $CS = \frac{1}{2}(40 - 22) \cdot 36 = 324$ and $PS = \frac{1}{2} \cdot 12 \cdot 36 = 216$; the DWL is the triangle $\frac{1}{2} \cdot 10 \cdot (48 - 36) = 60$ in both cases. The only difference is *who captures the rectangle* $360 = 10 \cdot 36$ between $P_s = 12$ and $P_d = 22$: with the tariff it is the State (revenue $T = 360$); with the quota it is the license holders (rent $R = 360$), typically foreign producers or the intermediaries who obtained the import license.

Key idea

Tariff and quota produce the same equilibrium price and quantity, the same CS and PS for domestic producers, and the same DWL. The only difference is distributional: the *rectangle* between P_s and P_d goes to the State (revenue) under the tariff, and to the license holders (private rent) under the quota.

Remark 13.5 (Japanese cars in the United States, 1981).

In 1981 Japan agreed to “voluntarily” limit exports of cars to the United States to 1.68 million units (*Voluntary Export Restraints*), as an alternative to a formal tariff that would have been prohibited by the GATT. The paradoxical result: the quota rent was captured by *Japanese producers* (Toyota, Honda, Nissan), which sold fewer cars but at significantly higher prices in the American market. The US government collected no revenue, American consumers paid higher prices, and Japanese producers earned extraordinary profits. Exactly the opposite of the declared political objective.

13.7 Subsidies

A *specific subsidy* s is a transfer from the State to the producer for each unit produced. The producer receives $P_s = P_d + s$: the price paid by the buyer plus the subsidy paid by the State. The effect on the supply curve is the mirror image of the tax: a *downward* shift by s , because for each quantity the producer accepts a price s lower.

Analytical example

With $s = 5$ in the reference market: $Q_s^S = 3(P_d + 5) = 3P_d + 15$. Imposing $Q^D = Q_s^S$:

$$80 - 2P_d = 3P_d + 15 \implies 65 = 5P_d \implies P_d^* = 13, P_s^* = 18, Q_s^* = 54.$$

Buyers pay 3 less per unit, producers receive 2 more per unit, and quantity rises from 48 to 54. Cost to the State: $s \cdot Q_s^* = 5 \cdot 54 = 270$.

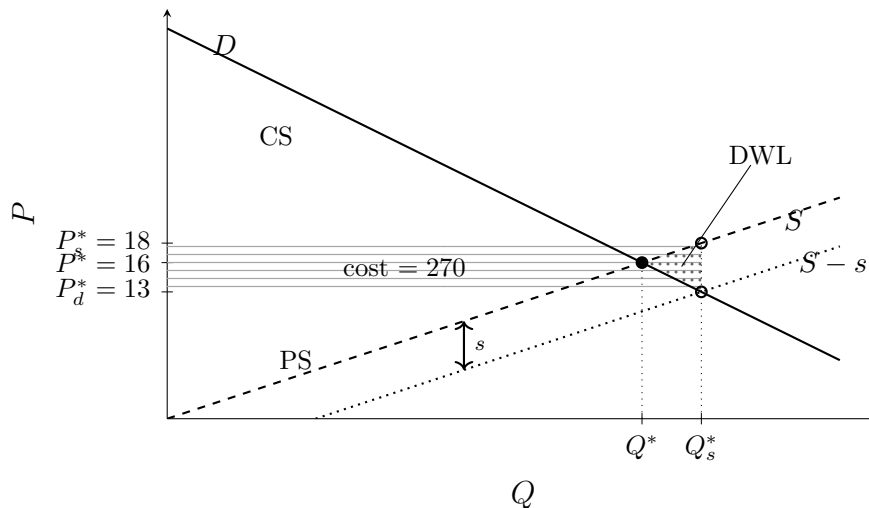


Figure 13.2: Specific subsidy $s = 5$. The equilibrium moves from $(Q^* = 48, P^* = 16)$ to $Q_s^* = 54$ with $P_d^* = 13$ and $P_s^* = 18$; supply shifts downward from S to $S - s$. Horizontal hatching: cost to the State $s \cdot Q_s^* = 270$; dotted: DWL, i.e., the part of the cost that turns into deadweight loss.

13.7.1 Why a subsidy generates DWL

Analytical example

With $s = 5$: $CS = \frac{1}{2} \cdot 54 \cdot (40 - 13) = 729$, $PS = \frac{1}{2} \cdot 54 \cdot 18 = 486$, public cost = 270, and

$$TS_{\text{net}} = CS + PS - \text{cost} = 729 + 486 - 270 = 945, \quad DWL = 960 - 945 = 15.$$

The DWL is also obtained directly from the triangle: $\frac{1}{2}(54 - 48) \cdot 5 = 15$.

The decomposition is instructive. Consumers and producers gain +153 and +102 respectively, for a private total of +255. The State spends 270. The net balance is $255 - 270 = -15$, exactly the DWL. Figure 13.3 shows how the cost rectangle is split: ΔCS and ΔPS are two *disjoint and adjacent* areas (separated by the line $P^* = 16$), not an area counted twice; the subsidy does not take surplus away from either party, it is the State that finances it.

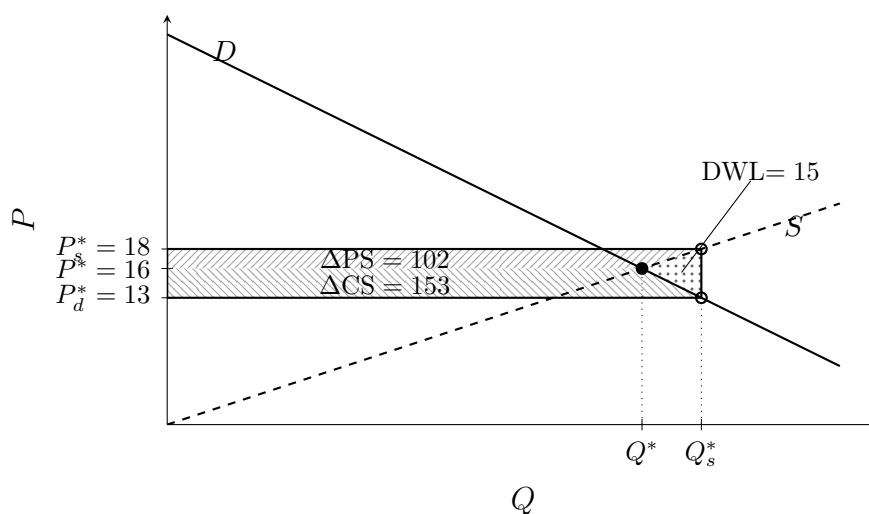


Figure 13.3: Decomposition of the subsidy cost rectangle. Public spending of 270 is split into three *disjoint* areas that tile it exactly: consumer gain $\Delta CS = 153$ (NW hatching), producer gain $\Delta PS = 102$ (NE hatching) and deadweight loss $DWL = 15$ (dotted). Only the last is value that vanishes: $270 = 153 + 102 + 15$.

Economic intuition

The subsidy pushes production *beyond* Q^* : for the units between $Q^* = 48$ and $Q_s^* = 54$, the marginal cost of production $P_S(Q)$ is greater than the consumer's willingness to pay $P_D(Q)$. Those “extra” units cost society more than they are worth. The trade takes place only because the State fills the gap with the subsidy—but the transfer exceeds the net social gain, and the difference is the DWL.

Key idea: Tax–subsidy symmetry

The tax *prevents* socially useful trades (units between Q_t^* and Q^* , where $P_D > P_S$): quantity too low. The subsidy *forces* socially useless trades (units between Q^* and Q_s^* , where $P_D < P_S$): quantity too high. Both distort Q away from Q^* , with the same DWL triangle. For taxes and subsidies of equal size ($t = s$) the DWL is identical in absolute value.

13.7.2 Applications of subsidies

Remark 13.6 (CAP direct payments).

With the MacSharry (1992) and Fischler (2003) reforms, the CAP replaced guaranteed minimum prices (price floors) with direct payments per hectare, decoupled since 2003 (in force from 2005): about 50 billion euros per year for the EU budget, equal to 30% of Community spending. They are subsidies *decoupled* from production (the payment does not depend on what the farmer produces), formally equivalent to lump-sum transfers per unit of land owned: less distortionary than classic guaranteed prices. The distributional effects, however, remain regressive (a large share of the payments goes to large landowners).

Remark 13.7 (Electric car ecobonus).

The Italian ecobonus of 2020–24 (a specific subsidy of up to 10,000 euros for the purchase of an electric car within price limits) is a paradigmatic case of a subsidy motivated by externalities (reduction of local emissions and of CO₂) and by infrastructure coordination (charging network). Two empirical observations: (i) pass-through to consumers was partial, with producers absorbing part of the subsidy by raising list prices (a phenomenon also documented for *Cash for Clunkers* in the US); (ii) there were “inframarginal” distortions (subsidies for purchases that would have happened anyway) and *timing distortion* (purchases brought forward from later years). The practical rule of Pigouvian theory: a subsidy motivated by externalities should be sized equal to the marginal external benefit. More generous amounts reproduce the DWL of our example.

13.8 Tax–subsidy comparison

	Equilibrium	Tax $t = 5$	Subsidy $s = 5$
P_d^*	16	19 (+3)	13 (−3)
P_s^*	16	14 (−2)	18 (+2)
Q^*	48	42 (−6)	54 (+6)
CS	576	441 (−135)	729 (+153)
PS	384	294 (−90)	486 (+102)
Revenue/Cost to State	0	+210	−270
Net TS	960	945	945
DWL	0	15	15

Besides the identical DWL, note that the subsidy costs the State more (270) than the tax raises (210), even though it produces the same effects on prices in absolute value. The reason is that the subsidy also pays for the “inframarginal” units (those that would have been produced anyway), whereas the tax hits only the units actually traded at the new equilibrium (fewer than at the old one). A subsidy of the same size as a tax is therefore intrinsically more costly for the State, and the opportunity cost falls on the rest of the economy (it must be financed with other taxes, which in turn have DWL).

13.9 Exercises

Exercise 13.1 (Specific tax).

Market with $Q^D = 80 - 2P$, $Q^S = 3P$. Equilibrium: $P^* = 16$, $Q^* = 48$, TS = 960. The

government imposes a specific tax $t = 10$ on the producer.

- Find P_d^* , P_s^* , Q_t^* .
- Compute CS, PS, revenue T and DWL. Verify $CS + PS + T + DWL = 960$.
- Compute the tax incidence (consumer and producer shares).
- Compare the DWL with that at $t = 5$ (DWL = 15). Is it consistent with $DWL \propto t^2$?

Exercise 13.2 (Equivalence Principle).

Same market, same tax $t = 5$, but levied on the buyer.

- Write the new demand curve as a function of P_s .
- Find P_d^* , P_s^* , Q_t^* .
- Compare with the case of the tax on the seller and comment on the equivalence of the areas (CS, PS, T , DWL).

Exercise 13.3 (Subsidy).

Same market. The government introduces a subsidy $s = 10$ to the producer.

- Find P_d^* , P_s^* , Q_s^* .
- Compute CS, PS, cost to the State and DWL.
- Does the subsidy increase net welfare ($TS = CS + PS - \text{cost}$)? Why does it generate DWL despite being a “free” transfer to producers?

Exercise 13.4 (Incidence for given elasticities).

In a market with $\varepsilon_D = -0.4$ and $\varepsilon_S = 0.6$ at equilibrium, the government introduces a small specific tax.

- Apply formula (13.1) to determine the shares of consumers and producers.
- If consumers are to bear at most 50% of the tax, which of the two elasticities must change, and in which direction?
- Using the empirical estimates from the box on excise duties, explain why excise duties on tobacco fall almost entirely on consumers.

Exercise 13.5 (Tax vs subsidy: comparison).

Same market $Q^D = 80 - 2P$, $Q^S = 3P$. Compare two scenarios:

- Specific tax $t = 5$ on the producer.
- Specific subsidy $s = 5$ to the producer.

- Compute ΔP_d and ΔP_s relative to $P^* = 16$ in the two cases.
- Verify that the DWL is identical in the two scenarios.
- Compute $\Delta CS + \Delta PS$ (total change in private surplus) in the two cases. Why are the values different even though the DWL is the same?

- d) If the State has a tight budget constraint, which of the two instruments is more costly to implement?

Exercise 13.6 (Tariff vs quota).

A country has domestic demand $Q^D = 120 - 4P$ and domestic supply $Q^S = 2P$. The world price is $P_w = 5$, and at this price the country imports the difference $Q^D(P_w) - Q^S(P_w)$.

- Compute imports at the world price.
- The government imposes a specific tariff $t = 5$. Find the new domestic price, CS, PS (domestic producers), revenue and DWL.
- Alternatively, the government imposes an import quota equal to the imports in part (b). Show that price, CS and PS are identical to the tariff case, but the State's revenue is zero: who captures the rent?

Exercise 13.7 (Optimal taxation: base vs rate).

The government must raise revenue $\bar{T} = 200$ and can tax two independent markets. In both, $\varepsilon_S \rightarrow \infty$ (perfectly elastic supply) and demands are linear with $|\varepsilon_D^A| = 0.5$ and $|\varepsilon_D^B| = 2.0$ at equilibrium. Reason about which market it is better to tax to minimize total DWL, recalling the heuristic formula $DWL \approx \frac{1}{2} \bar{T}^2 |\varepsilon_D| \varepsilon_S / (Q^* P^* \cdot (|\varepsilon_D| + \varepsilon_S))$, which for $\varepsilon_S \rightarrow \infty$ becomes $\frac{1}{2} \bar{T}^2 |\varepsilon_D| / (Q^* P^*)$.

Exam essentials

- Distinguish specific, ad valorem and lump-sum taxes; recognize why only the lump-sum tax does not distort Q^* .
- Given the market and the specific tax t , compute P_d^* , P_s^* , Q_t^* by imposing $Q^D(P_d) = Q^S(P_d - t)$.
- Compute CS, PS, revenue $T = t \cdot Q_t^*$ and DWL, verifying the identity $CS + PS + T + DWL = TS^*$.
- Determine tax incidence via $\Delta P_d/t$ and $|\Delta P_s|/t$, and via the elasticity formula $\varepsilon_S/(\varepsilon_S - \varepsilon_D)$.
- Apply the Equivalence Principle (Prop. 13.1): incidence is independent of who formally remits the tax.
- Recognize the relationship $DWL \propto t^2$ (Prop. 13.2) and apply it to the “*broad base, low rate*” argument.
- Analyze a subsidy s as the mirror image of the tax, showing that it generates DWL because it pushes production beyond Q^* to units with $P_S > P_D$.
- Distinguish a tariff (government revenue) from an import quota (private rent), recognizing their equivalence in price, quantity and DWL.

Questions and exercises

Multiple-choice questions

1. A price-taking firm has $C(Q) = Q^3 - 9Q^2 + 30Q + 20$ and faces $P = 15$. Which quantity maximizes profit?

- ☐ A $Q^* = 1$;
- ☐ B $Q^* = 3$;
- ☐ C $Q^* = 5$;
- ☐ D $Q^* = 6$.

2. Given $C(Q) = Q^2 + 10Q + 25$ (with $FC = 25$), the short-run shutdown threshold is the price:

- ☐ A $P = 0$;
- ☐ B $P = 10$;
- ☐ C $P = 20$;
- ☐ D $P = 25$.

3. Which of the following statements about the individual short-run supply curve of a price-taking firm is **FALSE**?

- ☐ A It coincides with the increasing part of MC above $AVC_{\min}$.
- ☐ B It shifts to the left if input costs increase.
- ☐ C It shifts if fixed costs increase.
- ☐ D At a price $P < AVC_{\min}$ the quantity supplied is $Q^s = 0$.

4. A firm has supply curve $Q^s = 2P - 6$ for $P \geq 3$. The elasticity of supply at $P = 10$ is:

- ☐ A $\varepsilon_S = 5/7$;
- ☐ B $\varepsilon_S = 1$;
- ☐ C $\varepsilon_S = 10/7$;

☐ $\varepsilon_S = 2$.

5. In a market with $Q^D = 100 - 5P$ and $Q^S = 5P$, at $P = 6$ we observe:

- ☐ competitive equilibrium;
- ☐ excess supply of 40 units;
- ☐ excess demand of 40 units;
- ☐ excess demand of 20 units.

6. In a constant-cost industry with *free entry*, a permanent increase in demand produces in the long run:

- ☐ P^* and Q^* both increase;
- ☐ P^* unchanged, Q^* increases, n^* increases;
- ☐ P^* increases, Q^* unchanged, n^* unchanged;
- ☐ P^* and Q^* unchanged, only n^* increases.

7. A price ceiling $\bar{P}$ and a price floor $\underline{P}$ that produce the same quantity traded $\tilde{Q} < Q^*$ generate:

- ☐ the same DWL and the same distribution between CS and PS;
- ☐ different DWL and a mirrored distribution between CS and PS;
- ☐ the same DWL but a mirrored distribution between CS and PS;
- ☐ different DWL but the same distribution between CS and PS.

8. Market $Q^D = 100 - 5P$, $Q^S = 5P$ (equilibrium $P^* = 10$, $Q^* = 50$). A price ceiling at $\bar{P} = 6$ produces a DWL equal to:

- ☐ 20;
- ☐ 40;
- ☐ 60;
- ☐ 80.

9. The share of the incidence of a specific tax borne by consumers is larger when:

- ☐ supply is more elastic than demand (in absolute value);
- ☐ demand is more elastic than supply (in absolute value);
- ☐ demand and supply have the same elasticity;
- ☐ the tax is legally levied on the seller.

10. A specific subsidy $s > 0$ to the producer generates DWL because:

- ☐ A it transfers surplus from producers to consumers;
- ☐ B it reduces the quantity traded below Q^* ;
- ☐ C it increases the quantity traded above Q^* , on units for which $P_S(Q) > P_D(Q)$;
- ☐ D it does not generate DWL, since it is a “free” transfer.

Exercises and open questions

1. A price-taking firm has $C(Q) = 2Q^2 + 10Q + 32$.

- a) Compute $MC(Q)$, $AVC(Q)$ and $AC(Q)$. Determine $AVC_{\min}$ (justifying the monotonicity of AVC) and $AC_{\min}$ with the condition $MC = AC$.
- b) Derive the short-run supply curve $Q^s(P)$ in analytical form, specifying its domain.
- c) For the prices $P \in \{8, 18, 26, 50\}$, build a table with Q^* , $\pi(Q^*)$, SR decision, LR decision, and classify each case into the four zones of Proposition 9.6.
- d) Determine the long-run break-even price and verify analytically that at that point $P = MC = AC$.

2. A competitive market has 50 firms of type A with $C_A(Q_A) = Q_A^2 + 4Q_A + 9$ and 30 firms of type B with $C_B(Q_B) = Q_B^2 + 10Q_B + 25$.

- a) Derive the individual supply curves $Q_A^s(P)$ and $Q_B^s(P)$, specifying their domains.
- b) Write the market supply curve $Q^s(P)$ in piecewise form and indicate the price at which the kink forms.
- c) Compute Q^s for $P \in \{6, 10, 20\}$.
- d) Compute the elasticity of market supply just before and just after the kink. On which side is the curve more elastic? Explain economically why.

3. Consider a market with $Q^D = 100 - 5P$ and $Q^S = 5P$.

- a) Compute the equilibrium (P^*, Q^*) , CS, PS and TS.
- b) Demand shifts to $Q^{D'} = 120 - 5P$. Compute the new equilibrium and recompute CS', PS', TS'.
- c) The firms in the industry all have technology $C(q) = q^2 + 2q + 16$. Compute P_{LR}^* , the efficient scale q^* and the number of firms n^* in the long run, before and after the demand shock.
- d) Comment on the difference between the effect of the shock in the SR (part (b)) and in the LR (part (c)) on the equilibrium price. What type of industry is implicit in the model?

4. Again in the market of Exercise 3 ($Q^D = 100 - 5P$, $Q^S = 5P$, equilibrium $(P^*, Q^*) = (10, 50)$), the government evaluates two alternative interventions:
- a) Price ceiling $\bar{P} = 6$: compute $\tilde{Q}$, any excess, CS, PS, DWL.
 - b) Price floor $\underline{P} = 14$: compute $\tilde{Q}$, any excess, CS, PS, DWL.
 - c) Compare the two interventions: same DWL? Why? How is the cost of the intervention distributed between consumers and producers in the two cases?
 - d) Suppose that, under the ceiling, rationing is random rather than efficient. What is a reasonable estimate of the actual CS? Comment on the difference with respect to the CS of part (a).
5. Again in the market $Q^D = 100 - 5P$, $Q^S = 5P$, the government compares a tax and a subsidy:
- a) It introduces a specific tax $t = 4$ on the producer. Compute P_d^* , P_s^* , Q_t^* , CS, PS, revenue T and DWL. Verify the conservation identity $CS + PS + T + DWL = TS^*$.
 - b) Compute the tax incidence (consumer and producer shares) and compare it with the result of the formula $\varepsilon_S/(\varepsilon_S - \varepsilon_D)$ evaluated at the initial equilibrium.
 - c) Alternatively, the government introduces a subsidy $s = 4$ to the producer. Compute P_d^* , P_s^* , Q_s^* , CS, PS, public cost and DWL.
 - d) Compare the two interventions: why is the DWL identical? Why, despite having the same magnitude of effects on prices (± 2), are the State's budget balances different in absolute value?

Part IV

Monopoly

Chapter 14

Monopoly

In the previous chapters we studied markets in which every firm is a *price-taker*: too small to influence the market price, it takes P^* as given. The First Welfare Theorem guarantees that the competitive equilibrium maximizes $TS = CS + PS$, and any distortion (taxes, administered prices) generates a deadweight loss $DWL > 0$.

In many real-world markets this structure does not apply. The polar opposite case is **monopoly**: a single firm controls supply, sets the price on its own, and faces market demand directly. In this chapter we study how it sets price and quantity, what the cost for social welfare is, and which economic policy instruments (taxes, regulation) change the equilibrium.

14.1 Features and sources of monopoly power

Definition 14.1 (Monopoly).

A **monopoly** is a market structure with a single firm producing a good with no close substitutes, protected by **barriers to entry** that prevent competition from other firms.

There are four essential features: a single seller with no direct competitors; the fact that the firm is a *price-maker*, i.e., it chooses the price or, equivalently, the quantity; barriers to entry that keep out potential entrants; and a downward-sloping demand curve faced by the firm, which coincides with market demand.

Definition 14.2 (Barriers to entry).

A **barrier to entry** is any structural, legal or strategic obstacle that prevents new firms from entering a market and eroding the profits of the incumbent firm.

Barriers to entry can be of different kinds. **Legal** barriers are established by the regulator: patents (typically twenty years of exclusivity, e.g., for drugs), licenses and administrative concessions (frequencies, lotteries), copyright. **Economic** barriers stem from technology or from control of essential resources: economies of scale in the presence of increasing returns, exclusive access to a non-replicable input (deposits, proprietary technological patents), or sunk fixed costs so high that they discourage entry. **Strategic** barriers result from deliberate choices by the incumbent: preemption of productive capacity, exclusive agreements with suppliers or distributors, advertising investments that raise entry costs for potential entrants.

Remark 14.1 (Types of monopoly).

Depending on the origin of the dominant barrier, it is useful to distinguish four types. A **legal**

monopoly arises from an exclusive grant (pharmaceutical patents, copyright). A **resource monopoly** arises when a firm controls almost all of a strategic input (De Beers for diamonds, OPEC for oil acting as a cartel). A **natural monopoly** emerges when technology yields a globally decreasing AC : in that case a single firm serves the entire market at lower cost than several firms (see §14.8). A **monopoly from innovation or network effects** stems from technological advantages or from positive consumption externalities (search engines, operating systems historically).

Key idea

Monopoly is not just a theoretical case: it is a market structure found in patented drugs, in infrastructure networks (water, railways, electricity distribution), and in digital platforms with network effects. Understanding how it works is the first step toward designing antitrust, regulatory and consumer-protection policies.

14.2 The monopolist's marginal revenue

Unlike the price-taking firm, the monopolist must lower the price to sell additional units: its demand curve is the entire, downward-sloping market demand curve. This radically changes the relationship between total revenue, marginal revenue and price.

Definition 14.3 (The monopolist's inverse demand).

The **inverse demand** $P(Q)$ gives the maximum price at which the market absorbs the quantity Q . For a monopolist, it coincides with market demand and is strictly decreasing: $P'(Q) < 0$.

Proposition 14.1 (The monopolist's marginal revenue). *For a monopolist with inverse demand $P(Q)$:*

$$TR(Q) = P(Q) \cdot Q, \quad MR(Q) = \frac{dTR}{dQ} = P(Q) + Q \cdot P'(Q).$$

Since $P'(Q) < 0$, we have $MR(Q) < P(Q)$ for every $Q > 0$.

Proof idea

By the product rule, $\frac{d}{dQ}(P \cdot Q) = P + Q \cdot dP/dQ$. The first term is the price (quantity effect: one more unit is sold at the current price); the second, negative, is the *price effect*: to sell $Q + 1$ units, the price must also be lowered on the Q inframarginal units that would have been sold anyway at the previous price.

Economic intuition

When the monopolist increases output by one unit, it obtains the price on the marginal unit but gives up part of the revenue on the inframarginal units. This is the key difference from perfect competition, where the firm is too small to influence the price: $P'(Q) = 0$ from its point of view, so $MR = P$ and there is no price effect.

Analytical example

Linear inverse demand $P(Q) = a - bQ$ with $a, b > 0$. Then:

$$TR(Q) = aQ - bQ^2, \quad MR(Q) = a - 2bQ.$$

The MR curve has the same intercept a as demand but *twice* the slope (in absolute value). Geometrically, MR crosses the horizontal axis at $Q = a/(2b)$, exactly halfway between the origin and the demand intercept $Q = a/b$.

Proposition 14.2 (Relationship between MR , P and the elasticity of demand). *For every $Q > 0$ at which demand is well defined:*

$$MR(Q) = P(Q) \left(1 - \frac{1}{|\varepsilon_D(Q)|} \right)$$

where $\varepsilon_D(Q) = (dQ/dP) \cdot (P/Q)$ is the price elasticity of demand.

Proof idea

Expanding the product: $MR = P + Q \cdot dP/dQ = P(1 + (Q/P) \cdot (dP/dQ)) = P(1 + 1/\varepsilon_D)$. Since $\varepsilon_D < 0$, we write $1/\varepsilon_D = -1/|\varepsilon_D|$ and obtain the form in the statement.

This formula has three practical implications. First, $MR > 0$ if and only if demand is *elastic* ($|\varepsilon_D| > 1$): in the inelastic region of the curve, an increase in Q reduces total revenue. Second, $MR = 0$ exactly at the point of unit elasticity, which corresponds to the maximum of $TR(Q)$. Third, $MR < 0$ in the inelastic region: the monopolist never sells in this region, because reducing output increases TR and (weakly) reduces costs, and hence increases profits.

Key idea

The monopolist **always operates in the elastic region of demand** ($|\varepsilon_D| > 1$). If it were in the inelastic region, reducing Q would increase total revenue and reduce costs: profit would rise strictly, contradicting optimality.

14.3 The monopolist's equilibrium: the $MR = MC$ rule

14.3.1 Setting up the problem

The monopolist chooses the quantity $Q \geq 0$ to maximize profit:

$$\max_{Q \geq 0} \pi(Q) = TR(Q) - TC(Q) = P(Q) \cdot Q - C(Q).$$

The technical difference from the price-taking firm's problem is that now P depends on Q , through the inverse demand.

14.3.2 First- and second-order conditions

Proposition 14.3 (The monopolist's FOC). *If $Q_m^* > 0$ is an interior maximum of the monopolist's profit, then it satisfies the first-order condition*

$$\left. \frac{d\pi}{dQ} \right|_{Q=Q_m^*} = MR(Q_m^*) - MC(Q_m^*) = 0 \quad \implies \quad \boxed{MR(Q_m^*) = MC(Q_m^*)}.$$

The equilibrium price is obtained from the inverse demand: $P_m^* = P(Q_m^*)$.

Proposition 14.4 (The monopolist's SOC). For Q_m^* to be a local maximum (and not a minimum) of profit, the second-order condition must hold:

$$\pi''(Q_m^*) = MR'(Q_m^*) - MC'(Q_m^*) < 0.$$

With linear demand (MR' a negative constant) and weakly increasing MC , the SOC is automatically satisfied at any solution of the FOC.

Key idea

The crucial difference from perfect competition is that under PC we have $MR = P$ (price-taker), so the FOC becomes $P = MC$. For the monopolist $MR < P$, so $MR = MC$ implies $P_m^* > MC(Q_m^*)$: the price is set **above** marginal cost. This *markup* is the source of market power.

14.3.3 A complete numerical example

Example 14.1 (Linear-quadratic monopoly).

Consider $P(Q) = 70 - Q$ and $C(Q) = Q^2 + 10Q + 50$.

Building the curves. The linear inverse demand has slope -1 and intercept 70 . Marginal revenue is $MR(Q) = 70 - 2Q$ (same intercept, twice the slope). Marginal cost is $MC(Q) = 2Q + 10$.

Monopoly quantity (FOC).

$$MR = MC \iff 70 - 2Q = 2Q + 10 \implies 4Q = 60 \implies Q_m^* = 15.$$

The SOC is satisfied: $MR'(Q) = -2 < MC'(Q) = 2$.

Monopoly price. From the inverse demand: $P_m^* = 70 - 15 = 55$. Marginal cost at the optimum is $MC(15) = 40$; the absolute *markup* is $P_m^* - MC = 15$, equal to 27.3% of the price.

Profit.

$$\pi_m = P_m^* \cdot Q_m^* - C(Q_m^*) = 55 \cdot 15 - (225 + 150 + 50) = 825 - 425 = 400.$$

Comparison with perfect competition. Under PC the same industry would have $P = MC$, i.e., $70 - Q = 2Q + 10 \implies Q_{pc} = 20$ and $P_{pc} = 50$. The monopoly therefore produces *less* ($15 < 20$) and charges *more* ($55 > 50$).

14.3.4 Graphical representation

Figure 14.1 shows the monopoly equilibrium for Example 14.1, compared with the competitive equilibrium.

Building the graph, step by step: (i) draw the inverse demand D and its “twin” MR with twice the slope; (ii) superimpose the MC curve; (iii) the intersection $MR \cap MC$ identifies Q_m^* on the horizontal axis; (iv) go up vertically from Q_m^* to the *demand curve* (not MR): that point gives P_m^* .

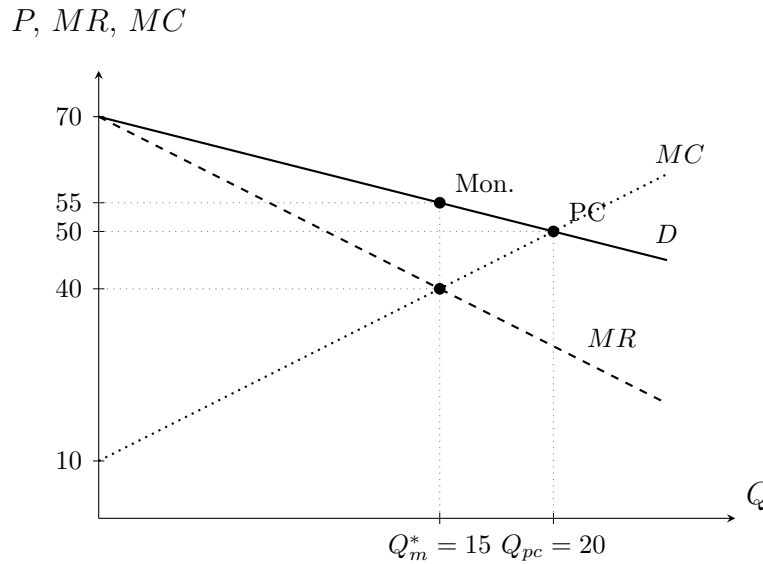


Figure 14.1: Monopoly and competitive equilibrium in Example 14.1. The monopoly quantity is determined by $MR = MC$ ($Q_m^* = 15$); the price $P_m^* = 55$ is read off the demand curve. The competitive point is obtained by imposing $P = MC$ ($Q_{pc} = 20$, $P_{pc} = 50$).

14.4 Welfare, surplus and deadweight loss

14.4.1 The three relevant areas

Under a uniform-price monopoly, there are three welfare areas, illustrated in Figure 14.2.

Definition 14.4 (Consumer surplus under monopoly).

CS_m is the area below the demand curve and above the price P_m^* between $Q = 0$ and $Q = Q_m^*$. With linear demand: $CS_m = \frac{1}{2} Q_m^* (P_{\max} - P_m^*)$, where $P_{\max} = P(0)$ is the vertical intercept of demand.

Definition 14.5 (Producer surplus (profit gross of fixed costs)).

$PS_m = \int_0^{Q_m^*} [P_m^* - MC(Q)] dQ$. When fixed costs are FC , the monopolist's net profit is $\pi_m = PS_m - FC$.

Definition 14.6 (Deadweight loss of monopoly).

The **deadweight loss** (DWL) of monopoly is the difference between total surplus under perfect competition and total surplus under monopoly:

$$DWL = TS_{pc} - TS_m = (CS_{pc} + PS_{pc}) - (CS_m + PS_m).$$

Proposition 14.5 (Triangle formula for the DWL of monopoly). *When D and MC are continuous curves and MC is (weakly) increasing:*

$$DWL = \int_{Q_m^*}^{Q_{pc}} [P(Q) - MC(Q)] dQ.$$

With linear D and linear MC , the integral is the area of a triangle:

$$DWL = \frac{1}{2} (Q_{pc} - Q_m^*) (P_m^* - MC(Q_m^*)).$$

Proof idea

The DWL is the sum, over all units not traded between Q_m^* and Q_{pc} , of the vertical distance between the marginal consumer's willingness to pay ($P(Q)$, read off demand) and the marginal cost of producing that unit ($MC(Q)$). With linear D and MC , the two curves are straight lines and the area is that of a triangle with base $Q_{pc} - Q_m^*$ and height $P_m^* - MC(Q_m^*)$.

Analytical example

Continuing Example 14.1: $Q_m^* = 15$, $P_m^* = 55$, $MC(15) = 40$, $Q_{pc} = 20$, $P_{pc} = 50$.

$$CS_m = \frac{1}{2} \cdot 15 \cdot (70 - 55) = 112.5, \quad PS_m = \int_0^{15} (55 - 2Q - 10) dQ = 675 - 225 = 450,$$

$$DWL = \frac{1}{2} \cdot 5 \cdot (55 - 40) = 37.5.$$

Check via the welfare difference: $TS_{pc} = 200 + 400 = 600$ and $TS_m = 112.5 + 450 = 562.5$, hence $DWL = 600 - 562.5 = 37.5$ (✓).

14.4.2 Transfer and destruction of value**Key idea**

Monopoly does *two* things, both relevant to the welfare assessment. First, it **redistributes** surplus from consumers to the producer (the rectangle $(P_m^* - P_{pc}) \cdot Q_m^*$, part of the monopolist's profit, was consumer surplus under PC). Second, it **destroys** value: the DWL triangle is value that is neither produced nor consumed. Antitrust policies are justified above all by the second dimension.

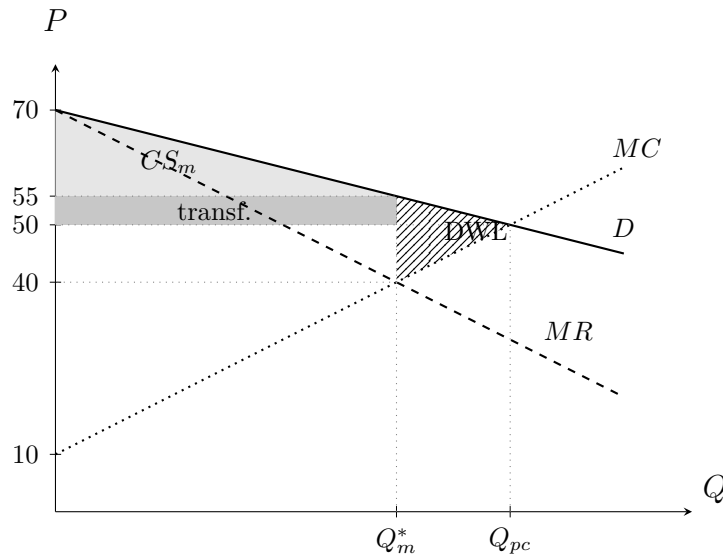


Figure 14.2: Consumer surplus under monopoly (CS_m , light triangle); transfer from consumers to the monopolist (intermediate rectangle); deadweight loss (hatched triangle) corresponding to the units $Q \in (Q_m^*, Q_{pc})$ not traded.

14.5 Demand shifts: the absence of a supply curve

Under perfect competition we derived a firm supply curve, which links price and quantity supplied independently of demand (for $P \geq AVC_{\min}$, $Q^s(P)$ is the increasing branch of MC). For monopoly this construction is *not* possible.

Proposition 14.6 (Nonexistence of the supply curve under monopoly). *For a monopolist there is, in general, no function $Q^s(P)$ linking the quantity produced to the price alone. The monopoly quantity depends jointly on the shape of demand (level and elasticity) and on costs.*

Proof idea

The monopolist's FOC is $MR(Q) = MC(Q)$, where MR depends on the inverse demand curve. Two different demand curves passing through the same point (Q^*, P^*) generally have different marginal revenues at Q^* , and hence induce different optima. The same price P^* can correspond to two different optimal quantities depending on which demand curve is chosen.

Economic intuition

The supply curve of a price-taking firm does not depend on demand because the firm takes P as given and chooses Q with $P = MC$. The monopolist, instead, must *reconstruct* at every instant the shape of its own MR curve, which depends on the elasticity of demand. Changing the elasticity, even holding the level fixed, shifts the optimum.

Analytical example

Let us show the argument concretely. Consider two demand curves that cross at $(Q, P) = (10, 30)$ with constant $MC = 10$.

Demand 1 (steeper): $P_1(Q) = 40 - Q$, $MR_1 = 40 - 2Q$. Optimum:
 $40 - 2Q = 10 \Rightarrow Q_1^* = 15$ and $P_1^* = 25$.

Demand 2 (flatter): $P_2(Q) = 35 - 0.5Q$, $MR_2 = 35 - Q$. Optimum:
 $35 - Q = 10 \Rightarrow Q_2^* = 25$ and $P_2^* = 22.5$.

The two curves *both* pass through $(10, 30)$, but the monopolist produces 15 or 25 depending on the shape. There is no single relationship $Q^s(P)$.

14.6 Specific tax on the monopolist

We introduce a specific tax $t > 0$ per unit produced, formally levied on the monopolist. The firm's effective marginal cost becomes $MC(Q) + t$, and the new FOC is

$$MR(Q) = MC(Q) + t.$$

Proposition 14.7 (Incidence of a specific tax under monopoly). *With linear demand $P(Q) = a - bQ$ and $MC(Q) = c + dQ$ ($d \geq 0$), the post-tax monopoly quantity is*

$$Q_m^t = \frac{a - c - t}{2b + d}$$

and the consumer price $P_m^t = a - bQ_m^t$ increases by

$$\Delta P_d = P_m^t - P_m^* = \frac{b}{2b + d} \cdot t \leq \frac{1}{2} \cdot t.$$

Proof idea

The FOC $MR = MC + t$ becomes $a - 2bQ = c + dQ + t$, hence $Q_m^t = (a - c - t)/(2b + d)$. Comparing with the pre-tax value $Q_m^* = (a - c)/(2b + d)$, $\Delta Q = -t/(2b + d)$ and $\Delta P_d = -b \cdot \Delta Q$, which gives the formula. For $d \geq 0$: $b/(2b + d) \leq b/(2b) = 1/2$.

Key idea

Unlike the competitive case, with weakly increasing MC and linear demand the incidence on consumers never exceeds 50%. The monopolist absorbs at least half of the tax, because its optimal choice is already above MC : part of the cost increase is absorbed by the existing markup, rather than passed on to prices.

Example

Continuing Example 14.1 ($P = 70 - Q$, $C = Q^2 + 10Q + 50$): with a tax $t = 6$ per unit, the new FOC is $70 - 2Q = 2Q + 10 + 6$, hence $Q_m^t = 13.5$. The price rises to $P_m^t = 56.5$, so $\Delta P_d = 1.5$: the incidence on consumers is $1.5/6 = 25\%$, and that on the monopolist is the remaining 75%. Tax revenue is $T = 6 \cdot 13.5 = 81$.

The additional deadweight loss, computed as a trapezoid between Q_m^t and Q_m^* with heights equal to $P - MC$ evaluated at Q_m^t and Q_m^* , is $\frac{1}{2}(15 - 13.5)(P_m^t - MC(Q_m^t) + P_m^* - MC(Q_m^*)) = \frac{1}{2}(1.5)(19.5 + 15) = 25.875$.

14.7 Multi-plant production

The monopolist may own several production plants with different technologies. If we assume it produces with n plants, each with cost function $C_i(Q_i)$, the problem becomes

$$\max_{Q_1, \dots, Q_n} \pi = P\left(\sum_{i=1}^n Q_i\right) \cdot \sum_{i=1}^n Q_i - \sum_{i=1}^n C_i(Q_i).$$

Proposition 14.8 (Allocation rule across plants). *At the optimum, the monopolist allocates output across plants so as to **equalize marginal costs**, and this common value coincides with aggregate marginal revenue:*

$$MR(Q^*) = MC_1(Q_1^*) = MC_2(Q_2^*) = \dots = MC_n(Q_n^*).$$

Proof idea

The FOCs with respect to each Q_i require $\partial\pi/\partial Q_i = 0$. Expanding: $MR(Q) - MC_i(Q_i) = 0$ for every i , where $MR(Q) = P(Q) + Q \cdot P'(Q)$ depends only on $Q = \sum_i Q_i$. Hence all the MC_i take the same value, and this common value is precisely MR .

Economic intuition

If one plant had a strictly lower MC than the others at the current allocation, it would be profitable to shift one unit from an “expensive” plant to a “cheap” one: total output stays unchanged (so MR is the same), but total cost falls. The optimal allocation is reached when all the MC_i are equal: no reallocation can reduce costs any further.

Analytical example

Two plants with $C_1(Q_1) = Q_1^2$ and $C_2(Q_2) = 2Q_2^2$, hence $MC_1 = 2Q_1$ and $MC_2 = 4Q_2$. Market demand $P(Q) = 60 - Q$.

Step 1—building the aggregate MC. The rule $MC_1 = MC_2$ implies $2Q_1 = 4Q_2$, so $Q_1 = 2Q_2$. Total quantity is $Q = Q_1 + Q_2 = 3Q_2$, so $Q_2 = Q/3$ and $Q_1 = 2Q/3$. Aggregate marginal cost (the common value $MC_1 = MC_2$) is $MC_{agg}(Q) = 2 \cdot (2Q/3) = 4Q/3$.

Step 2—overall FOC. $MR = MC_{agg}$:

$$60 - 2Q = \frac{4Q}{3} \implies 180 - 6Q = 4Q \implies Q^* = 18.$$

Step 3—allocation and price. $Q_2^* = 6$, $Q_1^* = 12$, $P^* = 60 - 18 = 42$. Check: $MC_1(12) = 24$ and $MC_2(6) = 24$ (✓).

Profit. $\pi = 42 \cdot 18 - (12^2 + 2 \cdot 6^2) = 756 - 216 = 540$.

14.8 Regulation and natural monopoly

14.8.1 When to regulate a monopoly

In the presence of a monopoly facing no competition, the regulator can intervene to reduce the *DWL* by bringing the equilibrium back toward the competitive level. The main instruments are the **price cap** (a ceiling on the price, $P \leq \bar{P}$) and $P = AC$ or $P = MC$ regulation (typically for natural monopoly).

Definition 14.7 (Natural monopoly).

A monopoly is called **natural** when, for every Q relevant to market demand, average cost $AC(Q)$ is strictly decreasing. In that case, a single firm produces at lower cost than under any division of the market among several firms.

Remark 14.2 (Sufficient condition).

A sufficient condition for natural monopoly is the presence of a fixed cost $F > 0$ together with constant or not too rapidly increasing MC : with $C(Q) = F + cQ$, $AC(Q) = F/Q + c$ is decreasing everywhere.

14.8.2 Three regimes compared

Consider a natural monopoly with $C(Q) = F + cQ$ and linear demand $P(Q) = a - bQ$. We compare:

- **Unregulated monopoly** ($MR = MC$): the firm maximizes profit. Low quantity, high price, $\pi > 0$, $DWL > 0$.
- **$P = AC$ regulation** (Ramsey average cost pricing): the regulator sets the price at average cost. The firm makes zero profit and covers its costs without needing a subsidy. Quantity is higher than under monopoly, but the *DWL* is still positive (because $P = AC > MC$).
- **$P = MC$ regulation** (first best): quantity is the competitive one and $DWL = 0$, but the firm operates at a loss equal to F : it needs a **subsidy** equal to the fixed costs.

Proposition 14.9 (Comparison of regimes in the linear case). *With $C(Q) = F + cQ$ and $P(Q) = a - bQ$, the three equilibrium quantities satisfy $Q_m^* < Q_{AC}^* < Q_{MC}^*$ and the three prices satisfy $P_m^* > P_{AC}^* > P_{MC}^*$. The $P = AC$ regime is a compromise solution: $DWL > 0$ but no subsidy is needed.*

Analytical example

$P(Q) = 50 - Q$, $C(Q) = 125 + 20Q$. Average cost $AC(Q) = 125/Q + 20$, decreasing for every $Q > 0$ ($AC'(Q) = -125/Q^2 < 0$): it is a natural monopoly.

Monopoly. $MR = 50 - 2Q$, $MC = 20$. $MR = MC$: $Q_m^* = 15$, $P_m^* = 35$,

$\pi_m = (35 - 20) \cdot 15 - 125 = 100$.

$P = AC$ (**Ramsey**). The equation $50 - Q = 125/Q + 20$ reduces to $Q^2 - 30Q + 125 = 0$, hence $Q \in \{5, 25\}$. Both solutions yield zero profit, but the regulator picks the larger one: at $Q = 25$ the price is lower (25 versus 45) and total surplus is higher, while the firm is indifferent because $\pi = 0$ in both cases. Hence $Q_{AC}^* = 25$, $P_{AC}^* = 25$, $\pi_{AC} = 0$.

$P = MC$ (**first best**). $50 - Q = 20 \Rightarrow Q_{MC}^* = 30$, $P_{MC}^* = 20$. Profit:

$(20 - 20) \cdot 30 - 125 = -125$. A subsidy of 125 is needed, which is exactly the fixed cost.

Consumer surplus (comparison). $CS_m = \frac{1}{2} \cdot 15 \cdot 15 = 112.5$; $CS_{AC} = \frac{1}{2} \cdot 25 \cdot 25 = 312.5$; $CS_{MC} = \frac{1}{2} \cdot 30 \cdot 30 = 450$.

Key idea

In natural monopoly the *first best* ($P = MC$) requires a subsidy equal to the fixed cost, because a firm with decreasing AC and $P = MC$ is necessarily making a loss. The $P = AC$ regime is a **second-best** solution: $DWL > 0$ but the regulator's budget is balanced. The choice depends on how costly (in terms of distortions) it is to finance the subsidy.

14.9 Exercises

Exercise 14.1 (Basic monopoly).

A monopolist faces the inverse demand $P(Q) = 60 - Q$ and has cost function $C(Q) = Q^2 + 12Q + 88$.

- Find $MR(Q)$ and $MC(Q)$, and derive Q_m^* from the FOC. Check the SOC.
- Compute P_m^* , $MC(Q_m^*)$ and the profit π_m .
- Find the quantity Q_{pc} and the price P_{pc} that would prevail under perfect competition with the same cost structure. Compute CS_m , CS_{pc} and the deadweight loss DWL .
- Verify the identity $TS_{pc} = CS_m + PS_m + DWL$, with PS_m computed by integrating $P_m^* - MC(Q)$ between 0 and Q_m^* .

Exercise 14.2 (Specific tax and incidence).

Same setup as Exercise 14.1. The government introduces a specific tax $t = 8$ per unit levied on the monopolist.

- Find the new monopoly quantity Q_m^t and the new price P_m^t .
- Compute the change $\Delta P_d = P_m^t - P_m^*$ and the incidence of the tax on consumers and on the monopolist. Verify analytically the statement of Proposition 14.7 for this case.
- Compute the tax revenue T and the new profit π_m^t .

- d) Compute the additional DWL due to the tax as a trapezoid between Q_m^t and Q_m^* , with heights $(P - MC)$ evaluated at the two endpoints.

Exercise 14.3 (Multi-plant monopolist).

A monopolist produces with two plants: $C_1(Q_1) = Q_1^2$ and $C_2(Q_2) = 2Q_2^2$. Demand is $P(Q) = 60 - Q$ with $Q = Q_1 + Q_2$.

- Build the aggregate marginal cost $MC_{agg}(Q)$ by imposing $MC_1(Q_1) = MC_2(Q_2)$ and solving for Q_1 and Q_2 as functions of Q .
- Find the optimal total quantity Q^* from the FOC $MR = MC_{agg}$ and the price P^* .
- Find the allocation (Q_1^*, Q_2^*) and verify numerically that $MC_1(Q_1^*) = MC_2(Q_2^*)$.
- Compute the monopolist's total profit (with no fixed costs).

Exercise 14.4 (Natural monopoly and regulation).

Demand $P(Q) = 50 - Q$, cost $C(Q) = 300 + 10Q$.

- Verify that average cost is decreasing everywhere: this is a natural monopoly.
- Compute the unregulated monopoly equilibrium: Q_m^* , P_m^* , π_m .
- Find the quantity and price under the $P = AC$ and $P = MC$ regimes. In the $P = MC$ case, compute the subsidy needed to cover costs.
- Compare the levels of CS in the three regimes and briefly discuss the *trade-off* between allocative efficiency and cost coverage.

Exam essentials

- Recognize the four features of monopoly and the main barriers to entry.
- Build $MR(Q)$ from the inverse demand: for linear D , MR has the same intercept and twice the slope.
- Apply the first-order rule $MR(Q_m^*) = MC(Q_m^*)$ and derive P_m^* from demand. Check the SOC.
- Compute CS_m , PS_m , π_m and DWL with the triangle formulas in the linear case.
- Recognize that the monopolist's supply curve *does not exist*: the quantity depends jointly on the shape of D and of MC .
- Compute the incidence of a specific tax t on the monopolist: consumers' share $\leq 50\%$ with linear D and weakly increasing MC .
- Solve a multi-plant problem by applying the rule $MR = MC_1 = \dots = MC_n$.
- Compare the three regimes $\{\text{Mon}, P = AC, P = MC\}$ in natural monopoly, recognizing that $P = MC$ requires a subsidy equal to F .

Chapter 15

Market power and price discrimination

In Chapter 14 we studied monopoly assuming that the firm charges a **uniform price** to all consumers. In this chapter we relax this assumption and introduce three natural extensions. First, we examine the **costs of market power**: beyond the *DWL*, the resources spent by firms to *obtain* or *defend* a monopoly position (*rent-seeking*). Second, we present the **Lerner index** as an operational measure of market power and show that it equals the inverse of the elasticity of demand. Third, we study **price discrimination**: the practice of charging different prices to different customers (or for different units of the same good), which changes both the monopolist's profit and the distribution of surplus.

15.1 Rent-seeking: the cost of the rent

The monopolist's positive profit is a *prize* that attracts resources. Firms are willing to spend to obtain or maintain monopoly positions through **lobbying**, litigation, aggressive *patent races*, and acquisitions of competitors. From a social point of view, these expenditures are wasted: they do not increase output or the quality of the good, but merely transfer a rent.

Definition 15.1 (Rent-seeking).

Rent-seeking is the expenditure of real resources (time, money, experts) aimed at obtaining or defending a position of economic rent, without creating social value.

15.1.1 The Tullock model with n contenders

Tullock (1967) proposed a simple model in which n identical firms compete for a rent of value V . Each firm i chooses the amount $x_i \geq 0$ to invest (lobbying, research, lawyers). The probability that firm i obtains the rent is proportional to its own expenditure, according to the *contest success function*

$$p_i(x_1, \dots, x_n) = \frac{x_i}{\sum_{j=1}^n x_j}.$$

The expected profit of firm i is therefore

$$\mathbb{E}[\pi_i] = \frac{x_i}{\sum_j x_j} \cdot V - x_i.$$

Proposition 15.1 (Symmetric Nash equilibrium in the Tullock model). *In the symmetric Nash equilibrium ($x_i = x^*$ for every i), individual and aggregate rent-seeking expenditures are*

$$x^* = V \cdot \frac{n-1}{n^2}, \quad X^* = n \cdot x^* = V \cdot \frac{n-1}{n}.$$

Rent dissipation, defined as the fraction X^*/V , is $(n-1)/n$, increasing in n , and tends to 1 as $n \rightarrow \infty$.

Proof idea

The FOC of each firm with respect to x_i , holding the others' x_j fixed:

$$\frac{\partial \mathbb{E}[\pi_i]}{\partial x_i} = \frac{\sum_j x_j - x_i}{\left(\sum_j x_j\right)^2} \cdot V - 1 = 0.$$

Imposing symmetry $x_j = x^*$ for every j : $\sum_j x_j = n x^*$, so $(n-1)x^*/(n x^*)^2 \cdot V = 1$, hence $x^* = V(n-1)/n^2$.

Key idea

The social loss from monopoly is not limited to the *DWL* triangle of Chapter 14: to it we must add the rent-seeking expenditure X^* , which grows with the number of contenders. When n is large, a fraction close to 100% of the rent is dissipated in unproductive activities. The **total social cost** of monopoly is therefore

$$\text{Social cost} = DWL + X^*.$$

Analytical example

Let $V = 200$ and $n = 5$ contenders. By Proposition 15.1:

$$x^* = 200 \cdot \frac{4}{25} = 32, \quad X^* = 5 \cdot 32 = 160, \quad \text{dissipation} = \frac{160}{200} = 80\%.$$

The expected profit of each firm is $\mathbb{E}[\pi_i] = V/n - x^* = 40 - 32 = 8$, positive (otherwise no one would participate), but much lower than the rent V/n that would obtain in the absence of competition.

Remark 15.1 (Limitations of the model).

The Tullock model is pedagogically powerful but has limitations: it assumes risk-neutral agents, a linear contest function and symmetry. In realistic settings, asymmetries between incumbents and entrants can reduce dissipation (smaller firms do not participate), while the presence of network or learning externalities can amplify the cost.

15.2 The Lerner index

The **Lerner index** (Lerner, 1934) is the standard measure of a firm's market power. It is defined as the *percentage markup* over marginal cost.

Definition 15.2 (Lerner index).

For a firm charging a uniform price P with marginal cost MC , the Lerner index is

$$L = \frac{P - MC}{P} \in [0, 1].$$

- $L = 0$: perfect competition ($P = MC$).

- $L \rightarrow 1$: the monopolist manages to extract almost all the value; this happens when MC is very low (e.g., software, digital goods) or $|\varepsilon_D|$ is close to 1.

Proposition 15.2 (Lerner and elasticity: the inverse elasticity rule). *In the uniform-price monopoly equilibrium, the Lerner index is exactly the inverse of the absolute value of the elasticity of demand evaluated at Q_m^* :*

$$L = \frac{P_m^* - MC(Q_m^*)}{P_m^*} = \frac{1}{|\varepsilon_D(Q_m^*)|}.$$

Proof idea

Recall the FOC $MR = MC$ and the formula for marginal revenue as a function of the elasticity (Prop. 14.2): $MC = MR = P(1 - 1/|\varepsilon_D|)$. Dividing both sides by P : $MC/P = 1 - 1/|\varepsilon_D|$, hence $(P - MC)/P = 1/|\varepsilon_D|$.

Economic intuition

The more elastic demand is, the less market power the monopolist has: if consumers react strongly to the price (high $|\varepsilon_D|$), a high markup makes sales collapse and becomes counterproductive. In the limit, $|\varepsilon_D| \rightarrow \infty$ (perfectly elastic demand) implies $L \rightarrow 0$: we recover perfect competition. The opposite case is $|\varepsilon_D| = 1$: $L = 1$, $P/MC \rightarrow \infty$ and $MR = 0$ (we are at the maximum of TR). The monopolist never sells where $|\varepsilon_D| < 1$, so $L \leq 1$ holds automatically.

Analytical example

Demand $P(Q) = 40 - Q$ with constant $MC = 10$. $MR = 40 - 2Q$, and $MR = MC$ gives $Q_m^* = 15$, $P_m^* = 25$. The Lerner index is

$$L = \frac{25 - 10}{25} = \frac{3}{5}.$$

Check via the elasticity: with $Q = 40 - P$, $dQ/dP = -1$, so $|\varepsilon_D| = |dQ/dP| \cdot P/Q = 1 \cdot 25/15 = 5/3$, and $1/|\varepsilon_D| = 3/5$ (✓).

Remark 15.2 (Lerner as an antitrust tool).

The Lerner index is used in antitrust analysis to quantify market power in specific markets. By estimating $|\varepsilon_D|$ from data (with structural demand methods), one can derive the implied L and compare it with observed prices: systematic deviations signal collusion or abuse of a dominant position.

15.3 Price discrimination: definition and conditions

Under a uniform price, the monopolist leaves consumers with a high willingness to pay a sizable surplus CS_m and forgoes selling to some consumers with low WTP (DWL). If it can charge different prices to different consumers (or for different units bought by the same consumer), both phenomena are mitigated: the profit increases.

Definition 15.3 (Price discrimination).

Price discrimination is the practice of charging different unit prices for the same good to different consumers, or for different units bought by the same consumer, in the absence of cost differences.

15.3.1 Feasibility conditions

For discrimination to be possible and profitable, three conditions must hold:

1. **Market power.** The firm must be a price-maker: under PC, any attempt to charge a higher price leads consumers to buy elsewhere. We need $|\varepsilon_D| < \infty$, i.e., demand that is not perfectly elastic.
2. **Information about preferences.** The seller must be able to distinguish (or induce consumers to self-select into) different groups. Direct identification (age, occupation, geography) or indirect (choice of good, time of purchase, quantity).
3. **No arbitrage.** Low-price consumers must not be able to resell to high-price consumers, otherwise market separation collapses. Non-resalable goods (services, named tickets, air travel) are the ideal candidates.

15.3.2 Pigou's three degrees

The classic taxonomy goes back to Pigou (1920).

- **First degree (perfect).** Each unit is sold at a price equal to the maximum willingness to pay for that unit. The monopolist captures *all* the surplus.
- **Second degree (nonlinear).** The price per unit depends on the *quantity* purchased or on the quality chosen: two-part tariffs, quantity discounts, menus of differentiated products.
- **Third degree (by group).** Consumers are divided into observable groups (students, seniors, geographic markets), and each group is charged a different uniform price.

We proceed in order of increasing realism: first first-degree discrimination (theoretical benchmark), then third-degree (the most common case in practice), and finally second-degree (the most sophisticated in terms of mechanism).

15.4 First-degree discrimination

15.4.1 Construction and properties

Under perfect price discrimination, the monopolist knows each consumer's maximum willingness to pay for each unit (and hence the entire inverse demand curve $P(Q)$), and charges $P(Q)$ as the price for the Q -th unit. The marginal revenue obtained by selling the Q -th unit is exactly $P(Q)$: there is no longer a *price effect* on the inframarginal units, because each is sold at its own price.

Proposition 15.3 (Quantity under first-degree discrimination). *Under monopoly with perfect price discrimination, $MR(Q) = P(Q)$ and the optimality rule becomes*

$$P(Q^I) = MC(Q^I),$$

which coincides with the competitive quantity $Q^I = Q_{pc}$. The DWL relative to perfect competition is zero.

Proposition 15.4 (Distribution of surplus under first-degree discrimination). *Under monopoly with perfect price discrimination, consumer surplus is $CS = 0$ and the entire market surplus is captured by the monopolist:*

$$PS^I = \int_0^{Q_{pc}} [P(Q) - MC(Q)] dQ, \quad CS^I = 0, \quad DWL^I = 0.$$

Key idea

First-degree discrimination is theoretically *efficient*: it produces the same quantity as perfect competition ($DWL = 0$). But it is *inequitable*: the entire surplus goes to the monopolist, $CS = 0$. It is the polar opposite of uniform-price monopoly: the same quantity as under competition, but a completely different distribution.

Analytical example

Recall Example 14.1 ($P = 70 - Q$, $MC = 2Q + 10$, $F = 50$). Under uniform-price monopoly: $Q_m^* = 15$, $P_m^* = 55$, $\pi_m = 400$, $CS_m = 112.5$, $DWL = 37.5$.

With first-degree discrimination: it sells up to $Q^I = Q_{pc} = 20$. Producer surplus gross of fixed costs is

$$PS^I = \int_0^{20} [(70 - Q) - (2Q + 10)] dQ = \int_0^{20} (60 - 3Q) dQ = 1,200 - 600 = 600.$$

Net profit: $\pi^I = 600 - 50 = 550$. Gain relative to uniform-price monopoly:

$550 - 400 = 150 = 112.5 + 37.5 = CS_m + DWL$. The monopolist captures the former consumer surplus *and* recovers the deadweight loss.

15.4.2 Applicability

First-degree discrimination is a theoretical benchmark: in practice, knowing the exact WTP of every single consumer is impossible. However, real cases come close: individually negotiated deals (used cars, real estate, B2B markets), online dynamic pricing algorithms that use personalized data (purchase history, searches). When it is approximated, the monopolist appropriates increasing shares of consumer surplus.

15.5 Third-degree discrimination

15.5.1 Setting up the problem

The monopolist identifies K groups of consumers with distinct inverse demands $P_k(Q_k)$ (for $k = 1, \dots, K$), assumes it can separate them at zero cost (no arbitrage), and charges a uniform price P_k within each group. The problem is

$$\max_{Q_1, \dots, Q_K} \pi = \sum_{k=1}^K P_k(Q_k) Q_k - C\left(\sum_{k=1}^K Q_k\right).$$

Proposition 15.5 (Optimal rule for third-degree discrimination). *At the optimum, marginal revenue in each market is equated to the (common) marginal cost:*

$$MR_1(Q_1^*) = MR_2(Q_2^*) = \dots = MR_K(Q_K^*) = MC(Q^*), \quad Q^* = \sum_{k=1}^K Q_k^*.$$

Proof idea

Setting $\partial\pi/\partial Q_k = 0$ for each k : $MR_k(Q_k) = MC(Q^*)$, where $Q^* = \sum_k Q_k$. All the MR_k are therefore equal to the same MC .

Proposition 15.6 (Third-degree rule in terms of Lerner). *If the monopolist applies the same Lerner rule in each market, then for every k :*

$$\frac{P_k^* - MC}{P_k^*} = \frac{1}{|\varepsilon_k(P_k^*)|}.$$

Consequently, P_k^* is **higher in markets with lower elasticity (in absolute value)**: consumers who are less responsive to price pay more.

Economic intuition

The rule is intuitive: shifting one unit of sales from one market to another stops being profitable when marginal revenues are identical. If in market A the marginal consumer adds €40 to revenue and in B only €30, it pays to sell one more unit in A . At the optimum, markets with more inelastic demand (“captive” consumers) pay a higher markup; those with elastic demand (easy alternatives) pay less.

Analytical example

Two separate markets: $P_1(Q_1) = 80 - Q_1$ and $P_2(Q_2) = 60 - Q_2$. Constant marginal cost $MC = 20$ (single plant).

Market 1: $MR_1 = 80 - 2Q_1 = 20 \Rightarrow Q_1^* = 30$, $P_1^* = 50$. $L_1 = 30/50 = 3/5$. Elasticity at (Q_1^*, P_1^*) : $|\varepsilon_1| = 50/30 = 5/3$, $1/|\varepsilon_1| = 3/5$ (✓).

Market 2: $MR_2 = 60 - 2Q_2 = 20 \Rightarrow Q_2^* = 20$, $P_2^* = 40$. $L_2 = 20/40 = 1/2$. Elasticity: $|\varepsilon_2| = 40/20 = 2$, $1/|\varepsilon_2| = 1/2$ (✓).

Total profit (excluding fixed costs): $\pi^{III} = (50 - 20) \cdot 30 + (40 - 20) \cdot 20 = 900 + 400 = 1,300$.

Comparison with a single price. Aggregating the two demands for $P \leq 60$:

$Q(P) = (80 - P) + (60 - P) = 140 - 2P$, hence $P(Q) = 70 - Q/2$ and $MR_{tot} = 70 - Q$.

From $70 - Q = 20$ we obtain $Q_u = 50$, $P_u = 45$ and profit $(45 - 20) \cdot 50 = 1,250$.

Discrimination yields a gain of $1,300 - 1,250 = 50$ relative to the single price.

15.5.2 Welfare effects

Welfare comparisons between a single price and third-degree discrimination are *ambiguous*. On the one hand, discrimination can expand the total quantity served (some markets, too elastic to be served at a single price, are reached): in that case it can reduce the *DWL*. On the other hand, it redistributes surplus across groups and from consumers to the monopolist. The empirical literature (e.g., textbooks in international markets, airline tickets) shows positive net welfare effects in many practical cases, especially when discrimination *opens up* markets that would otherwise not be served.

15.6 Second-degree discrimination: two-part tariffs and bundling

Under second-degree discrimination, the monopolist cannot observe the consumer’s “type,” but it can design **nonlinear pricing schemes** that induce self-selection. We study two main

instruments: the **two-part tariff** and **bundling**.

15.6.1 Two-part tariff: homogeneous consumers

Definition 15.4 (Two-part tariff).

A **two-part tariff** is a pricing scheme consisting of a *fixed fee* $T \geq 0$ (*entry fee*, paid only once to gain access to the good/service) and a *unit price* p charged on each unit consumed: $\text{Expenditure}(Q) = T + p \cdot Q$.

Proposition 15.7 (Optimal tariff with homogeneous consumers). *If consumers are identical, with individual demand $P(Q)$ and constant marginal cost MC , the optimal two-part tariff sets*

$$p^* = MC, \quad T^* = CS(p^*) = \int_0^{Q(MC)} [P(Q) - MC] dQ.$$

The monopolist extracts all consumer surplus through the fixed fee, while the unit price is exactly marginal cost: the quantity is efficient ($Q^I = Q_{pc}$).

Proof idea

Let $V(p)$ be the surplus each consumer obtains at unit price p : $V(p) = \int_0^{Q(p)} [P(Q) - p] dQ$, where $Q(p)$ is individual demand. The consumer participates if $T \leq V(p)$. The monopolist sets $T = V(p)$ (binding participation constraint) and maximizes $\pi(p) = V(p) + (p - MC) Q(p)$. The FOC with respect to p : $V'(p) + Q(p) + (p - MC) Q'(p) = 0$. By the envelope theorem, $V'(p) = -Q(p)$, so $(p - MC) Q'(p) = 0$, hence $p^* = MC$.

Economic intuition

With homogeneous consumers, the optimal two-part tariff “unbundles” the monopoly into two separate decisions: the quantity is the efficient one (because $p = MC$ does not distort the margin), and the entire consumer surplus is captured by the fixed fee. It is formally equivalent to first-degree discrimination: $DWL = 0$, $CS = 0$, the monopolist captures everything.

Analytical example

Individual demand $P(Q) = 30 - Q$, $MC = 10$. Optimal tariff: $p^* = 10$, individual quantity $Q(10) = 20$, surplus $V(10) = \frac{1}{2} \cdot 20 \cdot 20 = 200$, fixed fee $T^* = 200$. Profit per consumer: $T^* = 200$ (the margin on the unit price is zero).

15.6.2 Two-part tariff: heterogeneous consumers

If consumers have different willingness to pay, setting $T = V_H$ (the highest) excludes the low- V types. The problem becomes a *trade-off* between a **high unit price** (margin on units) and a **high fixed fee** (risk of excluding consumers).

Let the market consist of N_H consumers of type H with demand $P_H(Q) = a_H - Q$ and N_L consumers of type L with $P_L(Q) = a_L - Q$ ($a_H > a_L > MC$). The monopolist can choose among several strategies:

1. **Serve only the H types:** $p = MC$, $T = V_H(MC)$. Profit $\pi_A = N_H \cdot V_H(MC)$.

2. **Serve everyone, price at marginal cost:** $p = MC$, $T = V_L(MC) < V_H(MC)$. Profit $\pi_B = (N_H + N_L) \cdot V_L(MC)$.
3. **Serve everyone, price above marginal cost:** $p > MC$ (with $p < a_L$ so as not to exclude the L types), $T = V_L(p)$. The quantity distortion is offset by the margin on the H types' consumption.

Analytical example

Market with $N_H = 1$ consumer of type H ($P_H = 50 - Q$) and $N_L = 4$ consumers of type L ($P_L = 30 - Q$); $MC = 10$.

A—only H . $V_H(10) = \frac{1}{2} \cdot 40 \cdot 40 = 800$; $\pi_A = 1 \cdot 800 = 800$.

B—everyone, $p = MC$. $V_L(10) = \frac{1}{2} \cdot 20 \cdot 20 = 200$; $\pi_B = 5 \cdot 200 = 1,000$.

C—everyone, optimal unit price p . At price p , $Q_H = 50 - p$, $Q_L = 30 - p$, $V_L(p) = \frac{1}{2}(30 - p)^2$. Profit:

$$\pi_C(p) = (p - 10) \left[(50 - p) + 4(30 - p) \right] + 5 \cdot \frac{1}{2}(30 - p)^2.$$

Differentiating with respect to p and setting $= 0$: $70 - 5p = 0 \Rightarrow p^* = 14$, $T^* = \frac{1}{2} \cdot 16^2 = 128$. Substituting: $\pi_C = 1,040$.

Hence $\pi_C > \pi_B > \pi_A$: the optimal strategy is a unit price above MC with a moderate fixed fee. Sacrificing a little efficiency at the margin makes it possible to pocket higher margins on the H consumers without excluding the L types.

15.6.3 Bundling

Bundling is a form of second-degree discrimination in which the monopolist sells distinct goods in a *single package* at a price lower than the sum of the individual prices.

Definition 15.5 (Pure and mixed bundling).

We speak of **pure bundling** when the goods are sold *only* as a package and not separately; of **mixed bundling** when the consumer can choose between buying separately and buying the package.

Economic intuition

Bundling is profitable when the *correlation between consumers' willingness to pay for the two goods is negative or low*. In that case, the package “averages” the valuations: each consumer values the bundle similarly (even if valuations of the individual goods would be very heterogeneous), and the monopolist can charge a single price close to the common valuation, reducing the heterogeneity that would limit the price of the individual goods.

Example

Bundling: a numerical example. Two consumers (A and B) and two goods (1 and 2). Willingness to pay:

	Good 1	Good 2
Consumer A	80	30
Consumer B	30	80

Separate sales. For good 1, the monopolist can set $P_1 = 80$ (sells only to A, revenue 80) or $P_1 = 30$ (sells to both, revenue 60). Optimal: $P_1 = 80$, revenue 80. Similarly for good 2: $P_2 = 80$, revenue 80. **Total revenue from separate sales** = 160.

Pure bundling. Valuation of the bundle: A pays at most 110, B pays at most 110. Setting $P_{bundle} = 110$, both buy. **Total revenue** = 220.

The bundle increases revenue by 37.5% because it exploits the negative correlation between the valuations.

Remark 15.3 (Real-world cases).

Pure bundling: cable TV channel packages, software suites (Microsoft Office), restaurant set menus. Mixed bundling: cinemas (single ticket or season pass), magazines (single issue or subscription), public transport (single ticket or travel pass). The economic rationale is the same: to reduce the heterogeneity of valuations and increase surplus extraction.

15.7 Exercises

Exercise 15.1 (Rent-seeking in the Tullock model).

$n = 5$ firms compete for a rent $V = 200$ according to the Tullock model with a linear contest function.

- Write the FOC of the representative firm and derive the symmetric equilibrium expenditure x^* .
- Compute the aggregate expenditure X^* and the dissipation rate X^*/V .
- Compute the expected profit of each firm.
- Discuss how x^* , X^* and dissipation change as n varies (in particular as $n \rightarrow \infty$).

Exercise 15.2 (Lerner index and elasticity).

Demand $P(Q) = 40 - Q$, constant $MC = 10$.

- Find Q_m^* , P_m^* and the Lerner index L .
- Compute $|\varepsilon_D|$ at (Q_m^*, P_m^*) and verify the rule $L = 1/|\varepsilon_D|$.
- Keeping $MC = 10$, suppose demand becomes flatter: $P(Q) = 40 - 0.5Q$. Recompute L and compare. Is this consistent with the intuition that more elastic demand reduces market power?

Exercise 15.3 (First-degree discrimination).

Consider Exercise 14.1: demand $P(Q) = 60 - Q$ and cost $C(Q) = Q^2 + 12Q + 88$.

- a) Find the quantity Q^I and the profit π^I of the discriminating monopolist.
- b) Compute the gain relative to the uniform-price monopoly of Exercise 14.1.
- c) Verify that this gain is exactly equal to $CS_m + DWL$ of the uniform-price monopoly.
- d) Discuss: what changes for consumers? And for aggregate social welfare?

Exercise 15.4 (Third-degree discrimination).

A monopolist serves two separable markets with demands $P_1(Q_1) = 80 - Q_1$ and $P_2(Q_2) = 60 - Q_2$. Marginal cost is constant at $MC = 20$.

- a) Apply the rule $MR_k = MC$ in each market. Find (Q_1^*, P_1^*) and (Q_2^*, P_2^*) .
- b) Compute the Lerner indices L_1 and L_2 and the elasticities $|\varepsilon_1|$ and $|\varepsilon_2|$. Verify that the price is higher in the market with less elastic demand.
- c) Compute the total profit of the discriminating monopolist.
- d) Aggregate the two demands for $P \leq 60$ and find the optimal single price. Compare the profit with that of the previous part.

Exercise 15.5 (Two-part tariff: homogeneous consumers).

A monopolist serves N identical consumers, each with demand $P(Q) = 30 - Q$. Marginal cost is constant at $MC = 10$.

- a) Find the optimal two-part tariff (p^*, T^*) .
- b) Compute the quantity bought by each consumer and the profit per consumer.
- c) Compare individual consumer surplus under: (i) uniform-price monopoly, (ii) the two-part tariff. Comment.

Exercise 15.6 (Two-part tariff: heterogeneous consumers).

Market with one consumer of type H (individual demand $P_H = 50 - Q$) and four consumers of type L (individual demand $P_L = 30 - Q$). Constant $MC = 10$. The monopolist applies a two-part tariff (p, T) that is the same for all participants.

- a) **Strategy A: serve only the H type.** Find (p_A, T_A) and the profit π_A .
- b) **Strategy B: serve everyone, $p = MC$.** Find T_B (constrained by the participation of type L) and the profit π_B .
- c) **Strategy C: serve everyone, optimal $p > MC$.** Express π_C as a function of p (with $T = V_L(p)$). Maximize with respect to p , and find p_C^* , T_C^* and π_C^* .
- d) Compare the three strategies and discuss the *trade-off* between the margin on the unit price and the fixed fee.

Exam essentials

1. Compute the equilibrium of the symmetric Tullock model and the dissipation $X^*/V = (n - 1)/n$.
2. Define the Lerner index $L = (P - MC)/P$ and apply the rule $L = 1/|\varepsilon_D|$ at the uniform-price monopolist's optimum.
3. Recognize the three conditions for price discrimination: market power, separability of consumers, no arbitrage.
4. Compute price, quantity and surplus under first-degree discrimination: $Q^I = Q_{pc}$, $CS = 0$, $DWL = 0$, maximum PS .
5. Solve third-degree discrimination problems with K markets, applying $MR_k = MC$ in each and obtaining prices that differ according to the elasticities.
6. Build the optimal two-part tariff with homogeneous consumers ($p^* = MC$, $T^* = CS$) and analyze the trade-off with heterogeneous consumers.
7. Recognize bundling as a surplus-extraction tool when consumers' valuations of different goods are negatively correlated.

Questions and exercises

Multiple-choice questions

1. A monopolist has demand $P(Q) = 100 - Q$ and cost $C(Q) = Q^2 + 20Q + 100$. The equilibrium quantity is:

- ☐ A $Q_m^* = 10$;
- ☐ B $Q_m^* = 20$;
- ☐ C $Q_m^* = 25$;
- ☐ D $Q_m^* = 40$ (corresponds to PC).

2. In the previous question, the Lerner index at the monopolist's equilibrium is:

- ☐ A $L = 1/8$;
- ☐ B $L = 1/4$;
- ☐ C $L = 1/2$;
- ☐ D $L = 3/4$.

3. A monopolist with demand $P(Q) = 60 - Q$ and cost $C(Q) = Q^2 + 12Q + 88$ has $Q_m^* = 12$ and $P_m^* = 48$. The deadweight loss relative to perfect competition is:

- ☐ A $DWL = 12$;
- ☐ B $DWL = 18$;
- ☐ C $DWL = 24$;
- ☐ D $DWL = 36$.

4. A monopolist produces with two plants characterized by $MC_1(Q_1) \neq MC_2(Q_2)$. At the optimum, the allocation rule across plants is:

- ☐ A $MC_1(Q_1^*) = MC_2(Q_2^*) = P^*$;
- ☐ B $MC_1(Q_1^*) = MC_2(Q_2^*) = MR(Q^*)$, with $Q^* = Q_1^* + Q_2^*$;
- ☐ C $Q_1^* = Q_2^*$ (equal quantities);

- ☐ D each plant independently sets $MC_i = MR_i$, with MR_i computed on a share of demand.
5. A natural monopoly has demand $P = 50 - Q$ and cost $C(Q) = 125 + 20Q$. Under $P = AC$ (Ramsey) regulation, the quantity produced and the required subsidy are, respectively:
- ☐ A $Q = 15$, subsidy = 0;
- ☐ B $Q = 25$, subsidy = 0;
- ☐ C $Q = 30$, subsidy = 125;
- ☐ D $Q = 30$, subsidy = 0.
6. In the Tullock model with $V = 180$ and $n = 3$ contenders, total rent dissipation (X^*/V) is:
- ☐ A $1/3 \approx 33.3\%$;
- ☐ B $1/2 = 50\%$;
- ☐ C $2/3 \approx 66.7\%$;
- ☐ D 1 (complete dissipation).
7. A monopolist practices third-degree price discrimination between two markets with elasticities $|\varepsilon_1| = 2$ and $|\varepsilon_2| = 4$ and a common marginal cost MC . Which statement is **correct**?
- ☐ A $P_1 = P_2$ (the Lerner rule requires equal prices);
- ☐ B $P_1 > P_2$ because market 1 has less elastic demand;
- ☐ C $P_1 < P_2$ because market 1 has less elastic demand;
- ☐ D The sign of $P_1 - P_2$ cannot be determined without further information.
8. Which of the following statements about first-degree (perfect) price discrimination is **FALSE**?
- ☐ A The quantity produced coincides with that of perfect competition;
- ☐ B Consumer surplus is zero: $CS = 0$;
- ☐ C The deadweight loss relative to PC is zero: $DWL = 0$;
- ☐ D The unit price charged is uniform and equal to marginal cost.
9. A monopolist serves homogeneous consumers with individual demand $P(Q) = 30 - Q$ and has marginal cost $MC = 10$. The optimal two-part tariff sets the unit price and the fixed fee equal to, respectively:
- ☐ A $p = 10$, $T = 200$;

- ☐ B $p = 10, T = 400$;
☐ C $p = 20, T = 100$;
☐ D $p = 20, T = 200$.

10. A specific tax $t > 0$ levied on a monopolist with linear demand $P = a - bQ$ and weakly increasing marginal cost $MC = c + dQ$ ($d \geq 0$) generates, for consumers, a price increase ΔP_d such that:

- ☐ A $\Delta P_d = t$ (100% incidence on consumers);
☐ B $0 < \Delta P_d \leq t/2$;
☐ C $\Delta P_d > t$ (overshifting);
☐ D $\Delta P_d = 0$ (100% incidence on the monopolist).

Exercises and open questions

1. A monopolist faces the inverse demand $P(Q) = 80 - 2Q$ and has cost function $C(Q) = Q^2 + 32Q + 92$.

- Find $MR(Q)$ and $MC(Q)$. Derive Q_m^* , P_m^* , $MC(Q_m^*)$ and the profit π_m . Compute the Lerner index L and verify numerically the rule $L = 1/|\varepsilon_D|$.
- Find the quantity Q_{pc} and the price P_{pc} that would prevail under perfect competition with the same cost structure, and compute the deadweight loss DWL .
- The government introduces a specific tax $t = 12$ per unit levied on the monopolist. Find Q_m^t , P_m^t , the incidence of the tax on consumers and on the monopolist, and the tax revenue.
- Compute the post-tax profit π_m^t and the additional deadweight loss relative to the untaxed monopoly.

2. An industry is a natural monopoly with demand $P(Q) = 50 - Q$ and cost function $C(Q) = 300 + 10Q$.

- Verify formally that $AC(Q)$ is decreasing everywhere.
- Find the unregulated monopoly equilibrium: Q_m^* , P_m^* , π_m , CS_m .
- Find the quantity and price under the $P = AC$ regime (Ramsey average cost pricing), the required subsidy and CS_{AC} .
- Find the quantity and price under the $P = MC$ regime (first best), the required subsidy and CS_{MC} .
- Compare the three regimes: rank the quantities, prices and consumer surpluses. Discuss the trade-off between allocative efficiency (DWL) and cost coverage (need for a subsidy).

3. A monopolist faces the inverse demand $P(Q) = 70 - Q$ and has cost function $C(Q) = \frac{1}{2}Q^2 + 10Q + 50$.

- a) Find the uniform-price monopoly equilibrium: Q_m^* , P_m^* , $MC(Q_m^*)$, π_m , CS_m .
- b) Compute the Lerner index L and the elasticity $|\varepsilon_D|$ at (Q_m^*, P_m^*) . Verify numerically the rule $L = 1/|\varepsilon_D|$.
- c) Find the deadweight loss DWL relative to perfect competition.
- d) Now suppose the monopolist can practice first-degree price discrimination. Find Q^I , producer surplus PS^I and profit π^I .
- e) Verify that the monopolist's gain relative to the uniform-price case, $\pi^I - \pi_m$, is exactly equal to $CS_m + DWL$. Comment on the economic meaning of this decomposition.

Part V

Oligopoly

Chapter 16

Game theory

In the previous chapters we studied two polar market structures. In perfect competition the firm is a *price-taker*: too small to influence the price, it takes it as given. In monopoly the firm faces the entire market demand alone and has no rivals to worry about. In both cases a firm's decision does not depend on the decisions of other specific firms.

Most real markets, however, lie in between: a few players, each large enough to matter. The smartphone market is dominated by Apple, Samsung and Xiaomi; the European car market by Volkswagen, Stellantis, Renault and a few others; air routes are often served by two or three carriers; Italian large-scale retail is concentrated in a handful of chains. In these **oligopolistic** markets a new feature arises, absent both in perfect competition and in monopoly: **strategic interdependence**. Each firm's profit depends not only on its own choices but also on those of its competitors, and no firm can decide optimally without *anticipating* the others' moves.

To analyze strategic interdependence the tools seen so far are not enough. We need **game theory**, the branch of applied mathematics that studies the decisions of rational agents whose fates depend on one another. This chapter introduces its minimal toolkit—normal-form game, dominant strategy, Nash equilibrium, prisoner's dilemma, repeated game—which Chapter 17 will apply to the three classic oligopoly models.

16.1 The normal-form game

Definition 16.1 (Normal-form game).

A **normal-form game** is described by three elements: a set of **players** $i = 1, \dots, n$; for each player a set of **strategies** S_i , i.e., the actions available to them; and a **payoff function** $u_i(s_1, \dots, s_n)$ that assigns to each combination of strategies the outcome obtained by player i .

The analysis rests on three standard assumptions. Players are **rational**: they choose the strategy that maximizes their own payoff. They are **strategic**: in doing so they take into account that the others are also rational and will choose accordingly. The rules of the game—who plays, which strategies, which payoffs—are **common knowledge**: everyone knows them, everyone knows that everyone knows them, and so on.

When there are two players and few, discrete strategies, a game is represented by a **payoff matrix**: the rows are the strategies of player A, the columns those of player B, and each cell contains the ordered pair (u_A, u_B) . The convention, which we will follow systematically, is to write the row player's payoff first.

Example

Two firms, A and B, must decide whether to invest in an advertising campaign. Advertising steals customers from the rival, but if both advertise the effects cancel out and only the cost remains. Annual profits (millions of euros) are:

	B: Advertise	B: Don't advertise
A: Advertise	(3, 3)	(5, 1)
A: Don't advertise	(1, 5)	(4, 4)

If neither advertises both obtain 4; if both advertise, only 3; the firm that advertises while the rival abstains steals its customers and obtains 5 against 1.

16.2 Dominant strategy

The first solution concept is also the strongest. Sometimes a player has a strategy that is the best *whatever* the others do: in that case their choice requires no conjecture about the others' behavior.

Definition 16.2 (Dominant strategy).

A strategy s_i^* is **(strictly) dominant** for player i if it guarantees them a strictly higher payoff than any other of their strategies, for every possible combination of the opponents' strategies:

$$u_i(s_i^*, s_{-i}) > u_i(s_i, s_{-i}) \quad \forall s_i \neq s_i^*, \quad \forall s_{-i},$$

where s_{-i} denotes the strategy profile of all players except i .

Let us return to the advertising game and reason from the point of view of firm A, examining one column at a time. If B advertises (first column), A compares 3 (advertise) with 1 (don't advertise) and prefers to advertise. If B does not advertise (second column), A compares 5 with 4 and again prefers to advertise. Since "Advertise" is optimal in *both* scenarios, it is a dominant strategy for A. The game is symmetric, so the same holds for B.

Key idea

If a player has a dominant strategy, they will certainly play it: it is the rational choice regardless of what the others do. The limitation of the concept is that dominant strategies *do not always exist*. Most interesting games cannot be solved this way, and a more general solution criterion is needed: the Nash equilibrium.

16.3 Nash equilibrium

The Nash equilibrium, introduced by John Nash in 1950, is the central solution concept of non-cooperative game theory.

Definition 16.3 (Nash equilibrium).

A **Nash equilibrium** (NE) is a strategy profile $(s_1^*, \dots, s_n^*)$ such that no player can increase their own payoff by *unilaterally* changing their strategy, given the strategies of the others:

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \forall s_i, \quad \forall i.$$

The intuition is that in a Nash equilibrium each player is playing a *best response* to the strategies actually played by the others. Once the others' choices are observed, no one regrets their own: each would do exactly the same thing again. It is a condition of mutual consistency of expectations, not an agreement: players do not communicate, but in equilibrium each one's conjectures about the others turn out to be confirmed.

Finding the NE in the matrix: best-response highlighting

The practical method for identifying Nash equilibria in a 2×2 (or larger) matrix proceeds in three steps.

1. *For each column*, find the row player's maximum payoff and underline it: it is A's best response to that choice by B.
2. *For each row*, find the column player's maximum payoff and underline it: it is B's best response to that choice by A.
3. The cells in which *both* payoffs are underlined are the Nash equilibria: there each player is responding optimally to the other.

Let us apply the method to the advertising game. In the column "B: Advertise" A's maximum payoff is 3 (top row); in the column "B: Don't advertise" it is 5 (top row). In the row "A: Advertise" B's maximum payoff is 3 (left column); in the row "A: Don't advertise" it is 5 (left column). Underlining:

	B: Advertise	B: Don't advertise
A: Advertise	(<u>3</u> , <u>3</u>)	(<u>5</u> , 1)
A: Don't advertise	(1, <u>5</u>)	(4, 4)

The only cell with both payoffs underlined is (Advertise, Advertise): the Nash equilibrium is unique and yields (3, 3).

Remark 16.1 (Relationship between dominance and Nash).

If every player has a dominant strategy, the profile of dominant strategies is the unique Nash equilibrium—as in the advertising game, where "Advertise" is dominant for both. The converse is false: Nash equilibria also exist in games without dominant strategies. The Nash equilibrium is thus the *more general* concept; dominance is a special case that, when it applies, makes the prediction particularly robust.

16.4 The prisoner's dilemma

The advertising game has a property worth isolating. The equilibrium outcome (3, 3) is *worse for both* than the outcome (4, 4) that would arise if neither advertised. Yet (4, 4) is not an equilibrium: starting from there, each firm has an incentive to deviate unilaterally to "Advertise" ($5 > 4$). Individual rationality leads to a collectively suboptimal outcome. This structure has a name.

Definition 16.4 (Prisoner's dilemma).

A symmetric 2×2 game is a **prisoner's dilemma** if, denoting by T (temptation), R (reward

for mutual cooperation), P (punishment for mutual defection) and S (sucker's payoff) the four relevant payoffs, the following ordering holds:

$$T > R > P > S.$$

In this case “defect” is a dominant strategy, the unique NE is (Defect, Defect) with payoff P , and this equilibrium is Pareto-dominated by the outcome (Cooperate, Cooperate) with payoff R .

The name comes from the original formulation. Two suspects of a crime committed together are arrested and interrogated in separate cells, with no possibility of communicating. The prosecutor makes each of them the same offer: whoever confesses, accusing the accomplice, gets a reduced sentence; whoever stays silent cooperates with the accomplice. The payoffs, expressed in years in prison and therefore with a negative sign, are:

	Accomplice: Silent	Accomplice: Confess
Suspect: Silent	$(-1, -1)$	$(-10, 0)$
Suspect: Confess	$(0, -10)$	$(-5, -5)$

“Confess” is dominant: if the accomplice stays silent, confessing yields 0 against -1 ; if the accomplice confesses, it yields -5 against -10 . The equilibrium is therefore (Confess, Confess) with payoffs $(-5, -5)$, Pareto-dominated by (Silent, Silent) with payoffs $(-1, -1)$. In our notation $T = 0 > R = -1 > P = -5 > S = -10$.

Economic intuition

The prisoner's dilemma explains why perfectly rational agents can systematically fail to reach the outcome that both would prefer. The problem is neither irrationality nor poor information: it is the *structure* of incentives. Each, by making the optimal decision from their own point of view, imposes on the other a cost they do not take into account. It is the same logic as the negative externalities encountered in the chapters on welfare.

The prisoner's dilemma is not a curiosity: it is the basic model of a vast number of economic and social situations in which cooperation would be efficient but is not self-enforcing. We will meet it again in Chapter 17 as an exact description of the incentive to break a collusive agreement.

Remark 16.2 (The prisoner's dilemma in public goods and emissions).

The same structure describes the voluntary funding of a public good (each person is better off not contributing and enjoying the others' contributions, but if no one contributes the good is not produced) and the reduction of polluting emissions at the international level (each country is better off not reducing and letting the others reduce). In all these cases the non-cooperative outcome is inefficient, and institutions—contracts, sanctions, repeated interaction—are needed to move the system toward cooperation.

16.5 Repeated games

The prisoner's dilemma as we have presented it is a *one-shot* game: the players interact only once. But the firms in an oligopoly meet on the market repeatedly, every day, for years. If the same payoff matrix is played many times in a row by the same players, the analysis changes radically, because the prospect of future interactions creates a channel that the one-shot game lacks: the possibility of *punishing* tomorrow whoever deviates today.

16.5.1 Time horizon and discount factor

In a repeated game payoffs are realized in different periods $t = 0, 1, 2, \dots$ and must be made comparable. To this end we introduce the **discount factor** $\delta \in (0, 1)$: a payoff π obtained in period t is worth $\delta^t \pi$ when valued today. The parameter δ measures how much future periods weigh relative to the present: δ close to 1 describes *patient* players, who give the future almost the same weight as the present; δ close to 0 describes myopic players. The link with the interest rate r is $\delta = 1/(1 + r)$: low interest rates correspond to high patience.

The crucial distinction is between a finite and an infinite horizon. With a **finite horizon** of T periods known to all, cooperation does not hold: by backward induction, in the last period there is no future to protect and so everyone deviates; knowing this, already in the second-to-last period the future punishment is ineffective and everyone deviates; and so on back to the first period. With an **infinite horizon** (or an indefinite one, with no certain end date) there is no “last period” from which to start the induction: the threat of future punishment always remains credible, and cooperation *can* become sustainable. This is the case we analyze.

16.5.2 The tool: sum of a geometric series

Valuing today a stream of payoffs that repeats for infinitely many periods requires summing a geometric series. The fundamental formula, for $|\delta| < 1$, is

$$\sum_{t=0}^{\infty} \delta^t = 1 + \delta + \delta^2 + \delta^3 + \dots = \frac{1}{1 - \delta}.$$

From it follow the cases we will use, summarized in the following table.

Stream	Present value	When it is used
$\pi(1 + \delta + \delta^2 + \dots)$	$\frac{\pi}{1 - \delta}$	constant stream starting <i>today</i>
$\pi(\delta + \delta^2 + \delta^3 + \dots)$	$\frac{\delta \pi}{1 - \delta}$	constant stream starting <i>tomorrow</i>
$\pi(1 + \delta + \dots + \delta^{T-1})$	$\frac{1 - \delta^T}{1 - \delta} \pi$	constant stream over T periods

The second row is obtained from the first by multiplying by δ (the stream starts one period later); the third is the partial sum of the first T terms. These three results are all that is needed to analyze the sustainability of cooperation.

Key idea

Repetition of the game, by itself, is not enough to generate cooperation: with a finite horizon backward induction destroys it. What makes cooperation possible is the combination of an *infinite horizon* (no last period) and *sufficient patience* (δ high enough that future losses count more than the immediate gain from deviating). Chapter 17 translates this idea into the sustainability condition for a cartel.

16.6 Exercises

Exercise 16.1 (Dominant strategy and Nash equilibrium).

Two firms, A and B, simultaneously choose between an aggressive strategy (X) and an accommo-

dating one (Y). Profits (millions of euros) are:

	B: X	B: Y
A: X	(2, 2)	(8, 1)
A: Y	(1, 8)	(5, 5)

- Are there dominant strategies? For which players?
- Find the Nash equilibrium (or equilibria) using the best-response highlighting method.
- Is the equilibrium outcome Pareto-efficient? Is the game a prisoner's dilemma? Justify using the ordering of payoffs.

Exercise 16.2 (A game without dominant strategies).

Two TV broadcasters simultaneously choose the time slot for a flagship program: prime time (P) or late evening (S). The payoffs (audience share points) are:

	B: P	B: S
A: P	(2, 2)	(6, 4)
A: S	(4, 6)	(3, 3)

- Show that no player has a dominant strategy.
- Find all Nash equilibria in pure strategies.
- Comment: what kind of strategic problem does this game describe?

Exercise 16.3 (Geometric series and present value).

A firm expects to earn a constant profit $\pi = 50$ in every period, starting from the current period $t = 0$, over an infinite horizon. The discount factor is δ .

- Write the present value of the stream and compute it for $\delta = 0.8$ and for $\delta = 0.9$.
- Now suppose that the constant stream $\pi = 50$ starts only in period $t = 1$ (period $t = 0$ yields zero profit). Compute the present value for $\delta = 0.9$.
- Verify that the difference between the two present values computed with $\delta = 0.9$ is exactly the profit of period $t = 0$.

Exam essentials

1. Recognize the three elements of a normal-form game (players, strategies, payoffs) and build its 2×2 matrix.
2. Check whether a strategy is dominant by comparing its payoff with that of the alternatives, column by column (or row by row).
3. Find the Nash equilibria in pure strategies using the best-response highlighting method.
4. Recognize a prisoner's dilemma from the ordering $T > R > P > S$ and explain why the NE is Pareto-dominated.
5. Distinguish a finite horizon (backward induction $\Rightarrow$ no cooperation) from an infinite one (cooperation sustainable if δ is high).
6. Compute the present value of a constant stream using the geometric-series formulas, starting either today or tomorrow.

Chapter 17

Oligopoly models

Chapter 16 provided the language; this chapter applies it. We study three classic duopoly models, which differ in the *strategic variable* and in the *timing* of decisions. In the **Bertrand** model firms compete by setting *prices* simultaneously. In the **Cournot** model they compete by setting *quantities* simultaneously. In the **Stackelberg** model they compete in quantities but *sequentially*, with one firm moving first. Among the three we insert the analysis of **collusion**, which applies the repeated game of the previous chapter to establish when a cartel is self-enforcing.

To make comparisons immediate, all models are developed on the same reference market:

$$P = 30 - Q = 30 - q_1 - q_2, \quad MC_1 = MC_2 = 6, \quad FC = 0.$$

On this market the two benchmarks known from previous chapters are monopoly— $Q^M = 12$, $P^M = 18$, $\pi^M = 144$, obtained from $MR = 30 - 2Q = 6 = MC$ —and perfect competition, with $P = MC = 6$, $Q_{pc} = 24$ and zero profits. All oligopoly outcomes will lie between these two extremes.

17.1 Price competition: the Bertrand model

In 1883 Joseph Bertrand proposed a model in which firms compete on **price**. The assumptions are: two firms producing a **homogeneous** good with the same marginal cost $MC = 6$ and **unlimited capacity**; *simultaneous* choice of prices P_1 and P_2 ; consumers who buy entirely from the firm setting the lower price, with demand split equally in case of a tie.

17.1.1 The logic of undercutting and the equilibrium

Suppose firm 2 sets a price $P_2 > MC$. Firm 1 can set $P_1 = P_2 - \varepsilon$, with ε arbitrarily small: it thereby captures the *entire* market demand at a price only infinitesimally below P_2 , and earns a profit close to $(P_2 - MC)(30 - P_2)$, almost twice what it would earn by splitting the market in half at price P_2 . It therefore always pays to undercut the rival. But the reasoning is symmetric: firm 2 too has an incentive to set a price just below that of firm 1. As long as $P > MC$, each firm wants to cut its own price, and no configuration with $P > MC$ can be an equilibrium.

The spiral stops only at $P_1 = P_2 = MC = 6$. There no one can lower the price without selling below cost, and no one can raise it without losing *all* customers to the rival. It is the unique Nash equilibrium.

Proposition 17.1 (Bertrand equilibrium). *In the Bertrand duopoly with a homogeneous good, identical marginal costs MC and unlimited capacity, the unique Nash equilibrium entails*

$$P_1^* = P_2^* = MC, \quad \pi_1 = \pi_2 = 0.$$

Just two firms are enough to replicate the outcome of perfect competition.

The Bertrand paradox

The result is surprising: moving from one to two firms makes industry profit collapse from 144 (monopoly) to 0, and adding a third, fourth, n -th firm changes nothing further. This is the **Bertrand paradox**: the model predicts perfect competition already with a duopoly, whereas empirical evidence shows that real duopolies almost always maintain prices above marginal cost and positive profits.

17.1.2 Three ways out of the paradox

The paradox stems from three assumptions of the model; relaxing even just one of them restores positive profits.

Product differentiation. If the good is not perfectly homogeneous, consumers do not switch en masse at the slightest price difference: each firm retains its own residual demand even when charging a price above the rival's. In Hotelling's location model the distance between firms acts as a transport cost t borne by the consumer, and the equilibrium entails prices of the form $P^* = MC + t \cdot d$, increasing in the degree of differentiation d . Differentiation—spatial, by brand, by perceived quality—creates margins.

Capacity constraints. If no firm can serve the entire demand on its own at the price $P = MC$, the undercutting argument breaks down: a firm that lowers its price does not capture the whole market anyway, and the rival is left with a residual demand that it can serve at a price above marginal cost. The Edgeworth model shows that with constrained capacity an equilibrium in pure strategies often does not exist and prices fluctuate (the Edgeworth cycle). This is the typical case of capital-intensive industries with rigid capacity—steel, cement, electricity—where prices do not collapse to MC even with few players.

Repetition of the game. If firms interact repeatedly over an infinite horizon, the threat of future punishment can sustain prices above MC . This is the most important way out for theory and for antitrust policy, and we develop it in detail in the section on collusion.

17.2 Collusion and cartels

Definition 17.1 (Collusion).

Collusion occurs when the firms in an oligopoly coordinate their choices to maximize *joint* profit, behaving in effect like a monopolist that divides the market among them. An explicit collusive agreement is called a **cartel**.

On the reference market, the two colluding firms replicate the monopoly optimum: $Q^M = 12$, $P^M = 18$, $\pi^M = 144$. With an equal split each produces $q_i^{\text{coll}} = 6$ and earns $\pi_i^{\text{coll}} = 72$ —much more than the zero of the Bertrand duopoly.

17.2.1 The incentive to deviate

The cartel's problem is its instability. Suppose firms compete à la Bertrand and firm 2 abides by the agreement by setting the collusive price $P_2 = 18$. Firm 1, by setting $P_1 = 18 - \varepsilon$, takes the entire market from the rival: it sells $Q = 30 - 18 = 12$ units at a price close to 18 and earns

$$\pi_1^{\text{dev}} = (18 - 6) \cdot 12 = 144,$$

i.e., the entire monopoly profit, against the 72 from abiding by the agreement. The temptation to deviate is very strong, and by symmetry it applies to both: in a *one-shot* game collusion is not a Nash equilibrium. Here we recognize the structure of the prisoner's dilemma: the cooperative outcome (72, 72) is Pareto-superior to the non-cooperative equilibrium (0, 0), but it is not self-enforcing.

17.2.2 The grim trigger strategy and the sustainability condition

Repetition of the game offers a way out. Firms adopt a **punishment strategy** that makes deviation unprofitable. The simplest is the **grim trigger**: "I abide by the collusive price as long as the rival does too; as soon as I observe a deviation, I return to Bertrand competition *forever*, setting $P = MC$ and driving industry profits to zero in every future period".

With this strategy, firm 1 compares two scenarios, valued at period $t = 0$ over an infinite horizon.

If it *abides by* the agreement, it earns $\pi^{\text{coll}} = 72$ in every period:

$$V_{\text{coll}} = 72 + 72\delta + 72\delta^2 + \cdots = \frac{72}{1 - \delta}.$$

If it *deviates*, it earns $\pi^{\text{dev}} = 144$ today, but from tomorrow the punishment brings it back to zero profits forever:

$$V_{\text{dev}} = 144 + 0 \cdot \delta + 0 \cdot \delta^2 + \cdots = 144.$$

Collusion is sustainable when abiding by the agreement pays at least as much as deviating, $V_{\text{coll}} \geq V_{\text{dev}}$:

$$\frac{72}{1 - \delta} \geq 144 \iff 72 \geq 144(1 - \delta) \iff 144\delta \geq 72 \iff \delta \geq \delta^* = \frac{1}{2}.$$

Proposition 17.2 (Sustainability condition for collusion). *With a grim trigger strategy, collusion is sustainable as an equilibrium of the repeated game if and only if the discount factor satisfies $\delta \geq \delta^*$, where*

$$\delta^* = \frac{\pi^{\text{dev}} - \pi^{\text{coll}}}{\pi^{\text{dev}} - \pi^{\text{punish}}},$$

with π^{dev} the profit from deviation, π^{coll} the collusive profit and π^{punish} the profit in the punishment phase. In the Bertrand duopoly of the reference market, $\pi^{\text{dev}} = 144$, $\pi^{\text{coll}} = 72$, $\pi^{\text{punish}} = 0$ and hence $\delta^* = 1/2$.

Economic intuition

The condition $\delta \geq \delta^*$ has a precise economic reading: collusion holds if firms are *sufficiently patient*. Deviating brings forward a one-off gain ($\pi^{\text{dev}} - \pi^{\text{coll}}$) but sacrifices forever the difference between collusive profit and punishment profit. If the future weighs enough, the permanent loss exceeds the immediate gain. With ordinary interest rates δ is close to 0.95, well above $\delta^* = 1/2$: for real cartels the binding constraint is not patience, but the *observability of deviations* and the risk of sanctions by the antitrust authority.

Analytical example

Staying with the reference market ($\pi^{\text{coll}} = 72$, $\pi^{\text{dev}} = 144$, $\pi^{\text{punish}} = 0$, hence $\delta^* = 1/2$), we check sustainability for two values of the discount factor.

With $\delta = 0.4 < \delta^*$: $V_{\text{coll}} = 72/(1 - 0.4) = 120 < 144 = V_{\text{dev}}$. Deviating pays: collusion is *not* sustainable.

With $\delta = 0.7 > \delta^*$: $V_{\text{coll}} = 72/(1 - 0.7) = 240 > 144 = V_{\text{dev}}$. Abiding by the agreement pays: collusion *is* sustainable.

At $\delta = \delta^* = 1/2$ firms are exactly indifferent: $V_{\text{coll}} = 72/0.5 = 144 = V_{\text{dev}}$.

17.2.3 Number of firms and cartel fragility

The sustainability condition depends on the number n of firms participating in the cartel. Under Bertrand competition with n symmetric firms, collusion splits the monopoly profit ($\pi_i^{\text{coll}} = \pi^M/n$), deviation captures the entire market ($\pi^{\text{dev}} = \pi^M$) and punishment drives profits to zero ($\pi^{\text{punish}} = 0$). Substituting into the formula of Proposition 17.2:

$$\delta^*(n) = \frac{\pi^M - \pi^M/n}{\pi^M - 0} = 1 - \frac{1}{n} = \frac{n-1}{n}.$$

The threshold $\delta^*(n)$ is *increasing* in n : with $n = 2$ it equals $1/2$, with $n = 3$ it equals $2/3$, with $n = 4$ it equals $3/4$, and it tends to 1 as $n \rightarrow \infty$. The more firms participate in the cartel, the more patient they must be for the agreement to hold—and beyond a certain number collusion becomes effectively unsustainable. The result, which does not depend on the demand and cost parameters, formalizes a classic antitrust intuition: cartels are fragile in fragmented markets and robust in concentrated ones.

Remark 17.1 (The potato chip cartel: AGCM, 2026).

A recent case illustrates the model in an almost textbook fashion. On 28 April 2026 the Italian Competition Authority (AGCM) fined the three main Italian producers of private-label salty snacks—Amica Chips, Pata and Preziosi Food—a total of 23.3 million euros for a secret agreement, lasting more than eight years, to allocate supply contracts to large-scale retail chains. The mechanism was a customer-allocation scheme: in formally competitive tenders the firms submitted coordinated bids, letting the “designated” firm win for that chain. The mapping onto the model is direct: a weakly differentiated good (private-label potato chips), multi-year supply relationships (high discount factor, δ well above δ^*), and a cartel of $n = 3$ firms. Consistently with $\delta^*(3) = 2/3$, three patient firms manage to sustain the agreement. The tool that broke it was not the mathematics of sustainability but the *breakdown of observability*: the investigation started in 2024 from an anonymous whistleblower report, and the fines were reduced for the firms that cooperated under the leniency program.

Remark 17.2 (The leniency program as a prisoner’s dilemma).

The leniency program deserves a theoretical note. The antitrust authority reduces—even to zero—the fine for the firm that is first to report the cartel. In doing so it builds a *second prisoner’s dilemma*, superimposed on the one the cartel had solved: each firm knows that, if another reports first, whoever stays silent suffers the full fine. “Report before the other” becomes the dominant strategy of the new game. The authority does not merely punish cartels: it deliberately overturns their strategic structure, turning the cooperation among colluders into a new dilemma that makes it unstable.

17.3 Quantity competition: the Cournot model

The Bertrand paradox depends on having chosen price as the strategic variable. Augustin Cournot, as early as 1838, had proposed a model in which firms choose **quantity**, and the price emerges from the market to equate demand and total supply. The model describes well the markets in which the binding decision is *production capacity*, set in advance: steel, cement, electricity, raw materials, oil production quotas.

17.3.1 Reaction functions and equilibrium

The assumptions are: two firms that *simultaneously* choose quantities q_1 and q_2 of a **homogeneous** good; the price is determined by aggregate demand $P = 30 - q_1 - q_2$; each firm maximizes its own profit taking the rival's quantity as *given*. The profit of firm 1 is

$$\pi_1(q_1, q_2) = P q_1 - 6 q_1 = (30 - q_1 - q_2 - 6) q_1 = (24 - q_1 - q_2) q_1,$$

and symmetrically for firm 2. Maximizing with respect to q_1 , treating q_2 as a parameter, the first-order condition is

$$\frac{\partial \pi_1}{\partial q_1} = 24 - 2q_1 - q_2 = 0 \implies q_1^*(q_2) = 12 - \frac{q_2}{2}.$$

Definition 17.2 (Reaction function).

A firm's **reaction function** (or best-response function) is the quantity that maximizes its profit as a function of the rival's expected quantity. In the reference market:

$$q_1^*(q_2) = 12 - \frac{q_2}{2}, \quad q_2^*(q_1) = 12 - \frac{q_1}{2}.$$

The reaction function reads as follows: if the rival produces nothing ($q_2 = 0$), firm 1 becomes a monopolist and produces $q_1^* = 12 = q^M$; if the rival saturates the market, driving P down to MC ($q_2 = 24$), firm 1 is better off producing nothing. The slope $-1/2$ indicates that for each additional unit produced by the rival the firm responds by reducing its own output by half a unit: quantities are *strategic substitutes*.

In the Cournot equilibrium each firm is on its own reaction function and expectations are mutually correct. We therefore solve the system

$$\begin{cases} q_1 = 12 - \frac{q_2}{2} \\ q_2 = 12 - \frac{q_1}{2} \end{cases}$$

Substituting the second into the first:

$$q_1 = 12 - \frac{1}{2} \left(12 - \frac{q_1}{2} \right) = 6 + \frac{q_1}{4} \implies \frac{3}{4} q_1 = 6 \implies q_1^C = 8,$$

and by symmetry $q_2^C = 8$. Total quantity, price and profits follow:

$$Q^C = 16, \quad P^C = 30 - 16 = 14, \quad \pi_1^C = \pi_2^C = (14 - 6) \cdot 8 = 64.$$

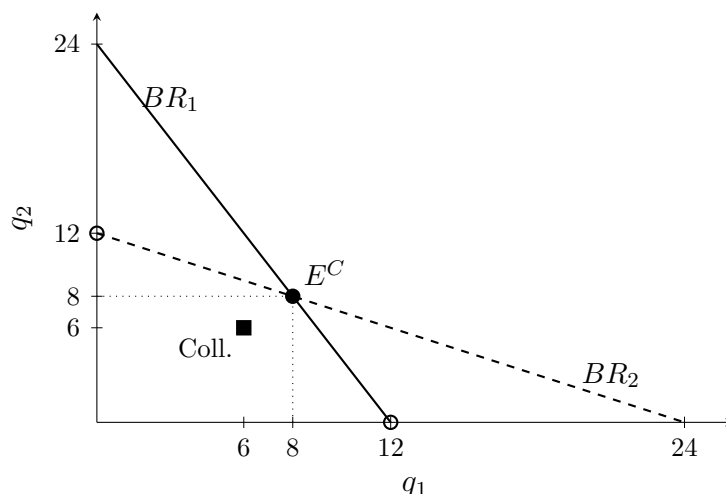


Figure 17.1: Cournot equilibrium in the reference market. The reaction functions BR_1 (solid) and BR_2 (dashed) intersect at $E^C = (8, 8)$. The points $(12, 0)$ and $(0, 12)$ correspond to single-firm monopoly; the point “Coll.” $(6, 6)$ to equal-split collusion, which does not lie on the reaction functions and is therefore not a Cournot equilibrium.

17.3.2 Comparison across structures and generalization to n firms

The Cournot outcome lies *between* monopoly and perfect competition: total quantity 16 (against 12 in monopoly and 24 in competition), price 14 (against 18 and 6), industry profit 128 (against 144 and 0). The Cournot duopoly does not collapse to perfect competition as the Bertrand one does: competing in quantities is less aggressive than competing in prices.

The result generalizes. With n identical firms, linear demand $P = a - bQ$ and marginal cost c , the symmetric Cournot equilibrium is

$$q_i^C = \frac{a - c}{(n + 1)b}, \quad Q^C = \frac{n(a - c)}{(n + 1)b}, \quad P^C = \frac{a + nc}{n + 1}.$$

For $n = 1$ we recover monopoly; as $n \rightarrow \infty$ total quantity tends to $(a - c)/b$ and price to c , i.e., to the perfectly competitive outcome. The number of firms works as a continuous measure of competitive intensity: more firms mean more output, a lower price and decreasing individual profits.

Bertrand versus Cournot

The two simultaneous-move models give opposite predictions for duopoly: Bertrand predicts $P = MC$ and zero profits already with two firms, Cournot an intermediate price with positive profits. The reason is the strategic variable. In Bertrand, by lowering the price slightly a firm captures *all* of the rival's market: competition is very fierce. In Cournot, the rival's quantity is given and each firm can only expand its own along its reaction function, without “stealing” the other's market: competition is softer. The choice of model depends on which variable, in the actual market, is really the binding one.

17.4 Sequential competition: the Stackelberg model

In the Bertrand and Cournot models firms decide simultaneously. In many markets, however, one firm moves *first* and does so *observably* and *irreversibly*: Heinrich von Stackelberg formalized

this situation in 1934. The firm that moves first is the **leader**; the one that observes and then reacts is the **follower**.

17.4.1 Backward induction and computing the equilibrium

The game is sequential and is solved by **backward induction**: we start from the last decision and work backward.

Second stage (follower). Having observed the quantity q_1 chosen by the leader, firm 2 maximizes its own profit. Its problem is identical to the Cournot one, so its reaction is the same reaction function:

$$q_2^*(q_1) = 12 - \frac{q_1}{2}.$$

First stage (leader). Firm 1 knows that the follower will react according to that function, and takes this into account by substituting it into its own profit *before* maximizing:

$$\pi_1 = (24 - q_1 - q_2) q_1 = \left(24 - q_1 - \left(12 - \frac{q_1}{2}\right)\right) q_1 = \left(12 - \frac{q_1}{2}\right) q_1.$$

The leader's first-order condition is

$$\frac{d\pi_1}{dq_1} = 12 - q_1 = 0 \implies q_1^S = 12,$$

and the follower reacts with $q_2^S = 12 - 12/2 = 6$. It follows that

$$Q^S = 18, \quad P^S = 30 - 18 = 12, \quad \pi_1^S = (12 - 6) \cdot 12 = 72, \quad \pi_2^S = (12 - 6) \cdot 6 = 36.$$

Proposition 17.3 (First-mover advantage). *In the Stackelberg duopoly the leader produces more than in Cournot and earns a higher profit, while the follower produces less and earns a lower profit: in the reference market $\pi_1^S = 72 > 64 = \pi_1^C$ and $\pi_2^S = 36 < 64 = \pi_2^C$. The first-mover advantage exists, however, only if the leader's commitment is observable and credible; without commitment, the only equilibrium remains the Cournot one.*

Economic intuition

Why does moving first pay? By committing irreversibly to a high quantity, the leader *forces* the follower to reduce its own: it “occupies” the market before the rival chooses. It is not an informational advantage—the follower knows nothing less—but a *commitment* advantage. The delicate point is precisely credibility: if the leader could revise q_1 after observing the follower's move, the commitment would vanish and we would be back to Cournot. The first-mover advantage is the economic value of irreversibility.

Remark 17.3 (Timing asymmetries in real markets).

The Stackelberg structure describes many markets. Tesla invested in electric-vehicle plants well ahead of the traditional carmakers, which later converted their own lines as followers. In air transport, low-cost carriers announce fleets and routes years in advance, leaving competitors the role of adapting. OPEC announces changes in production and non-OPEC producers react to the resulting price. In all these cases the first mover, if its commitment is credible, obtains the role—and the profit—of the leader.

17.4.2 Summary overview

The following table collects the four outcomes on the reference market $P = 30 - Q$, $MC = 6$, ordered by increasing total quantity.

Structure	q_1	q_2	Q	P	π_1	π_2
Monopoly / cartel	6	6	12	18	72	72
Cournot	8	8	16	14	64	64
Stackelberg	12	6	18	12	72	36
Bertrand / perfect comp.	—	—	24	6	0	0

Three regularities emerge. Total quantity rises, and hence price falls, moving from monopoly to Cournot to Stackelberg to Bertrand: each more competitive structure brings the outcome closer to perfect competition and improves consumer welfare. Industry profit, by contrast, decreases in the same direction. Finally, leader and follower have opposite preferences over timing: the leader prefers Stackelberg ($72 > 64$), the follower prefers Cournot ($64 > 36$).

17.5 Exercises

Exercise 17.1 (Symmetric Cournot).

Cournot duopoly with market demand $P = 50 - Q$ and identical marginal costs $MC_1 = MC_2 = 8$, zero fixed costs.

- Write the profit function and derive the reaction function of each firm.
- Find the Cournot equilibrium (q_1^C, q_2^C) , total quantity Q^C , price P^C and profits.
- Compute the monopoly outcome and the perfectly competitive outcome on the same market, and verify that the Cournot outcome lies between the two.

Exercise 17.2 (Cournot with asymmetric costs).

Cournot duopoly with demand $P = 100 - Q$. Firm 1 has $MC_1 = 10$, firm 2 has $MC_2 = 25$; zero fixed costs.

- Derive the two reaction functions.
- Find the equilibrium (q_1^C, q_2^C) , Q^C and P^C .
- Compute the profits of the two firms.
- Comment on the link between cost advantage, market share and profit.

Exercise 17.3 (Stackelberg and comparison with Cournot).

Duopoly in the industrial components market, with demand $P = 44 - Q$ and identical marginal costs $MC = 8$, zero fixed costs. Firm 1 is the Stackelberg leader: it chooses q_1 first, observably and irreversibly; firm 2 observes q_1 and then chooses q_2 .

- Using backward induction, derive q_1^S , q_2^S , Q^S and P^S .

- b) Compute the profits π_1^S and π_2^S .
- c) Solve for the Cournot equilibrium on the same market and compare it with the Stackelberg one: which firm prefers Stackelberg, which Cournot, and by how much?

Exercise 17.4 (Sustainability of a cartel).

Two firms compete à la Bertrand in a market with demand $P = 24 - Q$ and marginal cost $MC = 4$. They can collude by replicating the monopoly and splitting its profit equally; in case of deviation they adopt the grim trigger strategy (return to Bertrand forever).

- a) Compute the monopoly profit and the collusive profit per firm.
- b) Compute the profit from a single deviation (undercutting the collusive price).
- c) Determine the minimum discount factor δ^* that makes collusion sustainable, and verify it with the general formula.
- d) Is collusion sustainable with $\delta = 0.6$? And with $\delta = 0.45$?

Exam essentials

1. Derive the Bertrand equilibrium and explain the paradox: two firms suffice to obtain $P = MC$ with homogeneous goods and unlimited capacity.
2. List the three ways out of the paradox (differentiation, capacity, repetition) and the mechanism of each.
3. Build the Cournot reaction functions from the FOC and solve the system for the symmetric equilibrium.
4. Compute the Cournot equilibrium with asymmetric costs and interpret the cost–share–profit link.
5. Solve the Stackelberg model by backward induction and recognize the first-mover advantage.
6. Apply the condition $\delta \geq \delta^* = (\pi^{\text{dev}} - \pi^{\text{coll}})/(\pi^{\text{dev}} - \pi^{\text{punish}})$ to assess the sustainability of a cartel.
7. Rank monopoly, Stackelberg, Cournot and Bertrand by quantity, price, profit and welfare.

Questions and exercises

Multiple-choice questions

1. In a 2×2 game, a strategy s_i^* is strictly dominant for player i if:

- ☐ A it guarantees i the highest payoff in at least one column;
- ☐ B it is i 's best response whatever strategy the opponent plays;
- ☐ C it coincides with the opponent's dominant strategy;
- ☐ D it is part of at least one Nash equilibrium of the game.

2. Consider the payoff matrix (row payoff first):

	B: L	B: R
A: U	(3, 3)	(1, 4)
A: D	(4, 1)	(2, 2)

The Nash equilibrium in pure strategies is:

- ☐ A (U, L) , with payoffs (3, 3);
- ☐ B (D, R) , with payoffs (2, 2);
- ☐ C (D, L) , with payoffs (4, 1);
- ☐ D there is no equilibrium in pure strategies.

3. A symmetric 2×2 game is a prisoner's dilemma when the four relevant payoffs satisfy:

- ☐ A $R > T > P > S$;
- ☐ B $T > R > P > S$;
- ☐ C $T > R > S > P$;
- ☐ D $S > P > R > T$.

4. In the Bertrand duopoly with a homogeneous good, identical marginal costs equal to MC and unlimited capacity, the Nash equilibrium entails:

- ☐ A $P_1 = P_2 = MC$, zero profits;
- ☐ B $P_1 = P_2 = P^M$ (monopoly price), positive profits;
- ☐ C different prices, with the lower-price firm serving the market;
- ☐ D no equilibrium in pure strategies.

5. The “Bertrand paradox” refers to the fact that:

- ☐ A with differentiated goods prices rise above marginal cost;
- ☐ B two firms are enough to obtain the perfectly competitive outcome;
- ☐ C the Bertrand equilibrium never exists in pure strategies;
- ☐ D monopoly profit rises as the number of firms increases.

6. Cournot duopoly with $P = 40 - Q$ and $MC_1 = MC_2 = 4$. The equilibrium quantity of each firm is:

- ☐ A $q_i^C = 9$;
- ☐ B $q_i^C = 12$;
- ☐ C $q_i^C = 18$;
- ☐ D $q_i^C = 6$.

7. In the same market as the previous question ($P = 40 - Q$, $MC = 4$), the Cournot equilibrium price and the profit of each firm are, respectively:

- ☐ A $P^C = 16$, $\pi_i^C = 144$;
- ☐ B $P^C = 22$, $\pi_i^C = 162$;
- ☐ C $P^C = 16$, $\pi_i^C = 108$;
- ☐ D $P^C = 4$, $\pi_i^C = 0$.

8. In the Stackelberg duopoly (quantities, sequential moves) with identical marginal costs, compared with the Cournot equilibrium on the same market:

- ☐ A leader and follower produce the same quantity;
- ☐ B the leader produces more and earns a higher profit than the follower;
- ☐ C the follower, moving last, earns the higher profit;
- ☐ D the market price is higher than in Cournot.

9. In a cartel of n symmetric firms competing à la Bertrand, with grim trigger punishment (return to $P = MC$), the minimum discount factor that sustains collusion is:

- ☐ A $\delta^* = 1/n$, decreasing in n ;
- ☐ B $\delta^* = (n-1)/n$, increasing in n ;
- ☐ C $\delta^* = 1/2$, independent of n ;
- ☐ D $\delta^* = 1/(n+1)$, decreasing in n .

10. The *leniency* program of antitrust authorities is effective against cartels because:

- ☐ A it raises firms' discount factor, making them more patient;
- ☐ B it creates a new prisoner's dilemma in which "report first" is the dominant strategy;
- ☐ C it makes the good produced by the cartel perfectly homogeneous;
- ☐ D it guarantees the existence of a Nash equilibrium in pure strategies.

Exercises and open questions

1. Two airlines, A and B, operate on the same route and simultaneously choose whether to charge a *high* (H) or *low* (L) fare. Profits per season (millions of euros) are:

	B: H	B: L
A: H	(10, 10)	(2, 14)
A: L	(14, 2)	(5, 5)

- a) Determine any dominant strategies and the Nash equilibrium of the one-shot game. Verify that it is a prisoner's dilemma by stating the ordering $T > R > P > S$ explicitly.
- b) The two airlines operate on the route every season, over an infinite horizon, with discount factor δ and a grim trigger strategy (in case of deviation, low fare forever). Write the present value of cooperation and that of deviation for firm A.
- c) Determine the minimum discount factor δ^* that makes the high fare sustainable, and verify it with the general formula.

2. Cournot duopoly with market demand $P = 120 - 2Q$ and identical marginal costs $MC_1 = MC_2 = 24$, zero fixed costs.

- a) Write the profit function of firm 1 and derive the reaction functions.
- b) Find the Cournot equilibrium (q_1^C, q_2^C) , Q^C , P^C and the profits.
- c) Compute the monopoly outcome on the same market and compare total quantity, price and industry profit with the Cournot outcome.
- d) The firms consider a collusive agreement (equal split of the monopoly outcome). By how much would each firm's profit increase relative to Cournot?

3. Consider the same market as in the previous exercise ($P = 120 - 2Q$, $MC = 24$), but with firm 1 as the Stackelberg leader.

- a) Derive the Stackelberg equilibrium (q_1^S, q_2^S) , Q^S , P^S by backward induction.
- b) Compute the profits π_1^S and π_2^S .
- c) Compare with the Cournot equilibrium: by how much does each firm's profit change when moving from Cournot to Stackelberg? Which firm prefers which structure?
- d) Compare the price P^S with P^C : does the leader's first move benefit or harm consumers?

4. Three identical firms (a cartel with $n = 3$, inspired by the salty-snack case) operate in the market $P = 30 - Q$ with marginal cost $MC = 6$. They collude by replicating the monopoly and splitting its profit in equal shares. If one of them deviates, the other two revert to playing *the three-firm Cournot equilibrium* forever (grim trigger with Cournot punishment).

- a) Compute the collusive profit per firm.
- b) Compute each firm's profit in the three-firm Cournot equilibrium (punishment profit).
- c) Compute the profit from a single deviation: the deviating firm chooses its own optimal quantity while the other two continue to produce the collusive share.
- d) Determine the minimum discount factor δ^* that sustains the cartel and comment on its value compared with the two-firm case.

Part VI

Asymmetric Information

Chapter 18

Asymmetric Information

In the models studied so far the market operates with **perfect information**: all agents know the prices, qualities and characteristics of every good, as well as the intentions and actions of others. This assumption is convenient but rarely satisfied. The seller of a used car knows full well whether the engine has defects; the buyer does not. A loan applicant knows the prospects of his project; the bank does not. An insured person knows whether he smokes, practices extreme sports or has a family history of chronic illness; the insurance company does not. An employee knows how hard he really works; the employer can only observe the results.

In all these cases one party to the transaction has information that the other lacks. This asymmetry is not a mere imperfection: it can deeply distort exchange, drive precisely the most productive goods or workers out of the market, and in extreme cases cause entire markets to collapse. It is one of the most pervasive and least visible **market failures** of the modern economy.

This chapter introduces the two fundamental types of asymmetric information—adverse selection and moral hazard—analyzes their mechanisms using the benchmark theoretical models (Akerlof, Spence, Rothschild-Stiglitz, the principal-agent model), and discusses the available private and public remedies. The analysis is closely connected to contract theory, regulation and economic policy.

18.1 The two types of asymmetric information

The literature distinguishes two broad classes of asymmetric information according to *when* the informational advantage arises and *what* it concerns.

Definition 18.1 (Hidden information and adverse selection).

We speak of **hidden information** when one party knows, *before* the contract is signed, **its own characteristics** (type, quality, risk) that the other party cannot observe. The resulting problem is **adverse selection**: the worst types tend to prevail in the market, driving out the best types.

Definition 18.2 (Hidden action and moral hazard).

We speak of **hidden action** when one party takes, *after* the contract is signed, **actions** that the other party cannot observe or verify at zero cost. The resulting problem is **moral hazard**: an agent shielded from the consequences of his own actions tends to behave more riskily or less diligently.

Remark 18.1.

The timing distinction is crucial: in adverse selection the problem concerns who one is—an

attribute that predates the contract; in moral hazard the problem concerns what one does after the contract has been signed. In both cases the uninformed party (the **principal** in contract-theory terminology) cannot make optimal decisions, and the market fails to maximize social welfare.

Example

In the car insurance market both problems arise simultaneously.

Adverse selection (pre-contract): the company does not know the applicant's driving style, reflexes or attention to risk. Those who know they are dangerous drivers have a greater incentive to buy insurance, and to buy it for large amounts.

Moral hazard (post-contract): once covered by the policy, even careful drivers may become a little less attentive: they park in risky areas, drive a bit faster, neglect vehicle maintenance. The company does not observe every single action.

18.2 Adverse selection

18.2.1 Akerlof's model: the market for "lemons"

The seminal model of adverse selection is George Akerlof's, published in 1970 in the *Quarterly Journal of Economics* under the title "The Market for Lemons: Quality Uncertainty and the Market Mechanism." Akerlof received the Nobel Prize in Economics in 2001.

Setup. The used-car market contains two types of vehicles: *good* cars ("peaches") and *defective* cars ("lemons"). Sellers know the type of their own car; buyers do not. The subjective valuations are:

	Good car	Defective car
Value to the buyer	v_H	v_L
Value to the seller	c_H	c_L

with $v_H > c_H > v_L > c_L$: there are always gains from trade for both types. With **perfect information**, all types of car would be traded.

The information problem. The buyer cannot tell the type before purchase. If the proportion of good cars in the market is p , his expected willingness to pay is:

$$WTP = p \cdot v_H + (1 - p) \cdot v_L.$$

The seller of a good car sells only if $WTP \geq c_H$, i.e., if:

$$p \cdot v_H + (1 - p) \cdot v_L \geq c_H \quad \Longleftrightarrow \quad p \geq p^* = \frac{c_H - v_L}{v_H - v_L}.$$

Definition 18.3 (Adverse selection and critical threshold).

Adverse selection occurs when $p < p^*$: the price offered by the buyer is not enough to attract sellers of good cars. These withdraw, the actual proportion of good cars falls further, and the process feeds on itself until the **market for good cars collapses**.

Example 18.1 (Computing Akerlof's critical threshold).

$v_H = 12,000$, $c_H = 10,000$, $v_L = 6,000$, $c_L = 4,000$.

With $p = 1/2$: $WTP = \frac{1}{2} \cdot 12,000 + \frac{1}{2} \cdot 6,000 = 9,000 < 10,000 = c_H$.

Sellers of good cars do not sell. Only the bad ones remain; buyers, anticipating this, offer only $v_L = 6,000$. Sellers of bad cars accept ($6,000 > 4,000 = c_L$).

Outcome: the market for good cars collapses. Good cars remain unsold even though there are gains from trade. This is pure inefficiency due to asymmetric information.

The critical threshold is $p^* = (10,000 - 6,000)/(12,000 - 6,000) = 4/6 = 2/3$. For $p \geq 2/3$ the market works for both types; for $p < 2/3$ it collapses.

Key idea

Akerlof's mechanism is a **downward spiral** (*unraveling*): an average price that is too low drives out the good cars, lowers average quality, justifies an even lower price, drives out more good cars, and so on. The market does not converge to an efficient equilibrium but potentially to the total collapse of the high-quality segment. The welfare loss does not take the shape of a triangle (as in the deadweight loss of monopoly), but corresponds to the entire unrealized surplus on the good cars that are not traded.

18.2.2 Applications: insurance and credit

Insurance market. Akerlof's mechanism applies directly to private health insurance. The company does not observe the applicant's health status and offers a premium based on the average risk of the population. Healthy people find the premium too expensive and tend not to buy insurance; sick or high-risk people find the premium attractive and buy insurance en masse. The pool of insured people systematically deteriorates relative to the average: the company must raise the premium to cover its costs, which drives out even more healthy people, in a vicious circle that can lead to the collapse of the private market.

Remark 18.2 (The case for compulsory insurance).

The failure of the private insurance market due to adverse selection is one of the main economic arguments in favor of compulsory public health insurance (such as the Italian National Health Service (SSN) or US Medicare). Making participation compulsory eliminates the selection problem: all types enter the pool, the average risk corresponds to that of the population, and premiums can be set at levels that do not drive out the healthy.

Credit market. In the banking market the bank observes neither the quality of the borrower's project nor his financial discipline. It sets an interest rate r based on the average risk. Applicants with safe, low-return projects find the rate too high and give up; applicants with risky projects (high return if successful, high probability of failure) find the rate attractive and borrow anyway. The bank ends up with a portfolio overpopulated by risky borrowers.

Proposition 18.1 (Credit rationing (Stiglitz and Weiss, 1981)). *In the presence of adverse selection in the credit market, the bank cannot simply raise the interest rate to compensate for the higher risk: a higher rate further worsens the composition of the applicant pool, because it discourages safe projects and attracts risky ones. The equilibrium solution is often **quantity rationing of credit**: the bank does not lend to certain applicants even though they would be willing to pay a higher rate.*

18.2.3 The Rothschild-Stiglitz model (advanced)

Rothschild and Stiglitz (1976) formally analyze an insurance market with two types of insured—low risk (type L , accident probability p_L) and high risk (type H , $p_H > p_L$)—and competing companies. A contract is a pair (q, K) with premium q and indemnity K .

Contracts are represented in the (W_1, W_2) plane, where $W_1 = W_0 - q$ (wealth with no accident) and $W_2 = W_0 - L + K - q$ (wealth with an accident). The **fair-odds line** for a type with risk p has slope $-(1-p)/p$: along this line contracts are actuarially fair. With diminishing marginal utility, the insured prefers full insurance: the optimal point is at the intersection of the fair-odds line and the diagonal $W_1 = W_2$.

Proposition 18.2 (Rothschild-Stiglitz equilibrium). *In equilibrium (if it exists) companies offer two distinct contracts:*

- To type H : **full** insurance at the fair odds for p_H .
- To type L : **partial** insurance at the fair odds for p_L , with a reduced indemnity that serves as a separating device (H types do not find it worthwhile to choose it).

Type L is penalized by the information asymmetry: it obtains a contract with less coverage than in the first best. Type H is not penalized in this model (it already obtains full insurance). However, the separating equilibrium may fail to exist if the share of H types is low (L types numerous): in that case no set of contracts can be sustained as a Nash equilibrium under competition.

18.3 Moral hazard

18.3.1 Definition and structure

Definition 18.4 (Moral hazard).

Moral hazard occurs when a party, having signed a contract that partially or fully protects it from the negative consequences of its own actions, has an incentive to take actions that increase the probability or the size of such negative events, at the other party's expense.

The structure of the problem has three elements:

1. *Before the contract*: the parties agree on coverage (insurance, a wage, financing).
2. *After the contract*: the covered party changes behavior—reduces precaution, increases risk, lowers effort—because it does not bear the full consequences.
3. *The action is not observable* (or is observable only at prohibitive cost), so the other party cannot contract on it directly.

Remark 18.3 (Key difference from adverse selection).

In adverse selection the problem is *who you are*: a pre-existing attribute that the other party does not observe before the contract. In moral hazard the problem is *what you do*: an action taken after the contract that the other party does not observe. The distinction matters for remedies: adverse selection calls for type-revealing tools (signaling, screening); moral hazard calls for incentive-alignment tools (monitoring, risk sharing, bonuses).

Paradigmatic examples:

- *Car insurance*: once covered by a fully comprehensive policy, the driver has less incentive to park in safe places, drive cautiously, and carry out regular maintenance.
- *Employment relationship (principal-agent)*: the employer wants the employee to work hard; the employee, if paid a fixed wage, prefers to reduce (costly) effort to the minimum compatible with keeping the job.
- *Banking system (“too big to fail”)*: large systemic banks know that the government will not let them fail: the expectation of a bailout reduces the incentive to price risk correctly and to hold adequate reserves. The 2008 financial crisis is the emblematic case.
- *Health system*: patients covered by the SSN may demand more services than necessary; doctors paid per service may prescribe unnecessary tests.

18.3.2 A simple model of moral hazard

Consider an individual who can choose the level of precaution. Precaution reduces the probability of a harmful event, but has a private cost.

Example 18.2 (Insurance and the incentive to take precautions).
Accident probability as a function of precaution; damage: €1,000.

- *High precaution* (personal cost €100): accident probability = 10%.
- *Low precaution* (cost €0): accident probability = 40%.

Without insurance. The total expected cost is:

$$\text{High: } 0.10 \cdot 1,000 + 100 = 200 \quad \text{Low: } 0.40 \cdot 1,000 + 0 = 400.$$

The individual chooses high precaution ($200 < 400$).

With full insurance. The damage is entirely covered by the company. The individual chooses by minimizing only the personal cost: low precaution (cost 0 < cost 100). The accident probability rises to 40%.

Consequence for the company. The insurer anticipates this behavior and must raise the premium to cover the expected cost at 40% instead of 10%. If the premium is high enough, the exchange does not take place or takes place on inefficient terms. Moral hazard reduces the gains from insurance and in extreme cases eliminates them altogether.

18.3.3 The principal-agent model (advanced)

The **principal-agent model** is the canonical formalization of moral hazard in contractual relationships. The **principal** (the firm, the employer, the insurer) delegates to the agent (the manager, the employee, the insured) a task whose outcome depends both on the agent’s effort and on uncontrollable random factors.

Standard setup.

- Risk-neutral principal; risk-averse agent with utility $U(w) - c(e)$, where w is the wage, $e \in \{e_H, e_L\}$ is effort and $c_H > c_L = 0$.

- Output $\pi \in \{\pi_H, \pi_L\}$: probability p_H with effort e_H , probability $p_L < p_H$ with e_L .
- The agent has a reservation utility $\bar{u}$ (outside option).

The principal wants to induce high effort e_H . The optimal contract must satisfy two constraints:

$$\text{PC (Participation Constraint): } p_H U(w_H) + (1 - p_H) U(w_L) - c_H \geq \bar{u}$$

$$\text{IC (Incentive Constraint): } (p_H - p_L) [U(w_H) - U(w_L)] \geq c_H - c_L.$$

Proposition 18.3 (Insurance-incentives trade-off). *When effort is observable (first best), the principal can contract directly on e : the risk-averse agent is fully insured (fixed wage), and the principal bears all the risk. When effort is not observable (second best), the agent must be exposed to risk to give him incentives: the optimal contract sets $w_H > w_L$. The principal must pay the agent a higher expected wage than in the first best, i.e., a **risk premium** that compensates him for income variability (the agent still remains at the level $\bar{u}$): this extra cost is the real cost of the information asymmetry.*

The linear contract $w = \alpha + \beta \pi$ is the form most used in applications:

- $\beta = 0$: fixed wage, full insurance, no incentive $\Rightarrow$ low effort.
- $\beta = 1$: the agent “buys the firm,” maximum incentives, maximum risk.
- Optimum: $\beta^* \in (0, 1)$, increasing in the cost differential $\Delta c = c_H - c_L$, decreasing in the probability difference $\Delta p = p_H - p_L$ (an outcome that is more informative about effort relaxes the IC constraint) and in the agent’s risk aversion.

18.4 Remedies for asymmetric information

18.4.1 Signaling: the Spence model

Michael Spence (Nobel 2001) showed that the *informed* party can reduce the information asymmetry by sending a credible **signal**.

Definition 18.5 (Signaling).

Signaling is a mechanism through which the informed party takes an observable and costly action to credibly communicate its type to the other party. For the signal to be credible, it must satisfy the **differential cost** condition: the cost of the signal must be lower for high-quality types than for low-quality types.

The labor market model. Two types of workers: high productivity (H , marginal product w_H) and low productivity (L , marginal product $w_L < w_H$). Employers do not observe the type directly. The worker can acquire education at a cost c_H (for type H) or $c_L > c_H$ (for type L). The differential cost condition ($c_L > c_H$) is satisfied by assumption.

Separating equilibrium. There exists an equilibrium in which H graduates and earns w_H , and L does not graduate and earns w_L , if and only if the following **incentive-compatibility conditions** are simultaneously satisfied:

$$\text{IC}_H: w_H - c_H > w_L \quad (\text{graduating pays})$$

$$\text{IC}_L: w_H - c_L < w_L \quad (\text{imitating does not pay})$$

Key idea

Spence's radical claim is that education can work as a credible signal *even if it does not raise productivity*: what matters is not its educational content, but the fact that it is more costly to obtain for low-productivity types. In this case part of private and public spending on education is a pure **social cost of asymmetric information**, not an investment in human capital.

Exam essentials**How to check a separating equilibrium (Spence).**

1. Identify the wages offered to graduates (w_H) and to non-graduates (w_L).
2. Compute the net payoff of type H if it graduates ($w_H - c_H$) and if it does not (w_L). Check IC_H : $w_H - c_H > w_L$.
3. Compute the net payoff of type L if it imitates ($w_H - c_L$) and if it does not (w_L). Check IC_L : $w_H - c_L < w_L$.
4. If both ICs are satisfied, the separating equilibrium exists.

18.4.2 Screening

Definition 18.6 (Screening).

Screening is a mechanism through which the *uninformed* party designs a menu of contracts that induces the informed party to self-select, indirectly revealing its type through its contract choice.

The fundamental difference from signaling lies in who moves first: in signaling it is the informed party (the worker decides whether to graduate); in screening it is the uninformed party (the insurer builds the menu of policies).

Example

An insurance company does not know whether an applicant is high- or low-risk. It offers two policies:

- Policy A : low premium, high deductible (suited to low-risk types).
- Policy B : high premium, zero deductible (suited to high-risk types).

If the parameters are calibrated correctly (the self-selection IC conditions are satisfied), low-risk types choose A and high-risk types choose B . The company manages to differentiate premiums without directly observing the type.

Other examples: airfares with and without flexibility to change (the rigid, cheaper fare is chosen by tourists, who are less time-sensitive; the flexible, expensive one by business travelers), phone plans for high and low usage, optional extended warranties on electronic products.

18.4.3 Reputation, warranties and regulation

Reputation. In *repeated* markets, a high-quality firm has an incentive to invest in its reputation. A firm that behaves improperly loses its reputation and the future value of its brand: the cost of losing reputation is higher than the short-run gain from cheating customers. Reputation therefore works as a *commitment device* that makes honest behavior credible. Review systems such as those of Amazon, Airbnb and TripAdvisor aggregate dispersed information and make reputation observable at almost zero cost.

Warranties. A multi-year warranty is a credible signal of quality: only a producer genuinely confident in the quality of its product can offer it without incurring unsustainable losses from refund claims. A low-quality producer cannot imitate this strategy: the costs of honoring the warranty would be excessive.

Public regulation. When private mechanisms are insufficient, public intervention can correct information asymmetries:

- *Against adverse selection:* compulsory insurance (widens the pool), disclosure requirements (financial prospectuses, drug package leaflets), quality standards (certifications, safety rules).
- *Against moral hazard:* capital requirements (Basel III for banks), skin in the game (obligation to retain a share of the risk), inspections and controls, legal liability, minimum deductibles in health insurance policies.

Remark 18.4 (Regulation in turn creates asymmetries).

Regulation is not a free solution: it can generate *new* problems. Compulsory insurance eliminates adverse selection but may increase moral hazard (those who are covered use the service more). Bank capital requirements reduce the “too big to fail” moral hazard but may reduce the credit available. Case-by-case cost-benefit analysis is essential.

18.5 Exercises

Exercise 18.1 (Adverse selection and market collapse).

A market contains two types of goods: high quality (H) and low quality (L). Sellers know the type of their own good; buyers do not. Let v_H and v_L be the values to the buyer, and c_H and c_L the seller's opportunity costs, with $v_H > c_H > v_L > c_L$.

- a) Explain why, with symmetric information, both types of good would be traded. What is the condition for gains from trade to exist?
- b) Describe the adverse selection mechanism: how is the price offered by the buyer formed in the presence of asymmetric information? Why may high-quality sellers withdraw from the market?
- c) Illustrate the unraveling spiral: how does the collapse process feed on itself? What conditions stop the spiral before total collapse?
- d) Discuss in what sense the collapse of the market for high-quality goods is a market failure. Is it caused by “wrong” prices or by something else?

Exercise 18.2 (Spence signaling and separating conditions).

In a labor market with workers of type H (high productivity w_H) and type L (low productivity $w_L < w_H$), the worker can choose to acquire an educational qualification. The cost of the qualification is c_H for H types and c_L for L types.

- State the two incentive-compatibility conditions that must be satisfied for a separating equilibrium to exist. Explain intuitively what each condition guarantees.
- Explain why the differential cost condition ($c_L > c_H$) is necessary for the signal to be credible. What would happen if the two types bore the same cost of education?
- Discuss Spence's normative implication: in what sense can part of education spending be a social cost rather than an investment? What consequences does this observation have for the evaluation of policies expanding university education?
- Distinguish the separating equilibrium from the pooling equilibrium. Which of the two is preferable from the point of view of social efficiency?

Exercise 18.3 (Moral hazard and contract structure).

Consider a principal-agent relationship in which the principal wants to induce high effort from the agent.

- Explain why a fixed-wage contract cannot induce high effort in a moral hazard setting. Which assumption about the agent makes a fixed wage optimal under symmetric information (first best)?
- Describe the trade-off between insurance and incentives that emerges in the second best. In what sense is exposing the agent to risk necessary to induce effort, and at what cost to the principal?
- In the linear contract $w = \alpha + \beta \pi$, explain the economic meaning of the parameter β . What values of β are typically observed in settings with high vs. low observability of effort?
- Discuss why pre-2008 bank bonuses with unlimited upside and zero downside were a case of asymmetric moral hazard, and how subsequent regulatory reforms (CRD IV, Dodd-Frank) sought to remedy it.

Exercise 18.4 (Remedies: a comparison). a) Distinguish signaling and screening by indicating who moves first in each mechanism, and give a concrete example of each drawn from real markets other than education and car insurance.

- Explain why reputation works as a remedy for adverse selection in repeated markets but not in one-shot ones. What tools have digital platforms developed to extend reputation to occasional transactions as well?
- Discuss the statement: "Compulsory health insurance eliminates adverse selection but creates new moral hazard." In what sense are both parts of this statement correct?
- Is the deductible in insurance policies a screening tool or a tool for reducing moral hazard (or both)? Argue your answer, distinguishing pre-contract and post-contract effects.

Questions and exercises

Multiple-choice questions

1. Adverse selection is a problem caused by:

- ☐ A information hidden *after* the contract is signed;
- ☐ B information hidden *before* the contract is signed;
- ☐ C actions hidden *after* the contract is signed;
- ☐ D actions hidden *before* the contract is signed.

2. In Akerlof's model, an *unraveling spiral* occurs because:

- ☐ A the average quality of the market improves as the price offered rises;
- ☐ B the withdrawal of high-quality sellers reduces expected average quality, lowering the price offered and inducing further withdrawals;
- ☐ C buyers have no information about the type of car and therefore refuse to buy;
- ☐ D sellers of defective cars withdraw from the market to keep the price high.

3. The argument of Stiglitz and Weiss (1981) holds that banks facing adverse selection:

- ☐ A raise the interest rate to select the best borrowers;
- ☐ B lower the interest rate to attract more applicants;
- ☐ C ration credit quantitatively instead of raising the rate, because a higher rate worsens the composition of the pool;
- ☐ D eliminate the problem by requiring collateral from all borrowers.

4. The fundamental difference between signaling (Spence) and screening (Rothschild-Stiglitz) is:

- ☐ A the type of asymmetric information they address (adverse selection vs. moral hazard);
- ☐ B who moves first: the informed party in signaling, the uninformed party in screening;

- ☐ C the fact that signaling works only in labor markets, screening only in insurance markets;
- ☐ D the fact that signaling eliminates the asymmetry completely, while screening only reduces it partially.

5. For Spence, an educational signal is credible because:

- ☐ A education raises the productivity of type- H workers but not of type- L workers;
- ☐ B the cost of acquiring the signal is lower for H types than for L types, making imitation unprofitable for L ;
- ☐ C employers can directly verify the productivity of graduates;
- ☐ D the law makes certification of skills compulsory.

6. In a Spence separating equilibrium, the IC condition for type L requires that:

- ☐ A type L prefers graduating to not graduating;
- ☐ B type L 's net wage if it imitates ($w_H - c_L$) is lower than the wage without a degree (w_L);
- ☐ C type L is indifferent between graduating and not graduating;
- ☐ D the cost of education for type L equals that for type H .

7. Moral hazard in an insurance relationship arises because:

- ☐ A the insured hides information about his type before taking out the policy;
- ☐ B once covered against the risk, the insured reduces the precaution he would have taken without coverage;
- ☐ C the insurer sets premiums too high, excluding low risks;
- ☐ D the company does not know the applicant's risk type.

8. In the principal-agent model, the *Participation Constraint* (PC) ensures that:

- ☐ A the agent prefers high effort to low effort;
- ☐ B the agent receives at least his reservation utility, i.e., the outside option;
- ☐ C the principal obtains a non-negative profit;
- ☐ D the contract is actuarially fair.

9. In the linear contract $w = \alpha + \beta\pi$, the parameter $\beta = 0$ corresponds to:

- ☐ A maximum incentives and maximum risk for the agent;
- ☐ B a fixed wage: full insurance for the agent, no incentive to exert effort;

- ☐ C the agent buying the firm from the principal;
- ☐ D the first-best contract with observable effort.

10. Compulsory health insurance solves adverse selection because:

- ☐ A it completely eliminates moral hazard by encouraging prevention;
- ☐ B by forcing everyone to participate, it prevents the healthy from leaving the pool, stabilizing average risk;
- ☐ C it allows companies to directly observe the health status of the insured;
- ☐ D it lowers premiums below the actuarially fair level.

Open questions

1. In the Italian life insurance market, policy prices increase with the age of the insured. Analyze this pricing practice in light of the theory of adverse selection: is it a screening mechanism? Why is it more effective than a single premium for the whole population? What ethical and regulatory problems does it raise?
2. Explain the “too big to fail” phenomenon as a case of moral hazard. Identify: (a) who is the principal and who is the agent; (b) which action is not observable; (c) why the government bailout generates moral hazard. Discuss how the Basel III capital requirements and the bank resolution mechanism (BRRD) seek to reduce this problem.
3. Consider Spence’s claim that part of spending on university education is a pure cost of asymmetric information rather than a productive investment. (a) Explain the theoretical argument. (b) Discuss the empirical evidence that could distinguish Spence’s view from the human capital view (Becker). (c) What policy implications for public subsidies to university education emerge from each of the two views?
4. Digital platforms (Airbnb, Uber, Amazon Marketplace) have drastically reduced information problems in peer-to-peer markets. (a) Describe the concrete mechanisms through which each platform reduces adverse selection and moral hazard. (b) Has technology *eliminated* the information problem or only mitigated it? Cite cases in which the problems persist despite the platform. (c) Does the aggregation of detailed behavioral information create new information asymmetries? To whose advantage?
5. Analyze the deductible system in motor third-party liability (RC auto) policies as a tool that acts simultaneously on adverse selection (pre-contract) and on moral hazard (post-contract). Distinguish the two channels precisely and explain in what sense the deductible partially solves both. Discuss why a very high deductible would be socially inefficient even though it would further reduce the two information problems.

Appendix A

Basic Mathematics for Microeconomics

A.1 Basic derivatives

Function $f(x)$	Derivative $f'(x)$
c (constant)	0
x^n	nx^{n-1}
$\ln x$	$\frac{1}{x}$
e^x	e^x

Table A.1: Basic differentiation rules relevant to microeconomics.

Product:

$$(fg)' = f'g + fg'$$

Quotient:

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

Chain:

$$(f(g(x)))' = f'(g(x)) \cdot g'(x)$$

Analytical example

$$f(x) = x^2 \ln x$$

$$f'(x) = 2x \ln x + x^2 \cdot \frac{1}{x} = 2x \ln x + x.$$

A.2 Partial derivatives

When a function depends on several variables, the partial derivative measures the change with respect to a single variable, holding the others constant.

Definition A.1 (Partial derivative).

For $f(x, y)$:

$$\frac{\partial f}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}, \quad \frac{\partial f}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}.$$

Rule of thumb: to compute $\partial f / \partial x$, treat y as a constant and differentiate with respect to x as usual. Likewise for $\partial f / \partial y$.

Examples relevant to microeconomics:

- $U(x, y) = x^\alpha y^\beta$: $\frac{\partial U}{\partial x} = \alpha x^{\alpha-1} y^\beta = \frac{\alpha U}{x}$, $\frac{\partial U}{\partial y} = \beta x^\alpha y^{\beta-1} = \frac{\beta U}{y}$.
- $U(x, y) = \ln x + 2 \ln y$: $\frac{\partial U}{\partial x} = \frac{1}{x}$, $\frac{\partial U}{\partial y} = \frac{2}{y}$.
- $f(L, K) = AL^\alpha K^\beta$: $MPL = \frac{\partial f}{\partial L} = \alpha AL^{\alpha-1} K^\beta$, $MPK = \frac{\partial f}{\partial K} = \beta AL^\alpha K^{\beta-1}$.

A.3 Maxima and minima of functions of one variable

First-order necessary condition (FOC):

$$f'(x^*) = 0.$$

Second-order sufficient condition (SOC):

$$f''(x^*) < 0 \quad (\text{local maximum}), \quad f''(x^*) > 0 \quad (\text{local minimum}).$$

Key idea

The FOCs identify the *candidates* for the optimum. The SOC's determine the *nature* of the critical point (maximum, minimum, or inflection point).

Analytical example

$$\begin{aligned} \pi(q) &= 120q - 2q^2 - 200 \\ \pi'(q) &= 120 - 4q = 0 \quad \Rightarrow \quad q^* = 30. \quad \pi''(q) = -4 < 0 \quad (\text{maximum}). \end{aligned}$$

A.4 Constrained optimization and Lagrange multipliers

In microeconomics almost all optimization problems are *constrained*: the consumer maximizes $U(x, y)$ subject to $p_x x + p_y y = M$; the firm minimizes $wL + rK$ subject to $f(L, K) = \bar{Q}$. The general method is the **Lagrangian**.

Lagrange method (constrained maximum)

For $\max f(x, y)$ s.t. $g(x, y) = 0$, define:

$$\mathcal{L}(x, y, \lambda) = f(x, y) - \lambda g(x, y).$$

The first-order conditions (FOCs) are:

$$\frac{\partial \mathcal{L}}{\partial x} = 0, \quad \frac{\partial \mathcal{L}}{\partial y} = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda} = 0 \quad (\equiv g(x, y) = 0).$$

Interpretation of λ : from the FOC $\partial \mathcal{L} / \partial x = 0$ we obtain $\frac{\partial f}{\partial x} = \lambda \frac{\partial g}{\partial x}$. In the consumer's problem ($g = p_x x + p_y y - M$):

$$MU_x = \lambda p_x \quad \Rightarrow \quad \lambda = \frac{MU_x}{p_x} = \frac{MU_y}{p_y}.$$

λ measures the *marginal utility of income*: how much optimal utility increases if income M increases by one euro.

Example—Consumer with $U(x, y) = \ln x + 2 \ln y$:

$$\mathcal{L} = \ln x + 2 \ln y - \lambda(p_x x + p_y y - M).$$

FOCs:

$$\frac{1}{x} = \lambda p_x, \quad \frac{2}{y} = \lambda p_y, \quad p_x x + p_y y = M.$$

Dividing the first two: $\frac{y}{2x} = \frac{p_x}{p_y}$, i.e., $MRS = p_x/p_y$. Substituting into the constraint: $x^* = M/(3p_x)$, $y^* = 2M/(3p_y)$ (the consumer spends 1/3 of income on x and 2/3 on y).

Remark A.1 (Constrained second-order conditions).

For a constrained maximum in two variables one must check that the bordered Hessian matrix is negative definite. In the typical cases of the course (Cobb–Douglas, strictly concave functions) the SOC is automatically satisfied at an interior optimum: it is sufficient to check the FOCs.

A.5 Properties of powers and logarithms

Powers

- $x^a x^b = x^{a+b}$
- $\frac{x^a}{x^b} = x^{a-b}$
- $(x^a)^b = x^{ab}$
- $x^0 = 1; \quad x^{-a} = 1/x^a; \quad x^{1/2} = \sqrt{x}$

Logarithms

- $\ln(ab) = \ln a + \ln b$
- $\ln\left(\frac{a}{b}\right) = \ln a - \ln b$

- $\ln(a^b) = b \ln a$
- $\ln(e^x) = x; \quad e^{\ln x} = x$

Cobb–Douglas utility: Compute the marginal utilities of $U(x, y) = x^{0.5}y^{0.5}$:

$$MU_x = 0.5 x^{-0.5} y^{0.5} = \frac{1}{2} \sqrt{\frac{y}{x}}, \quad MU_y = 0.5 x^{0.5} y^{-0.5} = \frac{1}{2} \sqrt{\frac{x}{y}}.$$

$$MRS = MU_x / MU_y = y/x.$$

A.6 Homogeneous functions

Definition A.2 (Homogeneous function of degree k).

$f(x, y)$ is homogeneous of degree k if, for every $t > 0$:

$$f(tx, ty) = t^k f(x, y).$$

Practical test: multiply each argument by t , simplify, and read off the degree k .

Examples:

- $U(x, y) = x^\alpha y^\beta$: $U(tx, ty) = (tx)^\alpha (ty)^\beta = t^{\alpha+\beta} x^\alpha y^\beta = t^{\alpha+\beta} U$. Degree $k = \alpha + \beta$.
- Budget constraint: the set $B(\lambda M, \lambda p_x, \lambda p_y) = B(M, p_x, p_y)$ because multiplying everything by λ leaves the inequality $p_x x + p_y y \leq M$ unchanged. Marshallian demand is homogeneous of degree 0 in (M, p_x, p_y) .
- Cobb–Douglas production function: homogeneous of degree $\alpha + \beta$. If $\alpha + \beta = 1$: constant returns to scale.

A.7 Special products

- $(x + y)^2 = x^2 + 2xy + y^2$
- $(x - y)^2 = x^2 - 2xy + y^2$
- $x^2 - y^2 = (x - y)(x + y)$

Appendix B

Formula Sheet

This formula sheet collects, block by block, all the formulas and optimality conditions covered in the course. The conditions apply to *interior* solutions; always check non-negativity and, where necessary, check for a corner solution.

B.1 Consumer theory

Marginal utility and MRS

$$MU_x = \frac{\partial U}{\partial x}, \quad MU_y = \frac{\partial U}{\partial y}$$
$$MRS_{xy} = -\frac{dy}{dx}\bigg|_{\bar{U}} = \frac{MU_x}{MU_y}$$

MRS diminishing along the curve $\Leftrightarrow$ convex preferences (not equivalent to $\partial^2 U / \partial x^2 < 0$).

Budget constraint and consumer optimum

$$p_x x + p_y y = M \quad (\text{budget line; slope} = -p_x/p_y)$$

Interior optimum condition:

$$\boxed{MRS_{xy} = \frac{p_x}{p_y}} \quad \Leftrightarrow \quad \frac{MU_x}{p_x} = \frac{MU_y}{p_y}$$

Lagrangian: $\mathcal{L} = U(x, y) - \lambda(p_x x + p_y y - M)$

FOCs: $MU_x = \lambda p_x$, $MU_y = \lambda p_y$, $p_x x + p_y y = M$

$\lambda = MU_x/p_x = MU_y/p_y =$ marginal utility of income.

Three types of preferences

Type	Function	Optimum
Cobb-Douglas	$U = x^\alpha y^\beta$	$x^* = \frac{\alpha}{\alpha+\beta} \frac{M}{p_x}, y^* = \frac{\beta}{\alpha+\beta} \frac{M}{p_y}$
Perfect substitutes	$U = ax + by$	corner solution if $a/b \neq p_x/p_y$
Perfect complements	$U = \min\{ax, by\}$	$ax^* = by^*$; optimum at the kink $x^* = \frac{bM}{bp_x + ap_y}, y^* = \frac{aM}{bp_x + ap_y}$

Corner solution (quasi-linear and other cases)

If the interior FOCs yield $y^* < 0$ (or $x^* < 0$), the solution is a corner solution. Correct condition:

$$y^* = 0, \quad x^* = \frac{M}{p_x}, \quad \text{to be checked: } MRS(x^*, 0) \geq \frac{p_x}{p_y}$$

B.2 Individual demand and elasticities**Slutsky equation**

$$\underbrace{\frac{\partial x^D}{\partial p_x}}_{\text{total effect}} = \underbrace{\frac{\partial x^c}{\partial p_x}}_{\leq 0 \text{ (substitution)}} - \underbrace{x^D \frac{\partial x^D}{\partial M}}_{\text{income effect}}$$

Substitution effect always ≤ 0 ; income effect ≥ 0 depending on the type of good.

The three elasticities

$$\varepsilon_P = \frac{\partial Q}{\partial P} \cdot \frac{P}{Q} \quad \varepsilon_M = \frac{\partial Q}{\partial M} \cdot \frac{M}{Q} \quad \varepsilon_{xy} = \frac{\partial Q_x}{\partial P_y} \cdot \frac{P_y}{Q_x}$$

Price elasticity ε_P

$|\varepsilon_P| > 1$ elastic
 $|\varepsilon_P| < 1$ inelastic
 $|\varepsilon_P| = 1$ unit elastic

Income elasticity ε_M

$\varepsilon_M < 0$ inferior
 $0 < \varepsilon_M < 1$ necessity
 $\varepsilon_M > 1$ luxury

Cross-price elasticity

$\varepsilon_{xy} > 0$ substitutes
 $\varepsilon_{xy} < 0$ complements

Total revenue and ε_P : $TR = P \cdot Q$. $|\varepsilon_P| < 1 \Rightarrow P \uparrow \Rightarrow TR \uparrow$; $|\varepsilon_P| > 1 \Rightarrow P \uparrow \Rightarrow TR \downarrow$;
 $TR \text{ max} \Leftrightarrow |\varepsilon_P| = 1$.

B.3 Technology and production

Production function, MP and AP

$$Q = f(L, K), \quad MP_L = \frac{\partial Q}{\partial L}, \quad MP_K = \frac{\partial Q}{\partial K}, \quad AP_L = \frac{Q}{L}, \quad AP_K = \frac{Q}{K}$$

MP–AP rule: $MP_L > AP_L \Rightarrow AP_L \nearrow$; $MP_L < AP_L \Rightarrow AP_L \searrow$; $MP_L = AP_L \Rightarrow AP_L$ at its maximum.

Cobb–Douglas $Q = AL^\alpha K^\beta$:

$$MP_L = \alpha AL^{\alpha-1} K^\beta, \quad MP_K = \beta AL^\alpha K^{\beta-1}$$

Returns to scale

$$f(tL, tK) \begin{cases} > t f(L, K) & \text{IRS (increasing)} \\ = t f(L, K) & \text{CRS (constant)} \\ < t f(L, K) & \text{DRS (decreasing)} \end{cases}$$

Cobb–Douglas: $\alpha + \beta > 1 \Rightarrow \text{IRS}$; $\alpha + \beta = 1 \Rightarrow \text{CRS}$; $\alpha + \beta < 1 \Rightarrow \text{DRS}$.

Caution: SR vs LR

Diminishing marginal returns (SR, one fixed input) and returns to scale (LR, all inputs vary) are **orthogonal** concepts: they can coexist without contradiction.

B.4 Isoquants, MRTS and cost minimization

MRTS and isocost

$$MRTS_{LK} = -\frac{dK}{dL} \Big|_{\bar{Q}} = \frac{MP_L}{MP_K}$$

Isocost: $wL + rK = C \Rightarrow K = C/r - (w/r)L$ (slope = $-w/r$).

Optimality condition (cost min.):

$$\boxed{MRTS_{LK} = \frac{w}{r}} \Leftrightarrow \frac{MP_L}{w} = \frac{MP_K}{r}$$

Lagrangian: $\mathcal{L} = wL + rK - \lambda(f(L, K) - \bar{Q})$

FOCs: $w = \lambda MP_L$, $r = \lambda MP_K$; $\lambda = w/MP_L = r/MP_K = \text{marginal cost of production}$.

Special cases: perfect complements and substitutes

Technology	Isoquants	Optimum
$Q = \min\{aL, bK\}$	L-shaped	kink: $aL^* = bK^* = \bar{Q}$; $MRTS$ undefined
$Q = aL + bK$	parallel lines	corner sol.; $MRTS = a/b$ constant

With perfect complements $MRTS = w/r$ **does not apply**: the function is not differentiable at the kink; the optimum is the kink of the isoquant regardless of prices.

B.5 Short-run cost curves

SR cost structure

$$TC(Q) = FC + VC(Q)$$

$$AC(Q) = \frac{TC}{Q} = AFC(Q) + AVC(Q)$$

$$AFC(Q) = \frac{FC}{Q} \quad (\text{always decreasing})$$

$$MC(Q) = \frac{dTC}{dQ} = \frac{dVC}{dQ}$$

MC–AC rule: $MC < AC \Rightarrow AC \searrow$; $MC > AC \Rightarrow AC \nearrow$; $MC = AC \Rightarrow AC$ at its minimum.

The same rule holds for MC and AVC .

Minimum of average cost (efficient scale)

$$\frac{dAC}{dQ} = 0 \Leftrightarrow MC(Q^*) = AC(Q^*)$$

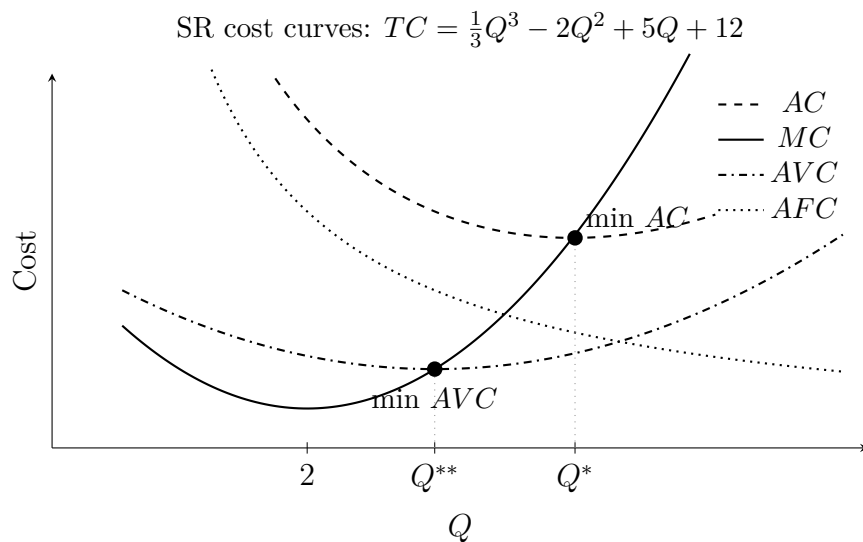
Example with $TC = F + cQ^2$:

$$MC = 2cQ, \quad AC = \frac{F}{Q} + cQ \Rightarrow Q^* = \sqrt{\frac{F}{c}}$$

For an SR AC , Q^* is the efficient scale of that plant; the minimum of the LAC is the **minimum efficient scale** (MES).

Effects of shocks: $F \uparrow$ $AC \uparrow$, MC unchanged $Q^* \uparrow$
 $c \uparrow$ $AC \uparrow$, $MC \uparrow$ $Q^* \downarrow$

Note: the minimum of AVC always comes before that of AC ($Q^{**} < Q^*$), because decreasing AFC shifts the minimum of AC to the right.



B.6 Long-run costs and the LAC curve

LAC curve: lower envelope

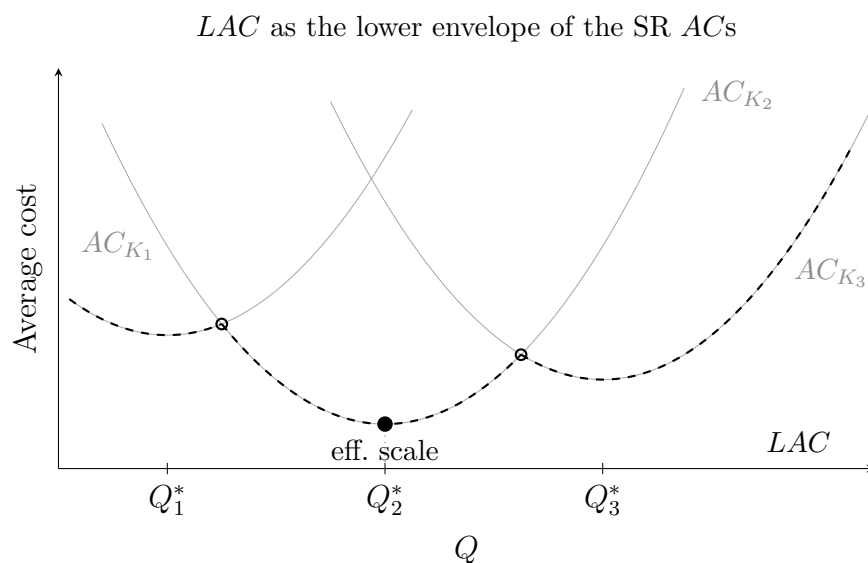
$$LAC(Q) = \min_K AC_{SR}(Q; K)$$

The LAC **cannot lie above** any SR AC curve: $LAC(Q) \leq AC_{SR}(Q; K)$ for every K . With a continuum of plant sizes it touches each SR AC at one point (tangency); with a finite number of plants each SR AC can form an entire segment of the LAC .

Decreasing $LAC \Leftrightarrow$ economies of scale (IRS)

Increasing $LAC \Leftrightarrow$ diseconomies of scale (DRS)

Minimum of $LAC =$ long-run efficient scale



Condition for the LAC to be a correct envelope

The LAC must satisfy $LAC(Q) \leq AC_K(Q)$ for every Q and for every K . If the AC_{SR} and the LAC are drawn as *independent* objects, the graph is almost always wrong: the curves must share the same functional parameters and the LAC must coincide with the AC_{SR} that dominates in each interval of Q .

B.7 Optimality conditions

Summary

Problem	Optimality condition	Notes
Consumer optimum	$MRS_{xy} = p_x/p_y$	$\Leftrightarrow MU_x/p_x = MU_y/p_y$
Cobb-Douglas	$x^* = \frac{\alpha}{\alpha+\beta} \frac{M}{p_x}$	constant expenditure share
Slutsky	$\partial x^D / \partial p_x = \partial x^c / \partial p_x - x^D \partial x^D / \partial M$	substitution eff. ≤ 0
Price elasticity	$\varepsilon_P = \frac{\partial Q}{\partial P} \cdot \frac{P}{Q}$	TR max if $ \varepsilon_P = 1$
MP-AP rule	$MP_L \gtrless AP_L \Rightarrow AP_L \nearrow \searrow$	max AP_L where $MP_L = AP_L$
MRTS	MP_L/MP_K	optimum: $MRTS = w/r$
Returns to scale (C-D)	$\alpha + \beta \gtrless 1$	IRS / CRS / DRS
Minimum of average cost	$MC(Q^*) = AC(Q^*)$	$Q^* = \sqrt{F/c}$ if $TC = F + cQ^2$
LR cost min.	$MRTS = w/r$	LAC = lower envelope

B.8 Profit maximization

Price-taking firm. It takes P as exogenously given. Total revenue $R(Q) = P \cdot Q$; marginal revenue $MR = P$.

Maximization problem: $\max_{Q \geq 0} \{ \pi(Q) = P \cdot Q - C(Q) \}$.

First- and second-order conditions:

$$\text{FOC: } P = MC(Q^*). \quad \text{SOC: } MC'(Q^*) > 0$$

(i.e., the FOC identifies a maximum only where MC is increasing; if $P = MC$ has multiple roots, select the one with $MC' > 0$.)

Profit as an area:

$$\pi(Q^*) = [P - AC(Q^*)] \cdot Q^*.$$

Sign of profit: $P > AC(Q^*) \Rightarrow \pi > 0$; $P = AC(Q^*) \Rightarrow \pi = 0$ (LR break-even); $P < AC(Q^*) \Rightarrow \pi < 0$.

Shutdown condition:

$$\text{SR: } P \geq AVC_{\min}. \quad \text{LR: } P \geq AC_{\min}.$$

The four zones (as P varies):

Zone	Condition	SR decision	LR decision
1	$P > AC_{\min}$	produce	stay ($\pi > 0$)
2	$P = AC_{\min}$	produce	indifferent (break-even)
3	$AVC_{\min} \leq P < AC_{\min}$	produce (loss, $\pi > -FC$)	exit
4	$P < AVC_{\min}$	shut down ($\pi = -FC$)	exit

Accounting vs economic profit:

$$\pi_{\text{acct}} = R - C_{\text{expl.}}; \quad \pi_{\text{econ}} = R - C_{\text{expl.}} - C_{\text{opp.}}$$

(economic profit includes the opportunity costs of all inputs; it is zero in LR equilibrium with free entry.)

B.9 Individual supply and market supply

SR individual supply curve: increasing portion of MC above $AVC_{\min}$:

$$Q^s(P) = \begin{cases} \text{solution of } P = MC(Q) \text{ with } MC' > 0 & P \geq AVC_{\min}, \\ 0 & P < AVC_{\min}. \end{cases}$$

An increase in FC *does not shift* $Q^s(P)$ ($MC = dVC/dQ$, independent of FC). An increase in variable costs (or a fall in productivity) shifts Q^s to the left.

Indirect profit function and the envelope theorem:

$$\Pi^*(P) = P \cdot Q^s(P) - C(Q^s(P)) \quad \implies \quad \frac{d\Pi^*}{dP} = Q^s(P)$$

(Hotelling's lemma: the derivative of the indirect profit function with respect to price yields optimal supply). Moreover $d^2\Pi^*/dP^2 = dQ^s/dP > 0$: Π^* is convex in P , equivalently Q^s is increasing (monotonicity of supply).

Market supply (horizontal summation). Given N firms with individual supplies $q_i^s(P)$:

$$Q^S(P) = \sum_{i=1}^N q_i^s(P).$$

If firms are heterogeneous with different $AVC_{\min}^i$, $Q^S(P)$ has *kinks* at the prices at which more and more types of firms enter production.

Supply elasticity:

$$\varepsilon_S = \frac{dQ^s}{dP} \cdot \frac{P}{Q^s}.$$

Special cases: line through the origin $\implies \varepsilon_S = 1$ at every point; vertical supply $\implies \varepsilon_S = 0$; horizontal supply $\implies \varepsilon_S \rightarrow \infty$. Determinants: availability of inputs, time horizon (LR more elastic than SR), technological substitutability.

B.10 Competitive equilibrium, welfare and the long run

Market equilibrium: pair (P^*, Q^*) such that $Q^D(P^*) = Q^S(P^*) = Q^*$. It is found by solving the system $Q^D(P) = Q^S(P)$.

Walrasian adjustment:

$$P < P^* \implies \text{excess demand} \implies P \uparrow; \quad P > P^* \implies \text{excess supply} \implies P \downarrow.$$

Stability: $\frac{d}{dP}[Q^S(P) - Q^D(P)]|_{P^*} > 0$.

Surplus and total welfare.

$$CS = \int_0^{Q^*} [P_D(q) - P^*] dq, \quad PS = \int_0^{Q^*} [P^* - P_S(q)] dq, \quad TS = CS + PS.$$

For linear curves: CS and PS are triangular areas, $CS = \frac{1}{2}Q^*(P_{\max}^D - P^*)$, $PS = \frac{1}{2}Q^*(P^* - P_{\min}^S)$.

First Fundamental Theorem of Welfare Economics. In a perfectly competitive market, the equilibrium (P^*, Q^*) maximizes TS. Any deviation from Q^* generates a deadweight loss (DWL).

Comparative statics (linear curves):

Shock	ΔQ^*	ΔP^*	
$D \uparrow$	$\uparrow$	$\uparrow$	
$D \downarrow$	$\downarrow$	$\downarrow$	
$S \uparrow$ (rightward)	$\uparrow$	$\downarrow$	
$S \downarrow$ (leftward)	$\downarrow$	$\uparrow$	
$D \uparrow, S \uparrow$	$\uparrow$ cert.	?	P^* ambiguous
$D \uparrow, S \downarrow$	?	$\uparrow$ cert.	Q^* ambiguous

Incidence of shocks. For a given shock: it falls on the *less* elastic curve. Inelastic supply $\Rightarrow$ large ΔP , small ΔQ .

LR equilibrium with free entry:

$$P_{LR}^* = AC_{\min} \iff \pi = 0, \quad q^* = \arg \min_q AC(q) \quad (\text{efficient scale}).$$

Number of firms: $n^* = Q_{\text{mkt}}^*/q^*$ (rounded to an integer). In the LR, a shock to D shifts Q_{mkt}^* and n^* , but P_{LR}^* remains anchored to $AC_{\min}$ if the industry is a constant-cost industry.

LR supply curve by input-cost regime:

Regime	$\partial w / \partial n$	Shape of LR curve
Constant costs	$= 0$	horizontal, $\varepsilon_S^{\text{LR}} = \infty$
Increasing costs	> 0	upward-sloping, $0 < \varepsilon_S^{\text{LR}} < \infty$
Decreasing costs	< 0	downward-sloping (rare)

B.11 Deadweight loss and administered prices

Deadweight loss (DWL).

$$DWL = TS(Q^*) - TS(\tilde{Q}) \geq 0.$$

Triangle formula (linear curves):

$$\text{DWL} = \frac{1}{2} (Q^* - \tilde{Q}) \cdot [P_D(\tilde{Q}) - P_S(\tilde{Q})].$$

General integral form:

$$\text{DWL}(\tilde{Q}) = \int_{\tilde{Q}}^{Q^*} [P_D(q) - P_S(q)] dq.$$

Price ceiling $\bar{P} < P^*$: quantity traded $\tilde{Q} = Q^S(\bar{P})$ (short side = supply). Excess demand $= Q^D(\bar{P}) - Q^S(\bar{P})$.

$$\begin{aligned} \text{CS}' &= \int_0^{\tilde{Q}} [P_D(q) - \bar{P}] dq, \\ \text{PS}' &= \int_0^{\tilde{Q}} [\bar{P} - P_S(q)] dq. \end{aligned}$$

ΔCS is positive only with efficient rationing (units go to buyers with the highest WTP); with random rationing CS can fall below its equilibrium level.

Price floor $\underline{P} > P^*$: quantity traded $\tilde{Q} = Q^D(\underline{P})$ (short side = demand). Excess supply $= Q^S(\underline{P}) - Q^D(\underline{P})$.

$$\text{CS}' = \int_0^{\tilde{Q}} [P_D(q) - \underline{P}] dq, \quad \text{PS}' = \int_0^{\tilde{Q}} [\underline{P} - P_S(q)] dq.$$

Ceiling/floor symmetry. For the same $\tilde{Q}$ (and hence the same $P_D(\tilde{Q}) - P_S(\tilde{Q})$), ceiling and floor produce the same DWL but mirror-image distributions between CS and PS.

Accounting identity. In the absence of public revenue/cost: $\text{CS} + \text{PS} + \text{DWL} = \text{TS}^*$.

When an intervention is binding: ceiling only if $\bar{P} < P^*$; floor only if $\underline{P} > P^*$. Otherwise, the constraint does not bind.

B.12 Taxes and subsidies

Types of tax.

- *Specific* (t per unit): S shifts up by t .
- *Ad valorem* ($\tau\%$ of the price): S rotates upward around the origin.
- *Lump-sum* (F): does not change MC ; $\text{DWL} = 0$.

Specific tax t on the producer. Post-tax supply curve: $Q_t^S(P_d) = Q^S(P_d - t)$. Equilibrium: $Q^D(P_d^*) = Q^S(P_d^* - t)$, with

$$P_d^* - P_s^* = t \quad (\text{tax wedge}).$$

Areas after the tax:

$$\begin{aligned} \text{CS} &= \frac{1}{2} Q_t^*(P_{\max}^D - P_d^*), & \text{PS} &= \frac{1}{2} Q_t^*(P_s^* - P_{\min}^S), \\ T &= t \cdot Q_t^* \quad (\text{revenue}), & \text{DWL} &= \frac{1}{2} (Q^* - Q_t^*) \cdot t. \end{aligned}$$

Accounting identity: $CS + PS + T + DWL = TS^*$.

Tax incidence.

$$\begin{aligned} \text{Cons. share} &= \frac{\Delta P_d}{t} = \frac{P_d^* - P^*}{t}, & \text{Prod. share} &= \frac{|\Delta P_s|}{t} = \frac{P^* - P_s^*}{t}, \\ \text{Cons. share} &= \frac{\varepsilon_S}{\varepsilon_S - \varepsilon_D}, & \text{Prod. share} &= \frac{-\varepsilon_D}{\varepsilon_S - \varepsilon_D}. \end{aligned}$$

Rule: the *less elastic* side bears the larger share. Extreme cases: $\varepsilon_D = 0$ or $\varepsilon_S \rightarrow \infty \Rightarrow 100\%$ on consumers; $\varepsilon_S = 0$ or $\varepsilon_D \rightarrow -\infty \Rightarrow 100\%$ on producers.

Equivalence Principle. Tax incidence does not depend on who formally remits the tax to the government, but only on the elasticities of D and S at equilibrium.

DWL–tax rate relationship (linear curves):

$$DWL(t) = \frac{1}{2} \cdot (Q^* - Q_t) \cdot t \propto t^2.$$

Doubling t *quadruples* the DWL. Policy corollary (*broad base, low rate*): it is more efficient to raise a given revenue with many small taxes on different bases than with a single high tax.

Ad valorem tax. For rate τ : $(1 - \tau)P_d = MC(Q)$, with $S_\tau(P_d) = Q^S((1 - \tau)P_d)$. *Local equivalence*: a specific tax t with equilibrium P_d^* is equivalent to an ad valorem tax $\tau^* = t/P_d^*$ (same equilibrium, same areas).

Tariffs and import quotas. For the same restriction of imports, a tariff and a quota produce the same domestic price, CS, PS and DWL. The only difference is *who collects* the rectangle between the world price and the domestic price: revenue to the government (tariff) vs rent to the license holders (quota).

Specific subsidy s to the producer. The mirror image of the tax: the supply curve shifts down by s . Equilibrium: $Q^D(P_d^*) = Q^S(P_d^* + s)$, with $P_s^* - P_d^* = s$. $Q_s^* > Q^*$. Areas:

$$\begin{aligned} CS &= \frac{1}{2}Q_s^*(P_{\max}^D - P_d^*), & PS &= \frac{1}{2}Q_s^*(P_s^* - P_{\min}^S), \\ \text{Public cost} &= s \cdot Q_s^*, & TS_{\text{net}} &= CS + PS - \text{Cost}, \\ DWL &= \frac{1}{2}(Q_s^* - Q^*) \cdot s. \end{aligned}$$

The subsidy creates DWL because it encourages the production of units with $P_S(Q) > P_D(Q)$: the social cost exceeds the social value. It is identical in magnitude to the DWL of a tax at the same rate ($t = s$) in the same market.

B.13 Monopoly

The monopolist's marginal revenue.

$$MR(Q) = P(Q) + Q \cdot P'(Q) < P(Q).$$

Linear demand $P = a - bQ$: $MR = a - 2bQ$ (same intercept, twice the slope; $MR = 0$ at $Q = a/(2b)$, half the market).

MR–elasticity relationship:

$$MR = P \left(1 - \frac{1}{|\varepsilon_D|} \right) \geq 0 \Leftrightarrow |\varepsilon_D| \geq 1.$$

The monopolist always operates on the elastic portion of demand.

Optimality condition:

$$MR(Q_m^*) = MC(Q_m^*), \quad P_m^* = P(Q_m^*) > MC(Q_m^*).$$

Comparison with perfect competition ($P = a - bQ$, constant $MC = c$):

	PC	Monopoly
Quantity	$Q_{pc} = (a - c)/b$	$Q_m^* = (a - c)/(2b) = Q_{pc}/2$
Price	$P_{pc} = c$	$P_m^* = (a + c)/2$
Profit	0	$\pi_m = (a - c)^2/(4b)$
DWL	0	$DWL_m = (a - c)^2/(8b)$

Lerner index (market power):

$$L = \frac{P_m^* - MC}{P_m^*} = \frac{1}{|\varepsilon_D|} \in [0, 1].$$

$L = 0$ under PC; L increases as $|\varepsilon_D|$ decreases.

Deadweight loss of monopoly (linear curves, constant $MC = c$):

$$DWL_m = \frac{1}{2}(Q_{pc} - Q_m^*)(P_m^* - c) = \frac{(a - c)^2}{8b}.$$

B.14 Oligopoly and game theory

Nash equilibrium. $(s_1^*, \dots, s_n^*)$ such that no player has an incentive to deviate unilaterally:

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \forall s_i, \forall i.$$

A *strictly dominant* strategy is optimal regardless of the others' choices.

Reference market: $P = a - bQ$, $MC_i = c$ for every i .

Bertrand (prices, simultaneous, homogeneous good)

$$P_1^* = P_2^* = c = MC, \quad \pi_i = 0.$$

Two firms are enough to replicate PC (*Bertrand paradox*).

Cournot (quantities, simultaneous)

Symmetric reaction function (duopoly): $q_i = \frac{a - c}{2b} - \frac{q_j}{2}$.

Symmetric equilibrium with n firms:

$$q_i^C = \frac{a - c}{(n + 1)b}, \quad Q^C = \frac{n(a - c)}{(n + 1)b}, \quad P^C = \frac{a + nc}{n + 1}, \quad \pi_i^C = \frac{(a - c)^2}{(n + 1)^2 b}.$$

For $n \rightarrow \infty$: $P^C \rightarrow c$, $\pi_i^C \rightarrow 0$ (PC). For $n = 1$: monopoly.

Stackelberg (quantities, sequential)

Leader (1) by backward induction:

$$q_1^S = \frac{a-c}{2b}, \quad q_2^S = \frac{a-c}{4b}, \quad Q^S = \frac{3(a-c)}{4b}, \quad P^S = \frac{a+3c}{4}.$$

$$\pi_1^S = \frac{(a-c)^2}{8b} > \pi_i^C > \pi_2^S = \frac{(a-c)^2}{16b}.$$

Leader advantaged (first-mover advantage); follower worse off relative to Cournot.

Welfare ranking (decreasing order of Q , increasing order of P): PC = Bertrand > Stackelberg > Cournot > Collusion/Monopoly.

Collusion in repeated games (grim trigger):

$$\delta \geq \delta^* = \frac{\pi^{\text{dev}} - \pi^{\text{coll}}}{\pi^{\text{dev}} - \pi^{\text{punish}}}.$$

Setting	π^{coll}	π^{dev}	π^{punish}	δ^*
Bertrand, $n = 2$	$(a-c)^2/(8b)$	$(a-c)^2/(4b)$	0	1/2
Cournot, $n = 2$	$(a-c)^2/(8b)$	$9(a-c)^2/(64b)$	$(a-c)^2/(9b)$	9/17

As n increases: δ^* rises, and collusion is harder to sustain.

B.15 Asymmetric information

Concept map:

Type	Timing	Problem	Failure	Remedy
Hidden information	pre-contract	who you are	Adverse selection	Signaling, screening
Hidden action	post-contract	what you do	Moral hazard	Incentives, monitoring

Akerlof: adverse selection. With a proportion p of high-quality goods ($v_H > c_H$) and $(1-p)$ of low-quality ones ($v_L > c_L$):

$$WTP = p v_H + (1-p) v_L.$$

The market for high-quality goods survives if and only if:

$$WTP \geq c_H \iff p \geq p^* = \frac{c_H - v_L}{v_H - v_L}.$$

For $p < p^*$: unraveling spiral until the high-quality segment collapses.

Spence: signaling. Separating equilibrium with types H (w_H) and L (w_L), education costs $c_H < c_L$:

$$\text{IC}_H: w_H - c_H > w_L \quad (\text{type } H \text{ gets a degree})$$

$$\text{IC}_L: w_H - c_L < w_L \quad (\text{type } L \text{ does not mimic})$$

The two ICs together are equivalent to $c_H < w_H - w_L < c_L$: they are compatible only if $c_L > c_H$ (differential cost condition).

Principal-agent model. The principal wants to induce high effort e_H (cost c_H , probability of success p_H). The optimal contract (w_H, w_L) must satisfy:

$$\mathbf{PC:} \quad p_H U(w_H) + (1 - p_H) U(w_L) - c_H \geq \bar{u}$$

$$\mathbf{IC:} \quad (p_H - p_L) [U(w_H) - U(w_L)] \geq c_H - c_L$$

First best (observable effort): $w_H^{FB} = w_L^{FB} = w^*$ (fixed wage; agent fully insured).

Second best (unobservable effort): $w_H^{SB} > w_L^{SB}$ (wage spread necessary). Risk premium = $E[w^{SB}] - w^* > 0$: real cost of the asymmetry borne by the principal.

Linear contract $w = \alpha + \beta\pi$:

- $\beta = 0$: fixed wage, full insurance, no incentive.
- $\beta = 1$: agent is residual claimant, maximum incentives, maximum risk.
- $\beta^* \in (0, 1)$: optimal, increasing in Δc , decreasing in Δp and in risk aversion.

Appendix C

Solutions to the Exercises

The solutions are organized chapter by chapter, in the same order in which the exercises appear in the text. For qualitative or graphical exercises a reasoned answer is provided; for numerical ones the complete procedure is shown.

Chapter 1 – Economic Method and Basic Tools

Exercise 1.1. On the PPF $B = 20 - A$, the opportunity cost of producing one more unit of A is the absolute value of the slope of the line, i.e. **1** unit of B . Geometrically: at the horizontal intercept ($A = 20$, $B = 0$) only A is produced; at the vertical intercept ($A = 0$, $B = 20$) only B . Giving up one unit of A frees the resources to produce exactly one unit of B .

Exercise 1.2.

$$f(x) = 4x^2 + 3x \quad \Rightarrow \quad f'(x) = 8x + 3.$$

Exercise 1.3.

$$\pi(q) = 60q - 6q^2 - 100 \quad \Rightarrow \quad \pi'(q) = 60 - 12q = 0 \quad \Rightarrow \quad q^* = 5.$$

SOC check: $\pi''(q) = -12 < 0$ (maximum). ✓

Exercise 1.4. A sunk cost has already been incurred and cannot be recovered: the future choice cannot change it. Including it in the cost-benefit calculation would lead to distorted decisions (e.g. continuing a loss-making project only because a lot has already been invested). The correct rule is: compare only *future* costs and benefits.

Exercise 1.5. PPF: $B = 12 - 0.25A$.

- Vertical intercept ($A = 0$): $B = 12$.
- Horizontal intercept ($B = 0$): $A = 12/0.25 = 48$.
- Slope: -0.25 . Interpretation: producing one more unit of A costs 0.25 units of B . Equivalently, 4 units of A cost 1 unit of B .

Chapter 2 – Preferences, Utility and MRS

Exercise 2.1. $U(x, y) = x^2y$. The marginal utilities are:

$$MU_x = \frac{\partial U}{\partial x} = 2xy, \quad MU_y = \frac{\partial U}{\partial y} = x^2.$$

Hence:

$$MRS = \frac{MU_x}{MU_y} = \frac{2xy}{x^2} = \frac{2y}{x}.$$

The MRS is decreasing in x and increasing in y : as the consumer increases x (holding y fixed), her willingness to give up y in exchange for x falls.

Exercise 2.2. Two functions represent the same preferences if and only if one is a monotonically increasing transformation of the other. We check that $V = f(U) = 2U + 5$ with $f' = 2 > 0$:

$$V(x, y) = 2xy + 5 = 2U(x, y) + 5.$$

Yes, V is a monotonically increasing transformation of U , so they represent the same preferences. In particular, $A \succcurlyeq B \iff U(A) \geq U(B) \iff V(A) \geq V(B)$.

Exercise 2.3. Convexity of preferences states that mixtures are at least as good as the extremes. This implies that, along an indifference curve, the more one consumes of a good (giving up some of the other), the less one is willing to give up further: the MRS diminishes. Formally, if U is strictly quasi-concave, the MRS is strictly decreasing in x along the indifference curve.

Exercise 2.4. PPF $B = 12 - 0.25A$:

(i) intercept on B : 12; intercept on A : 48.

(ii) slope = -0.25 : the opportunity cost of one unit of A is 0.25 units of B .

Exercise 2.5. By monotonicity, “more is better”: if x is added holding y fixed, well-being increases. Suppose, by contradiction, that an indifference curve has a positive slope at some point: moving along it up and to the right means having more of both goods, so by monotonicity the new bundle is strictly preferred to the old one, contradicting the assumption that they lie on the same curve. Hence the slope must be negative.

Exercise 2.6. $U(x, y) = x^{0.4}y^{0.6}$.

$$MU_x = 0.4x^{-0.6}y^{0.6}, \quad MU_y = 0.6x^{0.4}y^{-0.4}.$$

$$MRS = \frac{MU_x}{MU_y} = \frac{0.4}{0.6} \cdot \frac{y}{x} = \frac{2}{3} \cdot \frac{y}{x}.$$

Chapter 3 – Budget Constraint and Optimal Choice

Exercise 3.1. $M = 60$, $p_x = 10$, $p_y = 3$.

(i) Constraint: $10x + 3y = 60$; in explicit form $y = 20 - \frac{10}{3}x$. Vertical intercept: $y = 20$ (when $x = 0$). Horizontal intercept: $x = 6$ (when $y = 0$). Slope: $-10/3 \approx -3.33$.

(ii) Affordability check ($p_x x + p_y y \leq 60$):

- $(3, 10)$: $10 \cdot 3 + 3 \cdot 10 = 60 \leq 60$. **Affordable** (on the line).

- (2, 8): $10 \cdot 2 + 3 \cdot 8 = 44 \leq 60$. **Affordable** (interior).
- (5, 15): $10 \cdot 5 + 3 \cdot 15 = 95 > 60$. **Not affordable**.

(iii) The budget set is the triangle with vertices (0, 0), (6, 0), (0, 20).

Exercise 3.2. If income rises from M to $M' = 1.5M$ with prices unchanged, the new line is:

$$y = \frac{1.5M}{p_y} - \frac{p_x}{p_y}x.$$

The slope $-p_x/p_y$ is unchanged (prices do not change). Both intercepts increase by 50%: $M'/p_y = 1.5M/p_y$ and $M'/p_x = 1.5M/p_x$. The line shifts outward *in parallel*.

Exercise 3.3. $U(x, y) = x^{0.5}y^{0.5}$, $p_x = 10$, $p_y = 5$, $M = 100$.

Substitution method. From the constraint: $y = (100 - 10x)/5 = 20 - 2x$. Substituting: $\tilde{U}(x) = x^{0.5}(20 - 2x)^{0.5}$. Maximizing this is equivalent to maximizing $x(20 - 2x)$:

$$\frac{d}{dx}[x(20 - 2x)] = 20 - 4x = 0 \Rightarrow x^* = 5.$$

$$y^* = 20 - 2 \cdot 5 = 10.$$

Lagrange method.

$$\mathcal{L} = x^{0.5}y^{0.5} - \lambda(10x + 5y - 100).$$

FOC:

$$\frac{0.5y^{0.5}}{x^{0.5}} = 10\lambda, \quad \frac{0.5x^{0.5}}{y^{0.5}} = 5\lambda, \quad 10x + 5y = 100.$$

Dividing the first by the second: $y/x = 10/5 = 2$, i.e. $y = 2x$. Substituting into the constraint: $10x + 5(2x) = 100 \Rightarrow 20x = 100 \Rightarrow x^* = 5$, $y^* = 10$. $\lambda = \frac{0.5\sqrt{10/5}}{10} = \frac{\sqrt{2}}{20} \approx 0.0707$ (marginal utility of income).

Check: $MRS = y/x = 10/5 = 2 = p_x/p_y = 10/5$. ✓

Exercise 3.4. $U(x, y) = 2x + 3y$, $p_x = 5$, $p_y = 4$, $M = 100$.

(i) $MRS = MU_x/MU_y = 2/3$; $p_x/p_y = 5/4 = 1.25$.

(ii) Since $MRS = 2/3 < 5/4 = p_x/p_y$: the consumer values x relatively little compared with its price. It pays to spend everything on y : **corner solution** at y .

(iii) Optimal bundle: $x^* = 0$, $y^* = M/p_y = 100/4 = 25$.

Chapter 4 – Individual Demand, Comparative Statics and Elasticity

Exercise 4.1. $x^D(p, M) = 20 - 2p + 0.1M$, $p = 5$, $M = 100$.

(i) $x^D = 20 - 2(5) + 0.1(100) = 20 - 10 + 10 = 20$.

(ii) $\partial x^D / \partial p = -2$; $\partial x^D / \partial M = 0.1$.

(iii) Slutsky equation:

$$\frac{\partial x^c}{\partial p} = \frac{\partial x^D}{\partial p} + x^D \frac{\partial x^D}{\partial M} = -2 + 20 \cdot 0.1 = -2 + 2 = 0.$$

The substitution effect is zero: the income effect entirely accounts for the total price effect.

(iv) The total effect (-2) decomposes into: SE = 0 (at constant utility the compensated demand does not respond to the price) and IE = $-x^D \cdot \partial x^D / \partial M = -20 \cdot 0.1 = -2$ (the loss of real purchasing power reduces demand by 2 units).

Exercise 4.2. $Q = 100 - 2P$.

(i) $\varepsilon(P) = \frac{dQ}{dP} \cdot \frac{P}{Q} = -2 \cdot \frac{P}{100-2P}$.

(ii)

- $P = 10$: $Q = 80$, $\varepsilon = -2 \cdot 10/80 = -0.25$. $|\varepsilon| = 0.25 < 1$: **inelastic**.

- $P = 40$: $Q = 20$, $\varepsilon = -2 \cdot 40/20 = -4$. $|\varepsilon| = 4 > 1$: **elastic**.

(iii) $TR = P \cdot Q = P(100 - 2P) = 100P - 2P^2$. $\frac{dTR}{dP} = 100 - 4P = 0 \Rightarrow P^* = 25$. Check: $\varepsilon(25) = -2 \cdot 25/50 = -1$, so $|\varepsilon| = 1$. ✓

Exercise 4.3. $x^D(p_x, p_y, M) = 20 - 2p_x + p_y + 0.01M$.

(i) $\partial x^D / \partial p_y = 1 > 0$: the goods are **substitutes** (an increase in the price of y shifts demand toward x).

(ii) At $p_y = 3$, $p_x = 5$, $M = 100$:

$$x^D = 20 - 10 + 3 + 1 = 14.$$

$$\varepsilon_{xy} = \frac{\partial x^D}{\partial p_y} \cdot \frac{p_y}{x^D} = 1 \cdot \frac{3}{14} \approx 0.214.$$

Questions and Exercises – Consumer Theory

Multiple-choice questions

1. Answer C. The opportunity cost is the value of the best alternative forgone. A is not correct (the opportunity cost can differ from monetary expenditure, e.g. the work given up to attend university). B is wrong (a sunk cost is unrecoverable and must not influence future decisions). D is wrong (marginal analysis applies to continuous changes, not only to binary choices).

2. Answer B. The condition $f'(x^*) = 0$ is necessary (FOC); for a *local* maximum the SOC $f''(x^*) < 0$ is needed. A is wrong ($f'' > 0$ indicates a minimum). C and D do not satisfy the FOC.

3. Answer B. Transitivity states: if $A \succeq B$ and $B \succeq C$, then $A \succeq C$. A describes completeness. C confuses weak preference with strict preference. D is the representation theorem, which also requires other axioms.

4. Answer C. Under monotonicity, indifference curves are downward sloping: increasing x requires decreasing y to stay on the same curve. A is wrong (intersection violates transitivity). B is wrong (negative slope, not positive). D is wrong (straight lines correspond to perfect substitutes, a special case).

5. Answer B. At an interior optimum $MU_x/p_x = MU_y/p_y$ holds, i.e. the marginal utility per euro is equal for both goods. A inverts the price ratio. C is false in general ($MU_x = MU_y$ holds only if $p_x = p_y$). D describes corner solutions, not the interior optimum.

Exercises and open questions

Exercise 1.

(a) $V(x, y) = 2xy + 10 = 2U(x, y) + 10$. The function $f(u) = 2u + 10$ has $f' = 2 > 0$: it is a monotonically increasing transformation of U . Hence U and V represent the **same preferences**: $U(A) \geq U(B) \iff V(A) \geq V(B)$ for every pair of bundles.

(b) The four properties of indifference curves under monotonicity and convexity:

1. **Negative slope**: giving up one unit of x requires receiving y in compensation, otherwise one would be worse off (monotonicity).
2. **They do not intersect**: if they intersected, by transitivity we would have $U(A) = U(B)$ and $U(A) = U(C)$ but $U(B) \neq U(C)$, a contradiction.
3. **Convexity toward the origin**: mixtures are preferred to extremes (convexity of preferences); the MRS is decreasing.
4. **Higher curves to the north-east**: a bundle with more of both goods is strictly preferred (monotonicity).

(c) For $U(x, y) = xy$:

$$MU_x = y, \quad MU_y = x \quad \Rightarrow \quad MRS = \frac{MU_x}{MU_y} = \frac{y}{x}.$$

Interpretation: if $MRS = 3$ at a given point, the consumer is willing to give up 3 units of y to obtain 1 more unit of x while keeping utility unchanged.

Exercise 2. $U(x, y) = x^{0.5}y^{0.5}$, $p_x = 10$, $p_y = 5$, $M = 100$.

(a) *Substitution*. From the constraint: $y = (100 - 10x)/5 = 20 - 2x$. We maximize $\tilde{U}(x) = x^{0.5}(20 - 2x)^{0.5}$, equivalently $g(x) = x(20 - 2x) = 20x - 2x^2$:

$$g'(x) = 20 - 4x = 0 \Rightarrow x^* = 5, \quad y^* = 20 - 10 = 10.$$

(b) *Lagrange*. $\mathcal{L} = x^{0.5}y^{0.5} - \lambda(10x + 5y - 100)$. FOC: $0.5y^{0.5}/x^{0.5} = 10\lambda$ and $0.5x^{0.5}/y^{0.5} = 5\lambda$. Dividing: $y/x = 2 \Rightarrow y = 2x$. In the constraint: $10x + 10x = 100 \Rightarrow x^* = 5$, $y^* = 10$.

$\lambda = \frac{0.5\sqrt{10/5}}{10} = \frac{\sqrt{2}}{20} \approx 0.0707$: each additional euro of income raises optimal utility by about 0.0707.

(c) $MRS(5, 10) = y/x = 10/5 = 2$; $p_x/p_y = 10/5 = 2$. $MRS = p_x/p_y$. ✓

Exercise 3.

(a) The **PCC** is the locus of optimal bundles in (x, y) space as p_x varies (with p_y and M fixed): it is a curve in goods space. The **demand curve** projects these optimal points onto the (x, p_x) plane: it shows the quantity demanded of x for each value of p_x . The demand curve is *derived* from the PCC.

(b) The **ICC** is the locus of optimal bundles in (x, y) space as M varies (with p_x and p_y fixed). The **Engel curve** projects these points onto the (x, M) plane: it shows the quantity demanded of x for each income level (at fixed prices).

(c) If, as M rises, the optimal quantity of x decreases along the ICC, then $\partial x^D / \partial M < 0$: x is an **inferior good**.

(d) If the function $M(x)$ is concave, its inverse $x^D(M)$ is convex: the quantity demanded grows at an *increasing* rate as income rises ($\partial x^D / \partial M$ increases with M). For an Engel curve starting at the origin, convexity implies $\partial x^D / \partial M \geq x^D / M$, so the income elasticity $\varepsilon_M = (\partial x^D / \partial M) \cdot (M / x^D)$ is positive and greater than 1: x grows more than proportionally with income, behavior typical of **luxury goods**.

Chapter 5 – Technology and Production

Exercise 5.1. $Q = L^{0.5} K^{0.5}$.

(a)

$$MPL = \frac{\partial Q}{\partial L} = 0.5 L^{-0.5} K^{0.5} = \frac{1}{2} \sqrt{\frac{K}{L}}, \quad MPK = 0.5 L^{0.5} K^{-0.5} = \frac{1}{2} \sqrt{\frac{L}{K}}.$$

(b) Returns to scale:

$$f(tL, tK) = (tL)^{0.5} (tK)^{0.5} = t^{0.5+0.5} L^{0.5} K^{0.5} = t f(L, K).$$

$\alpha + \beta = 1$: **constant returns to scale**.

(c) $\partial MPL / \partial L = -0.25 L^{-1.5} K^{0.5} < 0$ for all $L, K > 0$: the MPL is strictly decreasing in L (with K fixed). ✓

Exercise 5.2. $Q = 6L^2 - L^3$ (short run).

(a)

$$MPL = \frac{dQ}{dL} = 12L - 3L^2, \quad APL = \frac{Q}{L} = 6L - L^2.$$

(b) $\frac{d(APL)}{dL} = 6 - 2L = 0 \Rightarrow L = 3$, with $APL(3) = 18 - 9 = 9$ (a maximum, since $\frac{d^2(APL)}{dL^2} = -2 < 0$). $MPL = APL \Rightarrow 12L - 3L^2 = 6L - L^2 \Rightarrow 2L(3 - L) = 0 \Rightarrow L = 3$ (for $L > 0$), with $MPL(3) = 36 - 27 = 9 = APL(3)$. ✓ The maximum of APL therefore coincides with the point $MPL = APL$, as implied by $\frac{d(APL)}{dL} = \frac{MPL - APL}{L}$: for $L < 3$, $MPL > APL$ (e.g. $L = 2$: $12 > 8$, APL rises); for $L > 3$, $MPL < APL$ (APL falls).

(c) $MPL = 0 \Rightarrow 3L(4 - L) = 0 \Rightarrow L^{TP} = 4$ (with $TP(4) = 96 - 64 = 32$).

Exercise 5.3. $Q = L^{0.7} K^{0.4}$.

$$\alpha + \beta = 0.7 + 0.4 = 1.1 > 1 \quad \Rightarrow \quad \text{increasing returns to scale.}$$

$$MPL = 0.7 L^{-0.3} K^{0.4}, \quad MPK = 0.4 L^{0.7} K^{-0.6}.$$

Exercise 5.4. $f(2L, 2K) = 180$ vs $2f(L, K) = 200$. Since $180 < 200$: **decreasing returns to scale**. Possible explanation: doubling all inputs creates coordination problems, bureaucracy, and congestion of facilities, which prevent output from doubling.

Chapter 6 – Isoquants, MRTS and Types of Technology

Exercise 6.1. $Q = L^{0.6} K^{0.4}$, point $(L, K) = (16, 9)$.

$$MPL = 0.6 L^{-0.4} K^{0.4}, \quad MPK = 0.4 L^{0.6} K^{-0.6}.$$

$$MRTS_{LK} = \frac{MPL}{MPK} = \frac{0.6}{0.4} \cdot \frac{K}{L} = \frac{3}{2} \cdot \frac{K}{L}.$$

At the point (16, 9):

$$MRTS_{LK} = \frac{3}{2} \cdot \frac{9}{16} = \frac{27}{32} \approx 0.844.$$

Interpretation: at that point, increasing L by 1 unit allows K to be reduced by about 0.844 units while keeping output Q constant.

Exercise 6.2. Suppose two isoquants Q_1 and $Q_2 > Q_1$ intersect at point P . Point P belongs to both isoquants, so $f(P) = Q_1$ and $f(P) = Q_2$ simultaneously, which is impossible since the production function assigns to each input a unique level of output. Contradiction: two isoquants cannot intersect.

Exercise 6.3.

- a) $Q = 3L + 5K$: **linear** isoquants ($MRTS = 3/5$ constant). The inputs are **perfect substitutes**: one can move from all K to all L with the same output.
- b) $Q = \min\{2L, K\}$: **L-shaped** isoquants with the kink on the line $K = 2L$. The inputs are **perfect complements**: they are needed in the fixed proportion $K/L = 2$.
- c) $Q = L^{0.5}K^{0.5}$: isoquants **convex** to the origin, an intermediate case. The inputs are substitutable but not perfectly.

Exercise 6.4. $Q = L^{0.5}K^{0.5}$, $(L, K) = (4, 9)$.

$$MRTS_{LK} = \frac{MPL}{MPK} = \frac{0.5 K^{0.5}/L^{0.5}}{0.5 L^{0.5}/K^{0.5}} = \frac{K}{L} = \frac{9}{4} = 2.25.$$

Interpretation: at that point, the firm is willing to reduce K by 2.25 units for each additional unit of L , keeping output unchanged. The MRTS is high because at the point (4, 9) capital is abundant relative to labor, so MPK is low and MPL is high.

Chapter 7 – Production Costs and Cost Curves

Exercise 7.1. $TC(Q) = 200 + 5Q + Q^2$.

(a) $FC = 200$; $VC(Q) = 5Q + Q^2$.

(b)

$$AFC = \frac{200}{Q}, \quad AVC = 5 + Q, \quad AC = \frac{200}{Q} + 5 + Q, \quad MC = 5 + 2Q.$$

(c) $AVC = 5 + Q$ is **linear and increasing** (not U-shaped in this case: variable costs are increasing from the start, with no initial decreasing phase).

(d) Minimum of AC :

$$\frac{dAC}{dQ} = -\frac{200}{Q^2} + 1 = 0 \quad \Rightarrow \quad Q^2 = 200 \quad \Rightarrow \quad Q^* = \sqrt{200} = 10\sqrt{2} \approx 14.14.$$

Check: $MC(Q^*) = 5 + 2\sqrt{200} = 5 + 20\sqrt{2}$; $AC(Q^*) = \frac{200}{\sqrt{200}} + 5 + \sqrt{200} = \sqrt{200} + 5 + \sqrt{200} = 5 + 20\sqrt{2}$. ✓

Exercise 7.2. $TC(Q) = F + cQ^2$, so $AC = F/Q + cQ$.

(a)

$$\frac{dAC}{dQ} = -\frac{F}{Q^2} + c = 0 \quad \Rightarrow \quad Q^2 = \frac{F}{c} \quad \Rightarrow \quad Q^* = \sqrt{\frac{F}{c}}.$$

(b) $F \uparrow \Rightarrow Q^* \uparrow$: higher fixed costs make it optimal to produce more to spread them over more units. $c \uparrow \Rightarrow Q^* \downarrow$: higher variable costs make it costly to expand production, so it pays to stop earlier.

Exercise 7.3 (True or False).

- True.** If $MC < AC$, the cost of the marginal unit is below the average: producing one more unit lowers the average. Formally: $\frac{d(AC)}{dQ} = \frac{MC-AC}{Q} < 0$.
- False.** MC intersects AC at the *minimum* of AC , not at the maximum. It is the point where $\frac{d(AC)}{dQ} = 0$, i.e. $MC = AC$.
- True.** $MC = dTC/dQ = dVC/dQ$: fixed costs do not vary at the margin, so they do not affect MC .
- True.** The same logic applies identically to AVC : $\frac{d(AVC)}{dQ} = \frac{MC-AVC}{Q}$, which is negative if and only if $MC < AVC$.

Exercise 7.4.

- Microchips: strongly decreasing LAC .** Design and fab costs are enormous (high F) but fixed; the marginal cost of producing one more chip is almost zero. Economies of scale dominate (TSMC, Intel).
- Individual consulting: increasing** or quickly constant LAC . The professional is the limiting factor; scaling up means hiring and coordinating others, with rising supervision costs and risks to the quality of the service.
- Large-scale retail: initially decreasing LAC** (centralized purchasing, logistics, IT), then it tends to level off or to rise when the chain becomes too large to manage efficiently.

Chapter 8 – Cost Minimization and Optimal Input Choice

Exercise 8.1. $w = 12$, $r = 24$, $C = 240$.

(a) $240 = 12L + 24K \Rightarrow K = 10 - \frac{1}{2}L$.

(b) Intercept on K : $K = 10$ (when $L = 0$). Intercept on L : $L = 20$ (when $K = 0$). Slope: $-w/r = -12/24 = -1/2$.

(c) With $w' = 18$ (holding $r = 24$ and $C = 240$ fixed): $K = 10 - \frac{18}{24}L = 10 - \frac{3}{4}L$. The intercept on L falls to $240/18 \approx 13.3$; the one on K stays at 10. The line *rotates inward* on the L axis: it becomes steeper (slope $-3/4$), reflecting the higher relative cost of labor.

Exercise 8.2. $MPL = 8$, $MPK = 4$, $w = 16$, $r = 24$.

(a)

$$MRTS = \frac{MPL}{MPK} = \frac{8}{4} = 2, \quad \frac{w}{r} = \frac{16}{24} = \frac{2}{3}.$$

Since $MRTS = 2 \neq 2/3 = w/r$: **we are not at the optimum.**

(b) Equivalent check:

$$\frac{MPL}{w} = \frac{8}{16} = 0.5 > \frac{MPK}{r} = \frac{4}{24} \approx 0.167.$$

One euro spent on L produces 0.5 additional units; one euro on K only 0.167. It pays to **increase L and reduce K** .

(c) Moving toward more L and less K : by diminishing returns, MPL falls; with less K available, MPK rises. The process continues until $MPL/w = MPK/r$.

Exercise 8.3. $Q = 2L^{0.5}K^{0.5}$, $w = 8$, $r = 2$, $\bar{Q} = 40$.

(a) $MRTS_{LK} = K/L$ (the constant 2 cancels out). Optimality condition:

$$\frac{K}{L} = \frac{w}{r} = \frac{8}{2} = 4 \Rightarrow K = 4L.$$

(b) Output constraint ($\alpha + \beta = 1$):

$$40 = 2L^{0.5}(4L)^{0.5} = 2 \cdot 2 \cdot L = 4L \Rightarrow L^* = 10, \quad K^* = 40.$$

(c) $C^* = 8 \cdot 10 + 2 \cdot 40 = 80 + 80 = 160$.

(d) $K^* = 40 \gg L^* = 10$: the mix is **capital-intensive**. Reason: $r = 2 \ll w = 8$, capital costs much less than labor, so the firm uses it to a much greater extent.

Exercise 8.4. Starting from Exercise 8.3 ($w = 8$, $r = 2$, $Q = 40$), now $w' = 2$.

(a) New optimal ratio: $K/L = w'/r = 2/2 = 1 \Rightarrow K = L$.

$$40 = 2L^{0.5}L^{0.5} = 2L \Rightarrow L^{**} = 20, \quad K^{**} = 20.$$

(b) $C^{**} = 2 \cdot 20 + 2 \cdot 20 = 80$. $C^{**} = 80 < C^* = 160$: the cost is halved. With a much lower w , the same output now costs much less because the input that has become cheaper (labor) is used much more.

(c) With $w = w' = r = 2$ the inputs have the same price: the optimal mix is $K = L$ (symmetric), and the firm prefers neither. Low-wage countries (high L , low w) tend to attract labor-intensive production; with high wages firms invest in automation to compensate.

Exercise 8.5. w rises from 10 to 14; $r = 20$; $Q = L^{0.7}K^{0.3}$.

$$\frac{K}{L} = \frac{\beta}{\alpha} \cdot \frac{w}{r} = \frac{0.3}{0.7} \cdot \frac{w}{r}.$$

$$\left. \frac{K}{L} \right|_{w=10} = \frac{3}{7} \cdot \frac{10}{20} \approx 0.214, \quad \left. \frac{K}{L} \right|_{w=14} = \frac{3}{7} \cdot \frac{14}{20} = 0.300.$$

(a) The ratio K/L increases from 0.214 to 0.300: the firm tends to substitute capital for labor.

(b) Who adapts easily: firms with flexible technology and access to credit to invest in automation; sectors in which L and K are good substitutes. Who suffers: SMEs with fixed plants; strongly labor-intensive sectors with little substitutability (e.g. personal care, specialized craftsmanship).

(c) The minimum cost C^* **increases**: paying more for an input means paying more to produce the same output. However, the increase is *dampened* compared with a firm that could not substitute inputs: the possibility of using more K in place of L limits the impact on total cost.

Questions and Exercises – Producer Theory

Multiple-choice questions

1. **Answer B.** The production function gives the *maximum* quantity of output obtainable from each combination of inputs. A describes the cost function. C is wrong: in the short run at least one input is fixed. D has no basis.
2. **Answer C.** For $Q = AL^\alpha K^\beta$: $f(tL, tK) = t^{\alpha+\beta} f(L, K)$. Returns are increasing if and only if $\alpha + \beta > 1$. A is wrong (what matters is the sum, not a comparison between the individual exponents). B describes constant returns. D is unrelated to the definition.
3. **Answer C.** APL is at its maximum when $MPL = APL$, since $\frac{d(APL)}{dL} = \frac{MPL - APL}{L} = 0$ implies $MPL = APL$. A is wrong ($MPL = 0$ is the maximum of TP). B is the condition under which APL is still *increasing*, not at its maximum. D makes no economic sense.
4. **Answer C.** AC is at its minimum where $\frac{d(AC)}{dQ} = \frac{MC - AC}{Q} = 0$, i.e. where $MC = AC$. A is wrong ($AFC = AVC$ has no implications for the minimum of AC). B is wrong. D is the condition for the minimum of MC , not of AC .
5. **Answer D.** A fixed-cost shock increases FC , and hence AFC and AC , but does not change $VC(Q)$, and consequently neither AVC nor MC changes. A, B and C are all wrong because those curves vary with a shock to FC .

Exercises and open questions

Exercise 1. $Q = L^{0.5} K^{0.5}$.

(a)

$$MPL = 0.5 L^{-0.5} K^{0.5}, \quad MPK = 0.5 L^{0.5} K^{-0.5}.$$

Diminishing: $\partial MPL / \partial L = -0.25 L^{-1.5} K^{0.5} < 0$; $\partial MPK / \partial K = -0.25 L^{0.5} K^{-1.5} < 0$. ✓

(b) $f(tL, tK) = (tL)^{0.5} (tK)^{0.5} = t \cdot L^{0.5} K^{0.5} = t f(L, K)$. $\alpha + \beta = 1$: **constant returns to scale**.

(c) $MRTS_{LK} = MPL / MPK = K / L$. At the point (4, 9): $MRTS = 9 / 4 = 2.25$. The firm can reduce K by 2.25 units for each additional unit of L without changing $\bar{Q}$. The $MRTS$ is high because at the point considered K is abundant relative to L ($K = 9 > L = 4$), so MPL is relatively high and MPK low.

(d) Along a convex isoquant, as more L (and less K) is used: MPL falls because of diminishing marginal returns to L , while MPK rises because less K is available. Since $MRTS = MPL / MPK$, the numerator falls and the denominator rises: $MRTS$ is strictly decreasing.

Exercise 2. $TC(Q) = 200 + 4Q + Q^2$.

(a) $FC = 200$; $VC(Q) = 4Q + Q^2$.

$$AFC = \frac{200}{Q}, \quad AVC = 4 + Q, \quad AC = \frac{200}{Q} + 4 + Q, \quad MC = 4 + 2Q.$$

(b) $\frac{dAC}{dQ} = -200/Q^2 + 1 = 0 \Rightarrow Q^* = \sqrt{200} = 10\sqrt{2} \approx 14.14$. Check: $MC(Q^*) = 4 + 20\sqrt{2}$; $AC(Q^*) = \frac{200}{\sqrt{200}} + 4 + \sqrt{200} = \sqrt{200} + 4 + \sqrt{200} = 4 + 20\sqrt{2}$. ✓

(c) $VC' = 1.5(4Q + Q^2) = 6Q + 1.5Q^2$, so:

$$MC' = 6 + 3Q, \quad AVC' = 6 + 1.5Q.$$

New minimum: $dAC'/dQ = -200/Q^2 + 1.5 = 0 \Rightarrow Q^{**} = \sqrt{200/1.5} = 20/\sqrt{3} \approx 11.55$.
 $Q^{**} < Q^*$: with higher variable costs it pays to produce less to minimize average cost.

(d) $MC = dTC/dQ = dVC/dQ$: fixed costs are constant, so their derivative with respect to Q is zero. In the short run, the firm cannot recover the FC already incurred (they are sunk): decisions about how much to produce depend only on MC and marginal revenue, not on the size of fixed costs.

Exercise 3.

(a) **Indifference curve vs isoquant.** Both describe a level set of a function: the indifference curve is the set $\{(x, y) : U(x, y) = \bar{U}\}$, the isoquant is $\{(L, K) : f(L, K) = \bar{Q}\}$. The geometric properties are identical (negative slope, no intersection, convexity). The substantive difference lies in the object: utility (subjective, ordinal) vs output (physical, measurable).

(b) **Budget constraint vs isocost.** The budget constraint $p_x x + p_y y = M$ defines the combinations of goods affordable given M . The isocost $wL + rK = C$ defines the combinations of inputs that cost exactly C . Same mathematical structure: a line with intercepts M/p_x , M/p_y (or C/w , C/r) and slope $-p_x/p_y$ (or $-w/r$). The difference: in the budget problem M is given and the optimal bundle is chosen; in the isocost problem C is the variable to be minimized and the isoquant is the constraint.

(c) **Optimality condition.** In both cases, at the optimum the slopes of the choice object and of the constraint coincide: $MRS = p_x/p_y$ for the consumer, $MRTS = w/r$ for the producer. The interpretation is analogous: the subjective trade-off (preferences or technology) must equal the market trade-off (prices of goods or of inputs). Equivalently, the marginal utility/product per euro spent must be equal for both goods/inputs.

The role of M vs $\bar{Q}$. In the consumer's problem, M fixes the size of the budget set: one chooses the highest indifference curve attainable given that budget (a *maximization* problem). In the producer's long-run problem, $\bar{Q}$ fixes the isoquant to work on: one looks for the lowest isocost that reaches it (a *minimization* problem). Geometrically, in the first case one moves toward higher indifference curves until the constraint is reached; in the second one moves toward lower isocosts until the constraint (the isoquant) is reached. The "direction of optimization" is opposite.

Chapter 9 – Profit Maximization

Exercise 1. $C(Q) = Q^3 - 6Q^2 + 15Q + 10$, $FC = 10$.

(a) $MC(Q) = 3Q^2 - 12Q + 15$; $AVC(Q) = Q^2 - 6Q + 15$; $AC(Q) = Q^2 - 6Q + 15 + 10/Q$.
 $AVC_{\min}$: $dAVC/dQ = 2Q - 6 = 0 \Rightarrow Q = 3$, $AVC(3) = 9 - 18 + 15 = 6$. Check: $MC(3) = 27 - 36 + 15 = 6 = AVC_{\min}$. ✓ Domain of SR supply: $P \geq 6$.

(b) For $P = 9$: $3Q^2 - 12Q + 15 = 9 \Rightarrow Q^2 - 4Q + 2 = 0$, with roots $Q = 2 \pm \sqrt{2}$. SOC: $MC'(Q) = 6Q - 12$.

- $Q = 2 - \sqrt{2} \approx 0.586$: $MC' < 0$ (profit minimum, discarded).

- $Q^* = 2 + \sqrt{2} \approx 3.414$: $MC' > 0$ (maximum).

With $Q^{*2} = 6 + 4\sqrt{2}$, $Q^{*3} = 20 + 14\sqrt{2}$:

$$\pi(Q^*) = -Q^{*3} + 6Q^{*2} - 6Q^* - 10 = 4\sqrt{2} - 6 \approx -0.34.$$

$\pi(0) = -FC = -10$. Since $P = 9 > AVC_{\min} = 6$, the firm produces in the SR.

(c) $AC_{\min}$ from $MC = AC$: $Q^3 - 3Q^2 - 5 = 0$. Numerically: $Q^{\min} \approx 3.43$, $AC_{\min} \approx 9.10$. Since $P = 9 < AC_{\min} \approx 9.10$, in the LR the firm exits.

(d) Case $P = 12$. $Q^2 - 4Q + 1 = 0 \Rightarrow Q = 2 \pm \sqrt{3}$. $Q^* = 2 + \sqrt{3} \approx 3.73$. With $Q^{*2} = 7 + 4\sqrt{3}$, $Q^{*3} = 26 + 15\sqrt{3}$: $\pi(Q^*) = 6\sqrt{3} \approx 10.39$. $P > AC_{\min}$: **zone 4**.
 Case $P = 4$. $P < AVC_{\min} = 6$: SR shutdown, $Q^* = 0$, $\pi = -10$. **Zone 1**.

Exercise 2. $C(Q) = Q^2 + 4Q + 25$, $FC = 25$.

(a) $MC = 2Q + 4$; $AVC = Q + 4$ linear and increasing $\Rightarrow AVC_{\min} = 4$. $AC = Q + 4 + 25/Q$; $AC_{\min}$ at $Q = 5$: $AC(5) = 14$. Check $MC(5) = 14$. ✓

(b) $Q^s(P) = (P - 4)/2$ for $P \geq 4$.

(c)

P	Q^*	π	SR	LR
6	1	-24	produces	exits
10	3	-16	produces	exits
14	5	0	produces	stays (break-even)
18	7	24	produces	stays ($\pi > 0$)

(d) LR break-even at $P = AC_{\min} = 14$, $Q^* = 5$, $MC(5) = AC(5) = 14$. ✓

Exercise 3. Martina's graphic design studio. Revenue $R = €120,000$, explicit costs $C_E = €74,000$, opportunity cost of labor $C_L = €36,000$, opportunity cost of capital $C_K = 0.03 \cdot 60,000 = €1,800$.

(a) $\pi_{\text{acct}} = 46,000$; $\pi_{\text{econ}} = 8,200$.

(b) $\pi_{\text{econ}} > 0$: it pays to continue.

(c) With $R = €105,000$: $\pi_{\text{econ}} = -6,800 < 0$: in the LR close down.

(d) In the SR with sunk FC : of the explicit costs, $VC = €50,000$ and hence $FC = 74,000 - 50,000 = €24,000$, against $R = €105,000$: $\pi_{\text{SR}}(\text{open}) = 105,000 - 50,000 - 24,000 - 36,000 - 1,800 = -€6,800$; $\pi_{\text{SR}}(\text{closed}) = -FC = -€24,000$ (the sunk rent is paid anyway). Since $-6,800 > -24,000$ (a difference of €17,200, independent of FC), it pays to stay open in the SR; in the LR, instead, it exits.

Exercise 4. $FC = 300$, $AVC_{\min} = 18$, $AC_{\min} = 30$, $P = 22$, $Q^* = 15$, $AC(Q^*) = 40$.

(a) $AVC_{\min} \leq P < AC_{\min}$: **zone 2**.

(b) $\pi(Q^*) = (22 - 40) \cdot 15 = -270$; $\pi(0) = -300$. $-270 > -300$: producing is correct.

(c) The FOC $P = MC(Q^*)$ does not depend on FC : $Q^* = 15$ unchanged. Only π changes: $\pi(Q^*) = -270 - 300 = -570$; $\pi(0) = -600$; it still produces.

(d) Fixed costs are sunk in the SR: they do not enter the marginal decision. As long as $P \geq AVC_{\min}$, each unit sold helps reduce the loss relative to shutting down. In the SR the firm minimizes its loss; in the LR it maximizes profit.

Chapter 10 – Firm Supply and Market Supply

Exercise 1. $C(Q) = 3Q^2 + 6Q + 12$, $FC = 12$.

(a) $MC = 6Q + 6$; $AVC = 3Q + 6$, $AVC_{\min} = 6$; $AC = 3Q + 6 + 12/Q$, $AC_{\min}$ at $Q = 2$: $AC(2) = 18$.

(b) $Q^s(P) = (P - 6)/6$ for $P \geq 6$.

(c)

P	Q^*	π	SR	LR
12	1	-9	produces	exits
18	2	0	produces	stays (break-even)
24	3	15	produces	stays
30	4	36	produces	stays

(d) $\varepsilon_S(P) = P/(P - 6)$. $\varepsilon_S(18) = 3/2 = 1.5$; $\varepsilon_S(30) = 5/4 = 1.25$. Decreasing in P : as the price rises, the firm moves away from the shutdown threshold and $\varepsilon_S \rightarrow 1^+$ as $P \rightarrow \infty$.

Exercise 2. $N_A = 60$ type-A firms with $C_A = Q_A^2$ ($MC_A = 2Q_A$, $AVC_{\min}^A = 0$); $N_B = 40$ type-B firms with $C_B = 2Q_B^2$ ($MC_B = 4Q_B$, $AVC_{\min}^B = 0$).

(a) $Q_A^s(P) = P/2$; $Q_B^s(P) = P/4$, both for $P \geq 0$.

(b) $Q^S(P) = 60 \cdot P/2 + 40 \cdot P/4 = 30P + 10P = 40P$; inverse $P = Q^S/40$.

(c) $Q^S(5) = 200$, $Q^S(10) = 400$, $Q^S(20) = 800$.

(d) With $N'_A = 120$: $Q^{S'} = 120 \cdot P/2 + 40 \cdot P/4 = 70P$; flatter in (Q, P) (slope $1/70$ instead of $1/40$). Elasticity at $P = 10$: in both cases $\varepsilon_S = 1$ (a line through the origin). What changes is the *scale* of supply, not its percentage responsiveness.

Exercise 3. $N_A = 80$ type-A firms ($MC_A = 2Q_A$, $AVC_{\min}^A = 0$); $N_B = 20$ type-B firms ($MC_B = Q_B$, $AVC_{\min}^B = 8$).

(a) $Q_A^s(P) = P/2$ for $P \geq 0$; $Q_B^s(P) = P$ for $P \geq 8$, 0 otherwise.

(b)

$$Q^S(P) = \begin{cases} 40P & 0 \leq P < 8, \\ 60P & P \geq 8. \end{cases}$$

At $P = 8$ the curve is *discontinuous*: $Q^S(8^-) = 320$, $Q^S(8^+) = 480$. The jump of $160 = 20 \cdot 8$ is due to the fact that each type-B firm starts producing with $Q_B = P_{\text{shut}} = 8 > 0$ (the FOC $P = MC_B = Q_B$ at $P = 8$ gives $Q_B = 8$).

(c) $Q^S(4) = 160$; $Q^S(8) = 480$ (upper branch); $Q^S(16) = 960$.

(d) Point elasticity: on both branches, since they are lines through the origin, $\varepsilon_S = 1$. The slope dQ^S/dP differs: 40 below the threshold, 60 above; the arc elasticity across the jump at $P = 8$ tends to $+\infty$. Market supply is infinitely elastic *across* the threshold, and strictly more price-sensitive (steeper slope) *above* the threshold.

Exercise 4. $C(Q) = Q^2 + 4Q + 9$, $Q^s(P) = (P - 4)/2$.

(a) $\Pi^*(P) = P \cdot (P - 4)/2 - ((P - 4)/2)^2 - 4 \cdot (P - 4)/2 - 9 = (P^2 - 8P - 20)/4 = P^2/4 - 2P - 5$.

(b) $d\Pi^*/dP = P/2 - 2 = (P - 4)/2 = Q^s(P)$. ✓ Hotelling's lemma.

(c) $d^2\Pi^*/dP^2 = 1/2 > 0$: Π^* is convex and Q^s strictly increasing (monotonicity of supply).

Exercise 5. $Q^s = 5P - 10$ for $P \geq 2$.

(a) $\varepsilon_S(P) = P/(P - 2)$. $\varepsilon_S(4) = 2$; $\varepsilon_S(10) = 5/4 = 1.25$.

(b) New $Q^{s'} = 5P - 15$ for $P \geq 3$. At $P = 10$: $Q^{s'} = 35$, $\varepsilon'_S = 5 \cdot 10/35 = 10/7 \approx 1.43$. It has increased: the curve is closer to the new shutdown threshold ($P = 3$ instead of $P = 2$), and ε_S is higher near the threshold.

(c) An increase in FC does not change MC (fixed costs do not depend on Q), so the SR supply curve remains unchanged: it coincides with the increasing portion of MC above $AVC_{\min}$, both of which are unchanged. The effect is merely a reduction in profit at every Q .

Chapter 11 – Competitive Equilibrium, Welfare and the LR

Exercise 1. $Q^D = 60 - 4P$, $Q^S = 16P$.

(a) $60 - 4P = 16P \Rightarrow 20P = 60 \Rightarrow P^* = 3$, $Q^* = 48$.

(b) $P = 1$: $Q^D = 56$, $Q^S = 16$. Excess demand = 40: P rises.

(c) $P = 5$: $Q^D = 40$, $Q^S = 80$. Excess supply = 40: P falls.

(d) Shock: $80 - 4P = 16P \Rightarrow P^{**} = 4$, $Q^{**} = 64$. Consistent with the table of shocks: $D \uparrow \Rightarrow P \uparrow, Q \uparrow$.

Exercise 2. $C(q) = q^2 + 6q + 25$, $Q^D = 120 - 2P$.

(a) $AC_{\min}$ from $MC = AC$: $2q + 6 = q + 6 + 25/q \Rightarrow q^* = 5$. $P_{LR}^* = AC(5) = 5 + 6 + 5 = 16$.

(b) $Q_{\text{mkt}}^* = Q^D(16) = 120 - 32 = 88$; $n^* = 88/5 = 17.6$ (say 18 firms). Check: $\pi = 16 \cdot 5 - (25 + 30 + 25) = 0$. ✓

(c) With $Q^{D'} = 160 - 2P$ (constant costs, P_{LR} unchanged): $P_{LR}^* = 16$, $Q_{\text{mkt}}^{**} = 160 - 32 = 128$, $n^{**} = 128/5 = 25.6$ (about 26 firms).

(d) Firms that entered: $n^{**} - n^* = 25.6 - 17.6 = 8$ firms.

Exercise 3. (*qualitative analysis*)

(a) Tax $t = 3$: MC rises, S shifts left. SR: $P \uparrow$, $Q \downarrow$. LR (constant costs): firms make $\pi < 0$, some exit; the new P_{LR}^* is higher (it reflects the new $AC_{\min}$, which includes the tax).

(b) Better technology: MC falls, S shifts right. SR: $P \downarrow$, $Q \uparrow$. LR: $\pi > 0$, firms enter; P_{LR}^* at the new, lower $AC_{\min}$.

(c) Income $\downarrow$: D shifts left. SR: $P \downarrow$, $Q \downarrow$. LR (constant costs): P_{LR}^* returns to $AC_{\min}$, firms exit.

(d) Key input $\uparrow$: MC and AC rise, S shifts left. SR: $P \uparrow$, $Q \downarrow$. LR: firms exit; P_{LR}^* at the new, higher $AC_{\min}$.

Exercise 4. $Q^D = 180 - 6P$, $Q^S = 12P$.

(a) $180 - 6P = 12P \Rightarrow P^* = 10$, $Q^* = 120$. Intercept of D : $P_{\max}^D = 30$. $CS = \frac{1}{2} \cdot 120 \cdot 20 = 1,200$; $PS = \frac{1}{2} \cdot 120 \cdot 10 = 600$; $TS = 1800$.

(b) Shock: $216 - 6P = 12P \Rightarrow P^{**} = 12$, $Q^{**} = 144$. New intercept: $P_{\max}^{D'} = 36$. $CS' = \frac{1}{2} \cdot 144 \cdot 24 = 1,728$; $PS' = \frac{1}{2} \cdot 144 \cdot 12 = 864$; $TS' = 2,592$.

(c) $\Delta CS = +528$, $\Delta PS = +264$, $\Delta TS = +792$. Ratio $528/264 = 2$: consumers gain twice as much as producers. The slope of inverse demand is $1/6$, that of inverse supply $1/12$: supply is twice as flat as demand. With linear curves $CS = \frac{1}{2}Q^2 \cdot \frac{1}{6}$ and $PS = \frac{1}{2}Q^2 \cdot \frac{1}{12}$: the gain is split in

proportion to the slopes of the inverse curves, $(1/6) : (1/12)$, i.e. 2:1 in favor of consumers (the steeper, less elastic side captures the larger share).

Exercise 5. (+30% demand shock in a constant-cost industry)

Short run (n fixed). The increase in demand creates, at the initial price P^* , excess demand. The price rises to a new equilibrium $P_{\text{SR}}^{**} > AC_{\min}$ and each firm moves along MC , producing $q^{**} > q^*$. In summary:

- $P_{\text{SR}}^{**} > P^*$ and $Q_{\text{mkt,SR}}^{**} > Q_{\text{mkt}}^*$;
- positive unit profits: $\pi = (P_{\text{SR}}^{**} - AC(q^{**})) \cdot q^{**} > 0$;
- PS_{SR} increases (movement along MC);
- CS_{SR} increases, evaluated on the new demand curve;
- n remains unchanged by definition of the short run.

Long run. Positive profits attract new firms: the SR market supply shifts right until it brings the price back to $P_{\text{LR}}^* = AC_{\min}$ (LR supply is perfectly elastic under the constant-cost assumption). In the new LR equilibrium:

- $P_{\text{LR}}^* = AC_{\min}$, *unchanged* relative to the initial equilibrium;
- $Q_{\text{mkt,LR}}^*$ increases by about 30% (in line with the demand shift evaluated at P_{LR}^*);
- n^* grows in the same proportion ($\approx +30\%$): each firm always produces at the efficient scale q^* ;
- zero long-run profits for every firm: $\pi_i = 0$;
- CS_{LR} increases: the entire welfare gain generated by the shock is captured by consumers;
- $PS_{\text{LR}} = 0$, measured on the horizontal LR supply at $P = AC_{\min}$ (both before and after the shock).

Summary. In a constant-cost industry the demand shock is absorbed entirely through *quantity*: in the LR Q_{mkt}^* and n^* change, not P^* . The additional surplus goes entirely to consumers.

Exercise 6. $Q^D = 300 - 10P$, with two supply scenarios.

Inelastic case ($Q^S = 100$ fixed): equilibrium at $300 - 10P = 100 \Rightarrow P^* = 20$, $Q^* = 100$. Shock $\Delta Q^D = +50$ (new $Q^{D'} = 350 - 10P$): $350 - 10P = 100 \Rightarrow P^{**} = 25$, $Q = 100$. $\Delta P^* = +5$.

Elastic case ($Q^S = 40P$): $300 - 10P = 40P \Rightarrow P^* = 6$, $Q^* = 240$. Shock: $350 - 10P = 40P \Rightarrow P^{**} = 7$, $Q = 280$. $\Delta P^* = +1$.

With inelastic supply the demand shock falls entirely on the price ($\Delta P = +5$, $\Delta Q = 0$); with elastic supply it falls mainly on quantity ($\Delta P = +1$, $\Delta Q = +40$). Consistent with the general rule: a shock falls on the less elastic curve.

Chapter 12 – Deadweight Loss and Administered Prices

Exercise 1. Market $Q^D = 80 - 2P$, $Q^S = 3P$, equilibrium $(16, 48)$. Price ceiling $\bar{P} = 12$.

(a) $Q^D(12) = 56$, $Q^S(12) = 36$. $\tilde{Q} = 36$; excess D = 20.

(b) $P_D(36) = 40 - 18 = 22$; $P_S(36) = 12$.

(c) $CS = \int_0^{36} (40 - q/2 - 12) dq = \left[28q - q^2/4\right]_0^{36} = 684$; $PS = \frac{1}{2} \cdot 36 \cdot 12 = 216$; $DWL = \frac{1}{2}(48 - 36)(22 - 12) = 60$. Check: $684 + 216 + 60 = 960$. ✓

(d) $\Delta CS = +108$, $\Delta PS = -168$, $\Delta TS = -60$. Producers lose more than consumers gain; the difference is the DWL. As always: CS rises only if rationing is efficient.

Exercise 2. Same market. Price floor $\underline{P} = 22$.

(a) $Q^D(22) = 36$, $Q^S(22) = 66$. $\tilde{Q} = 36$; excess S = 30.

(b) $P_D(36) = 22$ (the floor coincides with the price at which demand absorbs that quantity); $P_S(36) = 12$.

(c) $CS = \frac{1}{2} \cdot 36 \cdot 18 = 324$; $PS = \int_0^{36} (22 - q/3) dq = \left[22q - q^2/6\right]_0^{36} = 576$; $DWL = \frac{1}{2} \cdot 12 \cdot 10 = 60$. Check: $324 + 576 + 60 = 960$. ✓

(d) The DWL is identical to Exercise 1 because $\tilde{Q}$ and $P_D(\tilde{Q}) - P_S(\tilde{Q})$ are identical (ceiling–floor symmetry). Only the distribution changes: $\Delta CS = -252$, $\Delta PS = +192$.

Exercise 3. (*true/false with justification*)

(a) *False.* A ceiling helps only the consumers who *manage* to buy. Those who are rationed out lose all the CS they had at equilibrium. Aggregate CS may even fall below its equilibrium level if rationing is random. In addition, reduced quality and the black market further erode the theoretical benefits.

(b) *False; on the contrary, it usually creates unemployment.* A minimum wage $\underline{w} > w^*$ generates excess supply (more job seekers than jobs offered). Exception: a monopsony market, where a moderate minimum wage can increase employment (the opposite effect).

(c) *False.* DWL and distribution are distinct concepts. At the competitive equilibrium $DWL = 0$, but CS and PS are nonzero: there is a well-defined distribution of welfare. A lump-sum transfer from producers to consumers would have $DWL = 0$ while still redistributing.

(d) *True.* If $\bar{P} \geq P^*$, the market operates at $P^* \leq \bar{P}$: the ceiling does not bind, no transaction is constrained, and the equilibrium is identical to the case with no intervention. Symmetrically for a floor with $\underline{P} \leq P^*$.

Exercise 4. $L^D = 600 - 10w$, $L^S = 20w$, minimum wage $\underline{w} = 24$.

(a) $600 - 10w = 20w \Rightarrow w^* = 20$, $L^* = 400$. Inverse demand: $w_D(L) = (600 - L)/10$, intercept 60.

(b) $L^D(24) = 360$, $L^S(24) = 480$. $\tilde{L} = 360$; involuntary unemployment = 120.

(c) Firm surplus: $FS = \frac{1}{2} \cdot 360 \cdot (60 - 24) = 6,480$. Surplus of employed workers: $WS = \int_0^{360} (24 - L/20) dL = \left[24L - L^2/40\right]_0^{360} = 5,400$. $w_S(360) = 18$, so $DWL = \frac{1}{2}(400 - 360)(24 - 18) = 120$.

(d) Worker surplus at equilibrium: $WS_{eq} = \frac{1}{2} \cdot 400 \cdot 20 = 4,000$. With the minimum wage: $WS = 5,400 > 4,000$, a change of +1,400. In aggregate, workers gain. But the redistribution is

between *insiders* (the 360 employed, who earn a higher wage) and *outsiders* (the $400 - 360 = 40$ workers who were employed at $w^* = 20$ and now lose their jobs, and with them all of their surplus). Of the 120 involuntarily unemployed, the other 80 are new entrants attracted by the higher wage, who would not have worked at w^* .

Exercise 5. (*elasticity and the size of the DWL*)

The height of the DWL triangle is $P_D(\tilde{Q}) - P_S(\tilde{Q})$. With *inelastic* supply a small reduction in $\tilde{Q}$ makes $P_S(\tilde{Q})$ “drop” quickly (the curve is steep): the height grows a lot. With *elastic* supply the curve is flat and $P_S(\tilde{Q})$ changes little: the height is smaller. For the same deviation $Q^* - \tilde{Q}$, the DWL is **larger with inelastic supply**. Analogous reasoning applies to demand.

Exercise 6. (*DWL in the long run*)

The DWL of a ceiling (or floor) tends to grow in the transition from the SR to the LR for two main reasons:

(i) **Greater elasticity of demand in the LR.** In the LR consumers have time to substitute away from the good (look for alternatives, change habits): the demand curve becomes flatter. For the same price wedge, the reduction in the quantity traded is larger.

(ii) **Greater elasticity of supply in the LR.** In the LR producers can change productive capacity (build or dismantle plants, invest or exit). The supply curve becomes more elastic: a ceiling reduces investment in new capacity, a floor discourages the expansion of production. In both cases the quantity traded moves further away from Q^* .

The combination of these two factors produces a lower $\tilde{Q}$ (or one further from Q^*) and hence a larger DWL triangle.

Chapter 13 – Taxes and Subsidies

Exercise 1. Market $Q^D = 80 - 2P$, $Q^S = 3P$, equilibrium $(16, 48)$, $TS = 960$. Tax $t = 10$ on the producer.

(a) $Q_t^S(P_d) = 3(P_d - 10) = 3P_d - 30$. $80 - 2P_d = 3P_d - 30 \Rightarrow P_d^* = 22$, $P_s^* = 12$, $Q_t^* = 36$.

(b) $CS = \frac{1}{2} \cdot 36 \cdot (40 - 22) = 324$; $PS = \frac{1}{2} \cdot 36 \cdot 12 = 216$; $T = 10 \cdot 36 = 360$; $DWL = \frac{1}{2} (48 - 36) \cdot 10 = 60$. Check: $324 + 216 + 360 + 60 = 960$. ✓

(c) $\Delta P_d = 6$, $|\Delta P_s| = 4$. Consumer share $= 6/10 = 60\%$; producer share $= 4/10 = 40\%$. Identical to the tax $t = 5$: incidence depends on the elasticities at equilibrium, not on the tax rate.

(d) $DWL(t = 10)/DWL(t = 5) = 60/15 = 4 = (10/5)^2$. Consistent with $DWL \propto t^2$ (Prop. 13.2).

Exercise 2. Tax $t = 5$ on the buyer.

(a) The buyer pays P_s to the seller and t to the government: total cost $P_d = P_s + 5$. Demand as a function of P_s becomes $Q_t^D(P_s) = 80 - 2(P_s + 5) = 70 - 2P_s$.

(b) $70 - 2P_s = 3P_s \Rightarrow P_s^* = 14$, $P_d^* = 19$, $Q_t^* = 42$.

(c) Identical to the case of the tax on the seller: $CS = 441$, $PS = 294$, $T = 210$, $DWL = 15$. Incidence does not depend on who formally remits the tax (Prop. 13.1): the market shifts the burden of the tax along the slopes of the curves, with shares determined only by the elasticities.

Exercise 3. Subsidy $s = 10$.

(a) $Q_s^S(P_d) = 3(P_d + 10) = 3P_d + 30$. $80 - 2P_d = 3P_d + 30 \Rightarrow P_d^* = 10$, $P_s^* = 20$, $Q_s^* = 60$.

(b) $CS = \frac{1}{2} \cdot 60 \cdot 30 = 900$; $PS = \frac{1}{2} \cdot 60 \cdot 20 = 600$; $\text{cost} = 10 \cdot 60 = 600$; $TS_{\text{net}} = 900 + 600 - 600 = 900$. $DWL = 960 - 900 = 60 = \frac{1}{2}(60 - 48) \cdot 10$. ✓

(c) No, net TS falls from 960 to 900. The subsidy pushes production beyond $Q^* = 48$: for the units between 48 and 60, $P_S(Q) > P_D(Q)$ (the marginal cost exceeds willingness to pay). The government finances socially harmful trades: the government transfer exceeds the private gains by exactly 60.

Exercise 4. $\varepsilon_D = -0.4$, $\varepsilon_S = 0.6$.

(a) Consumer share $= \varepsilon_S / (\varepsilon_S - \varepsilon_D) = 0.6 / 1 = 60\%$; producer share $= 40\%$.

(b) To bring the consumer share to 50% we need $\varepsilon_S = |\varepsilon_D|$. Starting from $\varepsilon_S = 0.6$ and $|\varepsilon_D| = 0.4$, there are two routes: raise $|\varepsilon_D|$ to 0.6 (promotion of substitutes, advertising, regulation that facilitates substitution) or reduce ε_S to 0.4 (constraining the expansion of supply).

(c) The demand for tobacco is highly inelastic ($|\varepsilon_D| \approx 0.4$) because of physiological dependence and the absence of close substitutes; supply is very elastic ($\varepsilon_S \approx 5$ in the LR). Consumer share $= 5 / (5 + 0.4) \approx 93\%$: consumers bear almost the entire excise tax. This is why an increase in the excise tax translates into almost complete pass-through to the final price.

Exercise 5. Comparison of a tax $t = 5$ vs a subsidy $s = 5$.

(a) Tax: $\Delta P_d = +3$, $\Delta P_s = -2$. Subsidy: $\Delta P_d = -3$, $\Delta P_s = +2$. Mirror-image movements, identical in absolute value.

(b) Tax: $\tilde{Q} = 42$, $DWL = \frac{1}{2} \cdot 6 \cdot 5 = 15$. Subsidy: $\tilde{Q} = 54$, $DWL = \frac{1}{2} \cdot 6 \cdot 5 = 15$. Identical because the base and height of the triangle are the same: same distance from Q^* (6 units), same height $t = s = 5$.

(c) Tax: $\Delta CS + \Delta PS = -135 - 90 = -225$ (private agents lose 225, the government collects 210, $DWL = 15$). Subsidy: $\Delta CS + \Delta PS = +153 + 102 = +255$ (private agents gain 255, the government spends 270, $DWL = 15$). The absolute values differ because the position of the trapezoid (between P^* and P_d^*/P_s^*) relative to Q^* is different, and the increases/decreases in CS and PS are computed by integrating over $\tilde{Q} \leq Q^*$. The net balance “private agents + government” is instead always ± 15 , exactly the DWL .

(d) The subsidy costs the government 270, whereas the tax raises 210. The government must finance the subsidy with other taxes, which generate additional DWL : the opportunity cost of the subsidy is higher than that of an equivalent tax. Subsidies are therefore intrinsically more costly than taxes of the same size.

Exercise 6. $Q^D = 120 - 4P$, $Q^S = 2P$, world price $P_w = 5$.

(a) At $P = P_w = 5$: $Q^D(5) = 100$, $Q_{\text{dom}}^S(5) = 10$; imports $= 100 - 10 = 90$.

(b) Tariff $t = 5$: domestic price $= P_w + t = 10$. $Q^D(10) = 80$, $Q_{\text{dom}}^S(10) = 20$; imports $= 60$. $CS = \frac{1}{2} \cdot 80 \cdot (30 - 10) = 800$ (intercept of D : $P_{\text{max}}^D = 30$). $PS_{\text{dom}} = \frac{1}{2} \cdot 20 \cdot 10 = 100$. Revenue $T = t \cdot \text{imports} = 5 \cdot 60 = 300$. DWL : the sum of two triangles—(i) the production triangle (inefficient expansion of domestic production, Q_{dom}^S from 10 to 20): $\frac{1}{2} \cdot 10 \cdot 5 = 25$; (ii) the consumption triangle (reduction in demand, Q^D from 100 to 80): $\frac{1}{2} \cdot 20 \cdot 5 = 50$. Total $DWL = 75$.

(c) Quota equal to the imports in part (b): $\bar{Q}_{\text{imp}} = 60$. The domestic price such that $Q^D(P) - Q_{\text{dom}}^S(P) = 60$ is the solution of $120 - 4P - 2P = 60 \Rightarrow P = 10$, identical to the tariff. CS , PS_{dom} and DWL also coincide with part (b). The only difference is that the revenue rectangle ($T = 300$)

does not go to the government but becomes **quota rent** captured by the license holders (typically private importers or foreign producers who obtain the authorization). A perfect example of the economic equivalence of tariffs and quotas with a distributive difference.

Exercise 7. (*optimal taxation: which market to tax?*)

With perfectly elastic supply in both markets ($\varepsilon_S \rightarrow \infty$), the consumer share is 100% and the tax falls entirely on consumers. For a given revenue $\bar{T}$, the approximate DWL is proportional to $|\varepsilon_D|$ (formally: $DWL \propto \bar{T}^2 \cdot |\varepsilon_D| / (Q^* P^*)$). Hence, for the same revenue and the same $Q^* P^*$, it is better to tax the market with the *smaller* $|\varepsilon_D|$. Market A, with $|\varepsilon_D^A| = 0.5$, is less distortionary than B, with $|\varepsilon_D^B| = 2$: **tax A**. This is the logic of Ramsey's *inverse elasticity rule*: inelastic goods are efficient candidates for taxation. Note the classic *trade-off*: inelastic goods are often necessities (food, energy, medicines) and taxing them is regressive; the Ramsey rule must be balanced against distributive considerations.

Questions and Exercises – Perfectly Competitive Markets

Multiple-choice questions

1. **Answer C.** FOC $3Q^2 - 18Q + 15 = 0 \Rightarrow Q \in \{1, 5\}$. SOC $MC'(Q) = 6Q - 18$: $MC'(1) < 0$ (profit min), $MC'(5) > 0$ (max). A: wrong root. B: minimum of MC . D: does not satisfy the FOC.
2. **Answer B.** $AVC = Q + 10$ strictly increasing, $AVC_{\min} = 10$. SR shutdown threshold = $AVC_{\min}$. C is $AC_{\min} = 20$ (the LR threshold); D confuses the threshold with the level of FC .
3. **Answer C.** MC does not depend on FC ($MC = dVC/dQ$); the SR supply curve, being the increasing portion of MC above $AVC_{\min}$, does not shift. A, B, D are all correct properties.
4. **Answer C.** $Q^s(10) = 14$, $dQ^s/dP = 2$, $\varepsilon_S = 2 \cdot 10/14 = 10/7$.
5. **Answer C.** Equilibrium: $100 - 5P = 5P \Rightarrow P^* = 10$, $Q^* = 50$. At $P = 6 < P^*$: $Q^D(6) = 70$, $Q^S(6) = 30$, excess D = 40.
6. **Answer B.** In a constant-cost industry, $P_{LR}^* = AC_{\min}$ is determined only by technology and remains unchanged. A permanent demand shock increases Q_{mkt}^* and attracts new firms via free entry. D is incorrect because Q^* also increases, not only n^* .
7. **Answer C.** For the same $\tilde{Q}$, the base and height of the DWL triangle coincide, so the DWL is identical; only the way CS and PS are distributed between the two sides of the market changes (mirror images).
8. **Answer D.** Ceiling $\bar{P} = 6$: $Q^S(6) = 30 = \tilde{Q}$, excess D = 40. $P_D(30) = 20 - 30/5 = 14$, $P_S(30) = 6$. $DWL = \frac{1}{2}(50 - 30)(14 - 6) = \frac{1}{2} \cdot 20 \cdot 8 = 80$.
9. **Answer A.** The formula is consumer share = $\varepsilon_S / (\varepsilon_S - \varepsilon_D)$; it increases with the ratio $\varepsilon_S / |\varepsilon_D|$. When supply is more elastic than demand, consumers bear the larger share. D is incorrect by the Equivalence Principle.
10. **Answer C.** The subsidy pushes Q beyond Q^* . On the extra units (between Q^* and Q_s^*), $P_S(Q) > P_D(Q)$: each unit costs society more than it is worth. The government transfer exceeds the private gains, and the difference is the DWL. A is wrong (the subsidy does not transfer, it adds resources). B describes a tax, not a subsidy. D is the myth to be debunked.

Exercises and open questions

Exercise 1. $C(Q) = 2Q^2 + 10Q + 32$, $FC = 32$.

(a) $MC = 4Q + 10$, $AVC = 2Q + 10$ increasing $\Rightarrow AVC_{\min} = 10$. $AC = 2Q + 10 + 32/Q$, $AC_{\min}$ from $MC = AC$: $Q^{\min} = 4$, $AC_{\min} = 26$.

(b) $Q^s(P) = (P - 10)/4$ for $P \geq 10$.

(c)

P	Q^*	π	SR	LR	Zone
8	0	-32	shuts down	exits	1
18	2	-24	produces	exits	2
26	4	0	produces	stays (break-even)	3
50	10	168	produces	stays ($\pi > 0$)	4

(d) LR break-even at $P = AC_{\min} = 26$. $Q^* = 4$, $MC(4) = 4 \cdot 4 + 10 = 26 = AC(4)$. ✓

Exercise 2. Type-A firms ($MC_A = 2Q_A + 4$, $AVC_{\min}^A = 4$), type-B firms ($MC_B = 2Q_B + 10$, $AVC_{\min}^B = 10$).

(a) $Q_A^s(P) = (P - 4)/2$ for $P \geq 4$; $Q_B^s(P) = (P - 10)/2$ for $P \geq 10$.

(b) Horizontal aggregation with $N_A = 50$, $N_B = 30$:

$$Q^S(P) = \begin{cases} 25P - 100 & 4 \leq P < 10, \\ 40P - 250 & P \geq 10. \end{cases}$$

Kink at $P = 10$: $25 \cdot 10 - 100 = 150 = 40 \cdot 10 - 250$. ✓

(c) $Q^S(6) = 50$, $Q^S(10) = 150$, $Q^S(20) = 550$.

(d) At $P = 10$, $Q^S = 150$: $\varepsilon_S^- = 25 \cdot 10/150 = 5/3 \approx 1.67$; $\varepsilon_S^+ = 40 \cdot 10/150 = 8/3 \approx 2.67$. More elastic after the kink: the entry of type-B firms adds an extensive margin to the intensive margin of the type-A firms already active.

Exercise 3. $Q^D = 100 - 5P$, $Q^S = 5P$.

(a) $100 - 5P = 5P \Rightarrow P^* = 10$, $Q^* = 50$. $P_{\max}^D = 20$. $CS = \frac{1}{2} \cdot 50 \cdot 10 = 250$; $PS = \frac{1}{2} \cdot 50 \cdot 10 = 250$ (supply starts at the origin); $TS = 500$.

(b) $Q^{D'} = 120 - 5P$: $120 - 5P = 5P \Rightarrow P^{**} = 12$, $Q^{**} = 60$. $P_{\max}^{D'} = 24$. $CS' = \frac{1}{2} \cdot 60 \cdot 12 = 360$; $PS' = \frac{1}{2} \cdot 60 \cdot 12 = 360$; $TS' = 720$.

(c) With $C(q) = q^2 + 2q + 16$: $MC = 2q + 2$, $AC = q + 2 + 16/q$. $MC = AC \Rightarrow q = 16/q \Rightarrow q^* = 4$. $P_{LR}^* = AC(4) = 4 + 2 + 4 = 10$. *Original demand*: $Q_{\text{mkt},LR}^* = Q^D(10) = 100 - 50 = 50$; $n^* = 50/4 = 12.5$ (about 13 firms). *After the shock*: $Q_{\text{mkt},LR}^{**} = Q^{D'}(10) = 120 - 50 = 70$; $n^{**} = 70/4 = 17.5$ (about 18 firms).

(d) In the SR the price rises from 10 to 12 ($\Delta P^* = +2$); in the LR it returns to $P_{LR}^* = AC_{\min} = 10$, unchanged. The LR adjustment falls entirely on quantity and on the number of firms. Implicit assumption: a **constant-cost** industry (the technology $C(q)$ is common to all firms and does not depend on n).

Exercise 4. Same market, equilibrium (10, 50), $TS = 500$.

(a) Ceiling $\bar{P} = 6$: $Q^D(6) = 70$, $Q^S(6) = 30$. $\tilde{Q} = 30$, excess D = 40. $P_D(30) = 20 - 6 = 14$, $P_S(30) = 6$. CS = $\int_0^{30} (20 - q/5 - 6) dq = \left[14q - q^2/10\right]_0^{30} = 330$; PS = $\frac{1}{2} \cdot 30 \cdot 6 = 90$; DWL = $\frac{1}{2}(50 - 30)(14 - 6) = 80$. Check: $330 + 90 + 80 = 500$. ✓

(b) Floor $\underline{P} = 14$: $Q^D(14) = 30$, $Q^S(14) = 70$. $\tilde{Q} = 30$, excess S = 40. $P_D(30) = 14$, $P_S(30) = 6$. CS = $\frac{1}{2} \cdot 30 \cdot (20 - 14) = 90$; PS = $\int_0^{30} (14 - q/5) dq = \left[14q - q^2/10\right]_0^{30} = 330$; DWL = 80. Check: $90 + 330 + 80 = 500$. ✓

(c) The DWLs are identical because $\tilde{Q} = 30$ and the height of the triangle $P_D(\tilde{Q}) - P_S(\tilde{Q}) = 8$ are the same in the two cases. Mirror-image distribution: ceiling $\Delta\text{CS} = +80$, $\Delta\text{PS} = -160$; floor $\Delta\text{CS} = -160$, $\Delta\text{PS} = +80$.

(d) With random rationing, the 30 units go to 30 of the 70 consumers willing to pay at least $\bar{P} = 6$. The average WTP of a random buyer is $(20 + 6)/2 = 13$, so $\text{CS}_{\text{random}} \approx 30 \cdot (13 - 6) = 210$, significantly lower than the 330 with efficient rationing. In that case the ceiling not only reduces total TS (as is certain) but could also reduce CS below its equilibrium level (250).

Exercise 5. Same market.

(a) Tax $t = 4$: $Q_t^S = 5(P_d - 4) = 5P_d - 20$. $100 - 5P_d = 5P_d - 20 \Rightarrow P_d^* = 12$, $P_s^* = 8$, $Q_t^* = 40$. CS = $\frac{1}{2} \cdot 40 \cdot 8 = 160$; PS = $\frac{1}{2} \cdot 40 \cdot 8 = 160$; $T = 4 \cdot 40 = 160$; DWL = $\frac{1}{2} \cdot 10 \cdot 4 = 20$. Check: $160 + 160 + 160 + 20 = 500$. ✓

(b) $\Delta P_d = 2$, $|\Delta P_s| = 2$. Shares 50/50. At equilibrium: $\varepsilon_D = -5 \cdot 10/50 = -1$, $\varepsilon_S = 5 \cdot 10/50 = 1$. Formula: consumer share = $\varepsilon_S/(\varepsilon_S - \varepsilon_D) = 1/2 = 50\%$. ✓ Symmetric incidence because demand and supply have equal elasticity at equilibrium (the two curves are symmetric with respect to the equilibrium point).

(c) Subsidy $s = 4$: $Q_s^S = 5(P_d + 4) = 5P_d + 20$. $100 - 5P_d = 5P_d + 20 \Rightarrow P_d^* = 8$, $P_s^* = 12$, $Q_s^* = 60$. CS = $\frac{1}{2} \cdot 60 \cdot 12 = 360$; PS = $\frac{1}{2} \cdot 60 \cdot 12 = 360$; cost = $4 \cdot 60 = 240$; $\text{TS}_{\text{net}} = 360 + 360 - 240 = 480$; DWL = $500 - 480 = 20$. Check with the triangle formula: $\frac{1}{2}(60 - 50) \cdot 4 = 20$. ✓

(d) The DWL is identical because the base and height of the triangle are the same: $|Q - Q^*| = 10$ and height = $t = s = 4$ in both cases. The budget balances differ because the tax falls on the units actually traded at the new equilibrium ($Q_t^* = 40 < Q^*$), whereas the subsidy pays for all the units at the new equilibrium ($Q_s^* = 60 > Q^*$): for the same rate ± 4 , the number of taxed/subsidized units differs, and so, consequently, does the gross flow. In this example: $T = 160$ (revenue) vs cost = 240 (outlay): the ratio is $40/60 = 2/3 = Q_t^*/Q_s^*$.

Chapter 14 – Monopoly

Exercise 14.1. Demand $P(Q) = 60 - Q$, cost $C(Q) = Q^2 + 12Q + 88$.

(a) $MR(Q) = 60 - 2Q$ (same intercept as demand, twice the slope); $MC(Q) = 2Q + 12$. FOC: $60 - 2Q = 2Q + 12 \Rightarrow Q_m^* = 12$. SOC: $\pi'' = MR' - MC' = -2 - 2 = -4 < 0$ ✓.

(b) $P_m^* = 60 - 12 = 48$; $MC(12) = 36$ (absolute markup $P^* - MC = 12$, relative $12/48 = 25\%$); $\pi_m = 48 \cdot 12 - (144 + 144 + 88) = 576 - 376 = 200$.

(c) Perfect competition: $P = MC$ implies $60 - Q = 2Q + 12$, hence $Q_{pc} = 16$ and $P_{pc} = 44$. $CS_m = \frac{1}{2} \cdot 12 \cdot (60 - 48) = 72$; $CS_{pc} = \frac{1}{2} \cdot 16 \cdot 16 = 128$; $DWL = \frac{1}{2}(16 - 12)(48 - 36) = 24$.

(d) PS uses the price P_m^* charged on all the inframarginal units (not the demand curve): $PS_m = \int_0^{12} [P_m^* - MC(Q)] dQ = \int_0^{12} (48 - 2Q - 12) dQ = \int_0^{12} (36 - 2Q) dQ = 432 - 144 = 288$.

Profit net of $FC = 88$: $\pi_m = 288 - 88 = 200 \checkmark$. Similarly $PS_{pc} = \int_0^{16} (44 - 2Q - 12) dQ = \int_0^{16} (32 - 2Q) dQ = 512 - 256 = 256$, $\pi_{pc} = 256 - 88 = 168$, $TS_{pc} = CS_{pc} + PS_{pc} = 128 + 256 = 384$. Check: $TS_{pc} = CS_m + PS_m + DWL = 72 + 288 + 24 = 384 \checkmark$.

Exercise 14.2. Same setup, tax $t = 8$ on the monopolist.

(a) The effective MC becomes $MC(Q) + t = 2Q + 20$. New FOC: $60 - 2Q = 2Q + 20 \Rightarrow Q_m^t = 10$, $P_m^t = 60 - 10 = 50$.

(b) $\Delta P_d = P_m^t - P_m^* = 50 - 48 = 2$. Incidence on consumers: $2/8 = 25\%$. Incidence on the monopolist: the net price received per unit is $P_m^t - t = 50 - 8 = 42$, to be compared with $P_m^* = 48$: a change of -6 , i.e. $6/8 = 75\%$ on the monopolist. Sum $25\% + 75\% = 100\% \checkmark$. Analytical check with the Proposition ($P = a - bQ$, $MC = c + dQ$): here $b = 1$, $c = 12$, $d = 2$, so $\Delta P_d = b/(2b + d) \cdot t = \frac{1}{4} \cdot 8 = 2 \leq t/2 \checkmark$.

(c) Revenue $T = t \cdot Q_m^t = 8 \cdot 10 = 80$. After-tax profit: $\pi_m^t = 50 \cdot 10 - (100 + 120 + 88) - 80 = 500 - 308 - 80 = 112$.

(d) Additional DWL as a trapezoid between $Q_m^t = 10$ and $Q_m^* = 12$: $(P - MC)|_{Q^t} = 50 - (20 + 12) = 18$; $(P - MC)|_{Q^*} = 48 - (24 + 12) = 12$. Area: $\frac{1}{2} \cdot 2 \cdot (18 + 12) = 30$.

Exercise 14.3. Multi-plant: $C_1(Q_1) = Q_1^2$, $C_2(Q_2) = 2Q_2^2$, $P(Q) = 60 - Q$.

(a) $MC_1 = 2Q_1$, $MC_2 = 4Q_2$. The rule $MC_1 = MC_2$ gives $Q_1 = 2Q_2$. With $Q = Q_1 + Q_2 = 3Q_2$: $Q_2 = Q/3$, $Q_1 = 2Q/3$. Common value $MC_{agg}(Q) = MC_1(2Q/3) = 4Q/3$.

(b) $MR = 60 - 2Q$. FOC $MR = MC_{agg}$: $60 - 2Q = 4Q/3 \Rightarrow 180 - 6Q = 4Q \Rightarrow Q^* = 18$, $P^* = 60 - 18 = 42$.

(c) $Q_1^* = 12$, $Q_2^* = 6$. Check: $MC_1(12) = 24 = MC_2(6) \checkmark$.

(d) $\pi = 42 \cdot 18 - (144 + 72) = 756 - 216 = 540$.

Exercise 14.4. Natural monopoly: $P(Q) = 50 - Q$, $C(Q) = 300 + 10Q$.

(a) $AC(Q) = 300/Q + 10$, $AC'(Q) = -300/Q^2 < 0$ for all $Q > 0$: AC is strictly decreasing everywhere, hence a natural monopoly.

(b) $MR = 50 - 2Q$, $MC = 10$. $50 - 2Q = 10 \Rightarrow Q_m^* = 20$, $P_m^* = 30$. $\pi_m = (30 - 10) \cdot 20 - 300 = 400 - 300 = 100$.

(c) $P = AC$: $50 - Q = 300/Q + 10 \Rightarrow Q^2 - 40Q + 300 = 0$, hence $Q \in \{10, 30\}$. The economically sensible solution is $Q_{AC}^* = 30$, $P_{AC}^* = 20$, $\pi_{AC} = 0$ (the firm exactly covers its total costs). $P = MC$: $50 - Q = 10 \Rightarrow Q_{MC}^* = 40$, $P_{MC}^* = 10$; $\pi_{MC} = (10 - 10) \cdot 40 - 300 = -300$: required subsidy = 300 (exactly FC).

(d) CS under the three regimes: $CS_m = \frac{1}{2} \cdot 20 \cdot 20 = 200$; $CS_{AC} = \frac{1}{2} \cdot 30 \cdot 30 = 450$; $CS_{MC} = \frac{1}{2} \cdot 40 \cdot 40 = 800$. *Trade-off*: the *first best* $P = MC$ maximizes CS and eliminates DWL but requires a subsidy equal to the fixed costs; the $P = AC$ regime is a *second-best* solution in which the firm covers its costs ($\pi = 0$, no subsidy) but a positive DWL remains. The choice depends on the social cost (distortions) of the taxes needed to finance the subsidy.

Chapter 15 – Market Power and Price Discrimination

Exercise 15.1. Tullock with $V = 200$ and $n = 5$.

(a) Expected profit of firm i : $\mathbb{E}[\pi_i] = x_i V / \sum_j x_j - x_i$. FOC: $(\sum_j x_j - x_i) / (\sum_j x_j)^2 \cdot V = 1$. Symmetry $x_j = x^*$ for all j , $\sum_j x_j = nx^*$: $(n-1)x^* \cdot V / (nx^*)^2 = 1 \Rightarrow x^* = V(n-1)/n^2$. For $V = 200$, $n = 5$: $x^* = 200 \cdot 4/25 = 32$.

(b) $X^* = nx^* = 5 \cdot 32 = 160$. Dissipation: $X^*/V = 160/200 = 80\%$ (equivalent to $(n-1)/n = 4/5$).

(c) $\mathbb{E}[\pi_i] = V/n - x^* = 40 - 32 = 8$. Positive, so firms participate willingly.

(d) Dissipation $(n-1)/n$ is increasing in n and tends to 1 as $n \rightarrow \infty$. $X^* = V(n-1)/n$ is also increasing in n . Individual spending $x^* = V(n-1)/n^2$ rises up to $n = 2$ but then decreases ($n = 2$: 50; $n = 4$: 37.5; $n = 10$: 18); an increase in the number of contenders reduces the expected chance of winning and hence individual spending, but the total still grows.

Exercise 15.2. Demand $P = 40 - Q$, $MC = 10$.

(a) $MR = 40 - 2Q = 10 \Rightarrow Q_m^* = 15$, $P_m^* = 25$. $L = (25 - 10)/25 = 3/5$.

(b) $|\varepsilon_D| = |dQ/dP| \cdot P/Q = 1 \cdot 25/15 = 5/3$; $1/|\varepsilon_D| = 3/5 = L \checkmark$.

(c) Flatter demand $P = 40 - 0.5Q$, $MR = 40 - Q = 10 \Rightarrow Q = 30$, $P = 25$. $L = (25 - 10)/25 = 3/5$. Same price? Yes, but the quantity doubles. The Lerner index remains $3/5$ because at equilibrium the elasticity $|\varepsilon_D| = |dQ/dP| \cdot P/Q = 2 \cdot 25/30 = 5/3$ is the same. There is no contradiction with intuition: a *flatter* demand is not necessarily more *elastic* at the optimum. With $P = a - bQ$ and constant MC , $P_m^* = (a + MC)/2$ and $L = (a - MC)/(a + MC)$, independent of the slope b : L depends only on the intercept a . With a lower intercept, for example $P = 30 - 0.5Q$: $MR = 30 - Q = 10 \Rightarrow Q = 20$, $P = 20$, $|\varepsilon_D| = 2 \cdot 20/20 = 2$, $L = 1/2$ (lower). With a higher intercept, $P = 50 - 0.5Q$: $Q = 40$, $P = 30$, $|\varepsilon_D| = 2 \cdot 30/40 = 3/2$, $L = 2/3$ (higher): the analysis requires attention to the elasticity at the optimum, not only to the slope.

Exercise 15.3. First-degree discrimination with $P = 60 - Q$, $MC = 2Q + 12$, $FC = 88$.

(a) $Q^I = Q_{pc} = 16$ (sells up to $P = MC$). $PS^I = \int_0^{16} [(60-Q) - (2Q+12)] dQ = \int_0^{16} (48-3Q) dQ = 768 - 384 = 384$. Profit: $\pi^I = 384 - 88 = 296$.

(b) Uniform-price monopoly: $\pi_m = 200$. Gain: $\pi^I - \pi_m = 296 - 200 = 96$.

(c) From Exercise 14.1: $CS_m = 72$, $DWL = 24$. Sum: $72 + 24 = 96 = \pi^I - \pi_m \checkmark$. Decomposition: 72 is the old consumer surplus transferred to the monopolist, 24 is the DWL of the uniform-price monopoly recovered by producing up to Q_{pc} .

(d) For consumers: CS goes from 72 to 0 (-72). For aggregate welfare: $TS = CS + PS$ goes from $72 + 288 = 360$ to $0 + 384 = 384 \checkmark$. First-degree discrimination is *efficient* ($DWL = 0$) but *regressive*: the entire surplus goes to the producer.

Exercise 15.4. Third-degree discrimination: $P_1 = 80 - Q_1$, $P_2 = 60 - Q_2$, $MC = 20$.

(a) $MR_1 = 80 - 2Q_1 = 20 \Rightarrow Q_1^* = 30$, $P_1^* = 50$. $MR_2 = 60 - 2Q_2 = 20 \Rightarrow Q_2^* = 20$, $P_2^* = 40$.

(b) $L_1 = (50-20)/50 = 3/5$; $L_2 = (40-20)/40 = 1/2$. $|\varepsilon_1| = 1 \cdot 50/30 = 5/3 \Rightarrow 1/|\varepsilon_1| = 3/5 = L_1 \checkmark$. $|\varepsilon_2| = 1 \cdot 40/20 = 2 \Rightarrow 1/|\varepsilon_2| = 1/2 = L_2 \checkmark$. Market 1 has less elastic demand ($|\varepsilon_1| < |\varepsilon_2|$) and a higher price ($P_1 > P_2$), as predicted by the Lerner rule.

(c) $\pi^{III} = (50 - 20) \cdot 30 + (40 - 20) \cdot 20 = 900 + 400 = 1,300$.

(d) Aggregation for $P \leq 60$: $Q(P) = Q_1 + Q_2 = (80 - P) + (60 - P) = 140 - 2P$, hence $P(Q) = 70 - Q/2$ and $MR_{tot} = 70 - Q$. FOC: $70 - Q = 20 \Rightarrow Q_u = 50$, $P_u = 45$. Allocation: $Q_1 = 80 - 45 = 35$, $Q_2 = 60 - 45 = 15$, total 50 $\checkmark$. Single-price profit: $(45 - 20) \cdot 50 = 1,250$.

Discrimination yields $1,300 > 1,250$: gain = 50.

Exercise 15.5. Two-part tariff, homogeneous consumers: $P(Q) = 30 - Q$, $MC = 10$.

(a) By the Proposition on the optimal tariff with homogeneous consumers: $p^* = MC = 10$, $T^* = CS(p^*) = \frac{1}{2} \cdot 20 \cdot 20 = 200$.

(b) Individual quantity: $Q(10) = 30 - 10 = 20$. Profit per consumer: variable part $(p - MC)Q = 0$; fixed part $T^* = 200$. Total profit $N \cdot 200$.

(c) Under uniform-price monopoly: $MR = 30 - 2Q = 10 \Rightarrow Q_m = 10$, $P_m = 20$. $CS^{mon} = \frac{1}{2} \cdot 10 \cdot 10 = 50$ for each consumer. Under the two-part tariff: $CS^{2pp} = 0$ (the entire surplus is captured by T^*). The two-part tariff is maximally “unfavorable” for the individual consumer, but efficient for aggregate welfare (the quantity is the PC one).

Exercise 15.6. Two-part tariff, heterogeneous consumers: 1 consumer H ($P_H = 50 - Q$), 4 consumers L ($P_L = 30 - Q$), $MC = 10$.

(a) **Only H .** $p_A = MC = 10$, $T_A = V_H(10) = \frac{1}{2}(50 - 10)^2 = 800$. $\pi_A = 1 \cdot 800 = 800$.

(b) **Everyone**, $p = MC$. To avoid excluding the L types: $T_B \leq V_L(10) = \frac{1}{2}(30 - 10)^2 = 200$. Setting $T_B = 200$: $\pi_B = (1 + 4) \cdot 200 = 1,000$.

(c) **Everyone, optimal** $p > MC$. At unit price p (with $p < 30$ so as not to exclude the L types): $Q_H(p) = 50 - p$, $Q_L(p) = 30 - p$, $T(p) = V_L(p) = \frac{1}{2}(30 - p)^2$.

$$\pi_C(p) = (p - 10) \left[(50 - p) + 4(30 - p) \right] + 5 \cdot \frac{1}{2}(30 - p)^2 = (p - 10)(170 - 5p) + \frac{5}{2}(30 - p)^2.$$

Expanding: $\pi_C(p) = -\frac{5}{2}p^2 + 70p + 550$. FOC: $-5p + 70 = 0 \Rightarrow p_C^* = 14$. $T_C^* = \frac{1}{2}(30 - 14)^2 = 128$. Substituting: $\pi_C^* = -\frac{5}{2} \cdot 196 + 70 \cdot 14 + 550 = -490 + 980 + 550 = 1,040$.

(d) Comparison: $\pi_C^* = 1,040 > \pi_B = 1,000 > \pi_A = 800$. Optimal strategy: $p_C^* = 14$ (above MC) and $T_C^* = 128$ (below $V_L(MC)$). *Trade-off*: raising p above MC distorts quantities (an efficiency loss) but allows the firm to pocket a positive margin on the units sold, especially on the H types, who keep demanding more; at the same time T falls, because the participation constraint of the L types is $T \leq V_L(p)$ and $V_L(p)$ decreases in p . The trade-off is optimally resolved at $p_C^* = 14$.

Questions and Exercises – Monopoly and Market Power

Multiple-choice questions

1. **Answer B.** $MR = 100 - 2Q$, $MC = 2Q + 20$. $100 - 2Q = 2Q + 20 \Rightarrow Q_m^* = 20$, $P_m^* = 80$. Distractor D ($Q = 40$) corresponds to an incorrect PC quantity (it would be $Q_{pc} = (100 - 20)/3$, not 40).

2. **Answer B.** At $Q_m^* = 20$: $MC(20) = 60$, $P_m^* = 80$. $L = (80 - 60)/80 = 20/80 = 1/4$. Equivalently via $|\varepsilon_D| = 80/20 = 4$, $1/|\varepsilon_D| = 1/4$.

3. **Answer C.** From Exercise 14.1: $Q_m^* = 12$, $P_m^* = 48$, $MC(12) = 36$, $Q_{pc} = 16$. $DWL = \frac{1}{2}(16 - 12)(48 - 36) = \frac{1}{2} \cdot 4 \cdot 12 = 24$.

4. **Answer B.** The allocation rule (Multi-plant Proposition) requires $MC_1(Q_1^*) = \dots = MC_n(Q_n^*) = MR(Q^*)$, where MR is computed on aggregate quantity. Option A ($MC = P$) would apply to perfect competition.

5. **Answer B.** $P = AC$: $50 - Q = 125/Q + 20 \Rightarrow Q^2 - 30Q + 125 = 0 \Rightarrow Q \in \{5, 25\}$. Sensible

solution $Q = 25$, $P = 25$, $\pi = 0$ (no subsidy). Option C ($Q = 30$ with subsidy = 125) describes the $P = MC$ regime.

6. Answer C. $X^*/V = (n-1)/n = 2/3 \approx 66.7\%$. Equivalently: $x^* = 180 \cdot 2/9 = 40$, $X^* = 3 \cdot 40 = 120$, $120/180 = 2/3$.

7. Answer B. Lerner rule for each market: $P_k - MC = P_k/|\varepsilon_k|$, hence $P_k = MC/(1 - 1/|\varepsilon_k|)$. The function P_k is decreasing in $|\varepsilon_k|$: the market with the lower elasticity pays a higher price.

8. Answer D. Under first-degree discrimination, each unit is sold at the price $P(Q)$ given by demand (not uniform), so options A, B, C are correct. Option D, which states “a uniform price equal to MC ”, instead describes the two-part tariff with homogeneous consumers.

9. Answer A. By the Proposition on the optimal tariff with homogeneous consumers: $p^* = MC = 10$, $T^* = \frac{1}{2}(30 - 10)^2 = 200$.

10. Answer B. By the Proposition on tax incidence under monopoly with linear D and weakly increasing MC : $\Delta P_d = b/(2b + d) \cdot t$. Since $d \geq 0$, $b/(2b + d) \leq 1/2$ and hence $0 < \Delta P_d \leq t/2$. The incidence on consumers never exceeds 50%.

Exercises and open questions

Exercise 1. $P(Q) = 80 - 2Q$, $C(Q) = Q^2 + 32Q + 92$.

(a) $MR = 80 - 4Q$, $MC = 2Q + 32$. FOC: $80 - 4Q = 2Q + 32 \Rightarrow Q_m^* = 8$, $P_m^* = 64$, $MC(8) = 48$. $\pi_m = 64 \cdot 8 - (64 + 256 + 92) = 512 - 412 = 100$. $L = (64 - 48)/64 = 1/4$. $|\varepsilon_D| = \frac{1}{2} \cdot 64/8 = 4$, $1/|\varepsilon_D| = 1/4 = L$ ✓.

(b) PC: $80 - 2Q = 2Q + 32 \Rightarrow Q_{pc} = 12$, $P_{pc} = 56$. $DWL = \frac{1}{2}(12 - 8)(64 - 48) = \frac{1}{2} \cdot 4 \cdot 16 = 32$.

(c) Tax $t = 12$: $MC_{eff} = 2Q + 44$. $80 - 4Q = 2Q + 44 \Rightarrow Q_m^t = 6$, $P_m^t = 68$. $\Delta P_d = 68 - 64 = 4$, incidence on consumers = $4/12 = 1/3$. Monopolist's net price: $P^t - t = 68 - 12 = 56$, a change of -8 , incidence on the monopolist = $8/12 = 2/3$. Revenue $T = 12 \cdot 6 = 72$.

(d) $\pi_m^t = 68 \cdot 6 - (36 + 192 + 92) - 72 = 408 - 320 - 72 = 16$ (a profit loss of $100 - 16 = 84$, of which 72 goes to tax revenue and 12 to the reduction in the monopolist's private welfare). Additional DWL (trapezoid between $Q_m^t = 6$ and $Q_m^* = 8$): $(P - MC)|_{Q^t} = 68 - (12 + 32) = 24$; $(P - MC)|_{Q^*} = 64 - (16 + 32) = 16$. Area: $\frac{1}{2} \cdot 2 \cdot (24 + 16) = 40$.

Exercise 2. $P(Q) = 50 - Q$, $C(Q) = 300 + 10Q$.

(a) $AC(Q) = 300/Q + 10$, $AC'(Q) = -300/Q^2 < 0$ for all $Q > 0$: AC is strictly decreasing everywhere, hence a natural monopoly.

(b) $MR = 50 - 2Q$, $MC = 10$. $50 - 2Q = 10 \Rightarrow Q_m^* = 20$, $P_m^* = 30$. $\pi_m = (30 - 10) \cdot 20 - 300 = 400 - 300 = 100$. $CS_m = \frac{1}{2} \cdot 20 \cdot (50 - 30) = 200$.

(c) $P = AC$: $50 - Q = 300/Q + 10 \Rightarrow Q^2 - 40Q + 300 = 0 \Rightarrow Q \in \{10, 30\}$. Sensible solution $Q_{AC}^* = 30$, $P_{AC}^* = 20$, $\pi_{AC} = 0$ (no subsidy). $CS_{AC} = \frac{1}{2} \cdot 30 \cdot (50 - 20) = 450$.

(d) $P = MC$: $50 - Q = 10 \Rightarrow Q_{MC}^* = 40$, $P_{MC}^* = 10$, $\pi_{MC} = (10 - 10) \cdot 40 - 300 = -300$: required subsidy = 300 (equal to FC). $CS_{MC} = \frac{1}{2} \cdot 40 \cdot (50 - 10) = 800$.

(e) Ranking: $Q_m^* = 20 < Q_{AC}^* = 30 < Q_{MC}^* = 40$; $P_m^* = 30 > P_{AC}^* = 20 > P_{MC}^* = 10$; $CS_m = 200 < CS_{AC} = 450 < CS_{MC} = 800$. Trade-off: $P = MC$ is first best ($DWL = 0$, maximum CS) but requires a subsidy of 300; $P = AC$ is second best (residual $DWL > 0$ because $P = AC > MC$) but covers costs without a subsidy; the unregulated monopoly has both the maximum DWL and a transfer of CS to the monopolist. $P = AC$ regulation is the compromise

adopted in many sectors (utilities, historical fixed-line telephony).

Exercise 3. $P(Q) = 70 - Q$, $C(Q) = \frac{1}{2}Q^2 + 10Q + 50$.

(a) $MR = 70 - 2Q$, $MC = Q + 10$. FOC: $70 - 2Q = Q + 10 \Rightarrow Q_m^* = 20$, $P_m^* = 50$, $MC(20) = 30$. $\pi_m = 50 \cdot 20 - (\frac{1}{2} \cdot 400 + 200 + 50) = 1,000 - (200 + 200 + 50) = 550$. $CS_m = \frac{1}{2} \cdot 20 \cdot (70 - 50) = 200$.

(b) $L = (50 - 30)/50 = 2/5$. $|\varepsilon_D| = 1 \cdot 50/20 = 5/2$, $1/|\varepsilon_D| = 2/5 = L \checkmark$.

(c) PC: $70 - Q = Q + 10 \Rightarrow Q_{pc} = 30$, $P_{pc} = 40$. $DWL = \frac{1}{2}(30 - 20)(50 - 30) = \frac{1}{2} \cdot 10 \cdot 20 = 100$.

(d) First-degree discrimination: $Q^I = Q_{pc} = 30$. $PS^I = \int_0^{30} [(70 - Q) - (Q + 10)] dQ = \int_0^{30} (60 - 2Q) dQ = 1800 - 900 = 900$. Profit: $\pi^I = 900 - 50 = 850$.

(e) Gain: $\pi^I - \pi_m = 850 - 550 = 300$. Decomposition: $CS_m + DWL = 200 + 100 = 300 \checkmark$. Meaning: the discriminating monopolist pockets all the surplus that under uniform-price monopoly went to consumers ($CS_m = 200$, a transfer) and in addition recovers the deadweight loss ($DWL = 100$, value previously destroyed). First-degree discrimination is efficient in the allocative sense ($DWL = 0$, $Q = Q_{pc}$) but distributively highly unfavorable to consumers, who leave the market with $CS = 0$.

Chapter 16 – Game Theory

Exercise 16.1.

(a) For firm A: if B plays X , A compares 2 (playing X) with 1 (playing Y) and prefers X ; if B plays Y , A compares 8 with 5 and again prefers X . Hence X is a **dominant strategy for A**. The game is symmetric, so X is dominant for B as well.

(b) Best-response highlighting: in column X A's maximum is 2 (row X); in column Y it is 8 (row X); in row X B's maximum is 2 (column X); in row Y it is 8 (column X). The only cell with both payoffs underlined is (X, X) : **unique NE** $(X, X) = (2, 2)$, consistent with the dominant strategies.

(c) The outcome $(2, 2)$ is Pareto-dominated by $(Y, Y) = (5, 5)$, which, however, is not an equilibrium (from there each player would deviate to X : $8 > 5$). The four payoffs satisfy $T = 8 > R = 5 > P = 2 > S = 1$: **it is a prisoner's dilemma**. Individual rationality leads to the collectively worse outcome.

Exercise 16.2.

(a) For A: if B plays P , A prefers S ($4 > 2$); if B plays S , A prefers P ($6 > 3$). A's optimal choice changes with B's: A does *not* have a dominant strategy. By symmetry, neither does B.

(b) Best-response highlighting: in column P A's maximum is 4 (row S); in column S it is 6 (row P); in row P B's maximum is 4 (column S); in row S it is 6 (column P). There are two cells with both payoffs underlined: $(P, S) = (6, 4)$ and $(S, P) = (4, 6)$. **There are two Nash equilibria in pure strategies.**

(c) It is an *anti-coordination game*: the firms are better off when they choose *different* time slots (they do not compete head-on), but each prefers to be the one in prime time. The two NE are both efficient but distributively opposite; the game does not say which of the two will occur, and the strategic problem is precisely coordinating on one of the two.

Exercise 16.3.

- (a) Constant flow $\pi = 50$ from $t = 0$ over an infinite horizon: $V = 50/(1 - \delta)$. With $\delta = 0.8$: $V = 50/0.2 = 250$. With $\delta = 0.9$: $V = 50/0.1 = 500$.
- (b) Constant flow $\pi = 50$ starting from $t = 1$ (with zero at $t = 0$): $V = 50\delta/(1 - \delta)$. With $\delta = 0.9$: $V = 50 \cdot 0.9/0.1 = 45/0.1 = 450$.
- (c) Difference: $500 - 450 = 50$, exactly the profit of period $t = 0$ omitted in part (b). The check confirms the identity $\frac{\pi}{1-\delta} = \pi + \frac{\delta\pi}{1-\delta}$.

Chapter 17 – Oligopoly Models

Exercise 17.1. Cournot, $P = 50 - Q$, $MC = 8$.

- (a) $\pi_1 = (50 - q_1 - q_2)q_1 = (42 - q_1 - q_2)q_1$. FOC: $42 - 2q_1 - q_2 = 0 \Rightarrow q_1^*(q_2) = 21 - q_2/2$. By symmetry $q_2^*(q_1) = 21 - q_1/2$.
- (b) Substituting: $q_1 = 21 - \frac{1}{2}(21 - q_1/2) = 21 - 10.5 + q_1/4 \Rightarrow \frac{3}{4}q_1 = 10.5 \Rightarrow q_1^C = 14$, and $q_2^C = 14$. $Q^C = 28$, $P^C = 50 - 28 = 22$, $\pi_1^C = \pi_2^C = (22 - 8) \cdot 14 = 196$.
- (c) Monopoly: $MR = 50 - 2Q = 8 \Rightarrow Q^M = 21$, $P^M = 29$, $\pi^M = (29 - 8) \cdot 21 = 441$. Perfect competition: $P = MC = 8$, $Q_{pc} = 42$, $\pi = 0$. The Cournot outcome ($Q^C = 28$, $P^C = 22$) is intermediate: $Q^M = 21 < Q^C = 28 < Q_{pc} = 42$ and $P^M = 29 > P^C = 22 > P_{pc} = 8 \checkmark$.

Exercise 17.2. Asymmetric Cournot, $P = 100 - Q$, $MC_1 = 10$, $MC_2 = 25$.

- (a) $\pi_1 = (90 - q_1 - q_2)q_1$, FOC $90 - 2q_1 - q_2 = 0 \Rightarrow q_1^*(q_2) = 45 - q_2/2$. $\pi_2 = (75 - q_1 - q_2)q_2$, FOC $75 - q_1 - 2q_2 = 0 \Rightarrow q_2^*(q_1) = 37.5 - q_1/2$.
- (b) Substituting BR_2 into BR_1 : $q_1 = 45 - \frac{1}{2}(37.5 - q_1/2) = 45 - 18.75 + q_1/4 \Rightarrow \frac{3}{4}q_1 = 26.25 \Rightarrow q_1^C = 35$. Hence $q_2^C = 37.5 - 17.5 = 20$. $Q^C = 55$, $P^C = 100 - 55 = 45$.
- (c) $\pi_1^C = (45 - 10) \cdot 35 = 1,225$, $\pi_2^C = (45 - 25) \cdot 20 = 400$.
- (d) Firm 1, with the lower marginal cost, produces more than its rival (35 versus 20) and earns a profit more than three times as large (1,225 versus 400). In Cournot the cost advantage pays off twice: the efficient firm wins a larger market share and, at the common equilibrium price, earns a wider margin on each unit.

Exercise 17.3. Stackelberg, $P = 44 - Q$, $MC = 8$, firm 1 is the leader.

- (a) *Stage 2 (follower)*. $\pi_2 = (36 - q_1 - q_2)q_2$, FOC $36 - q_1 - 2q_2 = 0 \Rightarrow q_2^*(q_1) = 18 - q_1/2$. *Stage 1 (leader)*. Substituting into the leader's profit: $\pi_1 = (36 - q_1 - (18 - q_1/2))q_1 = (18 - q_1/2)q_1$; FOC $18 - q_1 = 0 \Rightarrow q_1^S = 18$. Follower's reaction: $q_2^S = 18 - 9 = 9$. Hence $Q^S = 27$, $P^S = 44 - 27 = 17$.
- (b) $\pi_1^S = (17 - 8) \cdot 18 = 162$, $\pi_2^S = (17 - 8) \cdot 9 = 81$.
- (c) Cournot in the same market: $BR_i = 18 - q_j/2$, hence $q_i^C = 12$, $Q^C = 24$, $P^C = 20$, $\pi_i^C = (20 - 8) \cdot 12 = 144$. Comparison: the leader prefers Stackelberg, with $\pi_1^S = 162 > 144 = \pi_1^C$ (a gain of +18); the follower prefers Cournot, with $\pi_2^C = 144 > 81 = \pi_2^S$ (a loss of -63 when moving to Stackelberg). The leader's credible commitment shifts surplus in its favor and penalizes the follower; the lower price ($P^S = 17 < 20 = P^C$) also benefits consumers.

Exercise 17.4. Bertrand cartel, $P = 24 - Q$, $MC = 4$.

- (a) Monopoly: $MR = 24 - 2Q = 4 \Rightarrow Q^M = 10$, $P^M = 14$, $\pi^M = (14 - 4) \cdot 10 = 100$. Equal-share collusion: $\pi_i^{\text{coll}} = 100/2 = 50$.

(b) By deviating à la Bertrand, the firm sets a price just below $P^M = 14$ and captures the entire market: it sells $Q = 24 - 14 = 10$ units with a margin of almost $14 - 4$, so $\pi^{\text{dev}} = (14 - 4) \cdot 10 = 100$ (the entire monopoly profit).

(c) Sustainability condition: $V_{\text{coll}} \geq V_{\text{dev}}$, i.e. $50/(1 - \delta) \geq 100$. Multiplying by $(1 - \delta) > 0$: $50 \geq 100(1 - \delta) \Rightarrow 100\delta \geq 50 \Rightarrow \delta \geq \delta^* = 1/2$. With the general formula: $\delta^* = (\pi^{\text{dev}} - \pi^{\text{coll}})/(\pi^{\text{dev}} - \pi^{\text{punish}}) = (100 - 50)/(100 - 0) = 1/2 \checkmark$.

(d) With $\delta = 0.6 > 1/2$ collusion is sustainable ($V_{\text{coll}} = 50/0.4 = 125 > 100 = V_{\text{dev}}$). With $\delta = 0.45 < 1/2$ it is not ($V_{\text{coll}} = 50/0.55 \approx 90.9 < 100$): it pays to deviate.

Questions and Exercises – Oligopoly

Multiple-choice questions

1. **Answer B.** A strategy is strictly dominant if it is the player's best response to *every* strategy of the opponent. Option A describes a condition that is too weak (dominance in a single column); options C and D confuse dominance with other concepts.
2. **Answer B.** For A: in column L the maximum is 4 (row D), in column R it is 2 (row D): D is dominant. For B: in row U the maximum is 4 (column R), in row D it is 2 (column R): R is dominant. The NE is $(D, R) = (2, 2)$. It is a prisoner's dilemma: $(U, L) = (3, 3)$ Pareto-dominates it but is not an equilibrium.
3. **Answer B.** By definition, a prisoner's dilemma requires $T > R > P > S$: the temptation to deviate beats mutual cooperation, which beats mutual conflict, which beats being betrayed. With this ordering "deviate" is dominant and the NE is Pareto-dominated.
4. **Answer A.** With a homogeneous good, identical costs and unlimited capacity, the only Nash equilibrium is $P_1 = P_2 = MC$ with zero profits: any higher price triggers undercutting. Option B (the monopoly price) is not an equilibrium because each firm would deviate.
5. **Answer B.** The Bertrand paradox is the fact that two firms are enough to obtain the perfectly competitive outcome ($P = MC$, zero profits), in contrast with the empirical evidence on real-world duopolies.
6. **Answer B.** With $P = 40 - Q$ and $MC = 4$: $\pi_1 = (36 - q_1 - q_2)q_1$, FOC $36 - 2q_1 - q_2 = 0$, BR $q_1 = 18 - q_2/2$. By symmetry, at equilibrium $q_i^C = (40 - 4)/3 = 12$.
7. **Answer A.** From $q_i^C = 12$: $Q^C = 24$, $P^C = 40 - 24 = 16$, $\pi_i^C = (16 - 4) \cdot 12 = 144$. Option D ($P = MC$) would correspond to Bertrand, not to Cournot.
8. **Answer B.** In the Stackelberg duopoly the leader, thanks to its observable and irreversible commitment, produces more than the follower and earns a higher profit ($\pi_1^S > \pi_2^S$); moreover $Q^S > Q^C$, so the price is *lower* than under Cournot (option D is wrong).
9. **Answer B.** With n symmetric Bertrand firms, $\pi^{\text{coll}} = \pi^M/n$, $\pi^{\text{dev}} = \pi^M$, $\pi^{\text{punish}} = 0$, hence $\delta^* = (\pi^M - \pi^M/n)/\pi^M = (n - 1)/n$, increasing in n : more firms make the cartel harder to sustain.
10. **Answer B.** The leniency program rewards whoever reports first, building a second prisoner's dilemma on top of the cartel: "report before the other" becomes the dominant strategy, and cooperation among cartel members is destabilized.

Exercises and open questions

Exercise 1. Airlines, fare H/L .

(a) For A: if B plays H , A prefers L ($14 > 10$); if B plays L , A prefers L ($5 > 2$). L is a **dominant strategy** for A, and by symmetry for B. One-shot NE: $(L, L) = (5, 5)$. Payoff ordering: $T = 14 > R = 10 > P = 5 > S = 2$, so **it is a prisoner's dilemma**: the cooperative outcome $(H, H) = (10, 10)$ Pareto-dominates it but is not an equilibrium.

(b) With grim trigger, evaluating at period $t = 0$:

$$V_{\text{coll}} = 10 + 10\delta + 10\delta^2 + \dots = \frac{10}{1 - \delta},$$

$$V_{\text{dev}} = 14 + 5\delta + 5\delta^2 + \dots = 14 + \frac{5\delta}{1 - \delta}.$$

By deviating, A gains 14 today; from tomorrow on the punishment brings both back to the low fare, with a profit of 5 forever.

(c) Sustainability: $V_{\text{coll}} \geq V_{\text{dev}}$. Multiplying by $(1 - \delta) > 0$:

$$10 \geq 14(1 - \delta) + 5\delta = 14 - 9\delta \implies 9\delta \geq 4 \implies \delta \geq \delta^* = \frac{4}{9} \approx 0.444.$$

Check with the general formula: $\delta^* = (\pi^{\text{dev}} - \pi^{\text{coll}})/(\pi^{\text{dev}} - \pi^{\text{punish}}) = (14 - 10)/(14 - 5) = 4/9$ ✓.

Exercise 2. Cournot, $P = 120 - 2Q$, $MC = 24$.

(a) $\pi_1 = (120 - 2q_1 - 2q_2 - 24)q_1 = (96 - 2q_1 - 2q_2)q_1$. FOC: $96 - 4q_1 - 2q_2 = 0 \Rightarrow q_1^*(q_2) = 24 - q_2/2$. By symmetry $q_2^*(q_1) = 24 - q_1/2$.

(b) Substituting: $q_1 = 24 - \frac{1}{2}(24 - q_1/2) = 24 - 12 + q_1/4 \Rightarrow \frac{3}{4}q_1 = 12 \Rightarrow q_1^C = 16$, and $q_2^C = 16$. $Q^C = 32$, $P^C = 120 - 64 = 56$, $\pi_1^C = \pi_2^C = (56 - 24) \cdot 16 = 512$.

(c) Monopoly: $MR = 120 - 4Q = 24 \Rightarrow Q^M = 24$, $P^M = 72$, $\pi^M = (72 - 24) \cdot 24 = 1,152$. Comparison: $Q^C = 32 > Q^M = 24$ (the duopoly produces more), $P^C = 56 < P^M = 72$ (lower price), industry profit $2 \cdot 512 = 1,024 < 1,152 = \pi^M$ (competition dissipates part of the monopoly profit).

(d) With collusion each firm would obtain $\pi^M/2 = 576$. Compared with Cournot (512), the gain per firm would be $576 - 512 = 64$.

Exercise 3. Stackelberg, $P = 120 - 2Q$, $MC = 24$, firm 1 is the leader.

(a) *Stage 2 (follower)*. $\pi_2 = (96 - 2q_1 - 2q_2)q_2$, FOC $96 - 2q_1 - 4q_2 = 0 \Rightarrow q_2^*(q_1) = 24 - q_1/2$. *Stage 1 (leader)*. Substituting into the leader's profit: $\pi_1 = (96 - 2q_1 - 2(24 - q_1/2))q_1 = (48 - q_1)q_1$; FOC $48 - 2q_1 = 0 \Rightarrow q_1^S = 24$. Follower's reaction: $q_2^S = 24 - 12 = 12$. Hence $Q^S = 36$, $P^S = 120 - 72 = 48$.

(b) $\pi_1^S = (48 - 24) \cdot 24 = 576$, $\pi_2^S = (48 - 24) \cdot 12 = 288$.

(c) Compared with Cournot ($q_i^C = 16$, $\pi_i^C = 512$): the leader goes from 512 to 576 (+64) and prefers Stackelberg; the follower goes from 512 to 288 (-224) and prefers Cournot. The first-mover advantage (+64 for the leader) is smaller than the follower's loss (-224): sequential moves expand total quantity.

(d) $P^S = 48 < P^C = 56$: total quantity is larger under Stackelberg ($Q^S = 36 > 32 = Q^C$), so the price is lower and consumers **benefit** from the leader moving first.

Exercise 4. Cartel with $n = 3$, $P = 30 - Q$, $MC = 6$, Cournot punishment.

(a) Collusion: the three firms replicate the monopoly $Q^M = 12$, $\pi^M = 144$, split equally; each produces $q_i^{\text{coll}} = 4$ and earns $\pi_i^{\text{coll}} = 144/3 = 48$.

(b) Three-firm Cournot: with $P = 30 - Q$ and $MC = 6$, $q_i^C = (30 - 6)/(3 + 1) = 6$, $Q^C = 18$, $P^C = 12$, and $\pi_i^C = (12 - 6) \cdot 6 = 36$. This is the punishment profit π^{punish} .

(c) Deviation: the deviating firm maximizes its own quantity while the other two stay at the collusive quota $4 + 4 = 8$. Its profit is $\pi = (30 - q - 8 - 6)q = (16 - q)q$, with FOC $16 - 2q = 0 \Rightarrow q^{\text{dev}} = 8$. Total quantity $8 + 8 = 16$, price $P = 14$, $\pi^{\text{dev}} = (14 - 6) \cdot 8 = 64$.

(d) Sustainability condition with the general formula:

$$\delta^* = \frac{\pi^{\text{dev}} - \pi^{\text{coll}}}{\pi^{\text{dev}} - \pi^{\text{punish}}} = \frac{64 - 48}{64 - 36} = \frac{16}{28} = \frac{4}{7} \approx 0.571.$$

The value $\delta^* = 4/7$ is higher than in the two-firm case with Bertrand punishment ($\delta^* = 1/2$): a larger cartel requires more patient firms. Cournot punishment, which leaves positive profits ($\pi^{\text{punish}} = 36 > 0$), also makes the threat less severe and raises the threshold further relative to a Bertrand punishment.

Chapter 18 – Asymmetric Information

Exercise 18.1. *Adverse selection and market collapse.*

(a) With *symmetric* information, both buyer and seller know the type of the good before the transaction. For a high-quality good there is a gain from trade if and only if the value to the buyer exceeds the seller's opportunity cost, i.e. $v_H > c_H$. Similarly, $v_L > c_L$ ensures that low-quality goods are also traded. Since both conditions hold by assumption, under symmetric information the market is efficient: all types are traded at any price $P_H \in (c_H, v_H)$ for the high types and $P_L \in (c_L, v_L)$ for the low types.

(b) With asymmetric information the buyer does not observe the type before purchase and must form expectations about it. If the expected proportion of high-quality goods is p , her expected willingness to pay is the weighted average $WTP = p v_H + (1 - p) v_L$. This is the only offer a rational, risk-neutral buyer will make. A high-quality seller accepts only if $WTP \geq c_H$, i.e. if the expected average price covers her opportunity cost. When $WTP < c_H$, high-quality sellers prefer not to sell the good: the market fails to set a price that simultaneously rewards type H and matches the buyer's expectations.

(c) The spiral is self-reinforcing, as follows. If the price offered is too low to attract high-quality sellers, they withdraw: the actual proportion of high-quality goods in the market falls from p toward zero. The buyer, observing or anticipating this selection, revises her WTP downward. A lower WTP drives out further high-quality sellers, reducing p again and lowering the price further. The spiral stops only if the initial proportion p exceeds the critical threshold $p^* = (c_H - v_L)/(v_H - v_L)$: in that case the WTP is already sufficient to retain high-quality sellers and the equilibrium with both types is stable. For $p < p^*$, the process leads to the complete collapse of the high-quality segment.

(d) Market failure in the Akerlof model is caused by the *lack of information*, not by wrong prices in the traditional sense. The price mechanism, which in perfect markets coordinates demand and supply efficiently, cannot perform this function when the price of a homogeneous good must aggregate unobservable heterogeneous types. The single market price is too coarse a signal: it does not convey information about quality. Efficient trades—those that would take place under

complete information—do not occur even though there is a positive potential surplus; this is the welfare loss caused by asymmetric information.

Exercise 18.2. *Spence signaling and separation conditions.*

(a) The two incentive-compatibility conditions are:

- IC_H : $w_H - c_H > w_L$. This ensures that high-productivity types actually find it worthwhile to obtain the degree: the additional wage earned with the degree ($w_H - w_L$) exceeds the cost of education c_H .
- IC_L : $w_H - c_L < w_L$. This ensures that low-productivity types do not find it worthwhile to mimic: the additional wage ($w_H - w_L$) does not compensate for the cost c_L , which by assumption is higher.

In a separating equilibrium both must hold simultaneously: no type has an incentive to deviate from its strategy.

(b) If $c_H = c_L = c$, the two ICs become $w_H - c > w_L$ and $w_H - c < w_L$, which contradict each other: they cannot both hold. Intuitively, if the cost of the signal is identical for both types, type L finds it worthwhile to mimic under exactly the same conditions under which type H finds it worthwhile to signal. The signal is not credible: any type L who gains from mimicking will do so, and employers, anticipating this, do not update their beliefs about productivity. The equilibrium collapses into a pooling equilibrium.

(c) Spence's point is that if education does not raise productivity, its only role is to sort types. Spending on education—both private (time, tuition fees, forgone income) and public (subsidies, university infrastructure)—generates no direct social return in terms of output: it is a pure cost of asymmetric information. This weakens the argument based on human capital externalities (Becker) for public subsidies to education: part of those subsidies might be financing pure signaling rather than productive training. Empirically, distinguishing the two hypotheses requires exogenous variation in access to education separate from its training content—instruments such as proximity to a university or academic credit without learning requirements can help.

(d) A *separating equilibrium* is a strategy profile in which the two types behave differently—type H workers graduate, type L workers do not—so that observable actions reveal the type and wages correspond to true productivity. A *pooling equilibrium*, instead, is a profile in which both types take the same action (for example, both graduate or neither does), employers receive no additional information and offer a weighted average wage. Separation is informationally richer but not necessarily more efficient from a social standpoint: if education does not raise productivity, the expenditure incurred by type H workers to distinguish themselves is pure cost; in that case a pooling equilibrium without education could be preferable in terms of aggregate welfare, provided that alternative mechanisms are found to reveal the type.

Exercise 18.3. *Moral hazard and contract structure.*

(a) With a fixed-wage contract w the agent obtains the same income regardless of output and hence regardless of effort. The cost of effort is private and positive: maximizing the net payoff $w - c(e)$ implies choosing the minimum effort e_L . The fixed contract is not *incentive-compatible* when effort is unobservable. In the *first best* (effort observable and contractible), the optimal contract directly specifies $e = e_H$, which makes the pay structure irrelevant on the incentive side. Since the principal is risk-neutral and the agent is risk-averse, the optimal first-best contract fully insures the agent: a fixed wage that guarantees him his reservation utility plus compensation for the cost of effort, with zero income variability.

(b) When effort is unobservable, the first-best fixed contract no longer works: the agent deviates to e_L . To induce e_H a wage spread $w_H > w_L$ is needed (a higher wage in case of success). But the risk-averse agent attaches a cost to being exposed to this variability: he must be compensated with a higher expected wage than in the first best. The principal must pay a *risk premium* that compensates the agent for the income variability borne for incentive purposes (the agent nonetheless remains at the reservation utility level $\bar{u}$). The principal's loss relative to the first best is the real cost of asymmetric information: the principal would like to contract directly on effort, but cannot, and must instead indirectly compensate for the cost of the risk transferred to the agent.

(c) β measures the share of additional output that the agent keeps: it is the sensitivity of pay to results. $\beta = 0$ is the fixed wage (no incentive); $\beta = 1$ means that the agent bears the risk entirely (as if he were the residual owner). In practice, a high β is observed where individual effort strongly affects the result and the latter is measurable (sales agents on commission, CEOs with stock options); a low β is observed where the result depends heavily on exogenous factors (university professors, public employees, team workers whose output is hard to attribute individually).

(d) A bonus with unlimited upside and zero downside creates an asymmetric β : the banker benefits fully from the gains in case of success, but the loss in case of failure is limited (in the worst case, the bonus is zero but the fixed salary remains and a new job can be found). This is equivalent to a call option on the firm's return, which systematically encourages high-variance strategies—high risk with a potentially high expected return—at the expense of the bank/the taxpayer. The regulatory response has been: CRD IV (EU) caps the variable bonus at 100% of fixed salary (200% with shareholder approval); clawback (ex post recovery of the bonus if the risks materialize); deferral of payment over a 3–5 year horizon (aligning incentives and risks over time). The Dodd-Frank Act (US) strengthened skin-in-the-game requirements on securitizations, requiring originators to retain 5% of the risk of the loans transferred.

Exercise 18.4. Remedies: a comparison.

(a) In *signaling* the informed party moves first, taking an observable and costly action to communicate its type. In *screening* the uninformed party moves first, designing a menu of contracts that induces the other party to self-select. Concrete examples: signaling—bond issuers (firms and sovereigns) voluntarily submit to the assessment of rating agencies and publish certified financial statements to signal their creditworthiness; pharmaceutical companies publish clinical trial data to signal the effectiveness of their drugs. Screening—airlines offer fares with and without the flexibility to make changes: time-sensitive business travelers choose the flexible, expensive ticket, tourists the cheap, rigid one; banks offer fixed-rate vs variable-rate mortgages with different spreads, letting customers reveal their risk tolerance.

(b) Reputation works as a credible commitment because whoever behaves improperly loses the future stream of profits tied to that reputation. In an indefinitely repeated market, the present value of this loss exceeds the short-run gain from cheating: the reputational grim-trigger mechanism becomes sustainable. In one-shot markets, by definition there is no “future” to protect: reputation has no deterrent value. Digital platforms extend reputation to occasional transactions as well by aggregating the reviews of many counterparties onto a single verified identity: the seller cannot escape her past simply by changing name or location, because the platform keeps the complete history and makes it visible to every new potential buyer.

(c) Compulsory insurance solves adverse selection because it forces all types—healthy and sick, low-risk and high-risk—to participate in the pool. The average risk of the pool corresponds to that of the overall population; premiums can be set at actuarially fair levels with respect to this average risk, without the upward pressure that drives out the healthy. However, coverage

expands the use of services: patients, not bearing the marginal cost of treatment, demand more of it than is socially optimal (excess demand); physicians, especially if paid fee-for-service, have an incentive to prescribe unnecessary tests. Two-sided moral hazard is an unavoidable cost of universal insurance, to be managed with cost sharing (co-payments, deductibles), gatekeeping by the general practitioner, and periodic audits of prescriptions.

(d) The deductible acts through both channels. *Pre-contract (screening)*: it is a self-selection device. Low-risk individuals prefer the policy with a high deductible and a low premium, because they know that the probability of having to pay the deductible is small; high-risk individuals prefer the policy with no deductible and a high premium, because they know they have a higher probability of an accident. If correctly calibrated, the deductible separates the types and allows the company to price risk more accurately. *Post-contract (reducing moral hazard)*: the insured personally bears the first part of the loss, which maintains a residual incentive to take precautions even after the contract is signed. A very high deductible would nonetheless be inefficient: the risk-averse insured would bear an amount of income variability close to that without insurance, wiping out the gains from insurance trade—that is, the contract’s ability to transfer risk from a risk-averse agent to a risk-neutral one (the diversified company). The optimal level of the deductible balances these two effects.

Questions and Exercises – Asymmetric Information

Multiple-choice questions

1. **Answer B.** Adverse selection is caused by information hidden *before* the contract (hidden information): one of the parties knows its own characteristics—type, quality, risk—which the other does not observe at the time of the transaction. Options A and C instead describe moral hazard (hidden action after the contract). Option D makes no economic sense: actions that can be hidden are by definition those taken, not those preceding the agreement.

2. **Answer B.** The unraveling spiral is the causal chain: withdrawal of high-quality sellers $\Rightarrow$ fall in the proportion $p \Rightarrow$ reduction in the buyer’s $WTP \Rightarrow$ further withdrawals. Option A describes the opposite of what happens. Option C goes too far: buyers at least have information on the distribution of types, even if they do not observe the individual type. Option D is the reverse: it is high-quality sellers (not low-quality ones) who withdraw, because the average price does not cover their opportunity cost.

3. **Answer C.** Stiglitz and Weiss (1981) show that raising the rate r worsens the composition of the pool: safe projects (which would accept a low rate) drop out, while risky projects (with a high return in case of success) remain. The equilibrium solution is quantity rationing: the bank does not lend to certain applicants even if they are willing to pay higher rates, because accepting them would worsen the expected returns of the portfolio. Option A is the naive (and wrong) answer to the problem. Option D may reduce risk ex post, but does not solve the ex ante selection problem described by Stiglitz and Weiss.

4. **Answer B.** The fundamental difference lies in the timing of the move: in signaling (Spence) the informed party (the worker) moves first by choosing the level of education; in screening (Rothschild-Stiglitz) the uninformed party (the insurer) moves first by constructing the menu of contracts. Both mechanisms address adverse selection (hidden information before the contract), so option A is wrong. Both apply to various markets, so option C is wrong. Neither completely eliminates the asymmetry, so option D is wrong.

5. **Answer B.** The signal is credible because its cost is differential: lower for type H than for

type L . This property makes mimicking unprofitable for type L (condition IC_L), making the association “graduate = type H ” credible. Option A is the human capital view (Becker), not that of Spence signaling, which explicitly does not require education to raise productivity. Options C and D describe alternative mechanisms (direct verification, regulation) that would make signaling unnecessary.

6. Answer B. The IC_L condition requires that type L not find it worthwhile to mimic type H : the net payoff from mimicking, $w_H - c_L$, must be lower than the payoff without a degree, w_L . Option A describes the opposite of IC_L : a type L who prefers to graduate is a type who mimics, and separation breaks down. Option C (indifference) is not sufficient to guarantee separation: a strict inequality is required. Option D would describe the situation in which the signal is not credible.

7. Answer B. Insurance moral hazard is a post-contract phenomenon: once covered against the risk, the insured change their behavior by reducing precaution, because they do not fully bear the consequences of the adverse event. Options A and D instead describe adverse selection (concealment of information before the contract). Option C describes an adverse selection problem on the insurance supply side, not behavioral moral hazard.

8. Answer B. The Participation Constraint ensures that the agent prefers accepting the contract to the outside option, i.e. that his net expected utility is $\geq \bar{u}$. It is the constraint that ensures the contract is individually rational. Option A instead describes the Incentive Constraint (IC). Options C and D are not concepts directly related to the PC.

9. Answer B. With $\beta = 0$ the wage is fixed ($w = \alpha$ regardless of π): the agent is fully insured against output variability and has no results-based incentive, which leads to low effort. Option A describes $\beta = 1$ (or a high β). Option C also describes $\beta = 1$ (the agent receives all output net of the fixed payment, equivalent to buying the firm). Option D is partially correct but misleading: in the first best with observable effort the fixed wage is used, but not because $\beta = 0$ is optimal for incentives—simply because effort is contracted directly.

10. Answer B. Compulsory insurance solves adverse selection by preventing low-risk types (the healthy) from voluntarily leaving the pool: everyone must participate, the average risk of the pool coincides with that of the population, and premiums stabilize at the actuarially fair level for the overall population. Option A is wrong: the mandate does not eliminate moral hazard; on the contrary, it can amplify it (universal coverage $\Rightarrow$ less incentive for prevention). Option C is wrong: the mandate does not require the company to observe individual health status. Option D is wrong: premiums are not necessarily below the actuarially fair level; they simply reflect the average risk of the entire population instead of that of the subset of voluntary insureds.

Open questions

1. Age-based life insurance rates and adverse selection.

Age-band pricing in life insurance policies is a *partial screening* mechanism: since age is observable by the company (identity document), the contract can be conditioned on this variable without requiring self-selection. In this sense it is not screening in the strict sense (Rothschild-Stiglitz), where the type is unobservable: it is rather a direct use of an observable but incomplete proxy for the actual risk. However, within each age band, adverse selection remains: the less healthy in each cohort insure more and for larger amounts. The average band premium solves inter-cohort selection but not intra-cohort selection.

Compared with a single premium for the entire population, age-based pricing is more efficient because it brings the premium closer to the actual actuarial cost of the insured: it reduces

cross-subsidies between the young (low risk) and the old (high risk), making the product more attractive to the young, who might otherwise leave the pool. From an ethical and regulatory standpoint, the question is whether age—like sex or genetic predispositions—is a characteristic on which it is legitimate to condition the price of an insurance product. European regulation (Gender Directive, 2012) banned the use of sex in insurance pricing (including life and motor third-party liability insurance), regarding it as discriminatory; age generally remains permitted. The tension is between actuarial efficiency (premiums proportional to individual risk) and intergenerational solidarity (pooling of risk between young and old).

2. “Too big to fail” as moral hazard.

(a) *Principal-agent structure.* The *principal* is the government (and ultimately the taxpayer), which bears the cost of bailouts. The *agent* is the bank (in the person of its management and shareholders), which makes portfolio and risk-management decisions.

(b) *Unobservable action.* The action that is not directly observable is the aggregate level of risk taken on by the bank: the composition of the portfolio, the use of financial leverage, the purchase of complex derivatives such as CDS, the correlation of exposures. The regulator can observe some indicators (leverage ratio, VaR) but not the entire probability distribution of losses, especially when risks are hidden off the balance sheet or embedded in complex financial structures.

(c) *Origin of moral hazard.* The implicit guarantee (or explicit, as in the case of banks with a government-backed rating) of a bailout in the event of systemic failure reduces the bank’s cost of debt capital: creditors accept lower returns because they know they are implicitly guaranteed. This lowers the cost of funding and encourages higher leverage and greater risk-taking—the institution internalizes profits in case of success and socializes losses in case of failure.

Regulatory remedies. Basel III (2010–2019) introduced stricter capital requirements (CET1 of at least 4.5% of RWA), the Liquidity Coverage Ratio (a 30-day liquidity buffer), the leverage ratio (3% of total assets), and a capital buffer for systemically important banks (G-SIB surcharge of up to 3.5%). The BRRD (Bank Recovery and Resolution Directive, EU 2014) introduced the bail-in mechanism: in the event of failure, shareholders, holders of subordinated bonds and deposits above €100,000 are hit before any public intervention, making the downside more symmetric and reducing the implicit subsidy of the state guarantee.

3. Spence vs. Becker and policy implications.

(a) *Spence’s argument.* In the Spence model education does not raise the worker’s marginal productivity. Yet, if the cost of education is sufficiently differential across types, it constitutes a credible signal: type *H* workers graduate to distinguish themselves, type *L* workers do not graduate because the cost exceeds the wage gain. The labor market works informationally, but spending on education generates no added value on the productive side: it is a pure transfer to educational institutions, financed by students and partly by the government, with no social return in terms of aggregate productivity.

(b) *Empirical evidence.* Human capital theory (Becker, Mincer) predicts a return to education tied to the content learned. Signaling theory predicts a return tied to the *credential* regardless of content. Empirical studies that can distinguish them: (i) *sheepskin effects*—analyses showing discontinuously higher returns upon obtaining the diploma relative to years of study, consistent with signaling; (ii) natural experiments that vary access to education holding content constant; (iii) international comparisons between education systems with a different emphasis on learning vs. the credential. The literature suggests that both mechanisms are present, in proportions that vary by level of education and institutional context.

(c) *Policy implications.* From Spence’s perspective, public subsidies to university education could sustain an inefficient level of spending on signaling: lowering the cost of the signal for

everyone also lowers c_H and c_L , but as long as the differential $c_L - c_H$ remains positive the separating equilibrium persists, simply at a lower social cost. In an extreme case, lowering costs enough could erode the sorting function of the degree and create credential inflation. From Becker's perspective, subsidies correct the imperfection of the student credit market (young people cannot offer collateral to invest in human capital) and internalize the positive externalities of education (more educated workers are more innovative, healthier, more civic-minded). The practical conclusion is that the evaluation of subsidies depends crucially on the relative importance of the two channels in the specific context.

4. *Digital platforms and asymmetric information.*

(a) *Concrete mechanisms.* Airbnb: two-way review systems (the host rates the guest and vice versa), identity verification via an ID document, photographs of the accommodation with minimum quality standards, a refund guarantee in case of discrepancy between the listing and reality. These tools reduce adverse selection (the buyer knows more about the quality of the accommodation and of the host) and moral hazard (host and guest know they will be rated after the stay, which gives them an incentive to behave properly during the stay). Uber: real-time rating of driver and passenger, continuous GPS tracking, automatic cashless payment (no bargaining over the price, eliminating the risk of opportunistic changes). GPS reduces the driver's moral hazard (routes longer than necessary) and the visibility of ratings reduces adverse selection on both drivers and passengers.

(b) *Persistent limits.* Platforms *mitigate* but do not eliminate asymmetries. Fake or purchased reviews distort reputation; systematic biases in ratings (e.g. the tendency not to leave negative reviews for fear of retaliation) make the average rating poorly discriminating; documented episodes of discrimination on Airbnb (differential acceptance based on name or profile photo) reveal asymmetries not captured by the average rating; in the platform-worker economy, information on the matching and pricing algorithm is asymmetric in the platform's favor.

(c) *New asymmetries.* The aggregation of behavioral data creates a new asymmetry, reversed with respect to the original one: the platform knows the user's preferences, history, willingness to pay and usage patterns in a far more granular way than the user knows the platform. This enables personalized dynamic pricing (surge pricing, user-specific prices), personalization of the offer and potentially third-degree or first-degree price discrimination. It is an information asymmetry in favor of the intermediary, not of the producer or consumer of the underlying good.

5. *The deductible as a dual instrument.*

Pre-contract channel (screening). The deductible is a contractual parameter that differentiates types along the dimension of expected risk. A low-risk individual knows that there is a low probability that she will actually have to pay the deductible: she therefore prefers a policy with a high deductible and a low premium, because the expected cost of the deductible is limited. A high-risk individual, instead, knows she has a high probability of an accident: she prefers a policy with a low (or zero) deductible and a higher premium, because she wants to be fully covered. If the company offers both policies with parameters calibrated on the self-selection IC conditions (Rothschild-Stiglitz), the types separate spontaneously without the company having to observe the risk profile directly: this is screening in the technical sense.

Post-contract channel (reducing moral hazard). Once the policy is signed, the deductible keeps a share of the loss borne by the insured. Even with insurance coverage, the individual personally bears the first F euros of each claim: precaution remains partially incentivized, because reducing the probability of an accident reduces the probability of having to pay the deductible. Partial coverage therefore leaves a residual incentive for prudence that full coverage would eliminate entirely.

Why a very high deductible is inefficient. At the extreme, a deductible equal to the entire amount of the loss is equivalent to having no insurance: the insured bears almost all of the income variability. But the gain from insurance trade derives precisely from the transfer of risk from a risk-averse agent (the insured) to a risk-neutral one (the company, diversified over the pool). If the deductible is very high, this transfer is minimal: the company bears almost no risk, and the insured remains almost fully exposed. The gains from risk pooling—the fundamental economic reason why insurance exists—are lost. The optimal level of the deductible balances the reduction in adverse selection and moral hazard against the efficiency loss in the allocation of risk.